MCK EDUCATION CENTER · MANDALAY

Grade 12 Mathematics Textbook

Notes · Examples · Exercises — Chapters 1–11

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CH.01 Complex Numbers

Notes · Examples · Exercises

NOTES

imaginary unit i

$i^2=-1$

Conjugate of z

$\bar{z}=x-y\,i=(x,-y)$

Division of Complex Numbers

$$z\,\bar{z} = (x+yi)(x-yi)$$
$$= x^2 + y^2$$

Let $z_1 = x_1 + y_1\,i$ and $z_2 = x_2 + y_2\,i$. We will calculate $\frac{z_1}{z_2}$,

$$\frac{z_1}{z_2} = \frac{z_1}{z_2}\cdot\frac{\bar{z}_2}{\bar{z}_2}$$
$$= \frac{(x_1 + y_1\,i)}{x_2 + y_2\,i}\cdot\frac{x_2 - y_2\,i}{x_2 - y_2\,i}$$
$$= \frac{x_1 x_2 + y_1 y_2 + (y_1 x_2 - x_1 y_2)i}{x_2^2 + y_2^2}$$
$$\frac{z_1}{z_2} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \frac{(y_1 x_2 - x_1 y_2)i}{x_2^2 + y_2^2}$$

absolute value of z , |z|=r

$$r=\sqrt{x^2+y^2},$$
$$x=r\cos\theta,\quad y=r\sin\theta$$
$$\cos\theta=\frac{x}{r},$$
$$\sin\theta=\frac{y}{r}$$

Trigonometric form of z

$z=r(\cos\theta+i\sin\theta)$

Polar form of z

$z=(r,\theta)$

Product Rule

$$z_1 z_2 = r_1(\cos\theta_1 + \text{isin}\,\theta_1)\, r_2(\cos\theta_2 + \text{isin}\,\theta_2)$$
$$= r_1 r_2\,((\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2$$
$$= r_1 r_2\,(\cos(\theta_1+\theta_2) + \text{isin}\,(\theta_1+\theta_2))$$
$$z_1 z_2 = r_1 r_2\,(\cos(\theta_1+\theta_2) + \text{isin}\,(\theta_1+\theta_2))$$

Multiplicative Inverse

$$z^{-1} = \frac{1}{x^2+y^2}(x - yi)$$
$$= \frac{1}{r^2}r(\cos\theta - \text{isin}\,\theta)$$
$$= \frac{1}{r}(\cos\theta - \text{isin}\,\theta)$$
$$z^{-1} = \frac{1}{r}(\cos(-\theta) + \text{isin}\,(-\theta))$$

Division Rule

$$\frac{z_1}{z_2} = z_1\,z_2^{-1}$$
$$= r_1(\cos\theta_1 + \text{isin}\,\theta_1)\,\frac{1}{r_2}(\cos(-\theta_2) + \text{isin}\,(-\theta_2))$$
$$\frac{z_1}{z_2} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + \text{isin}\,(\theta_1 - \theta_2))$$

Powers of complex numbers

$z^n = r^n\,(\cos n\theta + \text{isin}\,n\theta)$, where n is an integer

the roots of z

$w_k=\sqrt[n]{r}(\cos\frac{\theta+2k\pi}{n}+i\sin\frac{\theta+2k\pi}{n})$, k=0,1,2, ..., n-1

Section 1.1

EXAMPLE 1 Solve $x^2+2x+5=0$

SOLUTION

$$x^2+2x+5 = 0$$
$$x^2+2x+1 = -4$$
$$(x+1)^2 = -4$$
$$x+1 = \pm\sqrt{4}\,i$$
$$x+1 = \pm 2\,i$$
$$x = -1 \pm 2\,i$$

So there are two solution x=-1+2 i and x=-1-2 i.

EXAMPLE 2 Solve $x^2+2x+3=0$ and check your answers.

SOLUTION

$$x^2+2x+3 = 0$$
$$x^2+2x+1 = -2$$
$$(x+1)^2 = -2$$
$$x+1 = \pm\sqrt{2}\,i$$
$$x = -1 \pm \sqrt{2}\,i$$

So two solutions are $x=-1+\sqrt{2}\,i$ and $x=-1-\sqrt{2}\,i$.
For $x=-1+\sqrt{2}\,i$,

$$x^2+2x+3 = (-1+\sqrt{2}\,i)^2+2(-1+\sqrt{2}\,i)+3$$
$$= 1-2\sqrt{2}\,i+2i^2-2+2\sqrt{2}\,i+3$$
$$= 1-2\sqrt{2}\,i-2-2+2\sqrt{2}\,i+3$$
$$= 0$$

For $x=-1-\sqrt{2}\,i$,

$$x^2+2x+3 = (-1-\sqrt{2}\,i)^2+2(-1-\sqrt{2}\,i)+3$$
$$= 1+2\sqrt{2}\,i+2i^2-2-2\sqrt{2}\,i+3$$

$=1+2\sqrt{2}\,i-2-2-2\sqrt{2}\,i+3$

$=0$

EXERCISE 1.1

1.(a) Solve $x^2-6x+10=0$.

SOLUTION

$x^2-6x+10=0$

$x^2-6x+9=-1$

$(x-3)^2=-1$

$x-3=\pm i$

$x=3\pm i$

So two solutions are x=3+i and x=3-i.

1.(b) Solve $-2x^2+4x-3=0$

SOLUTION

$-2x^2+4x-3=0$

$2x^2-4x+3=0$

$x^2-2x+\frac{3}{2}=0$

$x^2-2x+1=-\frac{3}{2}+1$

$(x-1)^2=-\frac{1}{2}$

$x-1=\pm\frac{1}{\sqrt{2}}\,i$

$x=1\pm\frac{\sqrt{2}}{2}\,i$

1.(c) Solve $5x^2-2x+1=0$.

SOLUTION

$5x^2-2x+1=0$

$x^2-\frac{2}{5}x+\frac{1}{5}=0$

$(x^2-\frac{2}{5}x+(\frac{1}{5})^2)=-\frac{1}{5}+(\frac{1}{5})^2$

$(x-\frac{1}{5})^2=-\frac{1}{5}+\frac{1}{25}$

$=-\frac{4}{25}$

$x-\frac{1}{5}=\pm\frac{2}{5}\,i$

$x=\frac{1}{5}\pm\frac{2}{5}\,i$

So two solutions are $x=\frac{1}{5}+\frac{2}{5}\,i$ and $x=\frac{1}{5}-\frac{2}{5}\,i$.

1.(d) Solve $3x^2+7x+5=0$

SOLUTION

$3x^2+7x+5=0$

$x^2+\frac{7}{3}x+\frac{5}{3}=0$

$x^2+\frac{7}{3}x+(\frac{7}{6})^2=-\frac{5}{3}+(\frac{7}{6})^2$

$(x+\frac{7}{6})^2=-\frac{5}{3}+\frac{49}{36}$

$(x+\frac{7}{6})^2=-\frac{11}{36}$

$x+\frac{7}{6}=\pm\frac{\sqrt{11}}{6}\,i$

$x=-\frac{7}{6}\pm\frac{\sqrt{11}}{6}\,i$

So Two solutions are $x=-\frac{7}{6}+\frac{\sqrt{11}}{6}\,i$ and $x=-\frac{7}{6}-\frac{\sqrt{11}}{6}\,i$.

2.(a) Solve $x^2-2x+4=0$ and check your answer.

SOLUTION

$x^2-2x+1=-3$

$(x-1)^2=-3$

$x-1=\pm\sqrt{3}\,i$

$x=1\pm\sqrt{3}\,i$

Two solutions are $x=1+\sqrt{3}\,i$ and $x=1-\sqrt{3}\,i$.

Verification:

For $x=1+\sqrt{3}\,i$:

$x^2-2x+4=(1+\sqrt{3}\,i)^2-2(1+\sqrt{3}\,i)+4$

$=1+2\sqrt{3}\,i+3i^2-2-2\sqrt{3}\,i+4$

$=1+2\sqrt{3}\,i-3-2-2\sqrt{3}\,i+4$

$=0$

For $x=1-\sqrt{3}\,i$:

$x^2-2x+4=(1-\sqrt{3}\,i)^2-2(1-\sqrt{3}\,i)+4$

$=1-2\sqrt{3}\,i+3i^2-2+2\sqrt{3}\,i+4$

$=1-2\sqrt{3}\,i-3-2+2\sqrt{3}\,i+4$

$=0$

2.(b) Solve $x^2-4x+5=0$ and check your answer.

SOLUTION

$x^2-4x+4=-1$

$(x-2)^2=-1$

$x-2=\pm i$

$x=2\pm i$

So, two solutions are x=2+i and x=2-i.

For x=2+i:

$x^2-4x+5=(2+i)^2-4(2+i)+5$

$=4+4i+i^2-8-4i+5$

$=4+4i-1-8-4i+5$

$=0$

For x=2-i:

$x^2-4x+5=(2-i)^2-4(2-i)+5$

$=4-4i+i^2-8+4i+5$

$=4-4i-1-8+4i+5$

= 0

3 Find the value of i^n for every positive integer n , where $i^2=-1$, $i^3=i^2\ i$, $i^4=i^2\ i^2$, etc.

SOLUTION

- $i = i$
- $i^2 = -1$
- For n=3, $i^3=i^2\ i = -i$
- For n=4, $i^4=i^2 \cdot i^2 = (-1) \times (-1) = 1$
- For n=5, $i^5=i^4 \cdot i = (1) \cdot i = i$

The value of $i^n=i^{\text{the remainder when n is divided by 4}}$

Section 1.2

EXAMPLE 3 Compute (-2,3)(1,-2)+(1,1)(0,1).

SOLUTION

Method 1

(-2,3)(1,-2)+(1,1)(0,1) = (-2-(-6), 4+3)+(0-1,1+0)
= (4,7)+(-1,1)
= (3,8)

Method 2

(-2,3)(1,-2)+(1,1)(0,1) = (-2+3i)(1-2i)+(1+i)i
= (-2+4i+3i-6i^2)+(i+i^2)
= (-2+7i-6(-1))+(i-1)
= (4+7i)+(-1+i)
= 3+8i
= (3,8)

EXERCISE 1.2

1.2 - No. 1 Compute:

(a)(2,0)(3,5)+(3,-2)(0,1)
(b)(2,-5)(-1,0)+(1,0)(5,1)
(c)(-3,-2)(-2,-3)+(-2,-3)(-3,-2)
(d)(1,0)(0,1)+(0,1)(1,0)

SOLUTION

(a)(2,0)(3,5)+(3,-2)(0,1)
= (6-0, 10+0) + (0+2, 3+0)
= (6,10) + (2,3)
= (8,13)

(b)(2,-5)(-1,0)+(1,0)(5,1)
= (-2-0, 0+5) + (5-0, 1+0)
= (-2,5) + (5,1)
= (3,6)

(c)(-3,-2)(-2,-3)+(-2,-3)(-3,-2)
= (6-6, 9+4) + (6-6, 4+9)
= (0,13) + (0,13)
= (0,26)

(d)(1,0)(0,1)+(0,1)(1,0)
= (0-0, 1+0) + (0-0, 0+1)
= (0,1) + (0,1)
= (0,2)

1.2 - No. 2 Compute:

(a)(3+2i)(3-2i)+(-5+7i)(-1-i)
(b)(-1+i)(1-i)+(2+3i)
(c)(1+i)(1-i)+(-2+i)(-2+i)
(d)(3+2i)+(7-i)(-3+3i)

SOLUTION

(a)(3+2i)(3-2i)+(-5+7i)(-1-i)
= (9-6i+6i-4i^2)+ (5+5i-7i-7i^2)
= (9-4(-1))+(5-2i-7(-1))
= 13+(5-2i+7)
= 25-2i

(b)(-1+i)(1-i)+(2+3i)
= (-1+i+i-i^2)+(2+3i)
= (-1+2i+1)+(2+3i)
= 2+5i

(c)(1+i)(1-i)+(-2+i)(-2+i)
= (1-i+i-i^2)+(4-2i-2i+i^2)
= (1+1)+(3-4i)
= 5-4i

(d)(3+2i)+(7-i)(-3+3i)
= (3+2i)+(-21+21i+3i-3i^2)
= (3+2i)+(-21+24i+3)
= (3+2i)+(-18+24i)
= -15+26i

Section 1.3

EXAMPLE 4 Calculate $\frac{2+3i}{3+i}$.

SOLUTION

$$\frac{2+3i}{3+i} = \frac{2+3i}{3+i} \cdot \frac{3-i}{3-i}$$
$$= \frac{6-2i+9i-3i^2}{3^2+1^2}$$
$$= \frac{6+3+(9-2)i}{9+1}$$
$$= \frac{9+7i}{10}$$
$$= \frac{9}{10} + \frac{7}{10}i$$

INVERSE OF A COMPLEX NUMBER Calculate $\frac{1}{z}$ for z = x+yi, z ≠ 0 and verify that z ($\frac{1}{z}$) = 1.

SOLUTION

$$\frac{1}{z} = \frac{1}{x+yi} \cdot \frac{x-yi}{x-yi}$$
$$= \frac{x-yi}{x^2+y^2}$$

$= \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i$

Verification:

$z\left(\frac{1}{z}\right) = (x+yi)\left(\frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i\right)$

$= \frac{x^2 - xyi + xyi - y^2 i^2}{x^2+y^2}$

$= \frac{x^2+y^2}{x^2+y^2} + \frac{yx-xy}{x^2+y^2}i$

$= 1 + 0i$

$= 1$

EXERCISE 1.3

1.3 - No. 1 (a-f) Let $z_1=-2+3i$, $z_2=5+2i$. Compute:

(a) $z_1^2-2z_1+1$ (b) $3z_2^2+2z_2-1$ (c) $z_1\overline{z_2}+z_2\overline{z_1}$

(d) $\frac{1}{z_1}$ (e) $\frac{1}{z_2}$ (f) $\frac{1}{z_1z_2}$

SOLUTION

(a)$z_1^2-2z_1+1 = (-2+3i)^2-2(-2+3i)+1$

$= (4-12i+9i^2)+4-6i+1$

$= (4-12i-9)+5-6i$

$= -18i$

(b)$3z_2^2+2z_2-1 = 3(5+2i)^2+2(5+2i)-1$

$= 3(25+20i+4i^2)+10+4i-1$

$= 3(21+20i)+9+4i$

$= 72+64i$

(c)$z_1\overline{z_2}+z_2\overline{z_1} = (-2+3i)(5-2i)+(5+2i)(-2-3i)$

$= (-10+4i+15i-6i^2)+(-10-15i-4i-6i^2)$

$= (-4+19i)+(-4-19i)$

$= -8$

(d) $\frac{1}{z_1}$

$= \frac{1}{-2+3i} \cdot \frac{-2-3i}{-2-3i}$

$= \frac{-2-3i}{4+9}$

$= -\frac{2}{13} - \frac{3}{13}i$

(e) $\frac{1}{z_2}$

$= \frac{1}{5+2i} \cdot \frac{5-2i}{5-2i}$

$= \frac{5-2i}{25+4}$

$= \frac{5}{29} - \frac{2}{29}i$

(f) $\frac{1}{z_1z_2} = \frac{1}{(-2+3i)(5+2i)} = \frac{1}{-10-4i+15i+6i^2} = \frac{1}{-16+11i}$

$= \frac{1}{-16+11i} \cdot \frac{-16-11i}{-16-11i}$

$= \frac{-16-11i}{256+121}$

$= -\frac{16}{377} - \frac{11}{377}i$

1.3 - No. 1 (g-l) Compute:

(g) $\frac{z_1}{z_2}$ (h) $\frac{\overline{z_1}}{\overline{z_2}}$ (i) $\frac{z_2}{z_1}$

(j) $\left(\frac{z_2}{z_1}\right)$ (k) $\frac{\overline{z_1z_2}}{\overline{z_1}}$ (l) $\frac{z_2}{\overline{z_1}} + \frac{z_1}{\overline{z_2}}$

SOLUTION

(g) $\frac{z_1}{z_2}$

$= \frac{-2+3i}{5+2i} \cdot \frac{5-2i}{5-2i}$

$= \frac{-10+4i+15i+6}{29}$

$= -\frac{4}{29} + \frac{19}{29}i$

(h) $\frac{\overline{z_1}}{\overline{z_2}}$

$= \frac{-2-3i}{5-2i} \cdot \frac{5+2i}{5+2i}$

$= \frac{-10-4i-15i+6}{29}$

$= -\frac{4}{29} - \frac{19}{29}i$

(i) $\frac{z_2}{z_1}$

$= \frac{5+2i}{-2+3i} \cdot \frac{-2-3i}{-2-3i}$

$= \frac{-10-15i-4i+6}{13}$

$= -\frac{4}{13} - \frac{19}{13}i$

(j)Since $\frac{z_2}{z_1}$

$= -\frac{4}{13} - \frac{19}{13}i$, then $\overline{\left(\frac{z_2}{z_1}\right)}$

$= -\frac{4}{13} + \frac{19}{13}i$

(k)$\overline{z_1} = -2-3i$ and $\overline{z_2} = 5-2i$.

$\overline{z_1}z_2 = (-2-3i)(5+2i)$

$= -10-4i-15i-6i^2$

$= -10-19i+6$

$= -4-19i$

$z_1\overline{z_2} = (-2+3i)(5-2i)$

$= -10+4i+15i-6i^2$

$= -10+19i+6$

$= -4+19i$

Now, substitute these into the fraction:

$$\frac{\overline{z_1z_2}}{z_1\overline{z_2}} = \frac{-4-19i}{-4+19i}$$
$$= \frac{-4-19i}{-4+19i} \cdot \frac{-4-19i}{-4-19i}$$
$$= \frac{(-4-19i)^2}{16+361}$$
$$= \frac{16+152i+361i^2}{377}$$
$$= \frac{16-361+152i}{377}$$
$$= \frac{-345+152i}{377}$$
$$= -\frac{345}{377} + \frac{152}{377}i$$

(l) $\overline{z_1} = -2-3i$ and $\overline{z_2} = 5-2i$.

$$\frac{z_2}{\overline{z_1}} = \frac{5+2i}{-2-3i} \cdot \frac{-2+3i}{-2+3i}$$
$$= \frac{-10+15i-4i+6i^2}{(-2)^2-(3i)^2}$$
$$= \frac{-10+11i-6}{4+9}$$
$$= \frac{-16+11i}{13}$$

$$\frac{z_1}{\overline{z_2}} = \frac{-2+3i}{5-2i} \cdot \frac{5+2i}{5+2i}$$
$$= \frac{-10-4i+15i+6i^2}{5^2-(2i)^2}$$
$$= \frac{-10+11i-6}{25+4}$$
$$= \frac{-16+11i}{29}$$

$$\frac{z_2}{\overline{z_1}} + \frac{z_1}{\overline{z_2}} = \frac{-16+11i}{13} + \frac{-16+11i}{29}$$
$$= \frac{29(-16+11i)+13(-16+11i)}{377}$$
$$= \frac{(-464+319i)+(-208+143i)}{377}$$
$$= \frac{(-464-208)+(319i+143i)}{377}$$
$$= \frac{-672+462i}{377}$$
$$= -\frac{672}{377} + \frac{462}{377}i$$

1.3 - No. 2 Let $z_1=3-2i$, $z_2=-1+4i$. Show that:

(a) $\overline{z_1+z_2} = \overline{z_1}+\overline{z_2}$
(b) $\overline{z_1z_2} = \overline{z_1}\,\overline{z_2}$
(c) $\overline{\left(\frac{z_2}{z_1}\right)} = \frac{\overline{z_2}}{\overline{z_1}}$

SOLUTION

Given: $z_1=3-2i$, $z_2=-1+4i$

Then: $\overline{z_1}=3+2i$, $\overline{z_2}=-1-4i$

(a) To show: $\overline{z_1+z_2} = \overline{z_1}+\overline{z_2}$

L.H.S: $z_1+z_2 = (3-2i)+(-1+4i)$
$= 2+2i$
$\overline{z_1+z_2} = 2-2i$
R.H.S: $\overline{z_1}+\overline{z_2} = (3+2i)+(-1-4i)$
$= 2-2i$
$\therefore \overline{z_1+z_2} = \overline{z_1}+\overline{z_2}$ (Proven)

(b) To show: $\overline{z_1z_2} = \overline{z_1} \cdot \overline{z_2}$

L.H.S: $z_1z_2 = (3-2i)(-1+4i)$
$= -3+12i+2i-8i^2$
$= -3+14i+8$
$= 5+14i$
$\overline{z_1z_2} = 5-14i$
R.H.S: $\overline{z_1} \cdot \overline{z_2} = (3+2i)(-1-4i)$
$= -3-12i-2i-8i^2$
$= -3-14i+8$
$= 5-14i$
$\therefore \overline{z_1z_2} = \overline{z_1} \cdot \overline{z_2}$ (Proven)

(c) To show: $\overline{\left(\frac{z_2}{z_1}\right)} = \frac{\overline{z_2}}{\overline{z_1}}$

L.H.S: $\frac{z_2}{z_1} = \frac{-1+4i}{3-2i} \cdot \frac{3+2i}{3+2i}$
$= \frac{-3-2i+12i+8i^2}{9+4}$
$= \frac{-11+10i}{13}$
$= -\frac{11}{13} + \frac{10}{13}i$
$\overline{\left(\frac{z_2}{z_1}\right)} = -\frac{11}{13} - \frac{10}{13}i$
R.H.S: $\frac{\overline{z_2}}{\overline{z_1}} = \frac{-1-4i}{3+2i} \cdot \frac{3-2i}{3-2i}$
$= \frac{-3+2i-12i+8i^2}{9+4}$
$= \frac{-11-10i}{13}$
$= -\frac{11}{13} - \frac{10}{13}i$
$\therefore \overline{\left(\frac{z_2}{z_1}\right)} = \frac{\overline{z_2}}{\overline{z_1}}$ (Proven)

Section 1.4

EXAMPLE 5 Find the trigonometric form of the following complex numbers with $-\pi < \theta \le \pi$:

(a) $z = 1 + \sqrt{3}i$
(b) $z = -1 + i$
(c) $z = -\sqrt{3} - i$
(d) $z = -1$

SOLUTION

(a) $z = 1 + \sqrt{3}i$

$= (1, \sqrt{3})$

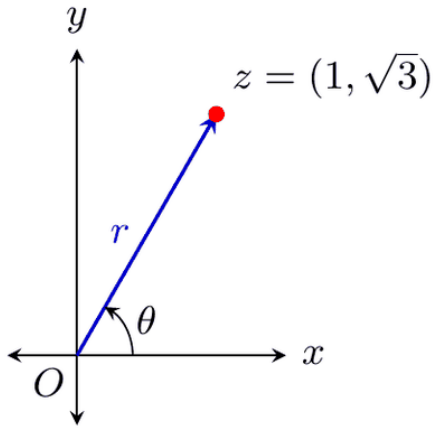
$r = \sqrt{1^2 + (\sqrt{3})^2}$

$= 2$

$\cos\theta = \frac{1}{2}, \quad \sin\theta = \frac{\sqrt{3}}{2}$

$\theta = \frac{\pi}{3}$

$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$



(b) $z = -1 + i$

$= (-1, 1)$

$r = \sqrt{(-1)^2 + 1^2}$

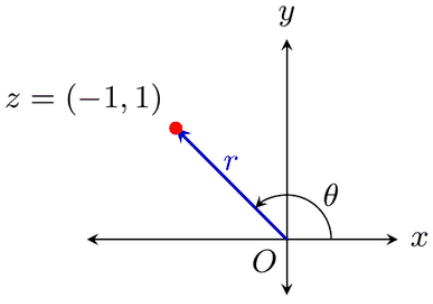
$= \sqrt{2}$

$\cos\theta = -\frac{1}{\sqrt{2}}, \quad \sin\theta = \frac{1}{\sqrt{2}}$

$\theta = \pi - \frac{\pi}{4}$

$= \frac{3\pi}{4}$

$z = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$



(c) $z = -\sqrt{3} - i$

$= (-\sqrt{3}, -1)$

$r = \sqrt{(-\sqrt{3})^2 + (-1)^2}$

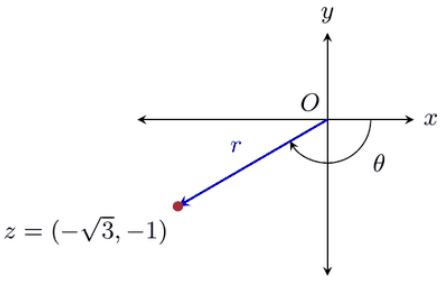
$= 2$

$\cos\theta = -\frac{\sqrt{3}}{2}, \quad \sin\theta = -\frac{1}{2}$

$\theta = -\pi + \frac{\pi}{6}$

$= -\frac{5\pi}{6}$

$z = 2\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$



(d) $z = -1$

$= (-1, 0)$

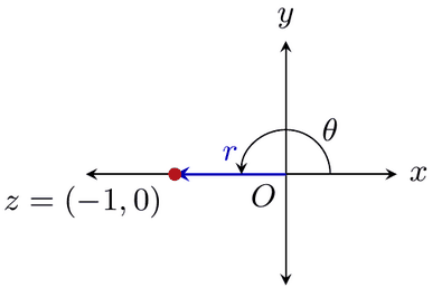
$r = \sqrt{(-1)^2 + 0^2}$

$= 1$

$\cos\theta = -1, \quad \sin\theta = 0$

$\theta = \pi$

$z = 1(\cos\pi + i\sin\pi)$



EXAMPLE 6 Given $z_1 = 1+\sqrt{3}i$ and $z_2 = -1+i$, find z_1z_2 by using trigonometric forms. Check your answer by direct multiplication.

SOLUTION

From Example 5:

$z_1 = 1+\sqrt{3}i$

$= 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$z_2 = -1+i$

$= \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

Using $z_1z_2 = r_1r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]$:

$z_1z_2 = 2\sqrt{2}\left[\cos\left(\frac{\pi}{3} + \frac{3\pi}{4}\right) + i\sin\left(\frac{\pi}{3} + \frac{3\pi}{4}\right)\right]$

$= 2\sqrt{2}\left(\cos\frac{13\pi}{12} + i\sin\frac{13\pi}{12}\right)$

Verification by direct multiplication:

$z_1z_2 = (1+\sqrt{3}i)(-1+i)$

$= -1+i-\sqrt{3}i+\sqrt{3}i^2$

$= -1-\sqrt{3}+(1-\sqrt{3})i$

EXAMPLE 7 Given $z = -\sqrt{3}-i$, using trigonometric form of z, find z^{-1}. Check your answer by showing that $zz^{-1}=1$.

SOLUTION

Trigonometric form (from Example 5c):

$$z = 2(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$$

Using $z^{-1} = \frac{1}{r}[\cos(-\theta) + i\sin(-\theta)]$:

$$z^{-1} = \frac{1}{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$$
$$= \frac{1}{2}(-\frac{\sqrt{3}}{2} + \frac{1}{2}i)$$
$$= -\frac{\sqrt{3}}{4} + \frac{1}{4}i$$

Verification:

$$zz^{-1} = (-\sqrt{3}-i)(-\frac{\sqrt{3}}{4} + \frac{1}{4}i)$$
$$= \frac{3}{4} - \frac{\sqrt{3}}{4}i + \frac{\sqrt{3}}{4}i - \frac{1}{4}i^2$$
$$= \frac{3}{4} + \frac{1}{4}$$
$$= 1$$

EXAMPLE 8 Given $z_1=1+\sqrt{3}i$ and $z_2=-1+i$, find $\frac{z_1}{z_2}$ by using trigonometric forms. Check your answer by direct calculation.

SOLUTION

Using $\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)]$:

$$\frac{z_1}{z_2} = \frac{2}{\sqrt{2}}[\cos(\frac{\pi}{3} - \frac{3\pi}{4}) + i\sin(\frac{\pi}{3} - \frac{3\pi}{4})]$$
$$= \sqrt{2}[\cos(-\frac{5\pi}{12}) + i\sin(-\frac{5\pi}{12})]$$

Direct calculation:

$$\frac{z_1}{z_2} = \frac{1+\sqrt{3}i}{-1+i} \cdot \frac{-1-i}{-1-i}$$
$$= \frac{-1-i-\sqrt{3}i-\sqrt{3}i^2}{1+1}$$
$$= \frac{-1+\sqrt{3}-(1+\sqrt{3})i}{2}$$
$$= \frac{\sqrt{3}-1}{2} - \frac{\sqrt{3}+1}{2}i$$

EXAMPLE 9 Given $z=1+\sqrt{3}i$, find (a) z^{10} and (b) z^{-10}.

SOLUTION

Trigonometric form: $z = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

(a) z^{10}:

$$z^{10} = 2^{10}(\cos\frac{10\pi}{3} + i\sin\frac{10\pi}{3})$$
$$= 1024(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3})$$
$$= 1024(-\frac{1}{2} - \frac{\sqrt{3}}{2}i)$$
$$= -512 - 512\sqrt{3}i$$

(b) z^{-10}:

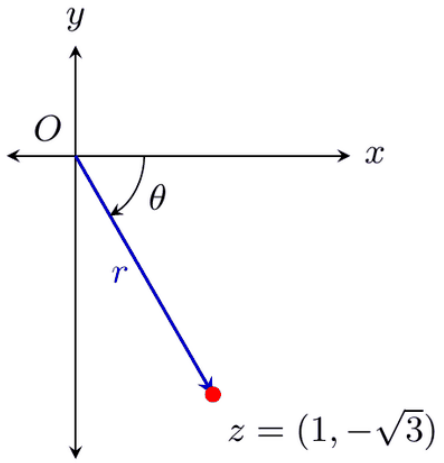
$$z^{-10} = \frac{1}{2^{10}}(\cos(-\frac{10\pi}{3}) + i\sin(-\frac{10\pi}{3}))$$
$$= \frac{1}{1024}(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$$
$$= \frac{1}{1024}(-\frac{1}{2} + \frac{\sqrt{3}}{2}i)$$
$$= -\frac{1}{2048} + \frac{\sqrt{3}}{2048}i$$

EXERCISE 1.4

1.4 - No. 1(a) Find the trigonometric form with $-\pi < \theta \le \pi$: $z = 1 - \sqrt{3}i$

SOLUTION

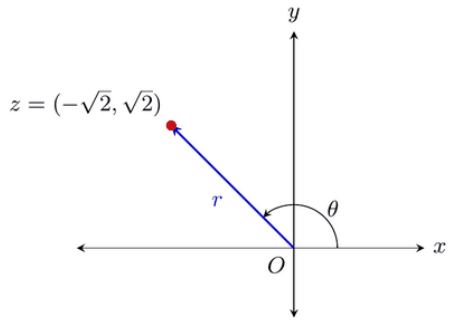
$$z = 1 - \sqrt{3}i$$
$$= (1, -\sqrt{3})$$
$$r = \sqrt{1^2 + (-\sqrt{3})^2}$$
$$= 2$$
$$\cos\theta = \frac{1}{2}, \quad \sin\theta = -\frac{\sqrt{3}}{2}$$
$$\theta = -\frac{\pi}{3}$$
$$\therefore z = 2[\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})]$$
$$= 2(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3})$$



1.4 - No. 1(b) Find the trigonometric form with $-\pi < \theta \le \pi$: $z = -\sqrt{2} + \sqrt{2}i$

SOLUTION

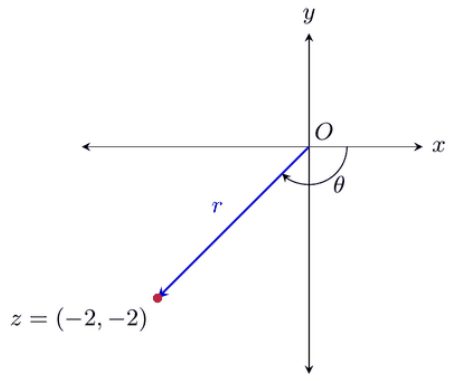
$$z = -\sqrt{2} + \sqrt{2}i$$
$$= (-\sqrt{2}, \sqrt{2})$$
$$r = \sqrt{(-\sqrt{2})^2 + (\sqrt{2})^2}$$
$$= 2$$
$$\cos\theta = -\frac{\sqrt{2}}{2}, \quad \sin\theta = \frac{\sqrt{2}}{2}$$
$$\theta = \frac{3\pi}{4}$$
$$\therefore z = 2(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$$



1.4 - No. 1(c) Find the trigonometric form with $-\pi < \theta \le \pi$: $z = -2 - 2i$

SOLUTION

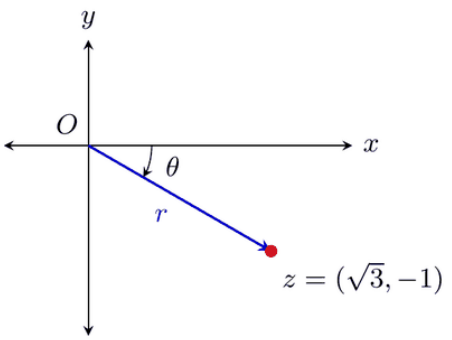
$$z = -2 - 2i$$
$$= (-2, -2)$$
$$r = \sqrt{(-2)^2 + (-2)^2}$$
$$= 2\sqrt{2}$$
$$\cos\theta = -\frac{1}{\sqrt{2}}, \quad \sin\theta = -\frac{1}{\sqrt{2}}$$
$$\theta = -\frac{3\pi}{4}$$
$$\therefore z = 2\sqrt{2}\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right]$$



1.4 - No. 1(d) Find the trigonometric form with $-\pi < \theta \le \pi$: $z = \sqrt{3} - i$

SOLUTION

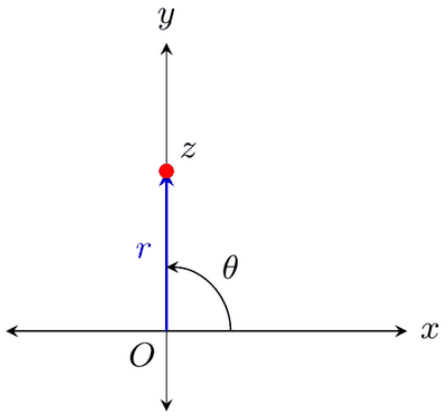
$$z = \sqrt{3} - i$$
$$= (\sqrt{3}, -1)$$
$$r = \sqrt{(\sqrt{3})^2 + (-1)^2}$$
$$= 2$$
$$\cos\theta = \frac{\sqrt{3}}{2}, \quad \sin\theta = -\frac{1}{2}$$
$$\theta = -\frac{\pi}{6}$$
$$\therefore z = 2\left[\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right]$$



1.4 - No. 1(e) Find the trigonometric form with $-\pi < \theta \le \pi$: $z = i$

SOLUTION

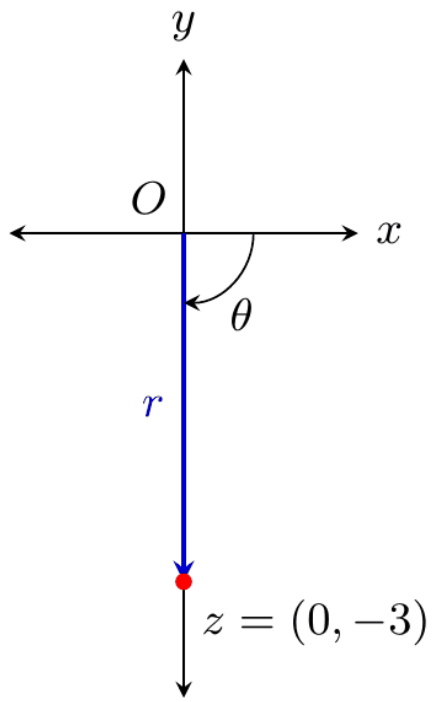
$$z = 0 + 1i$$
$$= (0, 1)$$
$$r = \sqrt{0^2 + 1^2}$$
$$= 1$$
$$\cos\theta = 0, \quad \sin\theta = 1$$
$$\theta = \frac{\pi}{2}$$
$$\therefore z = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$$



1.4 - No. 1(f) Find the trigonometric form with $-\pi < \theta \le \pi$: $z = -3i$

SOLUTION

$$z = 0 - 3i$$
$$= (0, -3)$$
$$r = \sqrt{0^2 + (-3)^2}$$
$$= 3$$
$$\cos\theta = 0, \quad \sin\theta = -1$$
$$\theta = -\frac{\pi}{2}$$
$$\therefore z = 3\left[\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right]$$



Exercise 1.4 - No. 2 Given $z_1 = 2 - 2\sqrt{3}i$, $z_2 = -1 - i$, find z_1z_2, z_1^{-1}, z_2^{-1}, $\frac{z_1}{z_2}$, $\frac{z_2}{z_1}$ using trigonometric forms.

SOLUTION

$z_1 = 2 - 2\sqrt{3}i$

$r_1 = 4, \cos\theta_1 = \frac{1}{2}, \sin\theta_1 = -\frac{\sqrt{3}}{2}$

$\theta_1 = -\frac{\pi}{3}$

$z_2 = -1 - i$

$r_2 = \sqrt{2}, \cos\theta_2 = -\frac{1}{\sqrt{2}}, \sin\theta_2 = -\frac{1}{\sqrt{2}}$

$\theta_2 = -\frac{3\pi}{4}$

(a)Product z_1z_2:

$z_1z_2 = r_1r_2[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)]$

$= 4\sqrt{2}[\cos(-\frac{13\pi}{12}) + i\sin(-\frac{13\pi}{12})]$

$= 4\sqrt{2}(\frac{-\sqrt{6}-\sqrt{2}}{4} + i\frac{\sqrt{2}-\sqrt{6}}{4})$

$= -(2\sqrt{3}+2) + (2-2\sqrt{3})i$

(b)Inverse z_1^{-1}:

$z_1^{-1} = \frac{1}{r_1}[\cos(-\theta_1) + i\sin(-\theta_1)]$

$= \frac{1}{4}(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

$= \frac{1}{4}(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$

$= \frac{1}{8} + \frac{\sqrt{3}}{8}i$

(c)Inverse z_2^{-1}:

$z_2^{-1} = \frac{1}{r_2}[\cos(-\theta_2) + i\sin(-\theta_2)]$

$= \frac{1}{\sqrt{2}}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

$= \frac{1}{\sqrt{2}}(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i)$

$= -\frac{1}{2} + \frac{1}{2}i$

(d)Division $\frac{z_1}{z_2}$:

$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)]$

$= 2\sqrt{2}(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12})$

$= 2\sqrt{2}(\frac{\sqrt{6}-\sqrt{2}}{4} + i\frac{\sqrt{6}+\sqrt{2}}{4})$

$= (\sqrt{3}-1) + (\sqrt{3}+1)i$

(e)Division $\frac{z_2}{z_1}$:

$\frac{z_2}{z_1} = \frac{r_2}{r_1}[\cos(\theta_2-\theta_1) + i\sin(\theta_2-\theta_1)]$

$= \frac{\sqrt{2}}{4}(\cos(-\frac{5\pi}{12}) + i\sin(-\frac{5\pi}{12}))$

$= \frac{\sqrt{2}}{4}(\frac{\sqrt{6}+\sqrt{2}}{4} - i\frac{\sqrt{6}-\sqrt{2}}{4})$

$= \frac{\sqrt{3}+1}{8} - \frac{\sqrt{3}-1}{8}i$

Exercise 1.4 - No. 3 Given $z = -2\sqrt{3} - 2i$, find (a) z^5 and (b) z^{-5}.

SOLUTION

$z = -2\sqrt{3} - 2i$

$= (3,-2)$

$r = 4$

$\cos\theta = -\frac{\sqrt{3}}{2}$

$\sin\theta = -\frac{1}{2}$

$\theta = -\frac{5\pi}{6}$

(a) $z^5 = r^5[\cos(5\theta) + i\sin(5\theta)]$

$= 4^5[\cos(-\frac{25\pi}{6}) + i\sin(-\frac{25\pi}{6})]$

$= 1024[\cos(-\frac{\pi}{6}) + i\sin(-\frac{\pi}{6})]$

$= 512\sqrt{3} - 512i$

(b) $z^{-5} = r^{-5}[\cos(-5\theta) + i\sin(-5\theta)]$

$= \frac{1}{4^5}[\cos(\frac{25\pi}{6}) + i\sin(\frac{25\pi}{6})]$

$= \frac{1}{1024}(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$

$= \frac{\sqrt{3}}{2048} + \frac{1}{2048}i$

Section 1.5

EXAMPLE 10 Find the cube roots of $z = -2 - 2i$.

SOLUTION

$$z = -2 - 2i$$
$$= (-2, -2)$$
$$r = \sqrt{(-2)^2 + (-2)^2}$$
$$= \sqrt{8}$$
$$= 2\sqrt{2}$$
$$\cos\theta = -\frac{2}{2\sqrt{2}}$$
$$= -\frac{1}{\sqrt{2}}, \quad \sin\theta = -\frac{2}{2\sqrt{2}}$$
$$= -\frac{1}{\sqrt{2}}$$
$$\theta = -\frac{3\pi}{4}$$
$$z = 2\sqrt{2}\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right]$$

The cube roots are: $w_k = \sqrt[3]{2}\left(\cos\frac{-\frac{3\pi}{4} + 2k\pi}{3} + i\sin\frac{-\frac{3\pi}{4} + 2k\pi}{3}\right), \quad k = 0, 1, 2$

When k=0:

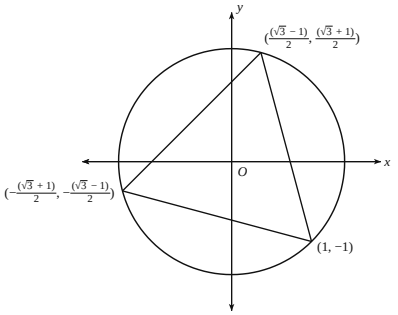
$$w_0 = \sqrt{2}\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)$$
$$= \frac{2}{2} - \frac{2}{2}i$$
$$= 1 - i$$

When k=1:

$$w_1 = \sqrt{2}\left(\frac{\sqrt{6}-\sqrt{2}}{4} + i\frac{\sqrt{6}+\sqrt{2}}{4}\right)$$
$$= \frac{\sqrt{12}-2}{4} + i\frac{\sqrt{12}+2}{4}$$
$$= \frac{\sqrt{3}-1}{2} + i\frac{\sqrt{3}+1}{2}$$

When k=2:

$$w_2 = \sqrt{2}\left(\frac{-\sqrt{6}-\sqrt{2}}{4} + i\frac{-\sqrt{6}+\sqrt{2}}{4}\right)$$
$$= \frac{-\sqrt{12}-2}{4} + i\frac{-\sqrt{12}+2}{4}$$
$$= \frac{-\sqrt{3}-1}{2} + i\frac{1-\sqrt{3}}{2}$$



EXAMPLE 11 Solve $z^6 = 1$.

SOLUTION

$$1 = (1, 0) = 1(\cos 0 + i\sin 0)$$
$$z^6 = 1(\cos 0 + i\sin 0)$$

The 6th roots are given by:

$$z = \sqrt[6]{1}\left(\cos\frac{0 + 2k\pi}{6} + i\sin\frac{0 + 2k\pi}{6}\right), \quad k = 0, 1, 2,$$

When k=0:

$$z = \cos 0 + i\sin 0$$
$$= 1$$

When k=1:

$$z = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$$
$$= \frac{1}{2} + i\frac{\sqrt{3}}{2}$$

When k=2:

$$z = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$$
$$= -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

When k=3:

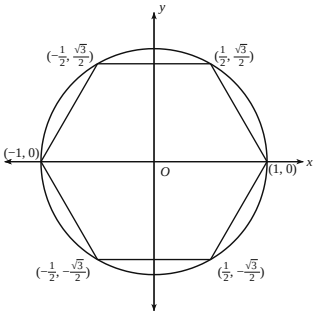
$$z = \cos\pi + i\sin\pi$$
$$= -1$$

When k=4:

$$z = \cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}$$
$$= -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

When k=5:

$$z = \cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}$$
$$= \frac{1}{2} - i\frac{\sqrt{3}}{2}$$



EXERCISE 1.5

1. (a) Find the square roots of $1+\sqrt{3}i$

SOLUTION

$$z = 1+\sqrt{3}i$$
$$= (1, \sqrt{3})$$
$$r = \sqrt{1^2+(\sqrt{3})^2}$$
$$= 2$$
$$\cos\theta = \frac{1}{2}$$
$$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta$$
$$= \frac{\pi}{3}$$
$$z = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$$

The square roots are:

$$w_k = \sqrt{2}\,(\cos\frac{\frac{\pi}{3}+2k\pi}{2} + i\sin\frac{\frac{\pi}{3}+2k\pi}{2})$$

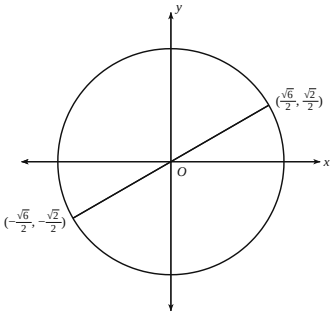
$k = 0, 1$

When k=0:

$$w_0 = \sqrt{2}\,(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$$
$$= \sqrt{2}\,(\frac{\sqrt{3}}{2} + \frac{1}{2}i)$$
$$= \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i$$

When k=1:

$$w_1 = \sqrt{2}\,(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6})$$
$$= \sqrt{2}\,(-\frac{\sqrt{3}}{2} - \frac{1}{2}i)$$
$$= -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i$$



1. (b) Find the square roots of i.

SOLUTION

$$z = i$$
$$= (0, 1)$$
$$r = 1$$
$$\cos\theta = 0$$
$$\sin\theta = 1$$
$$\theta = \frac{\pi}{2}$$
$$z = 1(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$$

The square roots are:

$$w_k = \sqrt{1}\,(\cos\frac{\frac{\pi}{2}+2k\pi}{2} + i\sin\frac{\frac{\pi}{2}+2k\pi}{2})$$

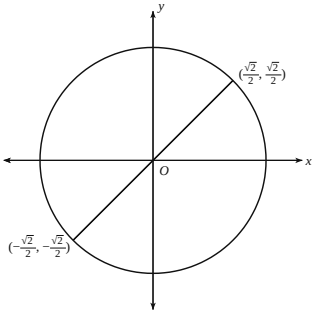
$k = 0, 1$

When k=0:

$$w_0 = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$$
$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$$

When k=1:

$$w_1 = \cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}$$
$$= -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$$



1. (c) Find the square roots of $-\sqrt{3}+i$

SOLUTION

$$z = -\sqrt{3}+i$$
$$= (-\sqrt{3}, 1)$$
$$r = 2$$
$$\cos\theta = -\frac{\sqrt{3}}{2}$$

$\sin\theta = \frac{1}{2}$

$\theta = \frac{5\pi}{6}$

$z = 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

The square roots are:

$w_k = \sqrt{2}\,(\cos\frac{\frac{5\pi}{6}+2k\pi}{2} + i\sin\frac{\frac{5\pi}{6}+2k\pi}{2})$, k=0, 1

$= \sqrt{2}\,(\cos\frac{5\pi+12k\pi}{12} + i\sin\frac{5\pi+12k\pi}{12})$

When k=0:

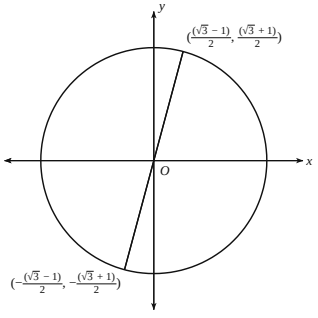
$w_0 = \sqrt{2}\,(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12})$

$= \frac{\sqrt{3}-1}{2} + \frac{\sqrt{3}+1}{2}i$

When k=1:

$w_1 = \sqrt{2}\,(\cos\frac{17\pi}{12} + i\sin\frac{17\pi}{12})$

$= \frac{1-\sqrt{3}}{2} - \frac{1+\sqrt{3}}{2}i$



1. (d) Find the square roots of $-1 - \sqrt{3}i$

SOLUTION

$z = -1 - \sqrt{3}i$

$= (-1, -\sqrt{3})$

$r = 2$

$\cos\theta = -\frac{1}{2}$

$\sin\theta = -\frac{\sqrt{3}}{2}$

$\theta = -\frac{2\pi}{3}$

$z = 2(\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3}))$

The square roots are:

$w_k = \sqrt{2}\,(\cos\frac{-\frac{2\pi}{3}+2k\pi}{2} + i\sin\frac{-\frac{2\pi}{3}+2k\pi}{2})$

k = 0, 1

When k=0:

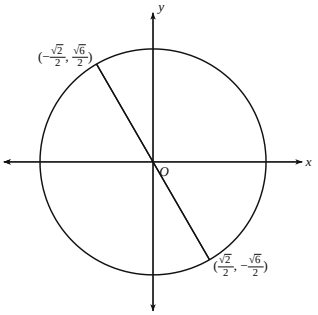
$w_0 = \sqrt{2}\,(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3}))$

$= \frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i$

When k=1:

$w_1 = \sqrt{2}\,(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

$= -\frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}i$



1. (e) Find the square roots of -i.

SOLUTION

$z = -i$

$= (0, -1)$

$r = 1$

$\cos\theta = 0$

$\sin\theta = -1$

$\theta = -\frac{\pi}{2}$

$z = 1(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}))$

The square roots are:

$w_k = \cos\frac{-\frac{\pi}{2}+2k\pi}{2} + i\sin\frac{-\frac{\pi}{2}+2k\pi}{2}$

k = 0, 1

When k=0:

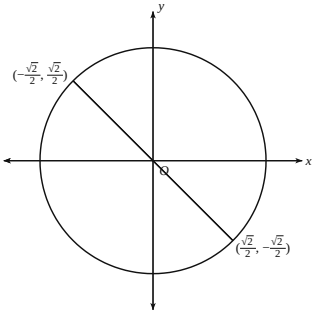
$w_0 = \cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})$

$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$

When k=1:

$w_1 = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}$

$= -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$



1. (f) Find the square roots of $\sqrt{3}$ - i.

SOLUTION

$$z = \sqrt{3} - i$$
$$= (\sqrt{3}, -1)$$
$$r = 2$$
$$\cos\theta = \frac{\sqrt{3}}{2}$$
$$\sin\theta = -\frac{1}{2} \Rightarrow \theta$$
$$= -\frac{\pi}{6}$$
$$z = 2\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$$

The square roots are:

$$w_k = \sqrt{2}\left(\cos\frac{-\frac{\pi}{6} + 2k\pi}{2} + i\sin\frac{-\frac{\pi}{6} + 2k\pi}{2}\right)$$

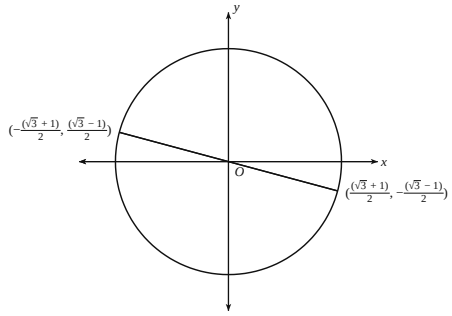
$k = 0, 1$

When k=0:

$$w_0 = \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$
$$= \frac{\sqrt{3}+1}{2} - \frac{\sqrt{3}-1}{2}i$$

When k=1:

$$w_1 = \sqrt{2}\left(\cos\frac{11\pi}{12} + i\sin\frac{11\pi}{12}\right)$$
$$= \frac{-\sqrt{3}-1}{2} + \frac{\sqrt{3}-1}{2}i$$



2.(a) (a) Find the cube roots of 1 + i.

SOLUTION

$$z = 1 + i$$
$$= (1, 1)$$
$$r = \sqrt{2}$$
$$\cos\theta = \frac{1}{\sqrt{2}}$$
$$\sin\theta = \frac{1}{\sqrt{2}} \text{ so } \theta = \frac{\pi}{4}$$
$$z = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

The cube roots are given by:

$$w_k = \sqrt[3]{\sqrt{2}}\left(\cos\frac{\frac{\pi}{4} + 2k\pi}{3} + i\sin\frac{\frac{\pi}{4} + 2k\pi}{3}\right)$$
$$= \sqrt[6]{2}\left(\cos\frac{\pi + 8k\pi}{12} + i\sin\frac{\pi + 8k\pi}{12}\right), k=0,1,2$$

When k=0:

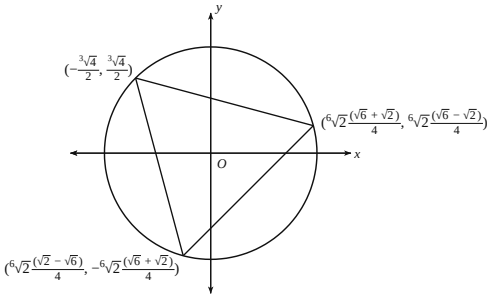
$$w_0 = \sqrt[6]{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$$
$$= \sqrt[6]{2}\left(\frac{\sqrt{6}+\sqrt{2}}{4} + i\frac{\sqrt{6}-\sqrt{2}}{4}\right)$$

When k=1:

$$w_1 = \sqrt[6]{2}\left(\cos\frac{9\pi}{12} + i\sin\frac{9\pi}{12}\right)$$
$$= \sqrt[6]{2}\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)$$

When k=2:

$$w_2 = \sqrt[6]{2}\left(\cos\frac{17\pi}{12} + i\sin\frac{17\pi}{12}\right)$$
$$= \sqrt[6]{2}\left(\frac{\sqrt{2}-\sqrt{6}}{4} - i\frac{\sqrt{6}+\sqrt{2}}{4}\right)$$



2. (b) Find the cube roots of i.

SOLUTION

$$z = i$$
$$= (0, 1)$$
$$r = 1$$
$$\cos\theta = 0$$
$$\sin\theta = 1 \text{ so } \theta = \frac{\pi}{2}$$
$$z = 1\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

The cube roots are given by:

$$w_k = \sqrt[3]{1}\left(\cos\frac{\frac{\pi}{2} + 2k\pi}{3} + i\sin\frac{\frac{\pi}{2} + 2k\pi}{3}\right)$$
$$w_k = \cos\frac{\pi + 4k\pi}{6} + i\sin\frac{\pi + 4k\pi}{6}$$

$k = 0, 1, 2$

When k=0:

$$w_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$$

$= \frac{\sqrt{3}}{2} + \frac{1}{2}i$

When k=1:

$w_1 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}$

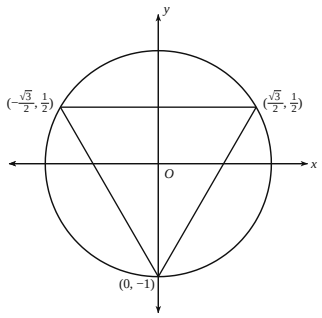
$= -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

When k=2:

$w_2 = \cos\frac{9\pi}{6} + i\sin\frac{3\pi}{2}$

$= 0 - i$

$= -i$



2. (c) Find the cube roots of -1 + i.

SOLUTION

$z = -1 + i$

$= (-1, 1)$

$r = \sqrt{2}$

$\cos\theta = -\frac{1}{\sqrt{2}}$

$\sin\theta = \frac{1}{\sqrt{2}}$ so $\theta = \frac{3\pi}{4}$

$z = \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

The cube roots are given by:

$w_k = \sqrt[3]{\sqrt{2}}\left(\cos\frac{\frac{3\pi}{4} + 2k\pi}{3} + i\sin\frac{\frac{3\pi}{4} + 2k\pi}{3}\right)$

$w_k = \sqrt[6]{2}\left(\cos\frac{3\pi + 8k\pi}{12} + i\sin\frac{3\pi + 8k\pi}{12}\right)$

$k = 0, 1, 2$

When k=0:

$w_0 = \sqrt[6]{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

$= \sqrt[6]{2}\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)$

When k=1:

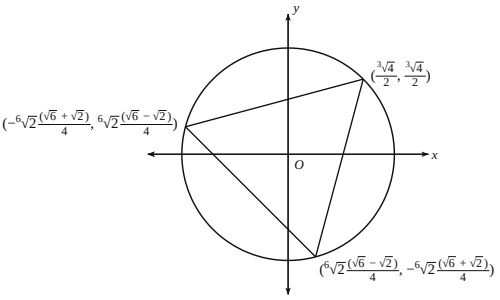
$w_1 = \sqrt[6]{2}\left(\cos\frac{11\pi}{12} + i\sin\frac{11\pi}{12}\right)$

$= \sqrt[6]{2}\left(-\frac{\sqrt{6}+\sqrt{2}}{4} + i\frac{\sqrt{6}-\sqrt{2}}{4}\right)$

When k=2:

$w_2 = \sqrt[6]{2}\left(\cos\frac{19\pi}{12} + i\sin\frac{19\pi}{12}\right)$

$= \sqrt[6]{2}\left(\frac{\sqrt{6}-\sqrt{2}}{4} - i\frac{\sqrt{6}+\sqrt{2}}{4}\right)$



2. (d) Find the cube roots of -1 - i.

SOLUTION

$z = -1 - i$

$= (-1, -1)$

$r = \sqrt{2}$

$\cos\theta = -\frac{1}{\sqrt{2}}$

$\sin\theta = -\frac{1}{\sqrt{2}}$ so $\theta = -\frac{3\pi}{4}$

$z = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$

The cube roots are given by:

$w_k = \sqrt[3]{\sqrt{2}}\left(\cos\frac{-\frac{3\pi}{4} + 2k\pi}{3} + i\sin\frac{-\frac{3\pi}{4} + 2k\pi}{3}\right)$

$w_k = \sqrt[6]{2}\left(\cos\frac{-3\pi + 8k\pi}{12} + i\sin\frac{-3\pi + 8k\pi}{12}\right)$

$k = 0, 1, 2$

When k=0:

$w_0 = \sqrt[6]{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

$= \sqrt[6]{2}\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)$

When k=1:

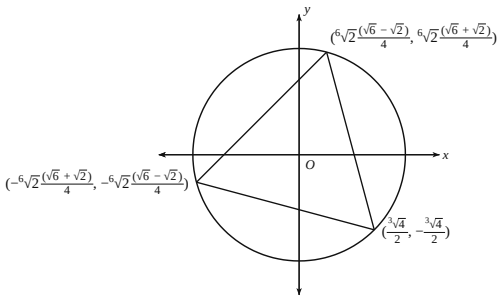
$w_1 = \sqrt[6]{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$

$= \sqrt[6]{2}\left(\frac{\sqrt{6}-\sqrt{2}}{4} + i\frac{\sqrt{6}+\sqrt{2}}{4}\right)$

When k=2:

$w_2 = \sqrt[6]{2}\left(\cos\frac{13\pi}{12} + i\sin\frac{13\pi}{12}\right)$

$= \sqrt[6]{2}\left(\frac{-\sqrt{6}-\sqrt{2}}{4} + i\frac{\sqrt{2}-\sqrt{6}}{4}\right)$



2. (e) Find the cube roots of -i.

SOLUTION

$$z = -i$$
$$= (0, -1)$$
$$r = 1$$
$$\cos\theta = 0$$
$$\sin\theta = -1 \text{ so } \theta = -\frac{\pi}{2}$$
$$z = 1\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$$

The cube roots are given by:

$$w_k = \sqrt[3]{1}\left(\cos\frac{-\frac{\pi}{2} + 2k\pi}{3} + i\sin\frac{-\frac{\pi}{2} + 2k\pi}{3}\right)$$
$$w_k = \cos\frac{-\pi + 4k\pi}{6} + i\sin\frac{-\pi + 4k\pi}{6}$$

$k = 0, 1, 2$

When k=0:

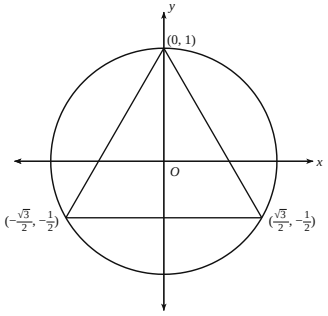
$$w_0 = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)$$
$$= \frac{\sqrt{3}}{2} - \frac{1}{2}i$$

When k=1:

$$w_1 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$$
$$= 0 + i$$
$$= i$$

When k=2:

$$w_2 = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}$$
$$= -\frac{\sqrt{3}}{2} - \frac{1}{2}i$$



2. (f) Find the cube roots of 1 - i.

SOLUTION

$$z = 1 - i$$
$$= (1, -1)$$
$$r = \sqrt{2}$$
$$\cos\theta = \frac{1}{\sqrt{2}}$$
$$\sin\theta = -\frac{1}{\sqrt{2}} \text{ so } \theta = -\frac{\pi}{4}$$
$$z = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$$

The cube roots are given by:

$$w_k = \sqrt[3]{\sqrt{2}}\left(\cos\frac{-\frac{\pi}{4} + 2k\pi}{3} + i\sin\frac{-\frac{\pi}{4} + 2k\pi}{3}\right)$$
$$w_k = \sqrt[6]{2}\left(\cos\frac{-\pi + 8k\pi}{12} + i\sin\frac{-\pi + 8k\pi}{12}\right)$$

$k = 0, 1, 2$

When k=0:

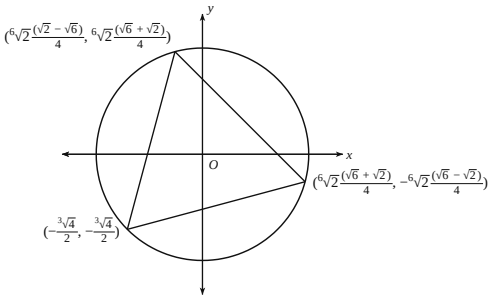
$$w_0 = \sqrt[6]{2}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$
$$= \sqrt[6]{2}\left(\frac{\sqrt{6}+\sqrt{2}}{4} - i\frac{\sqrt{6}-\sqrt{2}}{4}\right)$$

When k=1:

$$w_1 = \sqrt[6]{2}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$$
$$= \sqrt[6]{2}\left(\frac{\sqrt{2}-\sqrt{6}}{4} + i\frac{\sqrt{6}+\sqrt{2}}{4}\right)$$

When k=2:

$$w_2 = \sqrt[6]{2}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right)$$
$$= \sqrt[6]{2}\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right)$$



3. (a) Solve the equation $z^4 = -i$.

SOLUTION

$z^4 = -i$

$$-i = (0, -1) = 1\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$$
$$= 1\left(\cos\left(-\frac{\pi}{2} + 2k\pi\right) + i\sin\left(-\frac{\pi}{2} + 2k\pi\right)\right)$$
$$z = \sqrt[4]{1}\left(\cos\left(-\frac{\pi}{2} + 2k\pi\right) + i\sin\left(-\frac{\pi}{2} + 2k\pi\right)\right)^{1/}$$
$$z = 1\left(\cos\frac{-\frac{\pi}{2} + 2k\pi}{4} + i\sin\frac{-\frac{\pi}{2} + 2k\pi}{4}\right)$$

$k = 0, 1, 2, 3$

When k=0:

$$z = \cos\left(-\frac{\pi}{8}\right) + i\sin\left(-\frac{\pi}{8}\right)$$
$$= \frac{\sqrt{2+\sqrt{2}}}{2} - \frac{\sqrt{2-\sqrt{2}}}{2}\,i$$

When k=1:

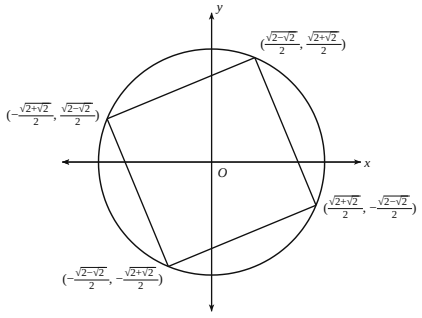
$$z = \cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}$$
$$= \frac{\sqrt{2-\sqrt{2}}}{2} + \frac{\sqrt{2+\sqrt{2}}}{2}\,i$$

When k=2:

$$z = \cos\frac{7\pi}{8} + i\sin\frac{7\pi}{8}$$
$$= -\frac{\sqrt{2+\sqrt{2}}}{2} + \frac{\sqrt{2-\sqrt{2}}}{2}\,i$$

When k=3:

$$z = \cos\frac{11\pi}{8} + i\sin\frac{11\pi}{8}$$
$$= -\frac{\sqrt{2-\sqrt{2}}}{2} - \frac{\sqrt{2+\sqrt{2}}}{2}\,i$$



3. (b) Solve the equation $z^4 = -1$.

SOLUTION

$z^4 = -1 = (-1, 0)$

$r = 1$

$\cos\theta = -1$

$\sin\theta = 0$ so $\theta = \pi$

$z^4 = 1\,[\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi)]$

$z = \sqrt[4]{1}\left(\cos\frac{\pi + 2k\pi}{4} + i\sin\frac{\pi + 2k\pi}{4}\right)$

$k = 0, 1, 2, 3$

The solutions are:

When k=0:

$$z = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$$
$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\,i$$

When k=1:

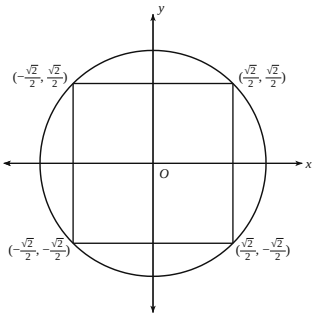
$$z = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}$$
$$= -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\,i$$

When k=2:

$$z = \cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}$$
$$= -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\,i$$

When k=3:

$$z = \cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}$$
$$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\,i$$



3. (c) Solve the equation $z^4 = -8 - 8\sqrt{3}i$.

SOLUTION

$z^4 = -8 - 8\sqrt{3}i = (-8, -8\sqrt{3})$

$r = \sqrt{(-8)^2 + (-8\sqrt{3})^2}$

$= 16$

$\cos\theta = -\frac{8}{16}$

$= -\frac{1}{2}, \quad \sin\theta = -\frac{8\sqrt{3}}{16}$

$= -\frac{\sqrt{3}}{2}$ so $\theta = -\frac{2\pi}{3}$

$z^4 = 16\left[\cos\left(-\frac{2\pi}{3} + 2k\pi\right) + i\sin\left(-\frac{2\pi}{3} + 2k\pi\right)\right]$

$z = \sqrt[4]{16}\left(\cos\frac{-\frac{2\pi}{3} + 2k\pi}{4} + i\sin\frac{-\frac{2\pi}{3} + 2k\pi}{4}\right)$

$z = 2\left(\cos\frac{-2\pi + 6k\pi}{12} + i\sin\frac{-2\pi + 6k\pi}{12}\right)$

$k = 0, 1, 2, 3$

The solutions are:

When k=0:

$$z = 2\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$$
$$= 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$$
$$= \sqrt{3} - i$$

When k=1:

$$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$
$$= 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$
$$= 1 + \sqrt{3}i$$

When k=2:

$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

$= 2\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$

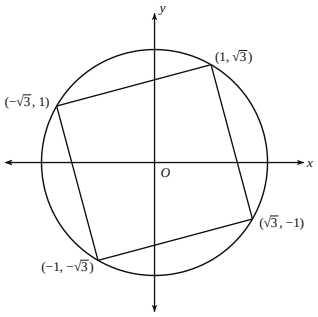
$= -\sqrt{3} + i$

When k=3:

$z = 2\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$

$= 2\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$

$= -1 - \sqrt{3}i$



3. (d) Solve the equation $z^6 = -1$.

SOLUTION

$z^6 = -1 = (-1, 0)$

$r = 1$

$\cos\theta = -1$

$\sin\theta = 0$ so $\theta = \pi$

$z^6 = 1\,[\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi)]$

$z = \sqrt[6]{1}\left(\cos\frac{\pi + 2k\pi}{6} + i\sin\frac{\pi + 2k\pi}{6}\right)$

$k = 0, 1, 2, 3, 4, 5$

The solutions are:

When k=0:

$z = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$

$= \frac{\sqrt{3}}{2} + \frac{1}{2}i$

When k=1:

$z = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$

$= 0 + i$

$= i$

When k=2:

$z = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}$

$= -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

When k=3:

$z = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}$

$= -\frac{\sqrt{3}}{2} - \frac{1}{2}i$

When k=4:

$z = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}$

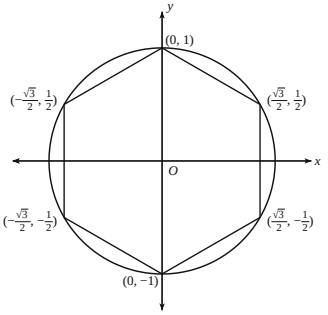
$= 0 - i$

$= -i$

When k=5:

$z = \cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}$

$= \frac{\sqrt{3}}{2} - \frac{1}{2}i$



CH.02 Mathematical Induction

Notes · Examples · Exercises

NOTES

Principle of Mathematical Induction

For each natural number n, let P(n) be a statement depending on n. Suppose that:

1. The statement is true for n=1.
2. For any natural number k, if the statement is true for n=k, then the statement is true for n=k+1.

Then the statement P(n) is true for all natural numbers n.

Two Steps of the Method

Initial step: Prove that the statement is true for n=1.
Inductive step: Assume that the statement is true for n=k, and use this assumption to prove that the statement is true for n=k+1.

Section 2.1

EXAMPLE 1 Use the mathematical induction principle to prove that $1+3+5+\ldots+(2n-1)=n^2$, for all natural numbers n.

SOLUTION

Let P(n) denote the statement $1+3+5+\ldots+(2n-1)=n^2$.

1. L.H.S. $=1$
   R.H.S. $=1^2$
   $=1$
   $\therefore$ L.H.S. $=$R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is

   $1+3+5+\ldots+(2k-1)=k^2$

   We will show that the statement is true for n=k+1, that is we have to show that

   $1+3+5+\ldots+(2k-1)+[2(k+1)-1]=(k+1)^2$.

   It is proved that:

   $1+3+5+\ldots+(2k-1)+[2(k+1)-1] = k^2+[2(k+1)-1]$
   $=k^2+2k+2-1$
   $=k^2+2k+1$
   $=(k+1)^2$.

   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

EXAMPLE 2 Use the mathematical induction principle to prove that $1+2+3+\ldots+n=\frac{n(n+1)}{2}$ for all natural numbers n.

SOLUTION

Let P(n) denote the statement $1+2+3+\ldots+n=\frac{n(n+1)}{2}$.

1. L.H.S. $=1$
   R.H.S. $=\frac{1(1+1)}{2}$
   $=\frac{1\times 2}{2}$
   $=1$
   $\therefore$ L.H.S. $=$R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is

   $1+2+3+\ldots+k=\frac{k(k+1)}{2}$

   We will show that the statement is true for n=k+1, that is we have to show that

   $1+2+3+\ldots+k+(k+1)=\frac{(k+1)(k+2)}{2}$.

   It is proved that:

   $1+2+3+\ldots+k+(k+1) = \frac{k(k+1)}{2}+(k+1)$
   $=\frac{k(k+1)+2(k+1)}{2}$
   $=\frac{(k+1)(k+2)}{2}$.

   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

EXAMPLE 3 Use the mathematical induction principle to prove that 3^n-1 is a multiple of 2 for all natural numbers n.

SOLUTION

Let P(n) denote the statement 3^n-1 is a multiple of 2.

1. For n=1, $3^1-1=2$ which is a multiple of 2.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is 3^k-1 is a multiple of 2. We will show that the statement is true for n=k+1, that is we have to show that $3^{(k+1)}-1$ is a multiple of 2. It is proved that:

   $3^{(k+1)}-1 = 3\cdot 3^k-1$
   $=(2+1)\cdot 3^k-1$
   $=2\cdot 3^k+3^k-1$.

   Since $2\cdot 3^k$ is a multiple of 2 and 3^k-1 is a multiple of 2, we can say that $2\cdot 3^k+3^k-1$ is also a multiple of 2. So, $3^{(k+1)}-1$ is also a multiple of 2. Therefore, the statement is true for n=k+1.

3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

EXAMPLE 4 Prove that $(ab)^n=a^nb^n$ for every natural number n.

SOLUTION

Let P(n) denote the statement $(ab)^n=a^nb^n$.

1. For n=1, L.H.S. $=(ab)^1=ab$, R.H.S. $=a^1b^1=ab$. L.H.S. = R.H.S.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is $(ab)^k=a^kb^k$ We will show that the statement is true for n=k+1, that is we have to show that $(ab)^{(k+1)}=a^{(k+1)}b^{(k+1)}$. It is proved that:

$$\begin{aligned}(ab)^{(k+1)} &=(ab)^k(ab)\\ &=(a^kb^k)(ab)\\ &=(a^ka)(b^kb)\\ &=a^{(k+1)}b^{(k+1)}.\end{aligned}$$

   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for every natural number n.

EXAMPLE 5 Prove that $1\cdot3+2\cdot3^2+3\cdot3^3+\ldots+n\cdot3^n=\frac{(2n-1)3^{n+1}+3}{4}$ for all natural numbers n by the use of the mathematical induction principle.

SOLUTION

Let P(n) denote the statement $1\cdot3+2\cdot3^2+3\cdot3^3+\ldots+n\cdot3^n=\frac{(2n-1)3^{n+1}+3}{4}$.

1. L.H.S. $=1\cdot3$
   $=3$

   R.H.S. $=\frac{(2(1)-1)3^{1+1}+3}{4}$
   $=\frac{(1)9+3}{4}$
   $=\frac{12}{4}$
   $=3$

   $\therefore$ L.H.S. =R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is

$$1\cdot3+2\cdot3^2+3\cdot3^3+\ldots+k\cdot3^k=\frac{(2k-1)3^{k+1}+3}{4}$$

   We will show that the statement is true for n=k+1, that is we have to show that

$$1\cdot3+2\cdot3^2+\ldots+(k+1)\cdot3^{k+1}=\frac{(2(k+1)-1)3^{(k+2)}+3}{4}$$

   It is proved that:

$$\begin{aligned}&1\cdot3+2\cdot3^2+\ldots+k\cdot3^k+(k+1)\cdot3^{k+1}\\ &=\frac{(2k-1)3^{k+1}+3}{4}+(k+1)\cdot3^{k+1}\\ &=\frac{(2k-1)3^{k+1}+3+4(k+1)3^{k+1}}{4}\\ &=\frac{(2k-1+4k+4)3^{k+1}+3}{4}\\ &=\frac{(6k+3)3^{k+1}+3}{4}\\ &=\frac{3(2k+1)3^{k+1}+3}{4}\\ &=\frac{(2k+1)3^{k+2}+3}{4}\\ &=\frac{(2(k+1)-1)3^{k+2}+3}{4}.\end{aligned}$$

   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

EXAMPLE 6 Use the mathematical induction principle to prove that a−b is a factor of a^n-b^n for all natural numbers n.

SOLUTION

Let P(n) denote the statement a−b is a factor of a^n-b^n.

1. For n=1, $a^1-b^1=a-b$. So, a−b is a factor of a^1-b^1.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is a−b is a factor of a^k-b^k. We will show that the statement is true for n=k+1, that is we have to show that a−b is a factor of $a^{k+1}-b^{k+1}$. It is proved that:

$$\begin{aligned}a^{k+1}-b^{k+1} &=a^{k+1}-a^kb+a^kb-b^{k+1}\\ &=a^k(a-b)+b(a^k-b^k).\end{aligned}$$

   Since a−b is a factor of $a^k(a-b)$ and a−b is a factor of a^k-b^k (by hypothesis), then a−b is also a factor of their sum $a^k(a-b)+b(a^k-b^k)$. So, a−b is a factor of $a^{k+1}-b^{k+1}$. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

EXAMPLE 7 Use the mathematical induction principle to prove that $4n<2^n$ for all natural numbers $n\geq5$.

SOLUTION

Let P(n) denote the statement $4n<2^n$.

1. For n=5, L.H.S. $=4(5)=20$, R.H.S. $=2^5=32$. L.H.S. < R.H.S.

The statement is true for n=5.

2. Assume that the statement is true for n=k, that is

$4k<2^k$.

We will show that the statement is true for n=k+1, that is we have to show that

$4(k+1)<2^{k+1}$.

It is proved that:

$4(k+1) = 4k+4$

$<2^k+4$ (since $4 < 4k$ for $k \geq 5$)

$<2^k+2^k$ (since $4k < 2^k$ by hypothesis)

$=2\cdot 2^k$

$=2^{k+1}$.

Hence $4(k+1)<2^{k+1}$. Therefore, the statement is true for n=k+1.

3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers $n \geq 5$.

EXERCISE 2.1

1. (a) Prove that $1+3+3^2+...+3^{n-1}=\frac{3^n-1}{2}$ for all natural numbers n by using the principle of mathematical induction.

SOLUTION

Let P(n) denote the statement $1+3+3^2+...+3^{n-1}=\frac{3^n-1}{2}$.

1. L.H.S. $=3^{1-1}$

$=3^0$

$=1$

R.H.S. $=\frac{3^1-1}{2}$

$=\frac{2}{2}$

$=1$

$\therefore$ L.H.S. =R.H.S.

The statement is true for n=1.

2. Assume that the statement is true for n=k, that is

$1+3+3^2+...+3^{k-1}=\frac{3^k-1}{2}$

We will show that the statement is true for n=k+1, that is we have to show that

$1+3+3^2+...+3^{k-1}+3^k=\frac{3^{k+1}-1}{2}$.

It is proved that:

$1+3+3^2+...+3^{k-1}+3^k = \frac{3^k-1}{2}+3^k$

$=\frac{3^k-1+2\cdot 3^k}{2}$

$=\frac{3(3^k)-1}{2}$

$=\frac{3^{k+1}-1}{2}$.

Therefore, the statement is true for n=k+1.

3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

1. (b) Prove that $1^3+2^3+3^3+...+n^3=(\frac{n(n+1)}{2})^2$ for all natural numbers n by using the principle of mathematical induction.

SOLUTION

Let P(n) denote the statement $1^3+2^3+3^3+...+n^3=(\frac{n(n+1)}{2})^2$.

1. L.H.S. $=1^3$

$=1$

R.H.S. $=(\frac{1(1+1)}{2})^2$

$=(\frac{1\times 2}{2})^2$

$=(1)^2$

$=1$

$\therefore$ L.H.S. =R.H.S.

The statement is true for n=1.

2. Assume that the statement is true for n=k, that is

$1^3+2^3+3^3+...+k^3=(\frac{k(k+1)}{2})^2$

We will show that the statement is true for n=k+1, that is we have to show that

$1^3+2^3+...+k^3+(k+1)^3=(\frac{(k+1)(k+2)}{2})^2$.

It is proved that:

$1^3+2^3+...+k^3+(k+1)^3 = \frac{k^2(k+1)^2}{4}+(k+1)^3$

$=(k+1)^2[\frac{k^2}{4}+(k+1)]$

$=(k+1)^2[\frac{k^2+4k+4}{4}]$

$=\frac{(k+1)^2(k+2)^2}{4}$

$=(\frac{(k+1)(k+2)}{2})^2$.

Therefore, the statement is true for n=k+1.

3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

1. (c) Prove that $2^3+4^3+6^3+...+(2n)^3=2n^2(n+1)^2$ for all natural numbers n by using the principle of

mathematical induction.

SOLUTION

Let P(n) denote the statement $2^3+4^3+6^3+...+(2n)^3=2n^2(n+1)^2$.

1. L.H.S. $=2^3$
   $=8$
   R.H.S. $=2(1)^2(1+1)^2$
   $=2\times1\times(2)^2$
   $=2\times4$
   $=8$
   $\therefore$ L.H.S. =R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is
   $2^3+4^3+6^3+...+(2k)^3=2k^2(k+1)^2$
   We will show that the statement is true for n=k+1, that is we have to show that
   $2^3+4^3+...+(2k)^3+(2(k+1))^3=2(k+1)^2(k+2)^2$.
   It is proved that:
   $2^3+4^3+...+(2k)^3+(2(k+1))^3 =2k^2(k+1)^2+8(k+1)^3$
   $=2(k+1)^2\,[k^2+4(k+1)]$
   $=2(k+1)^2\,[k^2+4k+4]$
   $=2(k+1)^2(k+2)^2$.
   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

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1. (d) Prove that $1\cdot2+2\cdot3+3\cdot4+...+n(n+1)=\frac{n(n+1)(n+2)}{3}$ for all natural numbers n by using the principle of mathematical induction.

SOLUTION

Let P(n) denote the statement $1\cdot2+2\cdot3+3\cdot4+...+n(n+1)=\frac{n(n+1)(n+2)}{3}$.

1. L.H.S. $=1\times2$
   $=2$
   R.H.S. $=\frac{1(1+1)(1+2)}{3}$
   $=\frac{1\times2\times3}{3}$
   $=\frac{6}{3}$
   $=2$
   $\therefore$ L.H.S. =R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is
   $1\cdot2+2\cdot3+...+k(k+1)=\frac{k(k+1)(k+2)}{3}$
   We will show that the statement is true for n=k+1, that is we have to show that
   $1\cdot2+...+k(k+1)+(k+1)(k+2)=\frac{(k+1)(k+2)(k+3)}{3}$.
   It is proved that:
   $1\cdot2+...+k(k+1)+(k+1)(k+2) =\frac{k(k+1)(k+2)}{3}+(k+1)(k+2)$
   $=(k+1)(k+2)\left[\frac{k}{3}+1\right]$
   $=\frac{(k+1)(k+2)(k+3)}{3}$.
   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

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1. (e) Prove that $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+...+\frac{1}{2^n}=1-\frac{1}{2^n}$ for all natural numbers n by using the principle of mathematical induction.

SOLUTION

Let P(n) denote the statement $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+...+\frac{1}{2^n}=1-\frac{1}{2^n}$.

1. L.H.S. $=\frac{1}{2}$
   R.H.S. $=1-\frac{1}{2^1}$
   $=1-\frac{1}{2}$
   $=\frac{1}{2}$
   $\therefore$ L.H.S. =R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is
   $\frac{1}{2}+\frac{1}{4}+...+\frac{1}{2^k}=1-\frac{1}{2^k}$
   We will show that the statement is true for n=k+1, that is we have to show that
   $\frac{1}{2}+...+\frac{1}{2^k}+\frac{1}{2^{k+1}}=1-\frac{1}{2^{k+1}}$.
   It is proved that:
   $\frac{1}{2}+...+\frac{1}{2^k}+\frac{1}{2^{k+1}} =(1-\frac{1}{2^k})+\frac{1}{2^{k+1}}$
   $=1-\frac{2}{2^{k+1}}+\frac{1}{2^{k+1}}$
   $=1-\frac{2-1}{2^{k+1}}$

$=1-\frac{1}{2^{k+1}}$.

Therefore, the statement is true for n=k+1.

3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

1. (f) Prove that $1+2+2^2+\ldots+2^{n-1}=2^n-1$ for all natural numbers n by using the principle of mathematical induction.

SOLUTION

Let P(n) denote the statement $1+2+2^2+\ldots+2^{n-1}=2^n-1$.

1. L.H.S. $=2^{1-1}$
   $=2^0$
   $=1$
   R.H.S. $=2^1-1$
   $=2-1$
   $=1$
   ∴ L.H.S. =R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is
   $1+2+2^2+\ldots+2^{k-1}=2^k-1$
   We will show that the statement is true for n=k+1, that is we have to show that
   $1+2+\ldots+2^{k-1}+2^k=2^{k+1}-1$.
   It is proved that:
   $1+2+\ldots+2^{k-1}+2^k=(2^k-1)+2^k$
   $=2(2^k)-1$
   $=2^{k+1}-1$.
   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

1. (g) Prove that $\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\ldots+\frac{1}{n(n+1)}=\frac{n}{n+1}$ for all natural numbers n by using the principle of mathematical induction.

SOLUTION

Let P(n) denote the statement $\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\ldots+\frac{1}{n(n+1)}=\frac{n}{n+1}$.

1. L.H.S. $=\frac{1}{1\times2}$
   $=\frac{1}{2}$
   R.H.S. $=\frac{1}{1+1}$
   $=\frac{1}{2}$
   ∴ L.H.S. =R.H.S.

   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is
   $\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\ldots+\frac{1}{k(k+1)}=\frac{k}{k+1}$
   We will show that the statement is true for n=k+1, that is we have to show that
   $\frac{1}{1\cdot2}+\ldots+\frac{1}{(k+1)(k+2)}=\frac{k+1}{k+2}$.
   It is proved that:
   $\frac{1}{1\cdot2}+\ldots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}=\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}$
   $=\frac{k(k+2)+1}{(k+1)(k+2)}$
   $=\frac{k^2+2k+1}{(k+1)(k+2)}$
   $=\frac{(k+1)^2}{(k+1)(k+2)}$
   $=\frac{k+1}{k+2}$.
   Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

2. Prove that 3 is a factor of 4^n-1 for all natural numbers n by using the mathematical induction.

SOLUTION

Let P(n) denote the statement 3 is a factor of 4^n-1.

1. For n=1, 3 is a factor of $4^1-1=3$.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is 3 is a factor of 4^k-1. We will show that the statement is true for n=k+1, that is we have to show that 3 is a factor of $4^{k+1}-1$. It is proved that:
   $4^{k+1}-1=4\cdot4^k-1$
   $=(3+1)\cdot4^k-1$
   $=3\cdot4^k+4^k-1$.
   Since 3 is a factor of $3\cdot4^k$ and 3 is a factor of 4^k-1, we can say that 3 is a factor of $3\cdot4^k+4^k-1$. So, 3 is a factor of $4^{k+1}-1$. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

3. Prove that n^3-n+3 is divisible by 3 for all natural numbers n by using the mathematical induction.

SOLUTION

Let P(n) denote the statement n^3-n+3 is divisible by 3.

1. For n=1, $1^3-1+3=3$ which is divisible by 3.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is k^3-k+3 is a multiple of 3. We will show that the statement is true for n=k+1, that is we have to show that $(k+1)^3-(k+1)+3$ is a multiple of 3. It is proved that:

$$\begin{aligned}(k+1)^3-(k+1)+3 &= k^3+3k^2+3k+1-k-1+3\\ &=(k^3-k+3)+3k^2+3k\\ &=\mathbf{(k^3-k+3)+3(k^2+k).}\end{aligned}$$

   Since k^3-k+3 is divisible by 3 and $3(k^2+k)$ is divisible by 3, we can say that $(k^3-k+3)+3(k^2+k)$ is also divisible by 3. So, $(k+1)^3-(k+1)+3$ is also divisible by 3. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

4. Prove that $3^{2n}-1$ is divisible by 8 for all natural numbers n by using the mathematical induction.

SOLUTION

Let P(n) denote the statement $3^{2n}-1$ is divisible by 8.

1. For n=1, $3^{2(1)}-1=8$ which is divisible by 8.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is $3^{2k}-1$ is divisible by 8. We will show that the statement is true for n=k+1, that is we have to show that $3^{2(k+1)}-1$ is divisible by 8. It is proved that:

$$\begin{aligned}3^{2(k+1)}-1 &= 3^{2k+2}-1\\ &=3^2\cdot 3^{2k}-1\\ &=9\cdot 3^{2k}-1\\ &=(8+1)\cdot 3^{2k}-1\\ &=\mathbf{8\cdot 3^{2k}+3^{2k}-1.}\end{aligned}$$

   Since $8\cdot 3^{2k}$ is a multiple of 8 and $3^{2k}-1$ is a multiple of 8, we can say that $8\cdot 3^{2k}+3^{2k}-1$ is also a multiple of 8. So, $3^{2(k+1)}-1$ is also a multiple of 8. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

5. Prove that $(n+1)^2<2n^2$ for all natural numbers $n\geq 3$ by using the mathematical induction.

SOLUTION

Let P(n) denote the statement $(n+1)^2<2n^2$.

1. For n=3, L.H.S. $=(3+1)^2=16$, R.H.S. $=2(3^2)=18$. L.H.S. < R.H.S.
   The statement is true for n=3.
2. Assume that the statement is true for n=k, that is $(k+1)^2<2k^2$. We will show that the statement is true for n=k+1, that is we have to show that $((k+1)+1)^2<2(k+1)^2$, which is $(k+2)^2<2(k+1)^2$. It is proved that:

$$\begin{aligned}(k+2)^2 &= k^2+4k+4\\ &=k^2+2k+1+2k+3\\ &=(k+1)^2+2k+3\\ &<2k^2+2k+3 \text{ (since } (k+1)^2<2k^2)\\ &<2k^2+4k+2 \text{ (since } 2k+3<4k+2 \text{ for } k\geq 3)\\ &=2(k^2+2k+1)\\ &=\mathbf{2(k+1)^2.}\end{aligned}$$

   Hence $(k+2)^2<2(k+1)^2$. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers $n\geq 3$.

6. Prove that $(2n+7)<(n+3)^2$ for all natural numbers n by using the mathematical induction.

SOLUTION

Let P(n) denote the statement $(2n+7)<(n+3)^2$.

1. For n=1, L.H.S. $=2(1)+7=9$, R.H.S. $=(1+3)^2=16$. L.H.S. < R.H.S.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is $(2k+7)<(k+3)^2$. We will show that the statement is true for n=k+1, that is we have to show that $(2(k+1)+7)<((k+1)+3)^2$, which is $(2k+9)<(k+4)^2$. It is proved that:

$$\begin{aligned}2k+9 &= (2k+7)+2\\ &<(k+3)^2+2\\ &=k^2+6k+9+2\\ &=k^2+6k+11\\ &<k^2+8k+16 \text{ (since } 6k+11<8k+16 \text{ for all } n\in\mathbb{N})\\ &=\mathbf{(k+4)^2.}\end{aligned}$$

   Hence $(2k+9)<(k+4)^2$. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

7. Prove that $x^{2n}-y^{2n}$ is divisible by x+y for all natural numbers n by using the mathematical induction.

SOLUTION

Let P(n) denote the statement $x^{2n}-y^{2n}$ is divisible by x+y.

1. For n=1, $x^{2(1)}-y^{2(1)}=x^2-y^2=(x-y)(x+y)$, which is divisible by x+y.
   The statement is true for n=1.
2. Assume that the statement is true for n=k, that is $x^{2k}-y^{2k}$ is divisible by x+y. We will show that the statement is true for n=k+1, that is we have to show that $x^{2(k+1)}-y^{2(k+1)}$ is divisible by x+y. It is proved that:

$$\begin{aligned} x^{2(k+1)}-y^{2(k+1)} &= x^{2k+2}-y^{2k+2} \\ &= x^{2k}\cdot x^2-y^{2k}\cdot y^2 \\ &= x^{2k}\cdot x^2-x^2\cdot y^{2k}+x^2\cdot y^{2k}-y^{2k}\cdot y^2 \\ &= x^2(x^{2k}-y^{2k})+y^{2k}(x^2-y^2) \\ &= \mathbf{x^2(x^{2k}-y^{2k})+y^{2k}(x-y)(x+y).} \end{aligned}$$

   Since $x^{2k}-y^{2k}$ is divisible by x+y (by hypothesis) and (x−y)(x+y) is divisible by x+y, we can say that $x^2(x^{2k}-y^{2k})+y^{2k}(x-y)(x+y)$ is also divisible by x+y. So, $x^{2(k+1)}-y^{2(k+1)}$ is divisible by x+y. Therefore, the statement is true for n=k+1.
3. Hence, by principle of mathematical induction, the statement P(n) is true for all natural numbers n.

CH.03 Analytical Solid Geometry

Notes · Examples · Exercises

NOTES

Coordinates of a point in space

A point P in space is determined by an ordered triple (x, y, z) of real numbers, measured along three mutually perpendicular axes (the x-axis, y-axis and z-axis) through the origin O.

Coordinate planes

The xy-plane (equation z=0), yz-plane (equation x=0) and zx-plane (equation y=0) are the three planes formed by pairs of axes. A plane parallel to the xy-plane has equation z=c; similarly x=a is parallel to the yz-plane and y=b is parallel to the zx-plane.

Directed values of a line segment

For a line segment PQ with $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$, the directed values ⟨l, m, n⟩ of PQ are $\langle x_2-x_1,\ y_2-y_1,\ z_2-z_1\rangle$. k is called a parameter, and (kl, km, kn) are also directed values of the same line for any real number k.

Skew lines

In space, a pair of lines that are neither parallel nor intersecting are called skew lines.

Plane through three points

A plane is determined by three points that are not on the same line. If A, B, C are such points with $\langle AB\rangle=\langle l_1,m_1,n_1\rangle$ and $\langle AC\rangle=\langle l_2,m_2,n_2\rangle$, the plane has Cartesian equation ax+by+cz=d.

Sphere

A sphere is the set of all points (x, y, z) whose distance from a fixed center (x_1, y_1, z_1) equals a fixed radius r.

Distance between two points

$$PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

Directed values of PQ

$\langle PQ\rangle=\langle l, m, n\rangle=\langle x_2-x_1,\ y_2-y_1,\ z_2-z_1\rangle$

Length of segment PQ

$$PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}=\sqrt{l^2+m^2+n^2}$$

Coordinates of point R on line PQ

$(x, y, z)=(x_1+kl,\ y_1+km,\ z_1+kn)$, for a real parameter k

Equation of a line (symmetric form)

$$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$$

Parallel lines

Two lines are parallel if and only if their directed values are multiples of each other by some real number.

cos∠PAQ

$$\cos\angle PAQ=\frac{l_1l_2+m_1m_2+n_1n_2}{\sqrt{l_1^2+m_1^2+n_1^2}\sqrt{l_2^2+m_2^2+n_2^2}}$$

Perpendicular lines

Two lines are perpendicular if and only if they intersect and $l_1l_2+m_1m_2+n_1n_2=0$ for any directed values $\langle l_1, m_1, n_1\rangle$ and $\langle l_2, m_2, n_2\rangle$ of the lines.

Cartesian form of a plane

ax+by+cz=d

where $a=m_1n_2-m_2n_1$, $b=n_1l_2-n_2l_1$, $c=l_1m_2-l_2m_1$, for two non-parallel directed values $\langle l_1,m_1,n_1\rangle$, $\langle l_2,m_2,n_2\rangle$ in the plane, and $d=ax_1+by_1+cz_1$ for a point (x_1,y_1,z_1) on the plane.

Line perpendicular to a plane

Any line with directed values ⟨ka, kb, kc⟩, for some parameter k, is perpendicular to the plane ax+by+cz=d.

Radius of a sphere

$$\sqrt{(x-x_1)^2+(y-y_1)^2+(z-z_1)^2}=r$$

Equation of a sphere

$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$$

with center (x_1, y_1, z_1) and radius r.

Section 3.1

EXAMPLE 1 Find the equation of the line through the point (−3, 5, 7) and perpendicular to

a. xy-plane
b. yz-plane
c. zx-plane.

Find the point of intersection of the line and plane.

SOLUTION

a. The equation of the line through the point (−3, 5, 7) and perpendicular to xy-plane is

   x=−3, y=5 or (−3, 5, z)

   The point of intersection of the line and xy-plane is (−3, 5, 0).

b. The equation of the line through the point (−3, 5, 7) and perpendicular to yz-plane is

y=5, z=7 or (x, 5, 7)

The point of intersection of the line and yz-plane is (0, 5, 7).

c. The equation of the line through the point (−3, 5, 7) and perpendicular to zx-plane is

x=−3,

z=7 or (−3, y, 7)

The point of intersection of the line and zx-plane is (−3, 0, 7).

EXERCISE 3.1

1. Find the equation of the plane containing the point (1, −2, 3) and parallel to

a. xy-plane
b. yz-plane
c. zx-plane.

SOLUTION

a. The equation of the plane through the point (1, −2, 3) and parallel to xy-plane is

z=3 or (x, y, 3)

b. The equation of the plane through the point (1, −2, 3) and parallel to yz-plane is

x=1 or (1, y, z)

c. The equation of the plane through the point (1, −2, 3) and parallel to zx-plane is

y=−2 or (x, −2, z)

2. Find the equation of the line through the point (2, 3, −4) and perpendicular to

a. xy-plane
b. yz-plane
c. zx-plane.

Find the point of intersection of the line and plane.

SOLUTION

a. The equation of the line through the point (2, 3, −4) and perpendicular to xy-plane is

x=2, y=3 or (2, 3, z)

The point of intersection of the line and xy-plane is (2, 3, 0).

b. The equation of the line through the point (2, 3, −4) and perpendicular to yz-plane is

y=3, z=−4 or (x, 3, −4)

The point of intersection of the line and yz-plane is (0, 3, −4).

c. The equation of the line through the point (2, 3, −4) and perpendicular to zx-plane is

x=2,

z=−4 or (2, y, −4)

The point of intersection of the line and zx-plane is (2, 0, −4).

3. Find the distance between the points (2, −3, 5) and (7, 5, −2).

SOLUTION

Let P(2, −3, 5) and Q(7, 5, −2)

$$PQ = \sqrt{(7-2)^2+(5-(-3))^2+(-2-5)^2}$$
$$= \sqrt{25+64+49}$$
$$= \sqrt{138}$$

4. Show that the points (−1, 2, 5), (1, 1, 6) and (0, 5, 6) form a right triangle.

SOLUTION

Let P(−1, 2, 5), Q(1, 1, 6) and R(0, 5, 6)

$$PQ = \sqrt{(1+1)^2+(1-2)^2+(6-5)^2}$$
$$= \sqrt{4+1+1}$$
$$= \sqrt{6}$$
$$QR = \sqrt{(0-1)^2+(5-1)^2+(6-6)^2}$$
$$= \sqrt{1+16+0}$$
$$= \sqrt{17}$$
$$PR = \sqrt{(0+1)^2+(5-2)^2+(6-5)^2}$$
$$= \sqrt{1+9+1}$$
$$= \sqrt{11}$$
$$PQ^2+PR^2 = 6+11$$
$$= 17$$
$$QR^2 = 17$$
$$\therefore PQ^2+PR^2 = QR^2$$

By Pythagoras theorem, ∴ △QPR is a right triangle.

5. Show that the points (1, −1, 2), (3, −2, 3) and (5, −3, 4) are collinear.

SOLUTION

Let A(1, −1, 2), B(3, −2, 3) and C(5, −3, 4)

$$AB = \sqrt{(3-1)^2+(-2+1)^2+(3-2)^2}$$
$$= \sqrt{4+1+1}$$
$$= \sqrt{6}$$
$$BC = \sqrt{(5-3)^2+(-3+2)^2+(4-3)^2}$$
$$= \sqrt{4+1+1}$$
$$= \sqrt{6}$$
$$AC = \sqrt{(5-1)^2+(-3+1)^2+(4-2)^2}$$
$$= \sqrt{16+4+4}$$
$$= \sqrt{24}$$
$$= 2\sqrt{6}$$

Therefore, AB=BC and AC=2AB (or 2BC). So the points A, B, C are collinear.

Section 3.2

EXAMPLE 2 Given P(1, 2, 3) and Q(3, 6, 5), find the coordinates of point R(x, y, z) on the line PQ with

respect to the point P and the following parameters

a. $k=\frac{1}{2}$

b. $k=2$

c. $k=-2$

SOLUTION

Given P(1, 2, 3) and Q(3, 6, 5), we have $\langle PQ\rangle=\langle l, m, n\rangle = \langle 2, 4, 2\rangle$.

a. $k=\frac{1}{2}$: $(x, y, z)=(1+\frac{1}{2}(2), 2+\frac{1}{2}(4), 3+\frac{1}{2}(2))=(2, 4, 4)$

b. $k=2$: $(x, y, z)=(1+2(2), 2+2(4), 3+2(2))=(5, 10, 7)$

c. **$k=-2$: $(x, y, z)=(1+(-2)(2), 2+(-2)(4), 3+(-2)(2))=(-3, -6, -1)$**

EXAMPLE 3 Given P(−1, 2, 3) and Q(3, 5, −2), determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.

a. $(1, \frac{7}{2}, \frac{1}{2})$

b. (7, 8, −7)

c. (−5, −1, 8)

d. (7, 8, −2)

SOLUTION

Given P(−1, 2, 3) and Q(3, 5, −2), we have $\langle PQ\rangle=\langle l, m, n\rangle=\langle 4, 3, -5\rangle$.

The equation of the line PQ is

$$\frac{x+1}{4}=\frac{y-2}{3}=\frac{z-3}{-5}$$

a. If $(x, y, z)=(1, \frac{7}{2}, \frac{1}{2})$, then

$$\frac{x+1}{4}=\frac{1+1}{4}=\frac{1}{2}$$

$$\frac{y-2}{3}=\frac{\frac{7}{2}-2}{3}=\frac{1}{2}$$

$$\frac{z-3}{-5}=\frac{\frac{1}{2}-3}{-5}=\frac{1}{2}$$

and so $\frac{x+1}{4}=\frac{y-2}{3}=\frac{z-3}{-5}$ for $(x, y, z)=(1, \frac{7}{2}, \frac{1}{2})$.

Therefore the point $(1, \frac{7}{2}, \frac{1}{2})$ is on the line PQ, corresponding parameter is $\frac{1}{2}$.

b. If (x, y, z)=(7, 8, −7), then

$$\frac{x+1}{4}=\frac{7+1}{4}=2$$

$$\frac{y-2}{3}=\frac{8-2}{3}=2$$

$$\frac{z-3}{-5}=\frac{-7-3}{-5}=2$$

and so $\frac{x+1}{4}=\frac{y-2}{3}=\frac{z-3}{-5}$ for (x, y, z)=(7, 8, −7).

Therefore the point (7, 8, −7) is on the line PQ, corresponding parameter is 2.

c. If (x, y, z)=(−5, −1, 8), then

$$\frac{x+1}{4}=\frac{-5+1}{4}=-1$$

$$\frac{y-2}{3}=\frac{-1-2}{3}=-1$$

$$\frac{z-3}{-5}=\frac{8-3}{-5}=-1$$

and so $\frac{x+1}{4}=\frac{y-2}{3}=\frac{z-3}{-5}$ for (x, y, z)=(−5, −1, 8).

Therefore the point (−5, −1, 8) is on the line PQ, corresponding parameter is −1.

d. If (x, y, z)=(7, 8, −2), then

$$\frac{x+1}{4}=\frac{7+1}{4}=2$$

$$\frac{y-2}{3}=\frac{8-2}{3}=2$$

$$\frac{z-3}{-5}=\frac{-2-3}{-5}=1$$

and so $\frac{x+1}{4}=\frac{y-2}{3}\neq\frac{z-3}{-5}$ for (x, y, z)=(7, 8, −2).

Therefore the point (7, 8, −2) is not on the line PQ.

EXAMPLE 4 Given P(2, 1, 3) and Q(6, −5, 3), determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.

a. (4, −2, 3)

b. (−2, 7, 3)

c. (10, −11, 3)

d. (1, 1, 3)

SOLUTION

Given P(2, 1, 3) and Q(6, −5, 3), we have $\langle PQ\rangle=\langle l, m, n\rangle=\langle 4, -6, 0\rangle$.

Then the coordinates of the point (x, y, z) on the line PQ are

$(x, y, z) = (2+4k, 1-6k, 3)$

This means that the line PQ is on the plane z=3.

a. If (x, y, z)=(4, −2, 3), then

$(4, -2, 3) = (2+4k, 1-6k, 3)$

we have $k=\frac{1}{2}$. Therefore the point (4, −2, 3) is on the line PQ with corresponding parameter $\frac{1}{2}$.

b. If (x, y, z)=(−2, 7, 3), then

(−2, 7, 3) = (2+4k, 1−6k, 3)

we have k=−1. Therefore the point (−2, 7, 3) is on the line PQ with corresponding parameter −1.

c. If (x, y, z)=(10, −11, 3), then

(10, −11, 3) = (2+4k, 1−6k, 3)

we have k=2. Therefore the point (10, −11, 3) is on the line PQ with corresponding parameter 2.

d. If (x, y, z)=(1, 1, 3), then

(1, 1, 3)

= (2+4k, 1−6k, 3)

There is no value k that satisfies this condition. Therefore the point (1, 1, 3) is not on the line PQ.

EXERCISE 3.2

1. Given P(3, 1, 5) and Q(−3, 7, −2), find the coordinates of point R(x, y, z) on the line PQ with respect to the point P and the following parameters.

a. $k=\frac{1}{2}$

b. k=3

c. k=−2

SOLUTION

Given P(3, 1, 5) and Q(−3, 7, −2), we have ⟨PQ⟩=⟨l, m, n⟩=⟨−6, 6, −7⟩.

$(x, y, z) = (x_1+kl, y_1+km, z_1+kn)$

a. $k=\frac{1}{2}$: $(x, y, z)=(3+\frac{1}{2}(-6), 1+\frac{1}{2}(6), 5+\frac{1}{2}(-7))=(0, 4, \frac{3}{2})$

b. k=3: (x, y, z)=(3+3(−6), 1+3(6), 5+3(−7))=(−15, 19, −16)

c. **k=−2: (x, y, z)=(3+(−2)(−6), 1+(−2)(6), 5+(−2)(−7))=(15, −11, 19)**

2. Given P(−2, 1, 3) and Q(4, 4, −3), determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding parameter with respect to the point P.

a. (6, 3, −6)

b. (6, 5, −5)

c. (−4, 0, 5)

d. (7, 8, −2)

SOLUTION

Given P(−2, 1, 3) and Q(4, 4, −3), we have ⟨PQ⟩=⟨l, m, n⟩=⟨6, 3, −6⟩.

The equation of the line PQ is

$$\frac{x+2}{6}=\frac{y-1}{3}=\frac{z-3}{-6}$$

a. If (x, y, z)=(6, 3, −6), then

$$\frac{x+2}{6}=\frac{6+2}{6}=\frac{4}{3}$$
$$\frac{y-1}{3}=\frac{3-1}{3}=\frac{2}{3}$$
$$\frac{z-3}{-6}=\frac{-6-3}{-6}=\frac{3}{2}$$

and so $\frac{x+2}{6}\neq\frac{y-1}{3}\neq\frac{z-3}{-6}$ for (x, y, z)=(6, 3, −6). Therefore the point (6, 3, −6) is not on the line PQ.

b. If (x, y, z)=(6, 5, −5), then

$$\frac{x+2}{6}=\frac{6+2}{6}=\frac{4}{3}$$
$$\frac{y-1}{3}=\frac{5-1}{3}=\frac{4}{3}$$
$$\frac{z-3}{-6}=\frac{-5-3}{-6}=\frac{4}{3}$$

and so $\frac{x+2}{6}=\frac{y-1}{3}=\frac{z-3}{-6}$ for (x, y, z)=(6, 5, −5). Therefore the point (6, 5, −5) is on the line PQ, corresponding parameter is $\frac{4}{3}$.

c. If (x, y, z)=(−4, 0, 5), then

$$\frac{x+2}{6}=\frac{-4+2}{6}=-\frac{1}{3}$$
$$\frac{y-1}{3}=\frac{0-1}{3}=-\frac{1}{3}$$
$$\frac{z-3}{-6}=\frac{5-3}{-6}=-\frac{1}{3}$$

and so $\frac{x+2}{6}=\frac{y-1}{3}=\frac{z-3}{-6}$ for (x, y, z)=(−4, 0, 5). Therefore the point (−4, 0, 5) is on the line PQ, corresponding parameter is $-\frac{1}{3}$.

d. If (x, y, z)=(7, 8, −2), then

$$\frac{x+2}{6}=\frac{7+2}{6}=\frac{3}{2}$$

$\frac{y-1}{3} = \frac{8-1}{3}$

$= \frac{7}{3}$

$\frac{z-3}{-6} = \frac{-2-3}{-6}$

$= \frac{5}{6}$

and so $\frac{x+2}{6} \neq \frac{y-1}{3} \neq \frac{z-3}{-6}$ for (x, y, z)=(7, 8, −2).

Therefore the point (7, 8, −2) is not on the line PQ.

3. Given P(3, 2, −1) and Q(4, 2, 5), determine whether or not the following points are on the line PQ. If the point is on the line PQ, find the corresponding real number with respect to the point P.

a. (5, 2, 11)

b. (2, 2, −7)

c. $(\frac{7}{2}, 2, 2)$

d. (6, 2, 10)

SOLUTION

Given P(3, 2, −1) and Q(4, 2, 5), we have ⟨PQ⟩=⟨l, m, n⟩ =⟨1, 0, 6⟩.

Then the coordinates of the point (x, y, z) on the line PQ are

(x, y, z) = (3+k, 2, −1+6k)

This means that the line PQ is on the plane y=2.

a. If (x, y, z)=(5, 2, 11), then

(5, 2, 11) = (3+k, 2, −1+6k)

we have k=2. Therefore the point (5, 2, 11) is on the line PQ with corresponding parameter 2.

b. If (x, y, z)=(2, 2, −7), then

(2, 2, −7) = (3+k, 2, −1+6k)

we have k=−1. Therefore the point (2, 2, −7) is on the line PQ with corresponding parameter −1.

c. If $(x, y, z)=(\frac{7}{2}, 2, 2)$, then

$(\frac{7}{2}, 2, 2) = (3+k, 2, -1+6k)$

we have $k=\frac{1}{2}$. Therefore the point $(\frac{7}{2}, 2, 2)$ is on the line PQ with corresponding parameter $\frac{1}{2}$.

d. If (x, y, z)=(6, 2, 10), then

(6, 2, 10)

= (3+k, 2, −1+6k)

There is no value k that satisfies this condition. Therefore the point (6, 2, 10) is not on the line PQ.

4. Find the points of intersection of the line joining the two points (2, 4, 5) and (3, 5, −4) with the following planes.

a. xy-plane

b. yz-plane

c. zx-plane

SOLUTION

Let P(2, 4, 5) and Q(3, 5, −4), we have ⟨PQ⟩=⟨l, m, n⟩ =⟨1, 1, −9⟩.

Let (x, y, z) be the point of intersection of the line joining two points P and Q.

$(x, y, z) = (x_1+kl, y_1+km, z_1+kn)$

=(2+k, 4+k, 5−9k)

a. On xy-plane, z=0: 5−9k=0 → $k=\frac{5}{9}$. The point of intersection is $(\frac{23}{9}, \frac{41}{9}, 0)$.

b. On yz-plane, x=0: 2+k=0 → k=−2. The point of intersection is (0, 2, 23).

c. On zx-plane, y=0: 4+k=0 → k=−4. The point of intersection is (−2, 0, 41).

Section 3.3

EXAMPLE 5 Given P(2, 1, 3), Q(6, −5, 4), R(2, 3, 4) and S(−1, 5, 4), determine whether the lines PQ and RS are parallel or skew or intersect.

SOLUTION

For P(2, 1, 3) and Q(6, −5, 4), ⟨PQ⟩=⟨4, −6, 1⟩.

For R(2, 3, 4) and S(−1, 5, 4), ⟨RS⟩=⟨−3, 2, −3⟩.

Since $\frac{4}{-3} \neq \frac{-6}{2} \neq \frac{1}{-3}$, directed values of ⟨PQ⟩ are not multiple of ⟨RS⟩. So the two lines are not parallel.

If a point (x, y, z) is on the lines PQ and RS, then

x =2+4s and x=2−3t

y =1−6s and y=3+2t

z =3+s and z=4−3t

for real numbers s and t. Solving the first two equations, $s=\frac{-3}{5}$ and $t=\frac{4}{5}$. But

$3+(\frac{-3}{5}) \neq 4-3(\frac{4}{5})$

these values of s and t do not satisfy the third equation, so the system has no solution and the lines do not intersect. Therefore the given lines are skew.

EXAMPLE 6 Given P(0, 0, 1), Q(3, 6, 4), R(0, 3, 1) and S(3, 0, 4), show that lines PQ and RS are perpendicular.

SOLUTION

Given P(0, 0, 1) and Q(3, 6, 4), ⟨PQ⟩=(3, 6, 3).

Given R(0, 3, 1) and S(3, 0, 4), ⟨RS⟩=(3, −3, 3).

And we have (3)(3)+(6)(−3)+(3)(3)=0.

If a point (x, y, z) is on the lines PQ and RS, then

$x = 3s$ and $x = 3t$

$y = 6s$ and $y = 3 - 3t$

$z = 1 + 3s$ and $z = 1 + 3t$

for real numbers s and t. Solving these equations, $s = \frac{1}{3}$ and $t = \frac{1}{3}$. The point of intersection is (1, 2, 2) and the two lines intersect.

Therefore the two lines are perpendicular.

EXAMPLE 7 Find the equation of the line passing through the point (−4, 7, −3) and perpendicular to the line

$(x, y, z) = (3 + 2k, -1 + 3k, 1 - k)$

Find also the point of intersection of two lines.

SOLUTION

Directed values of the given line are $\langle 2, 3, -1 \rangle$.

Directed values of the required line are

$\langle -4 - (3 + 2k), 7 - (-1 + 3k), -3 - (1 - k) \rangle = \langle -7 - 2k, 8 - 3k, -4 + k \rangle$

for some real number k. If the two lines are perpendicular, then

$2(-7 - 2k) + 3(8 - 3k) + (-1)(-4 + k) = 0$

$-14 - 4k + 24 - 9k + 4 - k = 0$

$14k = 14$

$k = 1$

So directed values of the required line are $\langle -9, 5, -3 \rangle$ and the equation of the line is

$(x, y, z) = (-4 - 9t, 7 + 5t, -3 - 3t)$

The point of intersection is

(x, y, z) = (5, 2, 0)

EXERCISE 3.3

1. Find $\cos\angle PAQ$ for the followings.

a. P(1, 2, −1), A(−2, 1, 5), Q(2, −1, 0)

b. P(0, 2, −3), A(2, −1, 5), Q(−2, 3, −1)

SOLUTION

a. P(1, 2, −1), A(−2, 1, 5), Q(2, −1, 0)

$\langle AP \rangle = \langle l_1, m_1, n_1 \rangle = \langle 3, 1, -6 \rangle$

$\langle AQ \rangle = \langle l_2, m_2, n_2 \rangle = \langle 4, -2, -5 \rangle$

$$\cos\angle PAQ = \frac{l_1 l_2 + m_1 m_2 + n_1 n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2}\,\sqrt{l_2^2 + m_2^2 + n_2^2}}$$

$$= \frac{3(4) + (1)(-2) + (-6)(-5)}{\sqrt{9 + 1 + 36}\,\sqrt{16 + 4 + 25}}$$

$$= \frac{40}{\sqrt{46}\,\sqrt{45}}$$

$$= \frac{40}{\sqrt{2070}}$$

$$= \frac{40}{3\sqrt{230}}$$

$$= \frac{4\sqrt{230}}{69}$$

b. P(0, 2, −3), A(2, −1, 5), Q(−2, 3, −1)

$\langle AP \rangle = \langle l_1, m_1, n_1 \rangle = \langle -2, 3, -8 \rangle$

$\langle AQ \rangle = \langle l_2, m_2, n_2 \rangle = \langle -4, 4, -6 \rangle$

$$\cos\angle PAQ = \frac{(-2)(-4) + 3(4) + (-8)(-6)}{\sqrt{4 + 9 + 64}\,\sqrt{16 + 16 + 36}}$$

$$= \frac{68}{\sqrt{77}\,\sqrt{68}}$$

$$= \frac{68}{\sqrt{5236}}$$

$$= \frac{68}{2\sqrt{1309}}$$

$$= \frac{34}{\sqrt{1309}}$$

2. Determine whether the lines PQ and RS are parallel or skew or intersect. If PQ and RS intersect, are they perpendicular?

a. P(1, 2, 3), Q(4, 5, 6), R(−2, 3, 5), S(4, 9, 11)

b. P(3, −1, −3), Q(2, −3, 1), R(3, −2, 5), S(−1, −2, 1)

c. P(4, −2, 5), Q(−2, 6, 1), R(−1, 1, 4), S(3, 3, 2)

d. P(−3, −1, 6), Q(−1, 3, 0), R(0, 6, 7), S(−4, −4, −1)

SOLUTION

a. P(1, 2, 3), Q(4, 5, 6), R(−2, 3, 5), S(4, 9, 11): $\langle PQ \rangle = \langle 3, 3, 3 \rangle$, $\langle RS \rangle = \langle 6, 6, 6 \rangle$. Since $\frac{3}{6} = \frac{3}{6} = \frac{3}{6}$, directed values of $\langle PQ \rangle$ are a multiple of $\langle RS \rangle$, so the two lines are parallel. Therefore PQ and RS do not intersect and are not perpendicular.

b. P(3, −1, −3), Q(2, −3, 1), R(3, −2, 5), S(−1, −2, 1): $\langle PQ \rangle = \langle -1, -2, 4 \rangle$, $\langle RS \rangle = \langle -4, 0, -4 \rangle$. Since $-\frac{1}{4} \neq -\frac{2}{0} \neq \frac{4}{-4}$, the lines are not parallel. Setting up the intersection system and solving the first two equations gives $s = \frac{1}{2}$, $t = \frac{1}{8}$, but these do not satisfy the third equation, so the system has no solution — the lines are skew, and since they do not intersect they are not perpendicular.

c. P(4, −2, 5), Q(−2, 6, 1), R(−1, 1, 4), S(3, 3, 2): $\langle PQ \rangle = \langle -6, 8, -4 \rangle$, $\langle RS \rangle = \langle 4, 2, -2 \rangle$. Since $-\frac{6}{4} \neq \frac{8}{2} \neq \frac{-4}{-2}$, the lines are not parallel. Solving the intersection system gives $s = \frac{1}{2}$, $t = \frac{1}{2}$, and these values satisfy the third equation too, so the lines intersect at (1, 2, 3).

$(-6)(4) + 8(2) + (-4)(-2) = -24 + 16 + 8 = 0$

Therefore the two lines are perpendicular.

d. P(−3, −1, 6), Q(−1, 3, 0), R(0, 6, 7), S(−4, −4, −1): $\langle PQ \rangle = \langle 2, 4, -6 \rangle$, $\langle RS \rangle = \langle -4, -10, -8 \rangle$. Since $\frac{2}{-4} \neq \frac{4}{-10} \neq \frac{-6}{-8}$, the lines are not parallel. Solving the intersection system gives $s = \frac{1}{2}$, $t = \frac{1}{2}$, and the

third equation is also satisfied, so the lines intersect at (−2, 1, 3).

$2(-4)+4(-1)+(-6)(-8)=-8-40+48=0$

Therefore the two lines are perpendicular.

3. Find the equation of the line passing through the point (8, −1, −10) and perpendicular to the line (x, y, z)= (1+2k, 2−k, 3−7k). Find also the point of intersection of two lines.

SOLUTION

Directed values of the given line are ⟨2, −1, −7⟩.

Directed values of the required line are

$\langle 8-(1+2k), -1-(2-k), -10-(3-7k)\rangle = \langle 7-2k, -3+k, -13+7k\rangle$

for some real number k. If the two lines are perpendicular, then

$2(7-2k)+(-1)(-3+k)+(-7)(-13+7k) = 0$

$14-4k+3-k+91-49k = 0$

$108-54k = 0$

$k = 2$

So directed values of the required line are ⟨3, −1, 1⟩ and the equation of the line is

$(x, y, z)=(8+3t, -1-t, -10+t)$

The point of intersection is

(x, y, z)=(5, 0, −11)

Section 3.4

EXAMPLE 8 Find the equation of the plane containing A(1, 0, 1), B(3, 6, 4) and C(−2, 3, 1).

SOLUTION

Given A(1, 0, 1) and B(3, 6, 4), $\langle AB\rangle=\langle l_1, m_1, n_1\rangle=\langle 2, 6, 3\rangle$.

Given A(1, 0, 1) and C(−2, 3, 1), $\langle AC\rangle=\langle l_2, m_2, n_2\rangle=\langle -3, 3, 0\rangle$.

$a=m_1n_2-m_2n_1 = 6(0)-3(3)$
$= -9$

$b=n_1l_2-n_2l_1 = 3(-3)-0(2)$
$= -9$

$c=l_1m_2-l_2m_1 = 2(3)-(-3)(6)$
$= 24$

$d=ax_1+by_1+cz_1 = -9(1)-9(0)+24(1)$
$= 15$

Therefore the equation of the plane is

$-9x-9y+24z=15$

(Note that we can use any point on the plane to calculate the value of d.)

EXAMPLE 9 Find the equation of the line passes through the point (−1, 3, 2) and perpendicular to the plane 3x−2y−z=3. Find the point of intersection of the line and the given plane.

SOLUTION

Directed values of the line perpendicular to the plane 3x−2y−z=3 are ⟨3, −2, −1⟩.

Then the equation of the line is

$$\frac{x+1}{3}=\frac{y-3}{-2}=\frac{z-2}{-1}$$

So coordinates of the points on the line are of the form

$x = -1+3s$

$y = 3-2s$

$z = 2-s$

If one of these points is on the plane, then

$3(-1+3s)-2(3-2s)-(2-s) = 3$

$-3+9s-6+4s-2+s = 3$

$14s = 14$

$s = 1$

Therefore the point of intersection is (2, 1, 1).

EXAMPLE 10 Find the equation of the plane containing the point (−1, 3, 2) and parallel to the plane 3x−2y−3z=2.

SOLUTION

Directed values of the line perpendicular to the plane 3x−2y−3z=2 are ⟨3, −2, −3⟩. This line is also perpendicular to the required plane, so the equation of the required plane is

$3x-2y-3z=d$

The point (−1, 3, 2) is on this plane, so

$3(-1)-2(3)-3(2) = d$
$d = -15$

The equation of the required plane is

$3x-2y-3z=-15$

EXERCISE 3.4

1. Find the equation of the plane containing

a. A(2, −5, 4), B(−5, 2, 4) and C(−2, 3, −1)

b. A(4, 2, −3), B(1, −2, 4) and C(−1, 0, 3)

SOLUTION

a. A(2, −5, 4), B(−5, 2, 4), C(−2, 3, −1): $\langle AB\rangle=\langle l_1, m_1, n_1\rangle = \langle -7, 7, 0\rangle$, $\langle AC\rangle=\langle l_2, m_2, n_2\rangle=\langle -4, 8, -5\rangle$.

$a=m_1n_2-m_2n_1 = 7(-5)-7(0)$
$= -35$

$b=n_1l_2-n_2l_1 = 0(-4)-(-5)(-7)$
$= -35$

$c=l_1m_2-l_2m_1 = (-7)(8)-(-4)(7)$
$= -56+28$
$= -28$

$d=ax_1+by_1+cz_1 = -35(2)+(-35)(-5)+(-28)(4)$
$= -70+175-112$
$= -7$

Therefore the equation of the plane is $-35x-35y-28z=-7$, that is $5x+5y+4z=1$.

b. A(4, 2, −3), B(1, −2, 4), C(−1, 0, 3): $\langle AB\rangle=\langle l_1, m_1, n_1\rangle$ $=\langle -3, -4, 7\rangle$, $\langle AC\rangle=\langle l_2, m_2, n_2\rangle=\langle -5, -2, 6\rangle$.

$$\begin{aligned} a=m_1n_2-m_2n_1 &= -4(6)-(-2)(7)\\ &= -24+14\\ &= -10\\ b=n_1l_2-n_2l_1 &= 7(-5)-6(-3)\\ &= -35+18\\ &= -17\\ c=l_1m_2-l_2m_1 &= (-3)(-2)-(-5)(-4)\\ &= 6-20\\ &= -14\\ d=ax_1+by_1+cz_1 &= (-10)(4)+(-17)(2)+(-14)(-3)\\ &= -40-34+42\\ &= -32 \end{aligned}$$

Therefore the equation of the plane is $-10x-17y-14z=-32$.

2. Find the equation of the line passing through the point (3, −2, −2) and perpendicular to the plane $-2x+3y-z=4$. Find the point of intersection of the line and the plane.

SOLUTION

Directed values of the line perpendicular to the plane $-2x+3y-z=4$ are $\langle -2, 3, -1\rangle$. Then the equation of the line is

$$\frac{x-3}{-2}=\frac{y+2}{3}=\frac{z+2}{-1}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= 3-2s\\ y &= -2+3s\\ z &= -2-s \end{aligned}$$

If one of these points is on the plane, then

$$\begin{aligned} -2(3-2s)+3(-2+3s)-(-2-s) &= 4\\ -6+4s-6+9s+2+s &= 4\\ 14s-10 &= 4\\ s &= 1 \end{aligned}$$

Therefore the point of intersection is (3−2s, −2+3s, −2−s)=(1, 1, −3).

3. Find the equation of the plane containing the point (2, 3, −1) and parallel to the plane $-2x+y+3z=6$.

SOLUTION

Directed values of the line perpendicular to the plane $-2x+y+3z=6$ are $\langle -2, 1, 3\rangle$. This line is also perpendicular to the required plane, so the equation of the required plane is

$$-2x+y+3z=d$$

The point (2, 3, −1) is on this plane, so

$$\begin{aligned} -2(2)+3+3(-1) &= d\\ d &= -4+3-3\\ &= -4 \end{aligned}$$

The equation of the required plane is $-2x+y+3z=-4$.

Section 3.5

EXAMPLE 11 Find the equation of the plane tangent to the sphere

$$(x-2)^2+(y-1)^2+(z+1)^2=14$$

at the point (3, 4, 1).

SOLUTION

Directed values of the line joining center C(2, 1, −1) of the sphere and the given point P(3, 4, 1) are $\langle CP\rangle=\langle 1, 3, 2\rangle$. Line CP is perpendicular to the tangent plane.

Thus the equation of the plane is

$$x+3y+2z=d$$

Since P(3, 4, 1) is on this plane, so we get

$$\begin{aligned} 3+3(4)+2(1) &= d\\ d &= 17 \end{aligned}$$

The equation of the plane is

$x+3y+2z=17$

EXAMPLE 12 Find the equation of the sphere with center (0, 1, 0) and touching the plane $x-2y+2z+5=0$.

SOLUTION

The line through the center (0, 1, 0) and perpendicular to the plane $x-2y+2z+5=0$ is

$$\frac{x}{1}=\frac{y-1}{-2}=\frac{z}{2}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= s\\ y &= 1-2s\\ z &= 2s \end{aligned}$$

If one of these points is on the plane, then

$$\begin{aligned} s-2(1-2s)+2(2s)+5 &= 0\\ s-2+4s+4s+5 &= 0\\ 9s &= -3\\ s &= -\frac{1}{3} \end{aligned}$$

Therefore the point of intersection is $P(-\frac{1}{3}, \frac{5}{3}, -\frac{2}{3})$. Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(-\frac{1}{3})^2+(\frac{5}{3}-1)^2+(-\frac{2}{3})^2}\\ &= \sqrt{\frac{1}{9}+\frac{4}{9}+\frac{4}{9}}\\ &= 1 \end{aligned}$$

Therefore the equation of the sphere is $x^2+(y-1)^2+z^2=1$.

EXAMPLE 13 Find the equation of a sphere that passes through the points (9, 0, 0), (3, 13, 5) and (11, 0, 10), given that its center lies on the yz-plane.

SOLUTION

The equation of the sphere with center (x_1, y_1, z_1) and radius r is

$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$$

Since the center lies on the yz-plane, $x_1=0$.

Substituting each given point,

$$81+y_1^2+z_1^2 = r^2 \quad ...(1)$$

$$9+(13-y_1)^2+(5-z_1)^2 = r^2 \quad ...(2)$$

$$121+y_1^2+(10-z_1)^2 = r^2 \quad ...(3)$$

From (1) and (3):

$$81+y_1^2+z_1^2 = 121+y_1^2+(10-z_1)^2$$

$$81+z_1^2 = 121+100-20z_1+z_1^2$$

$$0 = 140-20z_1$$

$$z_1 = 7$$

From (1) and (2):

$$81+y_1^2+7^2 = 9+(13-y_1)^2+(5-7)^2$$

$$130+y_1^2 = 9+169-26y_1+y_1^2+4$$

$$130 = 182-26y_1$$

$$y_1 = 2$$

Therefore $x_1=0$, $y_1=2$, $z_1=7$ and substituting into (1), $r^2=134$.

The equation of the sphere is

$$(x-0)^2+(y-2)^2+(z-7)^2=134$$

EXERCISE 3.5

1. Find the equation of sphere with center C and radius r.

a. C(1, −2, 4), r=3
b. C(2, 6, −3), r=2
c. C(2, 3, 5), r=5

SOLUTION

a. C(1, −2, 4), r=3:

$$(x-1)^2+(y+2)^2+(z-4)^2=3^2=9$$

b. C(2, 6, −3), r=2:

$$(x-2)^2+(y-6)^2+(z+3)^2=2^2=4$$

c. **C(2, 3, 5), r=5:**

$$(x-2)^2+(y-3)^2+(z-5)^2=5^2=25$$

2. Check whether the given point P lies inside, outside or on a sphere.

a. Center C(0,0,0), radius r=3, point P(1,1,1)
b. Center C(0,0,0), radius r=3, point P(2,1,2)
c. Center C(0,0,0), radius r=3, point P(10,10,10)

SOLUTION

a. $CP=\sqrt{1^2+1^2+1^2}=\sqrt{3}$. Since CP<r, the point P lies inside the sphere.
b. $CP=\sqrt{4+1+4}=\sqrt{9}=3$. Since CP=r, the point P lies on the sphere.
c. $CP=\sqrt{100+100+100}=\sqrt{300}=10\sqrt{3}$. **Since CP>r, the point P lies outside the sphere.**

3. Find the equation of the sphere on the join of (1, −1, 1) and (−3, 4, 5) as diameter.

SOLUTION

Let A=(1, −1, 1) and B=(−3, 4, 5).

$$\text{Diameter=AB} = \sqrt{(-3-1)^2+(4+1)^2+(5-1)^2}$$

$$= \sqrt{16+25+16}$$

$$= \sqrt{57}$$

$$\text{radius} = r = \frac{AB}{2} = \frac{\sqrt{57}}{2}$$

Center C is the midpoint of AB:

$$C=\left(\frac{1-3}{2}, \frac{-1+4}{2}, \frac{1+5}{2}\right)=\left(-1, \frac{3}{2}, 3\right)$$

The equation of the sphere is

$$(x+1)^2+\left(y-\frac{3}{2}\right)^2+(z-3)^2=\frac{57}{4}$$

4. Find the equation of the plane tangent to the sphere

$$(x+2)^2+(y-1)^2+(z+3)^2=27$$

at a point (3, 2, −2).

SOLUTION

Directed values of the line joining center C(−2, 1, −3) of the sphere and the given point P(3, 2, −2) are $\langle CP\rangle=\langle 5, 1, 1\rangle$. Line CP is perpendicular to the tangent plane.

Thus the equation of the plane is

$$5x+y+z=d$$

Since P(3, 2, −2) is on this plane, so we get

$$5(3)+2+(-2) = d$$

$$d = 15$$

The equation of the plane is

$$5x+y+z=15$$

5. Find the equation of the sphere with center (6, −7, −3) and touching the plane 4x−2y−z=17.

SOLUTION

The line through the center (6, −7, −3) and perpendicular to the plane 4x−2y−z=17 is

$$\frac{x-6}{4}=\frac{y+7}{-2}=\frac{z+3}{-1}$$

So coordinates of the points on the line are of the form $x=6+4s$, $y=-7-2s$, $z=-3-s$. If one of these points is on the plane, then

$4(6+4s)-2(-7-2s)-(-3-s) = 17$

$24+16s+14+4s+3+s = 17$

$21s+41 = 17$

$21s = -24$

$s = -\frac{8}{7}$

Therefore the point of intersection is $P=(\frac{10}{7}, -\frac{33}{7}, -\frac{13}{7})$. Thus we have the radius

$CP=\sqrt{(\frac{10}{7}-6)^2+(-\frac{33}{7}+7)^2+(-\frac{13}{7}+3)^2}=\frac{8\sqrt{21}}{7}$

Therefore the equation of the sphere is

$(x-6)^2+(y+7)^2+(z+3)^2=\frac{8^2 \cdot 21}{49}=\frac{192}{7}$

6. What is the equation of the sphere which passes through the points (3, 0, 2), (−1, 1, 1) and (2, −5, 4) and whose center lies on the plane $2x+3y+4z=6$?

SOLUTION

Let the equation of the sphere be

$x^2+y^2+z^2+Dx+Ey+Fz+G=0$, with center

$(-\frac{D}{2}, -\frac{E}{2}, -\frac{F}{2})$

Substituting the three given points:

$3D+2F+G = -13$ ...(1)

$-D+E+F+G = -3$ ...(2)

$2D-5E+4F+G = -45$ ...(3)

Since the center lies on $2x+3y+4z=6$:

$D+\frac{3}{2}E+2F=-6$ ...(4)

Solving equations (1)−(4) simultaneously gives $D=0$, $E=4$, $F=-6$, $G=-1$.

Therefore the equation of the sphere is

$x^2+y^2+z^2+4y-6z-1=0$

that is, $x^2+(y+2)^2+(z-3)^2=14$.

CH.04 Vector Algebra

Notes · Examples · Exercises

NOTES

Coordinates of a point in space

A point P in space is determined by an ordered triple (x, y, z) of real numbers, measured along three mutually perpendicular axes (the x-axis, y-axis and z-axis) through a fixed origin O.

Standard unit vectors

The standard unit vectors i, j, k in three dimensions are $\mathbf{i}=\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\mathbf{j}=\begin{pmatrix}0\\1\\0\end{pmatrix}$, $\mathbf{k}=\begin{pmatrix}0\\0\\1\end{pmatrix}$, representing the unit vectors in the positive x, y and z directions respectively.

Position vector

A vector whose initial point is the origin O is called a position vector. The position vector of a point P(a, b, c) is $\vec{OP}=\begin{pmatrix}a\\b\\c\end{pmatrix}=ai+bj+ck$, with magnitude $|\vec{OP}|=\sqrt{a^2+b^2+c^2}$.

Vector $\vec{AB}$ from two points

For points $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$, the vector $\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}x_2-x_1\\y_2-y_1\\z_2-z_1\end{pmatrix}$ is called the position vector of B relative to A, and $|\vec{AB}|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$.

Zero vector

The zero vector in three dimensions is $0=\begin{pmatrix}0\\0\\0\end{pmatrix}$.

Vector algebra (addition, subtraction, scalar multiplication)

For $\mathbf{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, $\mathbf{a}+\mathbf{b}=\begin{pmatrix}x_1+x_2\\y_1+y_2\\z_1+z_2\end{pmatrix}$, $\mathbf{a}-\mathbf{b}=\begin{pmatrix}x_1-x_2\\y_1-y_2\\z_1-z_2\end{pmatrix}$ and $k\mathbf{a}=\begin{pmatrix}kx_1\\ky_1\\kz_1\end{pmatrix}$ for any scalar k.

Equal vectors

Two vectors are equal if they have the same magnitude and direction. If $\mathbf{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, then $\mathbf{a}=\mathbf{b}$ if and only if $x_1=x_2$, $y_1=y_2$ and $z_1=z_2$.

Parallel vectors

Two vectors **a** and **b** are parallel if and only if $\mathbf{a}=k\mathbf{b}$ for some scalar k.

Unit vector

A vector with magnitude 1 is called a unit vector. For a nonzero vector **a**, the unit vector in the same direction of **a**, denoted â (â), is given by $\hat{a}=\frac{\mathbf{a}}{|\mathbf{a}|}$.

Collinear points

Points A, B, C are collinear if the vectors $\vec{AB}$ and $\vec{BC}$ (or $\vec{AC}$) are parallel and share a common point.

Scalar (dot) product

For $\mathbf{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, the scalar product is $\mathbf{a}\cdot\mathbf{b}=x_1x_2+y_1y_2+z_1z_2$, and also $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$, where θ is the angle between **a** and **b**.

Perpendicular vectors

Two nonzero vectors **a** and **b** are perpendicular if and only if $\mathbf{a}\cdot\mathbf{b}=0$.

Vector (cross) product

The vector product is defined only for three-dimensional vectors. For $\mathbf{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, $\mathbf{a}\times\mathbf{b}=\begin{pmatrix}y_1z_2-z_1y_2\\z_1x_2-x_1z_2\\x_1y_2-y_1x_2\end{pmatrix}$. The vector $\mathbf{a}\times\mathbf{b}$ is perpendicular to both **a** and **b**, with magnitude $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta$.

Direction vector of a line

A direction vector of a straight line is a vector parallel to the line.

Normal vector of a plane

A normal vector of a plane is a vector perpendicular to every line lying in the plane.

Magnitude of a position vector

$|\vec{OP}|=\sqrt{a^2+b^2+c^2}$ for $\vec{OP}=\begin{pmatrix}a\\b\\c\end{pmatrix}$

Unit vector

$\hat{a}=\frac{a}{|a|}$

Scalar (Dot) Product

$\mathbf{a}\cdot\mathbf{b}=x_1x_2+y_1y_2+z_1z_2=|\mathbf{a}||\mathbf{b}|\cos\theta$

Angle between two vectors

$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$

Vector (Cross) Product

For $\mathbf{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$,

$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}y_1z_2-z_1y_2\\z_1x_2-x_1z_2\\x_1y_2-y_1x_2\end{pmatrix}$

Area of a parallelogram

$\alpha(OACB)=|\mathbf{a}\times\mathbf{b}|$

for a parallelogram OACB with adjacent sides $\vec{OA}=\mathbf{a}$ and $\vec{OB}=\mathbf{b}$.

Area of a triangle

$\alpha(\Delta OAB)=\frac{1}{2}|\mathbf{a}\times\mathbf{b}|$

Vector equation of a line

$\mathbf{r}=\mathbf{a}+t\mathbf{b}$, $t\in\mathbb{R}$

where R is any point on the line with $\vec{OR}=\mathbf{r}$, A is a known fixed point on the line with $\vec{OA}=\mathbf{a}$, and $\mathbf{b}$ is the direction vector of the line.

Vector equation of a line (column form)

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}+t\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}$$

Parametric equations of a line

$x=a_1+tb_1$

$y=a_2+tb_2$

$z=a_3+tb_3$

where $t\in\mathbb{R}$ is the parameter.

Cartesian equation of a line

$$\frac{x-a_1}{b_1}=\frac{y-a_2}{b_2}=\frac{z-a_3}{b_3}$$

Vector equation of a plane (parametric form)

$\mathbf{r}=\mathbf{a}+t_1d_1+t_2d_2$

where A is a known point on the plane with $\vec{OA}=\mathbf{a}$, and d_1, d_2 are two non-parallel direction vectors lying in the plane.

Vector equation of a plane (normal form)

$\mathbf{r}\cdot\mathbf{n}=\mathbf{a}\cdot\mathbf{n}=d$

where $\mathbf{n}$ is a normal vector to the plane and A is a known point on the plane with $\vec{OA}=\mathbf{a}$.

Cartesian equation of a plane

$ax+by+cz=d$

where $\mathbf{n}=\begin{pmatrix}a\\b\\c\end{pmatrix}$ is the normal vector of the plane,

and $d=\mathbf{a}\cdot n$ for a point A on the plane with $\vec{OA}=\mathbf{a}$.

Section 4.1

EXAMPLE 1 Illustrate the points (a) A(0, 3, 0) (b) B(4, 0, 2) (c) C(−1, 2, 2) in a three-dimensional coordinate system.

SOLUTION

Each point is plotted by moving along the x, y and z axes by the amounts given by its coordinates from the origin O. For example, to plot B(4, 0, 2), move 4 units along the x-axis and 2 units along the z-axis (y=0).

EXAMPLE 2 If P is (−3, 1, 2) and Q is (1, −1, 3), find

a. $\vec{OP}$
b. $\vec{PQ}$
c. $|\vec{PQ}|$
d. $\vec{QP}$
e. $|\vec{QP}|$

SOLUTION

(a) $\vec{OP}=\begin{pmatrix}-3\\1\\2\end{pmatrix}$

(b) $\vec{PQ}=\vec{OQ}-\vec{OP}=\begin{pmatrix}1\\-1\\3\end{pmatrix}-\begin{pmatrix}-3\\1\\2\end{pmatrix}=\begin{pmatrix}1+3\\-1-1\\3-2\end{pmatrix}=\begin{pmatrix}4\\-2\\1\end{pmatrix}$

(c) $|\vec{PQ}|=\sqrt{4^2+(-2)^2+1^2}=\sqrt{21}$

(d) $\vec{QP}=\vec{OP}-\vec{OQ}=\begin{pmatrix}-3\\1\\2\end{pmatrix}-\begin{pmatrix}1\\-1\\3\end{pmatrix}=\begin{pmatrix}-3-1\\1+1\\2-3\end{pmatrix}=\begin{pmatrix}-4\\2\\-1\end{pmatrix}$

(e) $|\vec{QP}|=\sqrt{(-4)^2+2^2+(-1)^2}=\sqrt{21}$

EXAMPLE 3 If $\mathbf{p}=\begin{pmatrix}1\\-1\\4\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}-2\\0\\2\end{pmatrix}$, find:

a. $\mathbf{p}+\mathbf{q}$

b. $\mathbf{p}\frac{1}{2}\mathbf{q}$ (read: p minus one-half **q**)

c. $\frac{3}{2}\mathbf{q}-\mathbf{p}$

SOLUTION

(a) $\mathbf{p}+\mathbf{q}=\begin{pmatrix}1\\-1\\4\end{pmatrix}+\begin{pmatrix}-2\\0\\2\end{pmatrix}=\begin{pmatrix}1-2\\-1+0\\4+2\end{pmatrix}=\begin{pmatrix}-1\\-1\\6\end{pmatrix}$

(b) $\mathbf{p}\frac{1}{2}\mathbf{q}=\begin{pmatrix}1\\-1\\4\end{pmatrix}-\frac{1}{2}\begin{pmatrix}-2\\0\\2\end{pmatrix}=\begin{pmatrix}1+1\\-1-0\\4-1\end{pmatrix}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$

(c) $\frac{3}{2}\mathbf{q}-\mathbf{p}=\frac{3}{2}\begin{pmatrix}-2\\0\\2\end{pmatrix}-\begin{pmatrix}1\\-1\\4\end{pmatrix}=\begin{pmatrix}-3-1\\0+1\\3-4\end{pmatrix}=\begin{pmatrix}-4\\1\\-1\end{pmatrix}$

EXAMPLE 4 Find u and v given that $\mathbf{a}=\begin{pmatrix}-1\\-1\\u\end{pmatrix}$ is parallel to $\mathbf{b}=\begin{pmatrix}v\\2\\-2\end{pmatrix}$.

SOLUTION

Since **a** and **b** are parallel, $\mathbf{a}=k\mathbf{b}$ for some scalar k.

$$\begin{pmatrix}-1\\-1\\u\end{pmatrix}=k\begin{pmatrix}v\\2\\-2\end{pmatrix}$$

$-1=kv$, $-1=2k$ and $u=-2k$

From $-1=2k$, we get $k=\frac{1}{-2}$.

Thus $u=-2k=-2\times\frac{1}{-2}=1$, and from $-1=kv$, $v=\frac{-1}{k}=\frac{-1}{\frac{1}{-2}}$

$=2$.

Therefore u=1 and v=2.

EXAMPLE 5 ABCD is a parallelogram. A is (−1, 1, 1), B is (2, 0, −2), and D is (3, 1, 4). Find the coordinates of C.

SOLUTION

Let C be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}$ is parallel to $\vec{DC}$ and they have the same length and direction, so:

$$\vec{DC}=\vec{AB}$$

$$\vec{OC}-\vec{OD}=\vec{OB}-\vec{OA}$$

$$\begin{pmatrix}x\\y\\z\end{pmatrix}-\begin{pmatrix}3\\1\\4\end{pmatrix}=\begin{pmatrix}2\\0\\-2\end{pmatrix}-\begin{pmatrix}-1\\1\\1\end{pmatrix}$$

$$\begin{pmatrix}x-3\\y-1\\z-4\end{pmatrix}=\begin{pmatrix}3\\-1\\-3\end{pmatrix}$$

So $x-3=3\rightarrow x=6$, $y-1=-1\rightarrow y=0$, $z-4=-3\rightarrow z=1$.

Therefore C is (6, 0, 1).

EXAMPLE 6 (a) Find the unit vector in the same direction as $\mathbf{a}=\begin{pmatrix}2\\-2\\1\end{pmatrix}$. (b) Find a vector of magnitude 5 that is parallel to **a**.

SOLUTION

(a) $|\mathbf{a}|=\sqrt{2^2+(-2)^2+1^2}=3$

The unit vector of **a** is $\hat{a}=\frac{a}{|\mathbf{a}|}=\frac{1}{3}\begin{pmatrix}2\\-2\\1\end{pmatrix}=\begin{pmatrix}\frac{2}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix}$

(b) Let **b** be parallel to a with $|\mathbf{b}|=5$. Then $\mathbf{b}=5\hat{a}=\begin{pmatrix}\frac{10}{3}\\-\frac{10}{3}\\\frac{5}{3}\end{pmatrix}$ or $\mathbf{b}=-5\hat{a}=\begin{pmatrix}-\frac{10}{3}\\\frac{10}{3}\\-\frac{5}{3}\end{pmatrix}$.

EXAMPLE 7 Prove that A(8, 2, 2), C(20, 5, 5) and B(12, 3, 3) are collinear.

SOLUTION

$$\vec{AB}=\vec{OB}-\vec{OA}$$

$$=\begin{pmatrix}12\\3\\3\end{pmatrix}-\begin{pmatrix}8\\2\\2\end{pmatrix}$$

$$=\begin{pmatrix}4\\1\\1\end{pmatrix}$$

$$\vec{BC}=\vec{OC}-\vec{OB}$$

$$=\begin{pmatrix}20\\5\\5\end{pmatrix}-\begin{pmatrix}12\\3\\3\end{pmatrix}$$

$$=\begin{pmatrix}8\\2\\2\end{pmatrix}$$

$$=2\begin{pmatrix}4\\1\\1\end{pmatrix}$$

$$=2\cdot\vec{AB}$$

Therefore, $\vec{BC}$ is parallel to $\vec{AB}$. Since B is common to both, it follows that A, B and C are collinear.

EXERCISE 4.1

1. Let $\mathbf{a}=\begin{pmatrix}5\\-2\\-4\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}3\\-6\\1\end{pmatrix}$ and $\mathbf{c}=\begin{pmatrix}0\\7\\-1\end{pmatrix}$. Find the following vectors:

a. 3a

b. 4b

c. $\mathbf{a}-\mathbf{b}$

d. $\mathbf{b}+\mathbf{c}$

e. $2\mathbf{b}+\mathbf{c}$

f. $\mathbf{a}-2\mathbf{b}$

g. $\mathbf{a}+\mathbf{b}-2\mathbf{c}$

h. $3\mathbf{a}-\mathbf{b}+\mathbf{c}$

SOLUTION

a. $3\mathbf{a}=3\begin{pmatrix}5\\-2\\-4\end{pmatrix}=\begin{pmatrix}15\\-6\\-12\end{pmatrix}$

b. $4\mathbf{b}=4\begin{pmatrix}3\\-6\\1\end{pmatrix}=\begin{pmatrix}12\\-24\\4\end{pmatrix}$

c. $\mathbf{a}-\mathbf{b}=\begin{pmatrix}5\\-2\\-4\end{pmatrix}-\begin{pmatrix}3\\-6\\1\end{pmatrix}=\begin{pmatrix}5-3\\-2+6\\-4-1\end{pmatrix}=\begin{pmatrix}2\\4\\-5\end{pmatrix}$

d. $\mathbf{b}+\mathbf{c}=\begin{pmatrix}3\\-6\\1\end{pmatrix}+\begin{pmatrix}0\\7\\-1\end{pmatrix}=\begin{pmatrix}3\\1\\0\end{pmatrix}$

e. $2\mathbf{b}+\mathbf{c}=2\begin{pmatrix}3\\-6\\1\end{pmatrix}+\begin{pmatrix}0\\7\\-1\end{pmatrix}=\begin{pmatrix}6\\-12\\2\end{pmatrix}+\begin{pmatrix}0\\7\\-1\end{pmatrix}=\begin{pmatrix}6\\-5\\1\end{pmatrix}$

f. $\mathbf{a}-2\mathbf{b}=\begin{pmatrix}5\\-2\\-4\end{pmatrix}-2\begin{pmatrix}3\\-6\\1\end{pmatrix}=\begin{pmatrix}5\\-2\\-4\end{pmatrix}+\begin{pmatrix}-6\\12\\-2\end{pmatrix}=\begin{pmatrix}-1\\10\\-6\end{pmatrix}$

g. $\mathbf{a}+\mathbf{b}-2\mathbf{c}=\begin{pmatrix}5\\-2\\-4\end{pmatrix}+\begin{pmatrix}3\\-6\\1\end{pmatrix}-2\begin{pmatrix}0\\7\\-1\end{pmatrix}=\begin{pmatrix}8\\-8\\-3\end{pmatrix}+\begin{pmatrix}0\\-14\\2\end{pmatrix}=\begin{pmatrix}8\\-22\\-1\end{pmatrix}$

h. $3\mathbf{a}-\mathbf{b}+\mathbf{c}=3\begin{pmatrix}5\\-2\\-4\end{pmatrix}-\begin{pmatrix}3\\-6\\1\end{pmatrix}+\begin{pmatrix}0\\7\\-1\end{pmatrix}=\begin{pmatrix}15\\-6\\-12\end{pmatrix}+\begin{pmatrix}-3\\13\\-2\end{pmatrix}=\begin{pmatrix}12\\7\\-14\end{pmatrix}$

---

2. Given vectors $\mathbf{a}=\begin{pmatrix}1\\2\\7\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\4\\2\end{pmatrix}$ and $\mathbf{c}=\begin{pmatrix}-2\\p\\q\end{pmatrix}$.

a. Find the values of $\mathbf{p}$ and q such that $\mathbf{c}$ is parallel to $\mathbf{a}$.

b. Find the value of scalar k such that $\mathbf{a}+k\mathbf{b}$ is parallel to vector $\begin{pmatrix}0\\10\\23\end{pmatrix}$.

SOLUTION

(a) Since $\mathbf{c}$ is parallel to $\mathbf{a}$, $\mathbf{c}=k\mathbf{a}$ for some scalar k.

$$\begin{pmatrix}-2\\p\\q\end{pmatrix}=k\begin{pmatrix}1\\2\\7\end{pmatrix}$$

$$=\begin{pmatrix}k\\2k\\7k\end{pmatrix}$$

Comparing components: $k=-2$, so $\mathbf{p}=2k=-4$ and $\mathbf{q}=7k=-14$.

(b) Since $\mathbf{a}+k\mathbf{b}$ is parallel to $\begin{pmatrix}0\\10\\23\end{pmatrix}$, $\mathbf{a}+k\mathbf{b}=h\begin{pmatrix}0\\10\\23\end{pmatrix}$ for some scalar h.

$$\begin{pmatrix}1\\2\\7\end{pmatrix}+k\begin{pmatrix}-3\\4\\2\end{pmatrix}=h\begin{pmatrix}0\\10\\23\end{pmatrix}$$

$$\begin{pmatrix}1-3k\\2+4k\\7+2k\end{pmatrix}=\begin{pmatrix}0\\10h\\23h\end{pmatrix}$$

From the first component, $1-3k=0$, so $k=\frac{1}{3}$.

---

3. Points A, B, C and D have position vectors $\mathbf{a}=\begin{pmatrix}3\\-1\\1\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}5\\0\\3\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}7\\8\\-3\end{pmatrix}$ and $d=\begin{pmatrix}4\\3\\-2\end{pmatrix}$ respectively. Point E is the midpoint of $\vec{BC}$.

a. Find the position vector of E.

b. Show that ABED is a parallelogram.

SOLUTION

(a) $e=\frac{1}{2}(b+c)=\frac{1}{2}\left[\begin{pmatrix}5\\0\\3\end{pmatrix}+\begin{pmatrix}7\\8\\-3\end{pmatrix}\right]=\frac{1}{2}\begin{pmatrix}12\\8\\0\end{pmatrix}=\begin{pmatrix}6\\4\\0\end{pmatrix}$.

(b) $\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}5\\0\\3\end{pmatrix}-\begin{pmatrix}3\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\1\\2\end{pmatrix}$.

$\vec{DE}=\vec{OE}-\vec{OD}=\begin{pmatrix}6\\4\\0\end{pmatrix}-\begin{pmatrix}4\\3\\-2\end{pmatrix}=\begin{pmatrix}2\\1\\2\end{pmatrix}$.

Since $\vec{AB}=\vec{DE}$, $\vec{AB}$ is parallel to $\vec{DE}$ and equal in length, so ABED is a parallelogram.

---

4. Points A, B and C have position vectors $\mathbf{a}=\begin{pmatrix}2\\-1\\4\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}5\\1\\2\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}3\\1\\4\end{pmatrix}$ respectively. Find the position vector of point D such that ABCD is a parallelogram.

SOLUTION

Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}=\vec{DC}$, i.e. $\vec{OB}-\vec{OA}=\vec{OC}-\vec{OD}$.

$$\begin{pmatrix}5\\1\\2\end{pmatrix}-\begin{pmatrix}2\\-1\\4\end{pmatrix}=\begin{pmatrix}3\\1\\4\end{pmatrix}-\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$\begin{pmatrix}3\\2\\-2\end{pmatrix}=\begin{pmatrix}3\\1-y\\4-z\end{pmatrix}$

So $3-x=3\to x=0$; $1-y=2\to y=-1$; $4-z=-2\to z=6$.

Therefore D is (0, −1, 6).

5. K(1, −1, 0), L(4, −3, 7) and M(a, 2, b) are collinear. Find **a** and **b**.

SOLUTION

Since K, L, M are collinear, $\vec{KL}=h\vec{KM}$ for some scalar h, i.e. $\vec{OL}-\vec{OK}=h(\vec{OM}-\vec{OK})$.

$$\begin{pmatrix}4\\-3\\7\end{pmatrix}-\begin{pmatrix}1\\-1\\0\end{pmatrix}=h\left[\begin{pmatrix}a\\2\\b\end{pmatrix}-\begin{pmatrix}1\\-1\\0\end{pmatrix}\right]$$

$$\begin{pmatrix}3\\-2\\7\end{pmatrix}=h\begin{pmatrix}a-1\\3\\b\end{pmatrix}$$

Comparing the y-components: $-2=3h$, so $h=\frac{-2}{3}$.

From the x-component: $3=h(\mathbf{a}-1)$, so $\mathbf{a}-1=\frac{3}{h}=3\div\frac{-2}{3}=\frac{-9}{2}$, giving $\mathbf{a}=1+(\frac{-9}{2})=\frac{-7}{2}$.

From the z-component: $7=bh$, so $\mathbf{b}=\frac{7}{h}=7\div\frac{-2}{3}=\frac{-21}{2}$.

Section 4.2

EXAMPLE 8 Find the angle between the two vectors $\begin{pmatrix}3\\4\end{pmatrix}$ and $\begin{pmatrix}5\\-12\end{pmatrix}$.

SOLUTION

Let $\mathbf{a}=\begin{pmatrix}3\\4\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}5\\-12\end{pmatrix}$.

$|\mathbf{a}|=\sqrt{3^2+4^2}=5$ and $|\mathbf{b}|=\sqrt{5^2+(-12)^2}=13$.

$\mathbf{a}\cdot\mathbf{b}=\begin{pmatrix}3\\4\end{pmatrix}\cdot\begin{pmatrix}5\\-12\end{pmatrix}=3\times5+4\times(-12)=15-48=-33$.

$$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{-33}{5\times13}=\frac{-33}{65}$$

$\theta\approx120.5°$

EXAMPLE 9 Given points P(1, 0, −1), Q(2, 4, 1) and R(3, 5, 6), find ∠QPR.

SOLUTION

The angle between $\vec{PQ}$ and $\vec{PR}$ is given by θ where $\cos\theta=\frac{\vec{PQ}\cdot\vec{PR}}{|\vec{PQ}||\vec{PR}|}$.

$\vec{PQ}=\begin{pmatrix}2\\4\\1\end{pmatrix}-\begin{pmatrix}1\\0\\-1\end{pmatrix}=\begin{pmatrix}1\\4\\2\end{pmatrix}$, so $|\vec{PQ}|=\sqrt{1^2+4^2+2^2}=\sqrt{21}$.

$\vec{PR}=\begin{pmatrix}3\\5\\6\end{pmatrix}-\begin{pmatrix}1\\0\\-1\end{pmatrix}=\begin{pmatrix}2\\5\\7\end{pmatrix}$, so $|\vec{PR}|=\sqrt{2^2+5^2+7^2}=\sqrt{78}$.

$\vec{PQ}\cdot\vec{PR}=\begin{pmatrix}1\\4\\2\end{pmatrix}\cdot\begin{pmatrix}2\\5\\7\end{pmatrix}=(1)(2)+(4)(5)+(2)(7)=36$.

$$\cos\theta=\frac{36}{\sqrt{21}\times\sqrt{78}}$$

$\theta\approx27.2°$

EXAMPLE 10 Given that **a** and **b** are perpendicular vectors such that $|\mathbf{a}|=3$ and $|\mathbf{b}|=1$. Calculate $(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}+5\mathbf{b})$.

SOLUTION

Since **a** and **b** are perpendicular, $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}=0$.

$$\begin{aligned}(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}+5\mathbf{b})&=\mathbf{a}\cdot\mathbf{a}+5(\mathbf{a}\cdot\mathbf{b})-\mathbf{b}\cdot\mathbf{a}-5(\mathbf{b}\cdot\mathbf{b})\\&=\mathbf{a}\cdot\mathbf{a}-5(\mathbf{b}\cdot\mathbf{b})\\&=|\mathbf{a}|^2-5|\mathbf{b}|^2\\&=3^2-5\times1^2\\&=4\end{aligned}$$

EXAMPLE 11 Points A, B and C have position vectors $\mathbf{a}=k\begin{pmatrix}2\\-1\\1\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}3\\2\\-2\end{pmatrix}$ and $\mathbf{c}=\begin{pmatrix}1\\1\\4\end{pmatrix}$.

a. Find $\vec{BC}$.
b. Find $\vec{AB}$ in terms of k.
c. Find the value of k for which $\vec{AB}$ is perpendicular to $\vec{BC}$.

SOLUTION

(a) $\vec{BC}=\mathbf{c}-\mathbf{b}=\begin{pmatrix}1\\1\\4\end{pmatrix}-\begin{pmatrix}3\\2\\-2\end{pmatrix}=\begin{pmatrix}-2\\-1\\6\end{pmatrix}$.

(b) $\vec{AB}=\mathbf{b}-\mathbf{a}=\begin{pmatrix}3\\2\\-2\end{pmatrix}-k\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}3-2k\\2+k\\-2-k\end{pmatrix}$.

(c) Since $\vec{AB}$ is perpendicular to $\vec{BC}$, $\vec{AB}\cdot\vec{BC}=0$:

$$\begin{pmatrix}3-2k\\2+k\\-2-k\end{pmatrix}\cdot\begin{pmatrix}-2\\-1\\6\end{pmatrix}=0$$

$$-6+4k-2-k-12-6k=0$$

$$-3k=20$$

$$k=\frac{20}{-3}$$

EXERCISE 4.2

1. For $\mathbf{p}=\begin{pmatrix}3\\2\end{pmatrix}$, $\mathbf{q}=\begin{pmatrix}-1\\5\end{pmatrix}$ and $\mathbf{r}=\begin{pmatrix}-2\\4\end{pmatrix}$, find:

a. $\mathbf{q}\cdot\mathbf{p}$
b. $\mathbf{q}\cdot\mathbf{r}$
c. $\mathbf{q}\cdot(\mathbf{p}+\mathbf{r})$
d. $\mathbf{i}\cdot\mathbf{p}$

e. $\mathbf{q}\cdot\mathbf{j}$

f. $\mathbf{i}\cdot\mathbf{i}$

SOLUTION

a. $\mathbf{q}\cdot\mathbf{p}=\begin{pmatrix}-1\\5\end{pmatrix}\cdot\begin{pmatrix}3\\2\end{pmatrix}=(-1)(3)+(5)(2)=-3+10=7$

b. $\mathbf{q}\cdot\mathbf{r}=\begin{pmatrix}-1\\5\end{pmatrix}\cdot\begin{pmatrix}-2\\4\end{pmatrix}=(-1)(-2)+(5)(4)=2+20=22$

c. $\mathbf{q}\cdot(\mathbf{p}+\mathbf{r})=\begin{pmatrix}-1\\5\end{pmatrix}\cdot\begin{pmatrix}1\\6\end{pmatrix}=-1+30=29$ (since $\mathbf{p}+\mathbf{r}=\begin{pmatrix}1\\6\end{pmatrix}$)

d. $\mathbf{i}\cdot\mathbf{p}=\begin{pmatrix}1\\0\end{pmatrix}\cdot\begin{pmatrix}3\\2\end{pmatrix}=3$

e. $\mathbf{q}\cdot\mathbf{j}=\begin{pmatrix}-1\\5\end{pmatrix}\cdot\begin{pmatrix}0\\1\end{pmatrix}=5$

f. $\mathbf{i}\cdot\mathbf{i}=\begin{pmatrix}1\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\end{pmatrix}=1$

---

2. For $\mathbf{a}=\begin{pmatrix}2\\1\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-1\\1\\1\end{pmatrix}$ and $\mathbf{c}=\begin{pmatrix}0\\-1\\1\end{pmatrix}$, find:

a. $\mathbf{a}\cdot\mathbf{b}$

b. $\mathbf{b}\cdot\mathbf{a}$

c. $|\mathbf{a}|^2$

d. $\mathbf{a}\cdot\mathbf{a}$

e. $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$

f. $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$

SOLUTION

a. $\mathbf{a}\cdot\mathbf{b}=\begin{pmatrix}2\\1\\3\end{pmatrix}\cdot\begin{pmatrix}-1\\1\\1\end{pmatrix}=(-2)+1+3=2$

b. $\mathbf{b}\cdot\mathbf{a}=2$ (dot product is commutative)

c. $|\mathbf{a}|^2=2^2+1^2+3^2=14$

d. $\mathbf{a}\cdot\mathbf{a}=14$ (since $\mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$)

e. $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\begin{pmatrix}2\\1\\3\end{pmatrix}\cdot\begin{pmatrix}-1\\0\\2\end{pmatrix}=(-2)+0+6=4$ (since $\mathbf{b}+\mathbf{c}=\begin{pmatrix}-1\\0\\2\end{pmatrix}$)

f. $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}=2+\left[\begin{pmatrix}2\\1\\3\end{pmatrix}\cdot\begin{pmatrix}0\\-1\\1\end{pmatrix}\right]=2+(0-1+3)=2+2=4$

---

3. Find the angle between $\mathbf{m}$ and $\mathbf{n}$ if:

a. $\mathbf{m}=\begin{pmatrix}2\\-1\\-1\end{pmatrix}$ and $\mathbf{n}=\begin{pmatrix}-1\\3\\2\end{pmatrix}$

b. $\mathbf{m}=2\mathbf{j}-\mathbf{k}$ and $\mathbf{n}=\mathbf{i}+2\mathbf{k}$

SOLUTION

(a) $|\mathbf{m}|=\sqrt{4+1+1}=\sqrt{6}$, $|\mathbf{n}|=\sqrt{1+9+4}=\sqrt{14}$.

$\mathbf{m}\cdot\mathbf{n}=\begin{pmatrix}2\\-1\\-1\end{pmatrix}\cdot\begin{pmatrix}-1\\3\\2\end{pmatrix}=-2-3-2=-7.$

$$\cos\theta=\frac{\mathbf{m}\cdot\mathbf{n}}{|\mathbf{m}||\mathbf{n}|}=\frac{-7}{\sqrt{6}\cdot\sqrt{14}}$$

$\theta\approx 139.8°$

(b) $\mathbf{m}=2\mathbf{j}-\mathbf{k}=\begin{pmatrix}0\\2\\-1\end{pmatrix}$ and $\mathbf{n}=\mathbf{i}+2\mathbf{k}=\begin{pmatrix}1\\0\\2\end{pmatrix}$.

$|\mathbf{m}|=\sqrt{4+1}=\sqrt{5}$, $|\mathbf{n}|=\sqrt{1+4}=\sqrt{5}$.

$\mathbf{m}\cdot\mathbf{n}=(0)(1)+(2)(0)+(-1)(2)=-2.$

$$\cos\theta=\frac{-2}{\sqrt{5}\cdot\sqrt{5}}=\frac{-2}{5}$$

$\theta\approx 113.6°$

---

4. Find t if the given pair of vectors are (i) perpendicular (ii) parallel:

a. $\mathbf{p}=\begin{pmatrix}3\\t\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}-2\\1\end{pmatrix}$

b. $\mathbf{r}=\begin{pmatrix}t\\t+2\end{pmatrix}$ and $\mathbf{s}=\begin{pmatrix}t\\-4\end{pmatrix}$ (perpendicular case only)

c. $\mathbf{a}=\begin{pmatrix}0\\t+2\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}2-3t\\t\end{pmatrix}$

SOLUTION

(a)(i) Perpendicular: $\mathbf{p}\cdot\mathbf{q}=0\rightarrow\begin{pmatrix}3\\t\end{pmatrix}\cdot\begin{pmatrix}-2\\1\end{pmatrix}=0\rightarrow -6+t=0\rightarrow t=6.$

(a)(ii) Parallel: $\mathbf{p}=k\mathbf{q}\rightarrow\begin{pmatrix}3\\t\end{pmatrix}=k\begin{pmatrix}-2\\1\end{pmatrix}\rightarrow -2k=3\rightarrow k=\frac{-3}{2}$, and $t=k=\frac{-3}{2}$.

(b) Perpendicular: $\mathbf{r}\cdot\mathbf{s}=0\rightarrow\begin{pmatrix}t\\t+2\end{pmatrix}\cdot\begin{pmatrix}t\\-4\end{pmatrix}=0\rightarrow t^2-4(t+2)=0\rightarrow t^2-4t-8=0.$

Completing the square: $(t-2)^2=12$, so $t-2=\pm\sqrt{12}=\pm 2\sqrt{3}$, giving $t=2\pm 2\sqrt{3}$.

(c)(i) Perpendicular: $\mathbf{a}\cdot\mathbf{b}=0\rightarrow\begin{pmatrix}0\\t+2\end{pmatrix}\cdot\begin{pmatrix}2-3t\\t\end{pmatrix}=0\rightarrow t(t+2)=0\rightarrow t=0$ or $t=-2$.

(c)(ii) Parallel: $\mathbf{b}=k\mathbf{a}\rightarrow\begin{pmatrix}2-3t\\t\end{pmatrix}=k\begin{pmatrix}0\\t+2\end{pmatrix}\rightarrow 2-3t=0\rightarrow t=\frac{2}{3}$.

---

5. Find t if $\begin{pmatrix}3\\t\\-2\end{pmatrix}$ is perpendicular to $\begin{pmatrix}1-t\\-3\\4\end{pmatrix}$.

SOLUTION

Since the vectors are perpendicular, their dot product is zero:

$$\begin{pmatrix} 3 \\ t \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1-t \\ -3 \\ 4 \end{pmatrix} = 0$$

$3(1-t)+t(-3)+(-2)(4) = 0$

$3-3t-3t-8 = 0$

$-6t-5 = 0$

$t = \frac{-5}{6}$

6. ABCD is a parallelogram with $\vec{AB}$ parallel to $\vec{DC}$. Let $\vec{AB}=\mathbf{a}$ and $\vec{AD}=\mathbf{b}$.

a. Express $\vec{AC}$ and $\vec{BD}$ in terms of **a** and **b**.

b. Simplify $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{b}-\mathbf{a})$.

c. Hence show that if ABCD is a rhombus then its diagonals are perpendicular.

SOLUTION

(a) $\vec{AC}=\vec{AB}+\vec{BC}=\mathbf{a}+\mathbf{b}$, and $\vec{BD}=\vec{AD}-\vec{AB}=\mathbf{b}-\mathbf{a}$.

(b) $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{b}-\mathbf{a})=\mathbf{a}\cdot\mathbf{b}-\mathbf{a}\cdot\mathbf{a}+\mathbf{b}\cdot\mathbf{b}-\mathbf{b}\cdot\mathbf{a}=|\mathbf{b}|^2-|\mathbf{a}|^2$ (since $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$).

(c) In a rhombus, all sides are equal, so $\vec{AB}=\vec{AD}$, i.e. $|\mathbf{a}|=|\mathbf{b}|$, hence $|\mathbf{a}|^2=|\mathbf{b}|^2$.

The diagonals $\vec{AC}$ and $\vec{BD}$ satisfy $\vec{AC}\cdot\vec{BD}$ $=(\mathbf{a}+\mathbf{b})\cdot(\mathbf{b}-\mathbf{a})=|\mathbf{b}|^2-|\mathbf{a}|^2=0$.

Therefore $\vec{AC}$ is perpendicular to $\vec{BD}$, so the diagonals of a rhombus are perpendicular.

Section 4.3

EXAMPLE 12 Find the area of the parallelogram determined by the vectors $\mathbf{a}=\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}$.

SOLUTION

$$\mathbf{a}\times\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} 2\times(-1)-3\times4 \\ 3\times1-1\times(-1) \\ 1\times4-2\times1 \end{pmatrix} = \begin{pmatrix} -14 \\ 4 \\ 2 \end{pmatrix}$$

Now $|\mathbf{a}\times\mathbf{b}|=\sqrt{196+16+4}=\sqrt{216}$.

Therefore the area of the parallelogram is $\sqrt{216}$ unit².

EXAMPLE 13 Find the area of the triangle ABC with vertices A(1, −1, 3), B(0, 4, 1) and C(2, 7, 2).

SOLUTION

$$\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}-\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}=\begin{pmatrix} -1 \\ 5 \\ -2 \end{pmatrix}.$$

$$\vec{AC}=\vec{OC}-\vec{OA}=\begin{pmatrix} 2 \\ 7 \\ 2 \end{pmatrix}-\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}=\begin{pmatrix} 1 \\ 8 \\ -1 \end{pmatrix}.$$

$$\vec{AB}\times\vec{AC}=\begin{pmatrix} -1 \\ 5 \\ -2 \end{pmatrix}\times\begin{pmatrix} 1 \\ 8 \\ -1 \end{pmatrix}=\begin{pmatrix} 11 \\ -3 \\ -13 \end{pmatrix}.$$

$|\vec{AB}\times\vec{AC}|=\sqrt{11^2+(-3)^2+(-13)^2}=\sqrt{299}$.

Therefore, area of triangle ABC $=\frac{1}{2}|\vec{AB}\times\vec{AC}|=\frac{1}{2}\sqrt{299}$ unit².

EXAMPLE 14 (a) Calculate $\mathbf{a}\times\mathbf{b}$ when $\mathbf{a}=3\mathbf{i}+2\mathbf{j}+5\mathbf{k}$ and $\mathbf{b}=\mathbf{i}-4\mathbf{j}+2\mathbf{k}$.(b) Find a unit vector $\hat{\mathbf{n}}$ perpendicular to both **a** and **b**.

SOLUTION

(a) $\mathbf{a}\times\mathbf{b}=\begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}\times\begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix}=\begin{pmatrix} 2\times2-5\times(-4) \\ 5\times1-3\times2 \\ 3\times(-4)-2\times1 \end{pmatrix}=\begin{pmatrix} 24 \\ -1 \\ -14 \end{pmatrix}$.

(b) The vector $\mathbf{a}\times\mathbf{b}=\begin{pmatrix} 24 \\ -1 \\ -14 \end{pmatrix}$ is perpendicular to both **a** and **b**.

$|\mathbf{a}\times\mathbf{b}|=\sqrt{24^2+(-1)^2+(-14)^2}=\sqrt{773}$.

So a unit vector perpendicular to both **a** and **b** is

$$\hat{\mathbf{n}}=\frac{1}{\sqrt{773}}\begin{pmatrix} 24 \\ -1 \\ -14 \end{pmatrix}.$$

EXAMPLE 15 Given that $|\mathbf{a}|=4$, $|\mathbf{b}|=5$ and that **a** and **b** are perpendicular, evaluate $|(2\mathbf{a}-\mathbf{b})\times(\mathbf{a}+3\mathbf{b})|$.

SOLUTION

$(2\mathbf{a}-\mathbf{b})\times(\mathbf{a}+3\mathbf{b}) = 2(\mathbf{a}\times\mathbf{a})+6(\mathbf{a}\times\mathbf{b})-(\mathbf{b}\times\mathbf{a})-3(\mathbf{b}\times\mathbf{b})$

$=6(\mathbf{a}\times\mathbf{b})-(\mathbf{b}\times\mathbf{a})$ (since $\mathbf{a}\times\mathbf{a}=\mathbf{b}\times\mathbf{b}=0$)

$=6(\mathbf{a}\times\mathbf{b})+(\mathbf{a}\times\mathbf{b})$ (since $\mathbf{b}\times\mathbf{a}=-\mathbf{a}\times\mathbf{b}$)

$=7(\mathbf{a}\times\mathbf{b})$

So $|(2\mathbf{a}-\mathbf{b})\times(\mathbf{a}+3\mathbf{b})|=7|\mathbf{a}\times\mathbf{b}|=7|\mathbf{a}||\mathbf{b}|$ (since **a**, **b** are perpendicular, $\sin90°=1$)$=7\times4\times5=140$.

EXERCISE 4.3

1. Find a vector perpendicular to the following pair of vectors:

a. $\begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

b. $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix}$

SOLUTION

(a) Let $\mathbf{a}=\begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{b}=\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. A vector $\mathbf{a}\times\mathbf{b}$ is perpendicular to both **a** and **b**.

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix} 1\times3-1\times2 \\ 1\times1-3\times3 \\ 3\times2-1\times1 \end{pmatrix}=\begin{pmatrix} 1 \\ -8 \\ 5 \end{pmatrix}.$$

(b) Let $\mathbf{a}=\begin{pmatrix}3\\-1\\4\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-1\\1\\5\end{pmatrix}$. A vector $\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}(-1)(5)-4(1)\\4(-1)-3(5)\\3(1)-(-1)(-1)\end{pmatrix}=\begin{pmatrix}-9\\-19\\2\end{pmatrix}$.

---

2. Consider $\mathbf{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}1\\0\\-1\end{pmatrix}$:

a. Find $\mathbf{a}\times\mathbf{b}$.

b. Find $\sin\theta$ using $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta$.

SOLUTION

(a) $\mathbf{a}\times\mathbf{b}=\begin{pmatrix}2\\-1\\3\end{pmatrix}\times\begin{pmatrix}1\\0\\-1\end{pmatrix}=\begin{pmatrix}1\\5\\1\end{pmatrix}$.

(b) $|\mathbf{a}\times\mathbf{b}|=\sqrt{1+25+1}=\sqrt{27}$.

$|\mathbf{a}|=\sqrt{4+1+9}=\sqrt{14}$, $|\mathbf{b}|=\sqrt{1+0+1}=\sqrt{2}$.

$$\sin\theta=\frac{|\mathbf{a}\times\mathbf{b}|}{|\mathbf{a}||\mathbf{b}|}=\frac{\sqrt{27}}{\sqrt{14}\sqrt{2}}=\frac{\sqrt{27}}{\sqrt{28}}\approx0.982$$

Since $\mathbf{a}\cdot\mathbf{b}=(2)(1)+(-1)(0)+(3)(-1)=-1<0$, the angle θ is obtuse, so $\theta\approx180°-79.1°\approx100.9°$. (*Note: the source book states 74.6°, but since $\sin\theta$ alone does not determine whether the angle is acute or obtuse, and $\mathbf{a}\cdot\mathbf{b}$ is negative here, the correct obtuse angle is* $\approx100.9°$.)

---

3. Prove that for any two vectors $\mathbf{a}$ and $\mathbf{b}$, $|\mathbf{a}\times\mathbf{b}|^2+(\mathbf{a}\cdot\mathbf{b})^2=|\mathbf{a}|^2|\mathbf{b}|^2$.

SOLUTION

Since $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta$,

$$\begin{aligned}|\mathbf{a}\times\mathbf{b}|^2&=|\mathbf{a}|^2|\mathbf{b}|^2\sin^2\theta\\&=|\mathbf{a}|^2|\mathbf{b}|^2(1-\cos^2\theta)\\&=|\mathbf{a}|^2|\mathbf{b}|^2-|\mathbf{a}|^2|\mathbf{b}|^2\cos^2\theta\\&=|\mathbf{a}|^2|\mathbf{b}|^2-(\mathbf{a}\cdot\mathbf{b})^2\quad(\text{since }\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta)\end{aligned}$$

Therefore $|\mathbf{a}\times\mathbf{b}|^2+(\mathbf{a}\cdot\mathbf{b})^2=|\mathbf{a}|^2|\mathbf{b}|^2$.

---

4. Given points A, B and C with coordinates (3, −5, 1), (7, 7, 2) and (−1, 1, 3):

a. Calculate $\mathbf{p}=\vec{AB}\times\vec{AC}$ and $\mathbf{q}=\vec{BA}\times\vec{BC}$.

b. What can you say about vectors $\mathbf{p}$ and q?

SOLUTION

(a) $\vec{OA}=\begin{pmatrix}3\\-5\\1\end{pmatrix}$, $\vec{OB}=\begin{pmatrix}7\\7\\2\end{pmatrix}$, $\vec{OC}=\begin{pmatrix}-1\\1\\3\end{pmatrix}$.

$\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}4\\12\\1\end{pmatrix}$, $\vec{AC}=\vec{OC}-\vec{OA}=\begin{pmatrix}-4\\6\\2\end{pmatrix}$.

$\mathbf{p}=\vec{AB}\times\vec{AC}=\begin{pmatrix}4\\12\\1\end{pmatrix}\times\begin{pmatrix}-4\\6\\2\end{pmatrix}=\begin{pmatrix}(12)(2)-(1)(6)\\(1)(-4)-(4)(2)\\(4)(6)-(12)(-4)\end{pmatrix}=\begin{pmatrix}18\\-12\\72\end{pmatrix}$.

$\vec{BA}-\vec{AB}=\begin{pmatrix}-4\\-12\\-1\end{pmatrix}$, $\vec{BC}=\vec{OC}-\vec{OB}=\begin{pmatrix}-8\\-6\\1\end{pmatrix}$.

$\mathbf{q}=\vec{BA}\times\vec{BC}=\begin{pmatrix}-4\\-12\\-1\end{pmatrix}\times\begin{pmatrix}-8\\-6\\1\end{pmatrix}=\begin{pmatrix}-18\\12\\-72\end{pmatrix}$.

(b) **$\mathbf{q}=-\mathbf{p}$, so p and q have the same magnitude but opposite directions.**

---

5. The points A(3, 1, 2), B(−1, 1, 5) and C(7, 2, 3) are vertices of a parallelogram ABCD:

a. Find the coordinates of D.

b. Calculate the area of the parallelogram.

SOLUTION

(a) Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{DC}=\vec{AB}$, i.e. $\vec{OC}-\vec{OD}=\vec{OB}-\vec{OA}$.

$$\begin{pmatrix}7\\2\\3\end{pmatrix}-\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-1\\1\\5\end{pmatrix}-\begin{pmatrix}3\\1\\2\end{pmatrix}$$

$$\begin{pmatrix}7-x\\2-y\\3-z\end{pmatrix}=\begin{pmatrix}-4\\0\\3\end{pmatrix}$$

So x=11, y=2, z=0. Therefore D is (11, 2, 0).

(b) $\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}-4\\0\\3\end{pmatrix}$, $\vec{AD}=\vec{OD}-\vec{OA}=\begin{pmatrix}8\\1\\-2\end{pmatrix}$.

$\vec{AB}\times\vec{AD}=\begin{pmatrix}-4\\0\\3\end{pmatrix}\times\begin{pmatrix}8\\1\\-2\end{pmatrix}=\begin{pmatrix}-3\\16\\-4\end{pmatrix}$.

$|\vec{AB}\times\vec{AD}|=\sqrt{9+256+16}=\sqrt{281}$.

∴ the area of the parallelogram is $\sqrt{281}$ unit².

Section 4.4

EXAMPLE 16 Find the Cartesian equation of the line with vector equation $\mathbf{r}=\begin{pmatrix}1\\4\\-1\end{pmatrix}+t\begin{pmatrix}3\\2\\5\end{pmatrix}$.

SOLUTION

Since $\mathbf{r}=\mathbf{a}+t\mathbf{b}$, the parametric equations are:

$$x=1+3t\rightarrow t=\frac{x-1}{3}$$

$$y=4+2t\rightarrow t=\frac{y-4}{2}$$

$$z=-1+5t\rightarrow t=\frac{z+1}{5}$$

The Cartesian equation of the line is $\frac{x-1}{3}=\frac{y-4}{2}=\frac{z+1}{5}$.

EXAMPLE 17 Does the point A(3, −2, 2) lie on the line with equation $\frac{x+1}{2}=\frac{4-y}{3}=\frac{2z}{3}$?

SOLUTION

Substitute the coordinates of A(3, −2, 2) into each expression:

$\frac{x+1}{2}=\frac{3+1}{2}=3$

$\frac{4-y}{3}=\frac{4+2}{3}=2$

$\frac{2z}{3}=\frac{2\times2}{3}=\frac{4}{3}$

Since $2=2$ but $2\neq\frac{4}{3}$, the coordinates do not satisfy the Cartesian equation. Therefore the point does not lie on the line.

(Note: the source book prints the third expression as 3z/3, which is inconsistent with its own working of 2z/3; 2z/3 is used here as the correct term.)

EXAMPLE 18 Find the vector equation of the plane containing points M(2, 2, −2), N(1, −1, 3) and P(4, 0, 2).

SOLUTION

Let $\mathbf{a}=\vec{OM}=\begin{pmatrix}2\\2\\-2\end{pmatrix}$.

$\mathbf{d_1}=\vec{MN}=\vec{ON}-\vec{OM}=\begin{pmatrix}1\\-1\\3\end{pmatrix}-\begin{pmatrix}2\\2\\-2\end{pmatrix}=\begin{pmatrix}-1\\-3\\5\end{pmatrix}$.

$\mathbf{d_2}=\vec{MP}=\vec{OP}-\vec{OM}=\begin{pmatrix}4\\0\\2\end{pmatrix}-\begin{pmatrix}2\\2\\-2\end{pmatrix}=\begin{pmatrix}2\\-2\\4\end{pmatrix}$.

Therefore the vector equation of the plane is

$$\mathbf{r}=\begin{pmatrix}2\\2\\-2\end{pmatrix}+t_1\begin{pmatrix}-1\\-3\\5\end{pmatrix}+t_2\begin{pmatrix}2\\-2\\4\end{pmatrix}$$

EXAMPLE 19 Find the Cartesian equation of the plane containing points M(2, 2, −2), N(1, −1, 3) and P(4, 0, 2).

SOLUTION

The vector normal to the plane is $\mathbf{n}=\mathbf{d_1}\times\mathbf{d_2}$, using $\mathbf{d_1}=\begin{pmatrix}-1\\-3\\5\end{pmatrix}$ and $\mathbf{d_2}=\begin{pmatrix}2\\-2\\4\end{pmatrix}$ from Example 18.

$\mathbf{n}=\mathbf{d_1}\times\mathbf{d_2}$

$=\begin{pmatrix}-1\\-3\\5\end{pmatrix}\times\begin{pmatrix}2\\-2\\4\end{pmatrix}$

$=\begin{pmatrix}(-3)(4)-(5)(-2)\\(5)(2)-(-1)(4)\\(-1)(-2)-(-3)(2)\end{pmatrix}$

$=\begin{pmatrix}-2\\14\\8\end{pmatrix}$

This is a scalar multiple of $\begin{pmatrix}1\\-7\\-4\end{pmatrix}$ (dividing by −2), so we may take $\mathbf{n}=\begin{pmatrix}1\\-7\\-4\end{pmatrix}$.

$d=\mathbf{a}\cdot\mathbf{n}=\begin{pmatrix}2\\2\\-2\end{pmatrix}\cdot\begin{pmatrix}1\\-7\\-4\end{pmatrix}=2-14+8=-4.$

Therefore the Cartesian equation of the plane is $x-7y-4z=-4$.

(Check: substituting M(2,2,−2) gives 2−14+8=−4 ✓.)

EXAMPLE 20 Determine whether points A(3, −1, 4), B(2, 1, 1), C(4, 3, 1) and D(−3, 1, 4) lie in the same plane.

SOLUTION

The plane containing A, B, C has direction vectors $\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}-1\\2\\-3\end{pmatrix}$ and $\vec{AC}=\vec{OC}-\vec{OA}=\begin{pmatrix}1\\4\\-3\end{pmatrix}$.

A normal vector to this plane is $\mathbf{n}=\vec{AB}\times\vec{AC}=\begin{pmatrix}-1\\2\\-3\end{pmatrix}\times\begin{pmatrix}1\\4\\-3\end{pmatrix}=\begin{pmatrix}6\\-6\\-6\end{pmatrix}$, which is a scalar multiple of $\begin{pmatrix}1\\-1\\-1\end{pmatrix}$.

Using $\mathbf{n}=\begin{pmatrix}1\\-1\\-1\end{pmatrix}$, $d=\mathbf{n}\cdot\vec{OA}=\begin{pmatrix}1\\-1\\-1\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\4\end{pmatrix}=3+1-4=0$.

So the plane through A, B, C has equation $x-y-z=0$.

(Check: B: 2−1−1=0 ✓; C: 4−3−1=0 ✓.)

Substituting D(−3, 1, 4): $-3-1-4=-8\neq0$.

So, D does not lie on the same plane as A, B and C.

EXAMPLE 21 Find a vector equation of the plane containing the line $\mathbf{r}=\begin{pmatrix}-2\\1\\2\end{pmatrix}+t\begin{pmatrix}-1\\1\\1\end{pmatrix}$ and point A(3, −1, 2).

SOLUTION

Let $\vec{OA}=\begin{pmatrix}3\\-1\\2\end{pmatrix}$. The direction vector $\mathbf{d_1}=\begin{pmatrix}-1\\1\\1\end{pmatrix}$ of the given line lies in the plane.

B(−2, 1, 2) is a point on the line, hence on the plane. So a second direction vector in the plane is d_2= $\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}-2\\1\\2\end{pmatrix}-\begin{pmatrix}3\\-1\\2\end{pmatrix}=\begin{pmatrix}-5\\2\\0\end{pmatrix}$.

Therefore the vector equation of the plane is

$$\mathbf{r}=\begin{pmatrix}3\\-1\\2\end{pmatrix}+t_1\begin{pmatrix}-1\\1\\1\end{pmatrix}+t_2\begin{pmatrix}-5\\2\\0\end{pmatrix}$$

EXAMPLE 22 Vector $\mathbf{n}=\begin{pmatrix}2\\4\\-2\end{pmatrix}$ is perpendicular to the plane which contains point A(1, −5, 2).

a. Write an equation of the plane in the form $\mathbf{r}\cdot\mathbf{n}=d$.

b. Find the Cartesian equation of the plane.

SOLUTION

(a) The vector equation of the plane is $\mathbf{r}\cdot\mathbf{n}=\mathbf{a}\cdot\mathbf{n}$:

$$\mathbf{a}\cdot\mathbf{n}=\begin{pmatrix}1\\-5\\2\end{pmatrix}\cdot\begin{pmatrix}2\\4\\-2\end{pmatrix}=2-20-4=-22$$

So the vector equation of the plane is $\mathbf{r}\cdot\begin{pmatrix}2\\4\\-2\end{pmatrix}=-22$.

(b) With $\mathbf{r}=\begin{pmatrix}x\\y\\z\end{pmatrix}$:

$$\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}2\\4\\-2\end{pmatrix}=-22$$

$$2x+4y-2z=-22$$

The Cartesian equation of the plane is $2x+4y-2z=-22$.

EXERCISE 4.4

1. Find the vector equation of the line:

a. parallel to $\begin{pmatrix}2\\1\\3\end{pmatrix}$ and through the point (1, 3, −7).

b. through (0, 1, 2) and with direction vector i+j−2k

c. parallel to the x-axis and through the point (−2, 2, 2).

SOLUTION

(a) With $\mathbf{a}=\begin{pmatrix}1\\3\\-7\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}2\\1\\3\end{pmatrix}$, the vector equation is

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\3\\-7\end{pmatrix}+t\begin{pmatrix}2\\1\\3\end{pmatrix}$$

(b) With $\mathbf{a}=\begin{pmatrix}0\\1\\2\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}1\\1\\-2\end{pmatrix}$, the vector equation is

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}0\\1\\2\end{pmatrix}+t\begin{pmatrix}1\\1\\-2\end{pmatrix}$$

(c) A line parallel to the x-axis has direction vector $\begin{pmatrix}1\\0\\0\end{pmatrix}$. With $\mathbf{a}=\begin{pmatrix}-2\\2\\2\end{pmatrix}$, the vector equation is

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-2\\2\\2\end{pmatrix}+t\begin{pmatrix}1\\0\\0\end{pmatrix}$$

---

2. (a) Find the Cartesian equation of the line with parametric equations x=3t+1, y=4−2t, z=3t−1.(b) Find the unit vector in the direction of the line.

SOLUTION

(a) From the parametric equations, $t=\frac{x-1}{3}=\frac{y-4}{-2}=\frac{z+1}{3}$.

So the Cartesian equation of the line is $\frac{x-1}{3}=\frac{y-4}{-2}=\frac{z+1}{3}$.

(b) The vector equation is $\mathbf{r}=\begin{pmatrix}1\\4\\-1\end{pmatrix}+t\begin{pmatrix}3\\-2\\3\end{pmatrix}$, so the direction vector is $\mathbf{b}=\begin{pmatrix}3\\-2\\3\end{pmatrix}$.

$|\mathbf{b}|=\sqrt{3^2+(-2)^2+3^2}=\sqrt{22}$.

The unit direction vector is $\hat{b}=\frac{b}{|\mathbf{b}|}=\frac{1}{\sqrt{22}}\begin{pmatrix}3\\-2\\3\end{pmatrix}=\begin{pmatrix}\frac{3\sqrt{22}}{22}\\-\frac{\sqrt{22}}{11}\\\frac{3\sqrt{22}}{22}\end{pmatrix}$.

---

3. Find the equation of the plane:

a. with normal vector $\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and which passes through (−1, 2, 4).

b. perpendicular to the line joining points A(2, 3, 1) and B(5, 7, 2) and which passes through A.

c. containing A(3, 2, 1) and the line x=1+t, y=2−t, z=3+2t.

SOLUTION

(a) With $\mathbf{n}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\mathbf{a}=\begin{pmatrix}-1\\2\\4\end{pmatrix}$, $d=\mathbf{a}\cdot\mathbf{n}=\begin{pmatrix}-1\\2\\4\end{pmatrix}\cdot\begin{pmatrix}2\\-1\\3\end{pmatrix}=-2-2+12=8$.

The equation of the plane is $2x-y+3z=8$.

(b) A plane perpendicular to the line $\vec{AB}$ has normal vector equal to the direction of $\vec{AB}$: $\mathbf{n}=\vec{OB}-\vec{OA}=\begin{pmatrix}5\\7\\2\end{pmatrix}-\begin{pmatrix}2\\3\\1\end{pmatrix}=\begin{pmatrix}3\\4\\1\end{pmatrix}$.

$d=\vec{OA}\cdot\mathbf{n}=\begin{pmatrix}2\\3\\1\end{pmatrix}\cdot\begin{pmatrix}3\\4\\1\end{pmatrix}=6+12+1=19$.

The equation of the plane is $3x+4y+z=19$.

(Note: the source book instead computes n as the cross product of $\vec{OA}$ and $\vec{OB}$, giving $-x+y-z=0$ — but that vector is perpendicular to $\vec{AB}$ rather than parallel to it, so it does not satisfy "perpendicular to the line $\vec{AB}$". The normal of a plane perpendicular to a line must equal that line's direction vector, as used above.)

(c) The line $x=1+t$, $y=2-t$, $z=3+2t$ has vector form $\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+t\begin{pmatrix}1\\-1\\2\end{pmatrix}$, so direction vector $d_1=\begin{pmatrix}1\\-1\\2\end{pmatrix}$ lies in the required plane.

Point B(1, 2, 3) (from $t=0$) also lies on the plane, giving a second direction vector $d_2=\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}1\\2\\3\end{pmatrix}-\begin{pmatrix}3\\2\\1\end{pmatrix}=\begin{pmatrix}-2\\0\\2\end{pmatrix}$.

The vector equation of the plane is $\mathbf{r}=\begin{pmatrix}3\\2\\1\end{pmatrix}+t_1\begin{pmatrix}1\\-1\\2\end{pmatrix}+t_2\begin{pmatrix}-2\\0\\2\end{pmatrix}$.

---

4. Find the equation of the plane through A(−1, 2, 1), B(4, 1, 1) and C(2, 0, 3):

a. in vector form

b. in Cartesian form

SOLUTION

(a) Let $\mathbf{a}=\vec{OA}=\begin{pmatrix}-1\\2\\1\end{pmatrix}$. The direction vectors are:

$d_1=\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}4\\1\\1\end{pmatrix}-\begin{pmatrix}-1\\2\\1\end{pmatrix}=\begin{pmatrix}5\\-1\\0\end{pmatrix}$.

$d_2=\vec{AC}=\vec{OC}-\vec{OA}=\begin{pmatrix}2\\0\\3\end{pmatrix}-\begin{pmatrix}-1\\2\\1\end{pmatrix}=\begin{pmatrix}3\\-2\\2\end{pmatrix}$.

The vector equation of the plane is

$$\mathbf{r}=\begin{pmatrix}-1\\2\\1\end{pmatrix}+t_1\begin{pmatrix}5\\-1\\0\end{pmatrix}+t_2\begin{pmatrix}3\\-2\\2\end{pmatrix}$$

(b) A normal vector is $\mathbf{n}=d_1\times d_2=\begin{pmatrix}5\\-1\\0\end{pmatrix}\times\begin{pmatrix}3\\-2\\2\end{pmatrix}=\begin{pmatrix}-2\\-10\\-7\end{pmatrix}$.

$d=\mathbf{a}\cdot\mathbf{n}=\begin{pmatrix}-1\\2\\1\end{pmatrix}\cdot\begin{pmatrix}-2\\-10\\-7\end{pmatrix}=2-20-7=-25$.

So the Cartesian equation of the plane is $-2x-10y-7z=-25$, that is, $2x+10y+7z=25$.

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5. Find the Cartesian equation of the plane with vector equation $\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+t_1\begin{pmatrix}1\\1\\2\end{pmatrix}+t_2\begin{pmatrix}2\\-1\\5\end{pmatrix}$.

SOLUTION

A normal vector is $\mathbf{n}=d_1\times d_2=\begin{pmatrix}1\\1\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\5\end{pmatrix}=\begin{pmatrix}7\\-1\\-3\end{pmatrix}$.

$d=\mathbf{a}\cdot\mathbf{n}=\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}7\\-1\\-3\end{pmatrix}=7-2-9=-4$.

Therefore the Cartesian equation of the plane is $7x-y-3z=-4$.

CH.05 Permutations and Combinations

Notes · Examples · Exercises

NOTES

Permutation

A **permutation** is an arrangement of objects in a specific order. If there are n objects, then a permutation of any such r objects, with $r \le n$, is said to be a permutation of n objects taken r at a time. The number of permutations of n distinct objects taken r at a time is usually denoted by ${}^nP_r$.

Factorial

If n is a positive integer, n! (n factorial) stands for the product of the first n positive integers n, (n − 1), (n − 2), ..., 1. We define 0! to be 1.

Combination

A **combination** is a selection, in which the order does not matter, of objects from a collection. The number of combinations of n distinct objects taken r at a time is denoted by ${}^nC_r$.

Disjoint cases

Two cases of a counting problem are **disjoint** if they have no arrangement in common. If every possible arrangement falls into some of the cases, the Addition Principle can be applied.

The Multiplication Principle

Suppose a task can be performed in m ways, and no matter how the task has been performed another task can then be performed in n ways. Then the number of ways to perform these two tasks in succession is mn.

The principle can be extended to any finite number of successive tasks.

The Addition Principle

Suppose there are k pairwise disjoint cases, into which the elements in a counting problem fall. If there are n_1 elements in the first case, n_2 elements in the second case, n_3 elements in the third case, ..., and n_k elements in the kth case, then the number of elements in the problem is

$n_1 + n_2 + n_3 + \dots + n_k$

Factorial notation

If n is a positive integer,

$n! = n(n-1)(n-2)\dots 1$

$0! = 1$

Notice that

$n! = n \cdot (n-1)!$

$n! = n \cdot (n-1) \cdot (n-2)!$

Permutations nPr

If n and r are positive integers with $r \le n$,

${}^nP_r = n(n-1)(n-2)\dots(n-(r-1))$ (r factors)

We define ${}^nP_0 = 1$ for a nonnegative integer n.

Generally, for integers n and r with $0 \le r \le n$,

$${}^nP_r = \frac{n!}{(n-r)!}$$

Notice that ${}^nP_n = n!$.

Combinations nCr

Each combination of r objects gives rise to r! permutations, so ${}^nC_r \cdot r! = {}^nP_r$. If n and r are integers with $0 \le r \le n$, then

$${}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 1}$$

Properties of nCr

For integers n and r with $0 \le r \le n$,

${}^nC_1 = n$, ${}^nC_n = 1$ and ${}^nC_r = {}^nC_{n-r}$

Permutations with repetitions

Suppose we are given n objects, in which n_1 objects are of the first kind, n_2 objects are of the second kind, n_3 objects are of the third kind, ..., and n_r objects are of the rth kind, such that $n_1 + n_2 + \dots + n_r = n$. Then the number of permutations of all these n objects is

$$\frac{n!}{n_1!\, n_2! \dots n_r!}$$

The Exclusion Principle

Count what you are not interested in and subtract it from the total.

Number of subsets of a finite set

1. If a finite set contains n elements, then it has 2^n subsets.
2. If n is an integer with $n \ge 0$, then ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$.

Section 5.1

EXAMPLE 1 Suppose that there are 6 roads between town A and town B, and that 4 roads between town B and town C. Find the number of ways a person can drive from A to C by passing through B?

SOLUTION

There are 6 ways to drive from A to B, and for each such way there are 4 ways to drive from B to C.

So, the number of ways to drive from A to C through B is

$6 \times 4 = 24$

EXAMPLE 2 There are four blood types, namely A, B, AB and O. Blood can also be RH(+ve) or RH(−ve). A blood donor can be classified as either male or female. How many different possible ways can a donor have his or her blood labeled?

SOLUTION

There are 4 possibilities for the blood type, 2 possibilities for the RH factor and 2 possibilities for the gender of the donor.

So, the number of ways to label is

$4 \times 2 \times 2 = 16$

EXAMPLE 3 There are 3 picture nails on a wall. If there are 5 different pictures and each nail can hold only one picture, in how many different ways can the pictures be hung on all the nails?

SOLUTION

Any one of the 5 pictures can be chosen for the first nail, then any one from the remaining four for the second, and finally any one from the remaining three for the third.

The number of ways to choose the pictures for individual nails are 5, 4 and 3 respectively.

So, the number of ways to hang the pictures is

$5 \times 4 \times 3 = 60$

EXAMPLE 4 How many different numbers can be formed using the digits 3, 5, 6, 8 and 9 in such a way that the numbers contain two or three digits without any repetition?

SOLUTION

The two-digit numbers can be formed in $5 \times 4 = 20$ ways.

The three-digit numbers can be formed in $5 \times 4 \times 3 = 60$ ways.

So, there are $20 + 60 = 80$ different numbers.

EXAMPLE 5 How many integers, having digit 5 only once, are there between 0 and 100?

SOLUTION

The integers that we want to form, must be one of the followings:

i. One-digit integer, namely 5

ii. Two-digit integers having 5 as tens' digit (ones' digit may be anyone except 5)

iii. Two-digit integers having 5 as ones' digit (tens' digit may be anyone except 0 and 5)

For the first case, the number of integers = 1.

For the second case, the number of integers $= 1 \times 9 = 9$.

For the third case, the number of integers $= 8 \times 1 = 8$.

The required number of integers $= 1 + 9 + 8 = 18$.

EXAMPLE 6 Evaluate:

a. $\dfrac{7!}{4! \cdot 3!}$

b. $\dfrac{6! + 5! - 4!}{4!}$

SOLUTION

(a) $\dfrac{7!}{4! \cdot 3!} = \dfrac{7 \cdot 6 \cdot 5 \cdot 4!}{4! \cdot 3 \cdot 2 \cdot 1}$

$= 35$

(b) $\dfrac{6! + 5! - 4!}{4!}$

$= \dfrac{6 \cdot 5 \cdot 4! + 5 \cdot 4! - 4!}{4!}$

$= \dfrac{(30 + 5 - 1) \cdot 4!}{4!}$

$= 34$

EXAMPLE 7 Express $\dfrac{13 \cdot 12 \cdot 11}{3 \cdot 2}$ in factorial form.

SOLUTION

$\dfrac{13 \cdot 12 \cdot 11}{3 \cdot 2}$

$= \dfrac{13 \cdot 12 \cdot 11 \cdot 10!}{(3 \cdot 2 \cdot 1) \cdot 10!}$

$= \dfrac{13!}{3! \cdot 10!}$

EXERCISE 5.1

1. A store sells men's wear. It has 6 kinds of shirts, 4 kinds of pants and 3 kinds of coats. If a man wants to buy a shirt, a pant and a coat, in how many ways can this be done? (Assume that any choice meets his requirement.)

SOLUTION

There are 6 choices for a shirt, 4 choices for a pant and 3 choices for a coat.

The number of ways to choose $= 6 \times 4 \times 3 = 72$

72 ways

2. A television news director wishes to use three of the 7 news stories on an evening show. How many possible ways can the program be set up, if the three stories are to be classified as the lead story, the second story and the closing story?

SOLUTION

Total number of news stories = 7; numbers of stories to be use = 3.

There are 7 choices for the lead story, 6 choices for the second story and 5 choices for the closing story.

The number of ways that the program can be set up $= 7 \times 6 \times 5 = 210$

210 ways

3. If a garage door opener has a 10-digit keypad, containing 0, 1, 2, ..., 9, and the code to open the door must be a 4-digit code, how many codes are possible to create?

SOLUTION

Each of the 4 digits of the code can be any of the 10 digits 0, 1, 2, ..., 9.

The possible number of codes to create
$= 10 \times 10 \times 10 \times 10 = 10000$

10000 ways

4. There are 3 restaurants X, Y and Z, we can go for lunch, in a certain area. Suppose that no two restaurants have the same menu, and the numbers of choices for appetizer, main dish and dessert available are shown in the table. Find the number of ways to have a lunch, consisting of an appetizer, a main dish and a dessert.

Restaurants	No. of Appetizers	No. of Main dishes	No. of Desserts
X	4	10	2
Y	3	8	5
Z	5	12	3

SOLUTION

Restaurants	No. of Appetizers	No. of Main dishes	No. of Desserts
X	4	10	2
Y	3	8	5
Z	5	12	3

At restaurant X, we can get 4 types of Appetizers, 10 types of Main dishes and 2 types of Desserts.

The number of ways to have a lunch $= 4 \times 10 \times 2 = 80$

At restaurant Y, we can get 3 types of Appetizers, 8 types of Main dishes and 5 types of Desserts.

The number of ways to have a lunch $= 3 \times 8 \times 5 = 120$

At restaurant Z, we can get 5 types of Appetizers, 12 types of Main dishes and 3 types of Desserts.

The number of ways to have a lunch $= 5 \times 12 \times 3 = 180$

The required number of ways $= 80 + 120 + 180$
$= 380$

380 ways

5. A registration code consists of two of the 12 different capital letters A, B, C, ..., L, followed by one of the ten digits 0, 1, 2, ..., 9, for example ID5. How many codes are possible to generate:

a. if the repetition of the letters is allowed?
b. if the repetition of the letters is not allowed?
c. if two letters in the codes must be the same?
d. if two letters are different, but must be both vowels or both consonants?

SOLUTION

12 letters A, B, C, ..., L and 10 digits 0, 1, ..., 9. Code = letter, letter, digit.

(a) Repetition of the letters is allowed:

The number of codes $= 12 \times 12 \times 10 = 1440$

(b) Repetition of the letters is not allowed:

The number of codes $= 12 \times 11 \times 10 = 1320$

(c) The two letters must be the same:

The number of codes $= 12 \times 1 \times 10 = 120$

(d) The two letters are different, but both vowels or both consonants. There are 12 letters: 3 vowels (A, E, I) and 9 consonants.

Two vowels: $3 \times 2 \times 10 = 60$

Two consonants: $9 \times 8 \times 10 = 720$

The number of codes $= 60 + 720$
$= 780$

(a) 1440; (b) 1320; (c) 120; (d) 780

6.

a. Express $\frac{1}{5 \cdot 4 \cdot 3}$ in factorial form.

b. Evaluate $2 \cdot 9! + 8^2 \cdot 8!$.

SOLUTION

(a) $\frac{1}{5 \cdot 4 \cdot 3} = \frac{2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$

$= \frac{2!}{5!}$

(b) $2 \cdot 9! + 8^2 \cdot 8!$

$= 2 \cdot 9 \cdot 8! + 8^2 \cdot 8!$

$= (18 + 8^2) \cdot 8!$

$= 100 \cdot 40320$

$= 4032000$

(a) $\frac{2!}{5!}$; (b) 4032000

7. Prove that $\frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} = \frac{50}{7!}$.

SOLUTION

Prove that $\frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} = \frac{50}{7!}$.

L.H.S. $= \frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!}$

$= \frac{7 \cdot 6}{7 \cdot 6 \cdot 5!} + \frac{7}{7 \cdot 6!} + \frac{1}{7!}$

$= \frac{42}{7!} + \frac{7}{7!} + \frac{1}{7!}$

$= \frac{42 + 7 + 1}{7!}$

$= \frac{50}{7!}$ = R.H.S.

Hence proved.

Section 5.2

EXAMPLE 8 Evaluate ${}^{10}P_5 + {}^{10}P_0$.

SOLUTION

${}^{10}P_5 + {}^{10}P_0 = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 + 1$

$= 30240 + 1$

$= 30241$

EXAMPLE 9 Solve the equations for n.

a. ${}^{n}P_2 = 9n$

b. ${}^nP_3 = 12 \cdot {}^nP_2$

SOLUTION

$n \geq 2$ and

(a) ${}^nP_2 = 9n$

$n(n-1) = 9n$

$n - 1 = 9$

$n = 10$

$n \geq 3$ and

(b) ${}^nP_3 = 12 \cdot {}^nP_2$

$n(n-1)(n-2) = 12n(n-1)$

$n - 2 = 12$

$n = 14$

EXAMPLE 10 In how many ways can a president, a treasurer and a secretary for a committee be selected from a group of 15 people?

SOLUTION

The number of ways to select people for three different positions (ranks) is the number of permutations of 15 people taken 3 at a time, and hence it is

${}^{15}P_3 = 15 \cdot 14 \cdot 13$

${}^{15}P_3 = 2730$

EXAMPLE 11 In how many ways can all the letters of the word PENCIL be arranged, without repeating any letters?

SOLUTION

There are 6 distinct letters. So the number of ways to arrange the letters is

${}^6P_6 = 6!$

$= 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$= 720$

EXAMPLE 12 There are 2 buses, which have 5 and 4 vacant seats respectively, and 4 people at a bus stop. In how many ways can all these people be seated on either of the buses, but not both?

SOLUTION

The first bus has 5 vacant seats, so the number of ways to be seated there is

${}^5P_4 = 5 \cdot 4 \cdot 3 \cdot 2$

$= 120$

The second bus has 4 vacant seats, so the number of ways to be seated there is

$4! = 4 \cdot 3 \cdot 2 \cdot 1$

$= 24$

Therefore, the required number of ways is 120 + 24 = 144.

EXAMPLE 13 In how many ways can 6 different books be arranged along a line on a shelf if one of the books is a dictionary and it must be at one end?

SOLUTION

The dictionary must be fixed at the 1st or the 6th position in the arrangements.

If the dictionary is in the 1st position, the number of ways to arrange all the books in remaining positions is

$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$= 120$

If the dictionary is in the 6th position, the number of ways to arrange all the books in remaining positions is

$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$

$= 120$

So, the required number of ways = (1 × 120) + (120 × 1) = 240.

EXERCISE 5.2

1. Solve the equations for n.

a. ${}^nP_2 = 42$

b. ${}^nP_3 = 9 \cdot {}^nP_2$

SOLUTION

(a) ${}^nP_2 = 42$

$n(n-1) = 42 = 7 \cdot 6$

$n = 7 \quad (n \geq 2)$

(b) ${}^nP_3 = 9 \cdot {}^nP_2$

$n(n-1)(n-2) = 9 \cdot n(n-1) \quad (n \geq 3)$

$n - 2 = 9$

$n = 11$

(a) n = 7; (b) n = 11

2. A newspaper has 14 reporters available to cover 3 different stories. In how many ways can the reporters be assigned to cover the stories, if no reporter can be assigned to cover more than one story?

SOLUTION

A newspaper has 14 reporters. No reporter can be assigned to cover more than one story.

The number of ways to assign $= {}^{14}P_3$

$= 14 \times 13 \times 12$

$= 2184$

2184 ways

3. Suppose we have to make a signal by choosing 4 different flags out of 9 different coloured flags and arranging them in a row. How many different signals can we do?

SOLUTION

There are 9 different coloured flags and we choose 4 of them, arranging them in a row.

The number of different signals $= {}^9P_4$

$= 9 \times 8 \times 7 \times 6$

$= 3024$

3024 signals

4. The manager of 4 movie theaters is deciding which of 12 available movies to show. The theaters have different seating capacities. How many ways can he show 4 different movies in the theaters at the same time?

SOLUTION

There are 12 movies; the 4 theaters have different seating capacities, so order matters.

The number of ways to show $= {}^{12}P_4$

$= 12 \times 11 \times 10 \times 9$

$= 11880$

11880 ways

5. A classroom has two rows of eight seats each. There are 10 students, 5 of whom want to sit in the front row, 4 want to sit in the back row and the remaining student can sit in any seat. In how many ways can the students be seated?

SOLUTION

There are 10 students. Front row (8 seats): 5 students; back row (8 seats): 4 students; the remaining 1 student can sit in any of the remaining seats.

5 students can be seated in the front row in ${}^8P_5$ ways.

4 students can be seated in the back row in ${}^8P_4$ ways.

The remaining 1 student can be seated in the remaining 7 seats in ${}^7P_1$ ways.

Total number of ways $= {}^8P_5 \times {}^8P_4 \times {}^7P_1$

$= (8\times7\times6\times5\times4) \times (8\times7\times6\times5) \times 7$

$= 79027200$

79027200 ways

Section 5.3

EXAMPLE 14 In how many ways can a committee of 4 people be selected from a group of 10 people?

SOLUTION

The number of ways is

$${}^{10}C_4 = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1}$$

$${}^{10}C_4 = 210$$

EXAMPLE 15 Evaluate ${}^{21}C_1$, ${}^{21}C_{21}$, ${}^{21}C_{19}$ and ${}^{21}C_2$.

SOLUTION

$${}^{21}C_1 = 21$$

$${}^{21}C_{21} = \frac{21!}{21!(21-21)!} = \frac{21!}{21! \cdot 0!} = \frac{21!}{21! \cdot 1} = 1$$

$${}^{21}C_{19} = \frac{21!}{19!(21-19)!} = \frac{21!}{19!\,2!} = \frac{21 \cdot 20 \cdot 19!}{19! \cdot 2 \cdot 1} = 210$$

$${}^{21}C_2 = \frac{21!}{2!(21-2)!} = \frac{21!}{2!\,19!} = \frac{21 \cdot 20 \cdot 19!}{2 \cdot 1 \cdot 19!} = 210$$

We notice that ${}^{21}C_1 = 21$, ${}^{21}C_{21} = 1$ and ${}^{21}C_{19} = {}^{21}C_2 = {}^{21}C_{21-19}$.

EXAMPLE 16 A music class consists of 5 piano players, 7 guitarists and 4 violinists. A band of 1 piano player, 3 guitarists and 2 violinists must be chosen to play at a school concert. In how many ways can the band be chosen?

SOLUTION

From the music class,

1 piano player can be chosen in ${}^5C_1$ ways,

3 guitarists can be chosen in ${}^7C_3$ ways, and

2 violinists can be chosen in ${}^4C_2$ ways.

So, the number of ways to choose a band is

$${}^5C_1 \cdot {}^7C_3 \cdot {}^4C_2 = 5 \cdot \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} \cdot \frac{4 \cdot 3}{2 \cdot 1}$$

$${}^5C_1 \cdot {}^7C_3 \cdot {}^4C_2 = 1050$$

EXAMPLE 17 Suppose there are 4 black cars and 7 white cars. If all the cars are distinguishable, in how many ways can 3 cars of the same color be chosen?

SOLUTION

To get the same color, the 3 cars must be all black or all white.

Out of 4 black cars, three can be chosen in ${}^4C_3$ ways.

Out of 7 white cars, three can be chosen in ${}^7C_3$ ways.

So, the required number of ways is

$${}^4C_3 + {}^7C_3 = \frac{4 \cdot 3 \cdot 2}{3 \cdot 2 \cdot 1} + \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 4 + 35$$

$${}^4C_3 + {}^7C_3 = 39$$

EXAMPLE 18 There are 6 different books. In how many ways can the books be given to 3 children, if the youngest wants to receive 3 books, the elder 1 and the eldest 2 respectively?

SOLUTION

For the youngest child, the books can be chosen in
$^6C_3 = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20$ ways.

For the elder child, the books can be chosen in $^3C_1 = 3$ ways.

For the eldest child, the books can be chosen in $^2C_2 = 1$ way.

So, the required number of ways is $20 \cdot 3 \cdot 1 = 60$.

EXAMPLE 19 In how many ways can 4 fruits be selected out of 9 fruits, so as always to:

a. include the largest fruit? (Assume that such a largest fruit exists.)
b. exclude the smallest fruit? (Assume that such a smallest fruit exists.)

SOLUTION

(a) The largest one is included in every selection. The number of ways only depends on the choice of the other three from the remaining 8 fruits, and hence it is

$$^8C_3 = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$$

(b) Since the smallest one is excluded in every selection, we have to select 4 fruits from the remaining 8 fruits. So the required number of ways is

$$^8C_4 = \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 70$$

EXERCISE 5.3

1. Show that $^{12}C_5 \cdot {}^7C_4 = {}^{12}C_4 \cdot {}^8C_5$.

SOLUTION

Show that $^{12}C_5 \cdot {}^7C_4 = {}^{12}C_4 \cdot {}^8C_5$.

$$\text{L.H.S.} = {}^{12}C_5 \times {}^7C_4 = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \cdot \frac{7 \cdot 6 \cdot 5 \cdot 4}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1} \cdot \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = {}^{12}C_4 \cdot {}^8C_5 = \text{R.H.S.}$$

Hence proved. (Both sides = 27720)

2. There are 10 candle holders, which are fixed in different locations along a line and each can hold only one candle. In how many ways can 7 identical candles be put in these holders?

SOLUTION

There are 10 candle holders. 7 identical candles are put in 7 of them, so we only choose which 7 holders are used.

Number of ways in which 7 identical candles can be put $= {}^{10}C_7 = {}^{10}C_3$

$$= \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$

120 ways

3. There are 3 parts in a test. Each of the first two parts contains 5 questions, but the last part only 4. If a student must answer all from the first part, 4 and 3 questions from the second and the last parts respectively, in how many ways can this be done?

SOLUTION

There are 3 parts in a test: the first two parts have 5 questions each, the last part has 4 questions. The student must answer all 5 from the first part, 4 from the second part and 3 from the last part.

First part: $^5C_5 = 1$

Second part: $^5C_4 = {}^5C_1 = 5$

Last part: $^4C_3 = {}^4C_1 = 4$

The required number of ways $= 1 \times 5 \times 4 = 20$

20 ways

4. How many games can be played in a 9-team sport league if each team plays all other teams once?

SOLUTION

For each game, two teams are included.

The number of games can be played $= {}^9C_2$

$$= \frac{9 \cdot 8}{2 \cdot 1} = 36$$

36 games

5. How many lines are determined by 8 points, if no 3 such points are collinear? How many triangles are determined by these points?

SOLUTION

Since no 3 points are collinear, every pair of points determines a distinct line.

The number of lines determined by 8 points $= {}^8C_2$

$$= \frac{8 \times 7}{2 \times 1} = 28$$

We have 8 points and we have to choose 3 out of them.

The number of triangles $= {}^8C_3$

$$= \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$$

28 lines; 56 triangles

6. In how many ways can 4 fruits be selected out of 9 fruits, having different sizes, so as always to include the largest fruit and exclude the smallest fruit?

SOLUTION

Since the largest fruit is included and the smallest fruit is excluded for every selection, we need to choose 3 fruits out of the remaining 7 fruits.

The number of ways $= {}^7C_3$

$= \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1}$

$= 35$

35 ways

Section 5.4

EXAMPLE 20 In how many ways can a permutation of all the letters of the word EXCELLENCE be formed?

SOLUTION

In the word EXCELLENCE, there are 10 letters consisting of four E's, one X, two C's, two L's and one N. So the number of ways is

$$\frac{10!}{4!\,1!\,2!\,2!\,1!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4!}{4! \cdot 1 \cdot (2 \cdot 1) \cdot (2 \cdot 1) \cdot 1}$$

$$= 10 \cdot 9 \cdot 2 \cdot 7 \cdot 6 \cdot 5$$

$$= 37800$$

EXAMPLE 21 How many permutations are there of the letters of the word PROGRAM, if they do not end in: (a) 2 R's? (b) MAP?

SOLUTION

In the word PROGRAM, there are 7 letters consisting of one P, two R's, one O, one G, one A and one M, so the number of permutations of the letters is

$$\frac{7!}{2!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2!}{2!}$$

$$= 2520$$

(Hereafter the factor 1! may be omitted in the denominators of such fractions.)

(a) The number of permutations ending in two R's is $5! = 120$, since the two R's are fixed at the end.

The number of permutations not ending in two R's $= 2520 - 120 = 2400$.

(b) The number of permutations ending in MAP is

$$\frac{4!}{2!} = \frac{4 \cdot 3 \cdot 2!}{2!}$$

$$= 12$$

since MAP is fixed at the end, and there are two R's in the remaining 4 letters.

The number of permutations not ending in MAP = 2520 − 12 = 2508.

EXAMPLE 22 If A is a set containing 9 distinct elements, how many subsets of A contain

a. at most 2 elements?

b. at least 3 elements?

SOLUTION

(a) The number of subsets containing no element, 1 element and 2 elements are ${}^9C_0$, ${}^9C_1$ and ${}^9C_2$ respectively. So the number of subsets containing at most 2 elements is

$${}^9C_0 + {}^9C_1 + {}^9C_2 = 1 + 9 + \frac{9 \cdot 8}{2 \cdot 1}$$

$$= 1 + 9 + 36$$

$$= 46$$

(b) The number of all the subsets of A is $2^9 = 512$.

The number of subsets containing at least 3 elements = 512 − 46 = 466.

EXAMPLE 23 How many 4-digit even numbers, greater than 4000, can be formed using the digits 1, 2, 3, 4 and 5, without repeating any digit?

SOLUTION

Since the numbers are greater than 4000, the first digit must be 4 or 5.

Case 1. If the first digit is 4 and the last digit is 2:

First digit	Second digit	Third digit	Last digit
4	1 or 3 or 5	any one of the remaining two	2
1 way	3 ways	2 ways	1 way

the numbers can be formed in $1 \cdot 3 \cdot 2 \cdot 1 = 6$ ways.

Case 2. If the first digit is 5 and the last digit is 2 or 4:

First digit	Second digit	Third digit	Last digit
5	any one of three digits 1 or 3 or (4 or 2)	any one of the remaining two	2 or 4
1 way	3 ways	2 ways	2 ways

the numbers can be formed in $1 \cdot 3 \cdot 2 \cdot 2 = 12$ ways.

So, the number of 4-digit even numbers greater than 4000 is $6 + 12 = 18$.

EXAMPLE 24 In how many ways can 2 different chemistry books, 4 different mathematics books and 3 different physics books be arranged in a line on a shelf if

a. 2 chemistry books are to be placed on the left, 4 mathematics books in the middle and 3 physics books on the right?

b. books of the same subjects are together?

SOLUTION

(a) The number of ways to place 2 chemistry books in the left part is $2! = 2 \cdot 1 = 2$.

The number of ways to place 3 physics books in the right part is $3! = 3 \cdot 2 \cdot 1 = 6$.

The number of ways to place 4 mathematics books in the middle part is $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$.

The required number of ways $= 2 \cdot 24 \cdot 6 = 288$.

(b) The 3 subject-wise groups can be placed in $3! = 6$ ways, and in each such arrangement the books can be placed in 288 ways (as in part (a)).

The required number of ways to arrange books of the same subject together = 6 · 288 = 1728.

EXAMPLE 25 How many permutations of the letters S, U, N, D, A, Y are there if the two vowels are placed together?

SOLUTION

If the two vowels U and A unite to form a new single letter, then the number of permutations of the letters is $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$. Again, U and A can

be arranged in 2 different ways as UA and AU in each such permutation.

So the required number of permutations is 120 · 2 = 240.

EXAMPLE 26 Find the number of permutations of all the letters of the word INTERNET in such a way that there are exactly 4 letters between the two T's.

SOLUTION

The letters must be arranged in one of the three forms:

T ∗ ∗ ∗ ∗ T ∗ ∗ ∗ T ∗ ∗ ∗ ∗ ∗ T ∗ ∗ ∗ T ∗ ∗ ∗ ∗ T

in which the 6 letters (2 N's, 2 E's, 1 I and 1 R) other than 2 T's are to be put in star (∗) positions. For each such form, the number of ways to put the six letters in star (∗) positions is

$$\frac{6!}{2! \cdot 2!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2!}{2! \cdot 2 \cdot 1}$$
$$= 180$$

So the required number of permutations is 3 · 180 = 540.

EXERCISE 5.4

1. Find the number of permutations of the letters of each of the following words.

i. INFINITY
ii. PINEAPPLE
iii. ACCEPTABLE

SOLUTION

(i) INFINITY: 8 letters containing three I's, two N's, one F, one T and one Y.

The number of permutations of the letters $= \frac{8!}{3!\,2!}$

$= \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3!}{3!(2 \cdot 1)}$

$= 3360$

(ii) PINEAPPLE: 9 letters containing three P's, two E's, one I, one N, one A and one L.

The number of permutations of the letters $= \frac{9!}{3!\,2!}$

$= \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3!}{3!(2 \cdot 1)}$

$= 30240$

(iii) ACCEPTABLE: 10 letters containing two A's, two C's, two E's, one P, one T, one B and one L.

The number of permutations of the letters $= \frac{10!}{2!\,2!\,2!}$

$= \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2!}{2!(2 \cdot 1) \cdot (2 \cdot 1)}$

$= 453600$

(i) 3360; (ii) 30240; (iii) 453600

2. Passwords for a mobile payment system are to consist of 6 digits from a set of digits 0, 1, 2, ..., 9 and assume that there is no other restriction. How many such passwords have repeated digits?

SOLUTION

There are 10 digits (0, 1, 2, ..., 9).

The number of passwords

$= 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 10^6$

$= 1000000$

The number of passwords that have not repeated digits

$= {}^{10}P_6$

$= 10 \times 9 \times 8 \times 7 \times 6 \times 5$

$= 151200$

∴ The required number of passwords which have repeated digits

$= 1000000 - 151200$

$= 848800$

848800 passwords

3. A building has 10 windows in front. What is the number of signals that can be given, by having one or more of the windows open?

SOLUTION

There are 2 possible states for each window. [Each window can either be open or closed.]

The number of signals $= 2^{10}$

$= 1024$

The number of signals with all window closed $= 1$

∴ The required number of signals $= 1024 - 1$

$= 1023$

1023 signals

4. At a burger shop, 3 different types of buns, 7 types of cheeses and 5 types of vegetables are available. Assume that a burger may contain no cheese or no vegetable. In how many ways can different types of burgers be ordered if a person must have a bun, and may have at most 2 types of cheeses and any number of types vegetables.

SOLUTION

For 3 different types of buns, the number of ways:

$= {}^3C_1$

$= 3$

For 7 types of cheeses, the number of ways that at most 2 types of cheeses:

$= {}^7C_0 + {}^7C_1 + {}^7C_2$

$= 1 + 7 + 21$

$= 29$

For 5 types of vegetables, the number of ways (any number of types):

$= {}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$

$= 2^5$

$= 32$

∴ The required number of ways = 3 × 29 × 32
= 2784

2784 ways

5. How many different 4-digit codes can be formed using all the digits 0, 1, 2, 3 if

i. there is no restriction?
ii. repetition is not allowed?
iii. repetition is not allowed, and 0 is either the first or the last digit?

SOLUTION

Total there are 4 digits (0, 1, 2, 3), from which we have to form 4-digit numbers.

(i) If there is no restriction, each of the 4 places can be filled in 4 ways:

The number of different 4 digit codes = 4 × 4 × 4 × 4
= 256

(ii) If repetition is not allowed:

The number of different 4 digit codes = $^4P_4$
= 4!
= 4 × 3 × 2 × 1
= 24

(iii) If repetition is not allowed and 0 is either the first or the last digit.

Case 1: if the first digit is 0, the remaining places can be filled in 3, 2, 1 ways:

The number can be formed = 1 × 3 × 2 × 1
= 6

Case 2: if the last digit is 0:

The number can be formed = 3 × 2 × 1 × 1
= 6

∴ The number of different 4 digit codes can be forme
= 6 + 6
= 12

(i) 256; (ii) 24; (iii) 12

6. Using the letters of the word EQUATION without repetitions, how many 4-letter codes can be formed:

i. starting with T and ending with N?
ii. starting and ending with a consonant?
iii. with vowels only?
iv. if it contains 3 consonants?

SOLUTION

Total there are 8 letters (E, Q, U, A, T, I, O, N), from which we have to form 4-letter codes. Vowels: E, U, A, I, O (5); consonants: Q, T, N (3).

(i) Starting with T and ending with N: first place 1 way (T), last place 1 way (N), the two middle places from the remaining 6 letters in 6 ways, then 5 ways.

The number of 4-letter codes can be formed
= 1 × 6 × 5 × 1
= 30

(ii) Starting and ending with a consonant: the consonants are to occupy first and last place, which can be done in $^3P_2$ ways. We are left with 5 vowels and 1 consonant, i.e. 6 letters, of which two can be arranged in the middle in $^6P_2$ ways.

The number of different 4-letter codes = $^3P_2 \times {}^6P_2$
= (3 × 2) × (6 × 5)
= 180

(iii) With vowels only (E, U, A, I, O):

The number of 4-letter codes can be formed = $^5P_4$
= 5 × 4 × 3 × 2
= 120

(iv) If it contains 3 consonants (Q, T, N) and 1 vowel: choose the position of the vowel in 4 ways; for the other 3 places arrange the 3 consonants in 3, 2, 1 ways; the vowel can be any of 5 vowels.

The number of 4-letter codes can be formed
= 3 × 2 × 1 × 5
= 30

∴ The required number of codes = 4 × 30
= 120

(i) 30; (ii) 180; (iii) 120; (iv) 120

7. Three brothers and three sisters are lining up to be photographed. How many arrangements are there

i. with 3 sisters standing together?
ii. if brothers and sisters are in alternating positions?

SOLUTION

(i) With 3 sisters standing together: consider all 3 sisters as one bundle, then we have a total of 4 elements (3 brothers + 1 bundle), which can be arranged in 4! ways. In the sisters' case, the three sisters can be arranged in 3! ways.

The required number of arrangements = 4! × 3!
= (4·3·2·1) × (3·2·1)
= 24 × 6
= 144

(ii) If brothers and sisters are in alternating positions, there are two patterns: B G B G B G or G B G B G B.

B G B G B G
G B G B G B

For each pattern: 3 × 3 × 2 × 2 × 1 × 1 = 36

The required number of arrangements = 2 × 36
= 72

(i) 144; (ii) 72

8. How many permutations of the letters H, E, X, A, G, O, N are there if

i. the 3 vowels are placed together?
ii. the 3 vowels are not placed together?
iii. consonants and vowels do not appear alternately?

SOLUTION

The given word contains 3 vowels (E, A, O) and 4 consonants (H, X, G, N).

(i) The 3 vowels are placed together: taking 3 vowels as one letter and 4 more letters can be arranged in 5! ways. In the vowels' case, three vowels can be arranged in 3! ways.

The required number of permutations $= 5! \times 3!$
$= (5 \cdot 4 \cdot 3 \cdot 2 \cdot 1) \times (3 \cdot 2 \cdot 1)$
$= 120 \times 6$
$= 720$

(ii) The 3 vowels are not placed together:

The number of permutations $= {}^{7}P_{7} = 7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$
$= 5040$

The required number $= 5040 - 720$
$= 4320$

(vowels not together = total − vowels together)

(iii) Consonants and vowels do not appear alternately. Alternating is only possible as C V C V C V C.

C V C V C V C

The number of arrangements $= 4 \times 3 \times 3 \times 2 \times 2 \times 1 \times 1$
$= 144$

The required number of permutations $= 5040 - 144$
$= 4896$

(not alternate = total − alternate)

(i) 720; (ii) 4320; (iii) 4896

9. Find the number of permutations of all the letters of the word PENGUIN in such a way that there are exactly 3 letters between 2 N's.

SOLUTION

PENGUIN has 7 letters with two N's. Exactly 3 letters between the 2 N's:

(i) The letters must be arranged in one of the three forms (* = any of the other letters):

N * * * N * *
* N * * * N *
* * N * * * N

For each such form, the number of ways to put the 5 letters (P, E, G, U, I) in star (*) positions is 5! = 5·4·3·2·1 = 120.

∴ The required number of permutations $= 3 \times 120$
$= 360$

360

10. Find the number of permutations of all the letters of the word STRESSLESS in such a way that there are exactly 5 letters between:

i. 2 E's.
ii. 2 S's.

SOLUTION

STRESSLESS has 10 letters: 5 S's, 2 E's, T, R, L.

(i) Exactly 5 letters between the 2 E's. The letters must be arranged in one of the four forms:

E * * * * * E * * *
* E * * * * * E * *
* * E * * * * * E *
* * * E * * * * * E

For each such form, the 8 letters (5 S's, 1 T, 1 R, 1 L) other than 2 E's are to be put in star (*) positions. The number of ways is

$$\frac{8!}{5!} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{5!}$$
$$= 336$$

The required number of permutations $= 4 \times 336$
$= 1344$

(ii) Exactly 5 letters between 2 S's. The letters must be arranged in one of the four forms (S positions: 1&7, 2&8, 3&9, 4&10):

S * * * * * S * * *
* S * * * * * S * *
* * S * * * * * S *
* * * S * * * * * S

For each form, the 8 letters (3 S's, 2 E's, T, R, L) are put in star positions:

$$\frac{8!}{3!\,2!} = 3360$$

Some permutations are counted twice: those that fit two of the forms at the same time (for 1 and 2, 1 and 3, 1 and 4, 2 and 3, 2 and 4, 3 and 4 — 6 pairs). In each such case 4 S's are fixed and the remaining 6 letters (1 S, 2 E's, T, R, L) can be arranged in

$$\frac{6!}{2!} = 360$$

∴ The required number of permutations

$$= 4 \times \frac{8!}{3!\,2!} - 6 \times \frac{6!}{2!}$$
$$= 4(3360) - 6(360)$$
$$= 13440 - 2160$$
$$= 11280$$

(i) 1344; (ii) 11280

CH.06 Conic Sections

Notes · Examples · Exercises

NOTES

Conic Section

A conic section (or simply a conic) is a curve obtained by intersecting a plane with a double right-circular cone. The main conics are circles, ellipses, parabolas and hyperbolas, each satisfying the general equation $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$.

Circle

A circle is the set of all points in a plane that are equidistant from a fixed point, called the **center**, where the fixed distance is called the **radius**.

Parabola

A parabola is the set of all points (x, y) in the plane that are equidistant from a fixed line (the **directrix**) and a fixed point not on that line (the **focus**).

Vertex and Axis of a Parabola

The line through the focus and perpendicular to the directrix is the **axis** of the parabola. The point where the axis meets the parabola is the **vertex** — it lies exactly halfway between the focus and the directrix. A parabola is symmetric about its axis.

Latus Rectum

The line segment that passes through the focus and is parallel to the directrix is called the **latus rectum**. Its endpoints lie on the parabola, and if p is the distance from vertex to focus, the latus rectum has length 4p, extending 2p above and 2p below the axis.

Rotation of Axes

In a rotation of axes, the origin stays fixed while the x- and y-axes are rotated through an angle θ to new positions, denoted x′ and y′. Rotating axes lets us eliminate the xy-term from a conic's general equation.

General (Conic) Equation

$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$

Standard Form of the Equation of a Circle

$(x-h)^2+(y-k)^2=r^2$

Where center = (h, k) and radius = r.

General Form of the Equation of a Circle

$x^2+y^2+2gx+2fy+c=0$

Center = $(-g, -f)$, Radius = $\sqrt{g^2+f^2-c}$

Degenerate Cases of a Circle

For $x^2+y^2+2gx+2fy+c=0$:

1. If $g^2+f^2-c > 0$, the graph is a circle with center $(-g,-f)$ and radius $\sqrt{g^2+f^2-c}$.
2. If $g^2+f^2-c = 0$, the graph is a single point $(-g,-f)$.
3. If $g^2+f^2-c < 0$, the equation has no real solution and no graph.

Standard Forms of a Parabola — Vertex at (0,0)

Equation	Vertex	Focus	Directrix	Axis	Description
$y^2=4px$	(0,0)	(p,0)	x=−p	y=0	opens right
$y^2=-4px$	(0,0)	(−p,0)	x=p	y=0	opens left
$x^2=4py$	(0,0)	(0,p)	y=−p	x=0	opens up
$x^2=-4py$	(0,0)	(0,−p)	y=p	x=0	opens down

Standard Forms of a Parabola — Vertex at (h,k)

Equation	Vertex	Focus	Directrix	Axis	Description
$(y-k)^2=4p(x-h)$	(h,k)	(h+p,k)	x=h−p	y=k	opens right
$(y-k)^2=-4p(x-h)$	(h,k)	(h−p,k)	x=h+p	y=k	opens left
$(x-h)^2=4p(y-k)$	(h,k)	(h,k+p)	y=k−p	x=h	opens up
$(x-h)^2=-4p(y-k)$	(h,k)	(h,k−p)	y=k+p	x=h	opens down

Rotation of Axes Formulas

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

$x = x'\cos\theta - y'\sin\theta$

$y = x'\sin\theta + y'\cos\theta$

Eliminating the x′y′-Term (Rotation Angle & Invariants)

To remove the xy-term from $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ ($B\neq0$), rotate by θ where:

$$\cot 2\theta = \frac{A-C}{B} \quad (\theta=45° \text{ if } B(\cot 2\theta)=0, \text{ i.e. } A=C)$$

Rotation invariants:

1. $F = F'$
2. $A + C = A' + C'$
3. $B^2-4AC = B'^2-4A'C'$

Calculator tip: On the calculator, press **SHIFT** then **+** for Pol(x,y) and enter A−C,B (in that order) — it returns r and θ=2θ directly, without computing cos2θ by hand first. If B is negative, enter C−A,−B instead (flip both signs), so the angle comes back in the usual 0°–180° range.

Section 6.2

EXAMPLE 1 Find an equation of the circle of radius 4 centered at (−1, 2) in standard form.

SOLUTION

r=4, (h,k)=(−1,2)

$(x-h)^2+(y-k)^2=r^2$

The standard equation of the circle is

$(x+1)^2+(y-2)^2=16$

EXAMPLE 2 Find an equation of the circle with center (−1, 2) that passes through (3, −1).

SOLUTION

(h,k)=(−1,2), point on circle = (3,−1)

$r=\sqrt{(-1-3)^2+(2+1)^2}$

$=\sqrt{16+9}$

$=\sqrt{25}$

$=5$

$(x-h)^2+(y-k)^2=r^2$

The equation of the circle is

$(x+1)^2+(y-2)^2=25$

EXAMPLE 3 Find the center and radius of each of the following circles.

a. $(x-3)^2+(y-5)^2=16$

b. $(x+7)^2+(y-2)^2=9$

c. $x^2+y^2=1$

d. $x^2+(y-3)^2=49$

SOLUTION

Equation	Center (h,k)	Radius r
$(x-3)^2+(y-5)^2=16$	(3, 5)	4
$(x+7)^2+(y-2)^2=9$	(−7, 2)	3
$x^2+y^2=1$	(0, 0)	1
$x^2+(y-3)^2=49$	(0, 3)	7

EXAMPLE 4 Find the center and radius of the following circles.

a. $x^2+y^2-2x+10y+23=0$

b. $2x^2+2y^2-24x+47=0$

SOLUTION

(a) $x^2+y^2-2x+10y+23=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-2 \rightarrow g=-1$, $2f=10 \rightarrow f=5$, $c=23$

Center=(−g,−f)=(1,−5)

$r=\sqrt{g^2+f^2-c}$

$=\sqrt{1+25-23}$

$=\sqrt{3}$

(b) $2x^2+2y^2-24x+47=0 \rightarrow x^2+y^2-12x+\frac{47}{2}=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-12 \rightarrow g=-6$, $2f=0 \rightarrow f=0$, $c=\frac{47}{2}$

Center=(−g,−f)=(6,0)

$r=\sqrt{g^2+f^2-c}$

$=\sqrt{36-\frac{47}{2}}$

$=\sqrt{\frac{25}{2}}$

$=\frac{5\sqrt{2}}{2}$

EXAMPLE 5 Describe the graphs of the following equations.

a. $(x-1)^2+(y+4)^2=-9$

b. $(x-1)^2+(y+4)^2=0$

c. $x^2+y^2-4x+2y-44=0$

SOLUTION

(a) $r^2=-9<0$.

The right-hand side is negative, so **the equation has no graph**.

(b) $r^2=0 \rightarrow r=0, h=1, k=-4$.

The graph is the single point (1,−4).

(c) Comparing $x^2+y^2-4x+2y-44=0$ with $x^2+y^2+2gx+2fy+c=0$:

$2g=-4 \rightarrow g=-2$

$2f=2 \rightarrow f=1$

$c=-44$

$g^2+f^2-c=4+1+44=49>0$

The graph is a circle with center (2,−1) and radius 7.

EXERCISE 6.1

1.(a) Find the center and radius of each circle:$x^2+y^2=16$

SOLUTION

$x^2+y^2=16$

Comparing with $(x-h)^2+(y-k)^2=r^2$:

h=0, k=0, r=4

Center = (0,0), **Radius** r = 4.

1.(b) Find the center and radius of each circle:$(x-1)^2+(y-4)^2=25$

SOLUTION

$(x-1)^2+(y-4)^2=25$

Comparing with $(x-h)^2+(y-k)^2=r^2$:

h=1, k=4, r=5

Center = (1,4), **Radius** r = 5.

1.(c) Find the center and radius of each circle:$(x-3)^2+(y+5)^2=49$

SOLUTION

$(x-3)^2+(y+5)^2=49$

Comparing with $(x-h)^2+(y-k)^2=r^2$:

h=3, k=−5, r=7

Center = (3,−5), **Radius** r = 7.

1.(d) Find the center and radius of each circle:$(x+1)^2+y^2=1$

SOLUTION

$(x+1)^2+y^2=1$

Comparing with $(x-h)^2+(y-k)^2=r^2$:

h=−1, k=0, r=1

Center = (−1,0), **Radius** r = 1.

1.(e) Find the center and radius of each circle:$2x^2+2y^2-12x+8y-24=0$

SOLUTION

$2x^2+2y^2-12x+8y-24=0$

Dividing by 2: $x^2+y^2-6x+4y-12=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-6 \rightarrow g=-3$

$2f=4 \rightarrow f=2$

$c=-12$

Center = (−g,−f) = (3,−2)

$$\text{Radius } r=\sqrt{g^2+f^2-c}=\sqrt{9+4+12}=\sqrt{25}=5$$

1.(f) Find the center and radius of each circle:$x^2+y^2+x+y-\frac{1}{2}=0$

SOLUTION

$x^2+y^2+x+y-\frac{1}{2}=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=1 \rightarrow g=\frac{1}{2}$

$2f=1 \rightarrow f=\frac{1}{2}$

$c=-\frac{1}{2}$

Center = $(-g,-f)=(-\frac{1}{2},-\frac{1}{2})$

$$\text{Radius } r=\sqrt{g^2+f^2-c}=\sqrt{\frac{1}{4}+\frac{1}{4}+\frac{1}{2}}=\sqrt{1}=1$$

2.(a) Find the standard equation of the circle satisfying the given conditions.center (3, −2), radius = 4.

SOLUTION

Center (h,k)=(3,−2), radius r=4.

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x-3)^2+(y+2)^2=16$

2.(b) Find the standard equation of the circle satisfying the given conditions.center (2, 5), radius = $\sqrt{5}$.

SOLUTION

Center (h,k)=(2,5), radius $r=\sqrt{5}$.

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x-2)^2+(y-5)^2=5$

2.(c) Find the standard equation of the circle satisfying the given conditions.center (−4, 1), radius = 3.

SOLUTION

Center (h,k)=(−4,1), radius r=3.

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x+4)^2+(y-1)^2=9$

2.(d) Find the standard equation of the circle satisfying the given conditions.center (−4, 8), circle is tangent to the x-axis.

SOLUTION

Center (h,k)=(−4,8). Since the circle is tangent to the x-axis, the distance from the center to the x-axis is 8 units, so radius r=8.

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x+4)^2+(y-8)^2=64$

2.(e) Find the standard equation of the circle satisfying the given conditions.center (5, 8), circle is tangent to the y-axis.

SOLUTION

Center (h,k)=(5,8). Since the circle is tangent to the y-axis, the distance from the center to the y-axis is 5 units, so radius r=5.

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x-5)^2+(y-8)^2=25$

2.(f) Find the standard equation of the circle satisfying the given conditions.center (−3, −2), circle passes through the origin.

SOLUTION

Center (h,k)=(−3,−2). Since the circle passes through the origin (0,0):

$r=\sqrt{(0+3)^2+(0+2)^2}=\sqrt{9+4}$

$=\sqrt{13}$

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x+3)^2+(y+2)^2=13$

2.(g) Find the standard equation of the circle satisfying the given conditions.center (4, −5), circle passes through (1, 3).

SOLUTION

Center (h,k)=(4,−5). Since the circle passes through (1,3):

$r=\sqrt{(4-1)^2+(-5-3)^2}$

$=\sqrt{9+64}$

$=\sqrt{73}$

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x-4)^2+(y+5)^2=73$

2.(h) Find the standard equation of the circle satisfying the given conditions.diameter has end points (6, 1) and (−2, 3).

SOLUTION

Center (h,k) = midpoint of (6,1) and (−2,3) = $(\frac{6-2}{2}, \frac{1+3}{2})$ = (2,2).

Using (2,2) and (6,1):

$r=\sqrt{(2-6)^2+(2-1)^2}$

$=\sqrt{16+1}$

$=\sqrt{17}$

Standard equation of a circle is:

$(x-h)^2+(y-k)^2=r^2$

$(x-2)^2+(y-2)^2=17$

3.(a) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius.$x^2+y^2-6x+2y+9=0$

SOLUTION

$x^2+y^2-6x+2y+9=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-6 \rightarrow g$

$=-3$

$2f=2 \rightarrow f$

$=1$

$c=9$

If $g^2+f^2-c=9+1-9=1>0$,

Center $(-g,-f)=(3,-1)$, **radius** $r=1$.

The graph is a circle with center (3,−1) and radius 1.

3.(b) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius.$x^2+y^2-4x-6y+13=0$

SOLUTION

$x^2+y^2-4x-6y+13=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-4 \rightarrow g$

$=-2$

$2f=-6 \rightarrow f$

$=-3$

$c=13$

If $g^2+f^2-c=4+9-13=0$,

The graph is a single point (−g,−f) = (2,3).

3.(c) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius.$x^2+y^2+10y+26=0$

SOLUTION

$x^2+y^2+10y+26=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=0 \rightarrow g$

$=0$

$2f=10 \rightarrow f$

$=5$

$c=26$

If $g^2+f^2-c=25-26=-1<0$,

The equation has no graph.

3.(d) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius.$2x^2+2y^2-12x+8y-24=0$

SOLUTION

$2x^2+2y^2-12x+8y-24=0$

Dividing by 2: $x^2+y^2-6x+4y-12=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-6 \rightarrow g$

$=-3$

$2f=4 \rightarrow f$

$=2$

$c=-12$

If $g^2+f^2-c=9+4+12=25>0$,

Center $(-g,-f)=(3,-2)$, **radius** $r=\sqrt{25}=5$.

The graph is a circle with center (3,−2) and radius 5.

3.(e) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius.$x^2+y^2-10x-2y+29=0$

SOLUTION

$x^2+y^2-10x-2y+29=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g=-10 \rightarrow g$

$=-5$

$2f = -2 \rightarrow f$

$=-1$

$c = 29$

If $g^2+f^2-c=25+1-29=-3<0$,

The equation has no graph.

3.(f) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius. $16x^2+16y^2+40x+16y-7=0$

SOLUTION

$16x^2+16y^2+40x+16y-7=0$

Dividing by 16: $x^2+y^2+\frac{5}{2}x+y-\frac{7}{16}=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g = \frac{5}{2} \rightarrow g$

$=\frac{5}{4}$

$2f = 1 \rightarrow f$

$=\frac{1}{2}$

$c = -\frac{7}{16}$

If $g^2+f^2-c=\frac{25}{16}+\frac{1}{4}+\frac{7}{16}=\frac{9}{4}>0$,

Center $(-g,-f)=(-\frac{5}{4},-\frac{1}{2})$, **radius** $r=\sqrt{\frac{9}{4}}=\frac{3}{2}$.

The graph is a circle with center $(-\frac{5}{4},-\frac{1}{2})$ and radius $\frac{3}{2}$.

3.(g) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius. $\frac{x^2}{4}+\frac{y^2}{4}=1$

SOLUTION

$\frac{x^2}{4}+\frac{y^2}{4}=1$

$x^2+y^2=4$

$x^2+y^2-4=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g = 0 \rightarrow g$

$=0$

$2f = 0 \rightarrow f$

$=0$

$c = -4$

If $g^2+f^2-c=4>0$,

Center $(-g,-f)=(0,0)$, **radius** $r=\sqrt{4}=2$.

The graph is a circle with center (0,0) and radius 2.

3.(h) Determine whether each equation represents a circle, a point or no graph. If the equation represents a circle, find the center and radius. $2x^2+2y^2+8x+7=0$

SOLUTION

$2x^2+2y^2+8x+7=0$

Dividing by 2: $x^2+y^2+4x+\frac{7}{2}=0$

Comparing with $x^2+y^2+2gx+2fy+c=0$:

$2g = 4 \rightarrow g$

$=2$

$2f = 0 \rightarrow f$

$=0$

$c = \frac{7}{2}$

If $g^2+f^2-c=4+0-\frac{7}{2}=\frac{1}{2}>0$,

Center $(-g,-f)=(-2,0)$, **radius** $r=\sqrt{\frac{1}{2}}=\frac{\sqrt{2}}{2}$.

The graph is a circle with center (−2,0) and radius $\frac{\sqrt{2}}{2}$.

Section 6.3 (Vertex at Origin)

EXAMPLE 6 Sketch the graphs of the parabolas, showing the vertex, focus, directrix and end points of latus rectum of each equation.

a. $x^2=16y$

b. $y^2+8x=0$

SOLUTION

(a) Comparing $x^2=16y$ with $x^2=4py$:

$4p=16 \rightarrow p=4$

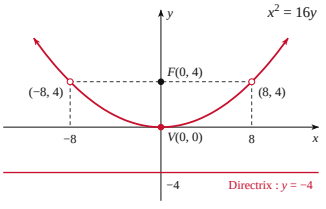
Vertex V $=(0,0)$

Focus F $=(0,4)$

Directrix : $y=-4$

End points of latus rectum $=(-2p,p)$ and $(2p,p)$

$=(-8,4)$ and $(8,4)$



(b) $y^2+8x=0 \rightarrow y^2=-8x$, comparing with $y^2=-4px$:

$4p=8 \rightarrow p=2$

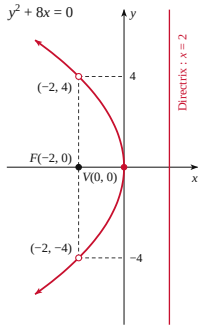
Vertex V $=(0,0)$

Focus F $=(-2,0)$

Directrix : $x=2$

End points of latus rectum $=(-p,-2p)$ and $(-p,2p)$

$=(-2,-4)$ and $(-2,4)$



EXAMPLE 7 Find an equation of the parabola that is symmetric about the y-axis with its vertex at the origin, and passes through (5, 2).

SOLUTION

Symmetric about the y-axis, vertex (0,0), opening upward: $x^2=4py$

At (5,2): $5^2=4p(2) \rightarrow 25=8p \rightarrow 4p=\frac{25}{2}$

$\therefore$ the equation is $x^2=\frac{25}{2}y$

EXERCISE 6.2

1. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y^2=8x$

SOLUTION

$y^2=8x$

Comparing with $y^2=4px$:

$4p=8 \rightarrow p=2$

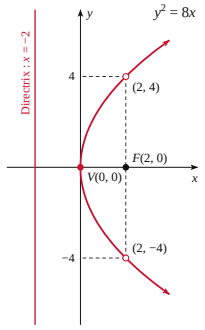
Vertex V =(0,0)

Focus F =(2,0)

Directrix : x=−2

End points of

latus rectum =(2,4) and (2,−4)



2. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y^2=-6x$

SOLUTION

$y^2=-6x$

Comparing with $y^2=-4px$:

$4p=6 \rightarrow p=\frac{3}{2}$

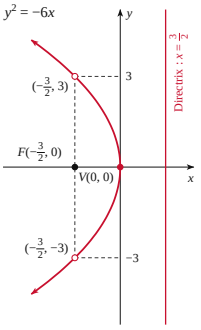
Vertex V =(0,0)

Focus F $=(-\frac{3}{2},0)$

Directrix : $x=\frac{3}{2}$

End points of

latus rectum $=(-\frac{3}{2},3)$ and $(-\frac{3}{2},-3)$



3. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$x+y^2=0$

SOLUTION

$x+y^2=0 \rightarrow y^2=-x$

Comparing with $y^2=-4px$:

$4p=1 \rightarrow p=\frac{1}{4}$

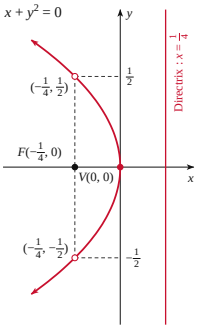
Vertex V =(0,0)

Focus F $=(-\frac{1}{4},0)$

Directrix : $x=\frac{1}{4}$

End points of

latus rectum $=(-\frac{1}{4},\frac{1}{2})$ and $(-\frac{1}{4},-\frac{1}{2})$



4. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y=x^2$

SOLUTION

$y=x^2 \rightarrow x^2=y$

Comparing with $x^2=4py$:

$4p=1 \rightarrow p=\frac{1}{4}$

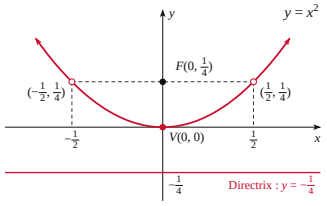
Vertex V =(0,0)

Focus F $=(0,\frac{1}{4})$

Directrix : $y=-\frac{1}{4}$

End points of

latus rectum $=(-\frac{1}{2},\frac{1}{4})$ **and** $(\frac{1}{2},\frac{1}{4})$



5. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y=-x^2$

SOLUTION

$y=-x^2 \rightarrow x^2=-y$

Comparing with $x^2=-4py$:

$4p=1 \rightarrow p=\frac{1}{4}$

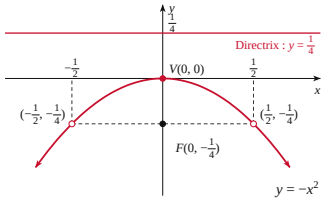
Vertex V $=(0,0)$

Focus F $=(0,-\frac{1}{4})$

Directrix : $y=\frac{1}{4}$

End points of

latus rectum $=(-\frac{1}{2},-\frac{1}{4})$ **and** $(\frac{1}{2},-\frac{1}{4})$



6. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y=3x^2$

SOLUTION

$y=3x^2 \rightarrow x^2=\frac{1}{3}y$

Comparing with $x^2=4py$:

$4p=\frac{1}{3} \rightarrow p=\frac{1}{12}$

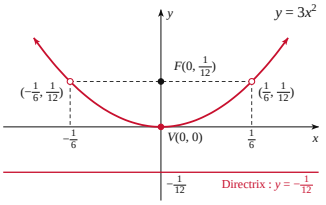
Vertex V $=(0,0)$

Focus F $=(0,\frac{1}{12})$

Directrix : $y=-\frac{1}{12}$

End points of

latus rectum $=(-\frac{1}{6},\frac{1}{12})$ **and** $(\frac{1}{6},\frac{1}{12})$



7. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y=-3x^2$

SOLUTION

$y=-3x^2 \rightarrow x^2=-\frac{1}{3}y$

Comparing with $x^2=-4py$:

$4p=\frac{1}{3} \rightarrow p=\frac{1}{12}$

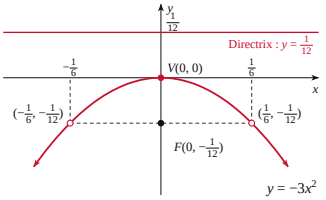
Vertex V $=(0,0)$

Focus F $=(0,-\frac{1}{12})$

Directrix : $y=\frac{1}{12}$

End points of

latus rectum $=(-\frac{1}{6},-\frac{1}{12})$ **and** $(\frac{1}{6},-\frac{1}{12})$



8. In Exercise 1-8, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$x^2+6y=0$

SOLUTION

$x^2+6y=0 \rightarrow x^2=-6y$

Comparing with $x^2=-4py$:

$4p=6 \rightarrow p=\frac{3}{2}$

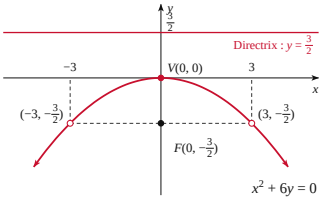
Vertex V $=(0,0)$

Focus F $=(0,-\frac{3}{2})$

Directrix : $y=\frac{3}{2}$

End points of

latus rectum $=(-3,-\frac{3}{2})$ **and** $(3,-\frac{3}{2})$



9. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Vertex (0,0), focus

(3,0).

SOLUTION

Opens right (focus right of vertex). $p=3$. $y^2=4px \rightarrow y^2=12x$

10. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Vertex (0,0), focus (0,−4)

SOLUTION

Opens down. $p=4$. $x^2=-4py \rightarrow x^2=-16y$

11. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Vertex (0,0), directrix $x=7$.

SOLUTION

Directrix right of vertex → opens left, $p=7$. $y^2=-4px \rightarrow y^2=-28x$

12. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Vertex (0,0), directrix $y=\frac{1}{2}$.

SOLUTION

Directrix above vertex → opens down, $p=\frac{1}{2}$.

$x^2=-4py \rightarrow x^2=-2y$

13. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Focus (0,−3), directrix $y=3$.

SOLUTION

Opens down, $p=3$. $x^2=-4py \rightarrow x^2=-12y$

14. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Focus (6,0), directrix $x=-6$.

SOLUTION

Opens right, $p=6$. $y^2=4px \rightarrow y^2=24x$

15. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Vertex (0,0); symmetric about the x-axis; passes through (2,2).

SOLUTION

Opens right: $y^2=4px$. At (2,2): $4=8p \rightarrow 4p=2$. $y^2=2x$

16. In Exercise 9-16, find an equation for the parabola satisfying the given conditions.Vertex (0,0); symmetric about the y-axis; passes through (−1,3).

SOLUTION

Opens up: $x^2=4py$. At (−1,3): $1=12p \rightarrow 4p=\frac{1}{3}$. $x^2=\frac{1}{3}y$

Section 6.3 (Vertex at (h,k))

EXAMPLE 8 Write the given equation $y^2-12x-6y-3=0$ in standard form. Sketch its graph showing the vertex, focus, directrix and end points of latus rectum.

SOLUTION

$$\begin{aligned} y^2-6y &= 12x+3 \\ y^2-6y+9 &= 12x+3+9 \\ (y-3)^2 &= 12x+12 \\ &= 12(x+1) \end{aligned}$$

Comparing with $(y-k)^2=4p(x-h)$:

$$\begin{aligned} k &= 3 \\ 4p &= 12 \rightarrow p \\ &= 3 \\ h &= -1 \end{aligned}$$

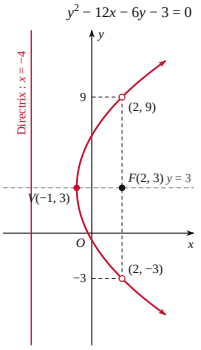
Vertex V $=(h,k)=(-1,3)$

Focus F $=(h+p,k)=(2,3)$

Directrix : $x=h-p=-4$

End points of latus rectum $=(h+p,k+2p)$ and $(h+p,k-2p)$

$=(2,9)$ and $(2,-3)$



EXAMPLE 9 Write the given parabola equation $x^2-2x+8y-23=0$ in standard form. Sketch its graph showing the vertex, focus, directrix and end points of latus rectum.

SOLUTION

$$\begin{aligned} x^2-2x &= -8y+23 \\ x^2-2x+1 &= -8y+23+1 \\ (x-1)^2 &= -8y+24 \\ &= -8(y-3) \end{aligned}$$

Comparing with $(x-h)^2=-4p(y-k)$:

$$\begin{aligned} h &= 1 \\ k &= 3 \\ 4p &= 8 \rightarrow p \\ &= 2 \end{aligned}$$

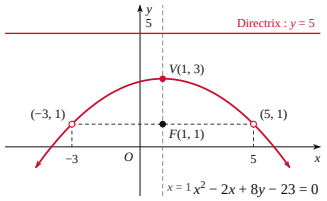
Vertex V $=(h,k)=(1,3)$

Focus F $=(h,k-p)=(1,1)$

Directrix : $y=k+p=5$

End points of latus rectum $=(h-2p,k-p)$ and $(h+2p,k-p)$

$=(-3,1)$ and $(5,1)$



EXAMPLE 10 Find the standard form of the equation of the parabola with vertex (1, 2) and focus at (4, 2).

SOLUTION

Focus is right of vertex on a horizontal line → opens right.

h =1

k =2

Focus (h+p,k)=(4,2) → 1+p=4 → p=3

$(y-k)^2=4p(x-h)$ → $(y-2)^2=12(x-1)$

EXAMPLE 11 Find the general form of the equation of the parabola with vertex (2, 1) and focus (2, 4).

SOLUTION

Focus is above vertex on a vertical line → opens up.

h =2

k =1

Focus (h,k+p)=(2,4) → 1+p=4 → p=3

$(x-h)^2=4p(y-k)$ → $(x-2)^2=12(y-1)$

$x^2-4x+4=12y-12$ → $x^2-4x-12y+16=0$ **(general form)**

EXERCISE 6.3

1. In Exercise 1-6, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$x^2+4x+4y=0$

SOLUTION

$x^2+4x+4y=0$ → $(x+2)^2=-4(y-1)$

Comparing with $(x-h)^2=-4p(y-k)$:

h =−2

k =1

p =1

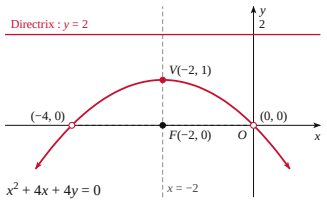
Vertex V =(−2,1)

Focus F =(−2,0)

Directrix : y=2

End points of

latus rectum =(0,0) and (−4,0)



2. In Exercise 1-6, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$(x-1)^2+8(y+1)=0$

SOLUTION

$(x-1)^2+8(y+1)=0$ → $(x-1)^2=-8(y+1)$

Comparing with $(x-h)^2=-4p(y-k)$:

h =1

k =−1

p =2

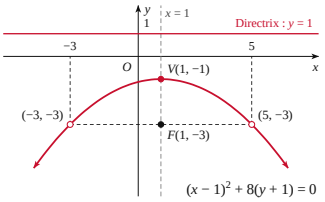
Vertex V =(1,−1)

Focus F =(1,−3)

Directrix : y=1

End points of

latus rectum =(5,−3) and (−3,−3)



3. In Exercise 1-6, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$(x+5)+(y-1)^2=0$

SOLUTION

$(x+5)+(y-1)^2=0$ → $(y-1)^2=-(x+5)$

Comparing with $(y-k)^2=-4p(x-h)$:

h =−5

k =1

4p =1 → p

$=\frac{1}{4}$

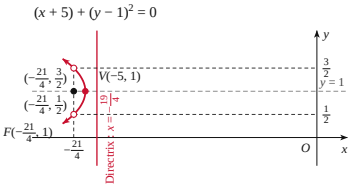
Vertex V =(−5,1)

Focus F $=(-\frac{21}{4},1)$

Directrix : $x=-\frac{19}{4}$

End points of

latus rectum $=(-\frac{21}{4},\frac{3}{2})$ and $(-\frac{21}{4},\frac{1}{2})$



4. In Exercise 1-6, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y^2+6y+8x+25=0$

SOLUTION

$y^2+6y+8x+25=0$ → $(y+3)^2=-8(x+2)$

Comparing with $(y-k)^2=-4p(x-h)$:

h =−2

k =−3

p =2

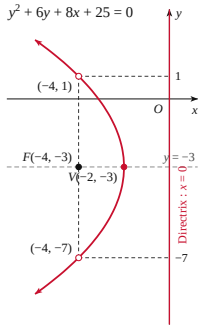
Vertex V =(−2,−3)

Focus F =(−4,−3)

Directrix : x=0

End points of

latus rectum =(−4,1) and (−4,−7)



5. In Exercise 1-6, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$x^2+4x+6y-2=0$

SOLUTION

$x^2+4x+6y-2=0 \rightarrow (x+2)^2=-6(y-1)$

Comparing with $(x-h)^2=-4p(y-k)$:

h =−2

k =1

$p = \frac{3}{2}$

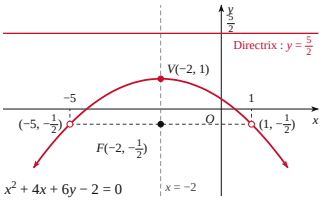
Vertex V =(−2,1)

Focus F $=(-2,-\frac{1}{2})$

Directrix : $y=\frac{5}{2}$

End points of

latus rectum $=(1,-\frac{1}{2})$ and $(-5,-\frac{1}{2})$



6. In Exercise 1-6, (a) sketch the parabola, (b) show the focus, vertex, directrix and end points of latus rectum.$y^2+x+y=0$

SOLUTION

$y^2+x+y=0 \rightarrow (y+\frac{1}{2})^2=-(x-\frac{1}{4})$

Comparing with $(y-k)^2=-4p(x-h)$:

$h = \frac{1}{4}$

$k = -\frac{1}{2}$

4p =1 → p

$= \frac{1}{4}$

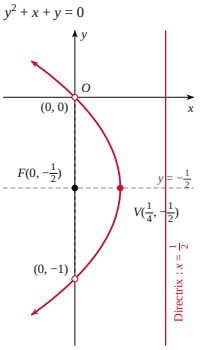
Vertex V $=(\frac{1}{4},-\frac{1}{2})$

Focus F $=(0,-\frac{1}{2})$

Directrix : $x=\frac{1}{2}$

End points of

latus rectum =(0,0) and (0,−1)



7. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Vertex (4,−5), focus (1,−5).

SOLUTION

Horizontal axis, focus left of vertex → opens left.
h=4,k=−5. h−p=1 → p=3.

The equation of the parabola is $(y+5)^2=-12(x-4)$.

8. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Vertex (5,1), focus (5,3).

SOLUTION

Vertical axis, focus above vertex → opens up.
h=5,k=1. k+p=3 → p=2.

The equation of the parabola is $(x-5)^2=8(y-1)$.

9. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Vertex (1,1), directrix y=−2.

SOLUTION

Directrix below vertex → opens up. h=1,k=1. k−p=−2 → p=3.

The equation of the parabola is $(x-1)^2=12(y-1)$.

10. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Vertex (2,−1), directrix x=−1.

SOLUTION

Directrix left of vertex → opens right. h=2,k=−1.
h−p=−1 → p=3.

The equation of the parabola is $(y+1)^2=12(x-2)$.

11. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Focus (−1,4), directrix x=5.

SOLUTION

Focus left of directrix → opens left. k=4, h−p=−1, h+p=5 → h=2, p=3.

The equation of the parabola is $(y-4)^2=-12(x-2)$.

12. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Focus (6,0), directrix x=−6.

SOLUTION

Focus right of directrix → opens right. k=0, h+p=6, h−p=−6 → h=0, p=6.

The equation of the parabola is $y^2=24x$.

13. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Axis x=0 passes through (2,−1) and (−4,5).

SOLUTION

Since the axis of symmetry is x=0 (the y-axis), the vertex lies on the y-axis, so h=0. The equation of the parabola is $x^2=4p(y-k)$.

Substitute (2,−1):

$$\begin{aligned} 2^2 &= 4p(-1-k) \\ 4 &= 4p(-1-k) \\ 1 &= p(-1-k) \quad ...(1) \end{aligned}$$

Substitute (−4,5):

$$\begin{aligned} (-4)^2 &= 4p(5-k) \\ 16 &= 4p(5-k) \\ 4 &= p(5-k) \quad ...(2) \end{aligned}$$

Divide equation (2) by (1):

$$\begin{aligned} \frac{4}{1} &= \frac{p(5-k)}{p(-1-k)} \\ 4(-1-k) &= 5-k \\ -4-4k &= 5-k \\ -3k &= 9 \\ k &= -3 \end{aligned}$$

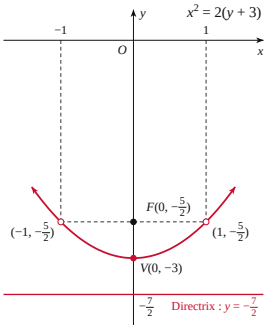
Substitute k=−3 into (1):

$$\begin{aligned} 1 &= p(-1-(-3)) \\ 1 &= 2p \rightarrow 4p \\ &= 2 \end{aligned}$$

The equation of the parabola is:

$$x^2 = 2(y-(-3))$$

$$\mathbf{x^2 = 2(y+3)}$$



14. In Exercise 7-14, find an equation for the parabola satisfying the given conditions.Axis y=0 passes through (3,2) and (2,−3).

SOLUTION

Since it opens to the left and the axis is y=0, the vertex is (h,0). The equation is $(y-k)^2=-4p(x-h)$. Since k=0, we have $y^2=-4p(x-h)$.

Substitute (3,2):

$$\begin{aligned} 2^2 &= -4p(3-h) \\ 4 &= -4p(3-h) \\ 1 &= -p(3-h) \quad ...(1) \end{aligned}$$

Substitute (2,−3):

$$\begin{aligned} (-3)^2 &= -4p(2-h) \\ 9 &= -4p(2-h) \quad ...(2) \end{aligned}$$

Divide equation (2) by (1):

$$\begin{aligned} \frac{9}{1} &= \frac{-4p(2-h)}{-p(3-h)} \\ 9 &= \frac{4(2-h)}{3-h} \\ 9(3-h) &= 4(2-h) \\ 27-9h &= 8-4h \\ -5h &= -19 \rightarrow h \\ &= \frac{19}{5} \end{aligned}$$

Substitute $h=\frac{19}{5}$ into (1):

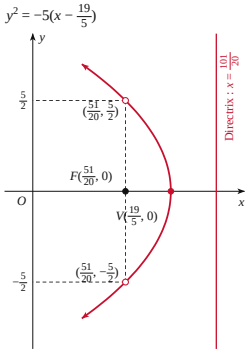
$$\begin{aligned} 1 &= -p(3-\frac{19}{5}) \\ 1 &= -p(\frac{15}{5}-\frac{19}{5}) \\ 1 &= -p(-\frac{4}{5}) \\ 1 &= \frac{4}{5}p \rightarrow p \\ &= \frac{5}{4} \end{aligned}$$

$$\therefore -4p=-4(\frac{5}{4})=-5$$

The equation of the parabola is:

$$y^2 = -5(x-\frac{19}{5})$$

$$\mathbf{y^2 = -5x+19}$$



Section 6.4

EXAMPLE 12 Find the new coordinates of the point (1, 2) if the coordinate axes are rotated through an angle of θ=30°.

SOLUTION

$x=1, y=2, \cos30°=\frac{\sqrt{3}}{2}, \sin30°=\frac{1}{2}$

$x' = x\cos\theta + y\sin\theta$

$= 1\cdot\frac{\sqrt{3}}{2} + 2\cdot\frac{1}{2}$

$= \frac{\sqrt{3}}{2} + 1$

$y' = -x\sin\theta + y\cos\theta$

$= -1\cdot\frac{1}{2} + 2\cdot\frac{\sqrt{3}}{2}$

$= -\frac{1}{2} + \sqrt{3}$

$\therefore$ **new coordinates** $= \left(\frac{\sqrt{3}}{2}+1, -\frac{1}{2}+\sqrt{3}\right)$

EXAMPLE 13 Rotate the coordinate axes to remove the xy-term of the equation $3x^2-2\sqrt{3}xy+y^2+2x+2\sqrt{3}y=0$ in x′y′-coordinates. Then sketch the graph.

SOLUTION

$A=3, B=-2\sqrt{3}, C=1$

Discriminant: $B^2-4AC=12-12=0$ (Parabola)

$\cot2\theta = \frac{A-C}{B}$

$= \frac{2}{-2\sqrt{3}}$

$= -\frac{1}{\sqrt{3}}$

Since $\cot2\theta=-\frac{1}{\sqrt{3}}$, we have $2\theta=120°$, so $\theta=60°$.

$\cos2\theta=-\frac{1}{2}$

$\sin\theta = \sqrt{\frac{1-\cos2\theta}{2}}$

$= \sqrt{\frac{1-(-\frac{1}{2})}{2}}$

$= \sqrt{\frac{3}{4}}$

$= \frac{\sqrt{3}}{2}$

$\cos\theta = \sqrt{\frac{1+\cos2\theta}{2}}$

$= \sqrt{\frac{1+(-\frac{1}{2})}{2}}$

$= \sqrt{\frac{1}{4}}$

$= \frac{1}{2}$

The rotation formulas are:

$x = x'\cos\theta - y'\sin\theta$

$= \frac{1}{2}x' - \frac{\sqrt{3}}{2}y'$

$y = x'\sin\theta + y'\cos\theta$

$= \frac{\sqrt{3}}{2}x' + \frac{1}{2}y'$

Substituting these into the original equation $(\sqrt{3}x-y)^2+2(x+\sqrt{3}y)=0$:

$[\sqrt{3}(\frac{1}{2}x'-\frac{\sqrt{3}}{2}y')-(\frac{\sqrt{3}}{2}x'+\frac{1}{2}y')]^2+2[(\frac{1}{2}x'-\frac{\sqrt{3}}{2}y')+\sqrt{3}(\frac{\sqrt{3}}{2}x'+\frac{1}{2}y')]=0$

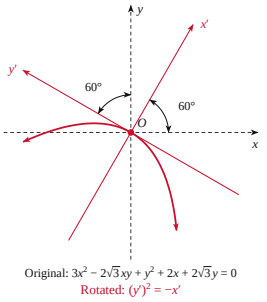
$(\frac{\sqrt{3}}{2}x'-\frac{3}{2}y'-\frac{\sqrt{3}}{2}x'-\frac{1}{2}y')^2+2(\frac{1}{2}x'-\frac{\sqrt{3}}{2}y'+\frac{3}{2}x'+\frac{\sqrt{3}}{2}y')=0$

$(-2y')^2+2(2x') = 0$

$4(y')^2+4x' = 0$

$(y')^2 = -x'$

The new conic is a parabola that opens to the left with vertex (0,0) in the x′y′-system. Since $\sin\theta=\frac{\sqrt{3}}{2}$, the angle of rotation is θ=60°.



EXAMPLE 14 Rotate the coordinate axes to remove the xy-term of the equation $16x^2-24xy+9y^2+100x-200y+100=0$ in x′y′-coordinates. Then sketch the graph.

SOLUTION

$A=16, B=-24, C=9$

Discriminant: $B^2-4AC=576-576=0$ (Parabola)

$\cot2\theta = \frac{A-C}{B}$

$= \frac{16-9}{-24}$

$= -\frac{7}{24}$

Since $\cot2\theta=-\frac{7}{24}$ (2θ in Quadrant II), we have $\cos2\theta=-\frac{7}{25}$.

$$\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1-(-\frac{7}{25})}{2}} = \sqrt{\frac{32}{50}} = \frac{4}{5}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1+(-\frac{7}{25})}{2}} = \sqrt{\frac{18}{50}} = \frac{3}{5}$$

The rotation formulas are:

$$x = x'\cos\theta - y'\sin\theta = \frac{3}{5}x' - \frac{4}{5}y'$$

$$y = x'\sin\theta + y'\cos\theta = \frac{4}{5}x' + \frac{3}{5}y'$$

Since $16x^2-24xy+9y^2=(4x-3y)^2$ and $100x-200y=100(x-2y)$, the original equation can be written as $(4x-3y)^2+100(x-2y)+100=0$.

Substituting the rotation formulas:

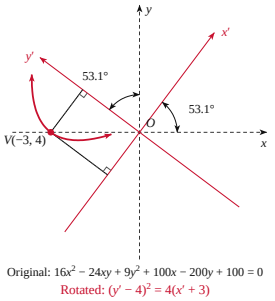
$$4x-3y = 4(\frac{3}{5}x' - \frac{4}{5}y') - 3(\frac{4}{5}x' + \frac{3}{5}y') = \frac{12}{5}x' - \frac{16}{5}y' - \frac{12}{5}x' - \frac{9}{5}y' = -5y'$$

$$x-2y = (\frac{3}{5}x' - \frac{4}{5}y') - 2(\frac{4}{5}x' + \frac{3}{5}y') = \frac{3}{5}x' - \frac{4}{5}y' - \frac{8}{5}x' - \frac{6}{5}y' = -x' - 2y'$$

So the equation becomes:

$$(-5y')^2+100(-x'-2y')+100 = 0$$
$$25(y')^2-100x'-200y'+100 = 0$$
$$(y')^2-4x'-8y'+4 = 0$$
$$(y')^2-8y'+16 = 4x'-4+16$$
$$(y'-4)^2 = 4(x'+3)$$

The new conic is a parabola opening right with vertex $(h',k')=(-3,4)$ in the $x'y'$-plane. Since $\sin\theta=\frac{4}{5}$, the rotation angle is $\theta\approx 53.1°$.



EXERCISE 6.4

1. Rotate the coordinate axes to remove the xy-term. Then sketch the graph. $x^2+2\sqrt{3}xy+3y^2+2\sqrt{3}x-2y=0$

SOLUTION

$A=1, B=2\sqrt{3}, C=3$

$$\cot 2\theta = \frac{A-C}{B} = \frac{1-3}{2\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

$\therefore 2\theta=120° \rightarrow \theta=60°$

The rotation formulas are:

$$x = x'\cos\theta - y'\sin\theta = \frac{1}{2}x' - \frac{\sqrt{3}}{2}y'$$

$$y = x'\sin\theta + y'\cos\theta = \frac{\sqrt{3}}{2}x' + \frac{1}{2}y'$$

Since $x^2+2\sqrt{3}xy+3y^2=(x+\sqrt{3}y)^2$ and $2\sqrt{3}x-2y=2(\sqrt{3}x-y)$, the equation becomes $(x+\sqrt{3}y)^2+2(\sqrt{3}x-y)=0$.

Substituting the rotation formulas:

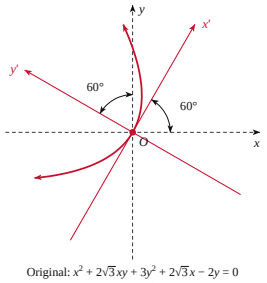
$$x+\sqrt{3}y = (\frac{1}{2}x' - \frac{\sqrt{3}}{2}y') + \sqrt{3}(\frac{\sqrt{3}}{2}x' + \frac{1}{2}y') = \frac{1}{2}x' - \frac{\sqrt{3}}{2}y' + \frac{3}{2}x' + \frac{\sqrt{3}}{2}y' = 2x'$$

$$\sqrt{3}x-y = \sqrt{3}(\frac{1}{2}x' - \frac{\sqrt{3}}{2}y') - (\frac{\sqrt{3}}{2}x' + \frac{1}{2}y') = \frac{\sqrt{3}}{2}x' - \frac{3}{2}y' - \frac{\sqrt{3}}{2}x' - \frac{1}{2}y' = -2y'$$

So the equation becomes:

$$(2x')^2+2(-2y') = 0$$
$$4(x')^2-4y' = 0$$
$$(x')^2 = y'$$

The new conic is a parabola that opens up with vertex $(0,0)$ in the $x'y'$-system. Since $\sin\theta=\frac{\sqrt{3}}{2}$, $\theta=60°$.



2. Rotate the coordinate axes to remove the xy-term. Then sketch the graph. $9x^2-24xy+16y^2-80x-60y+100=0$

SOLUTION

$A=9,\ B=-24,\ C=16$

$$\cot 2\theta = \frac{A-C}{B} = \frac{9-16}{-24} = \frac{7}{24}$$

Since $\cot 2\theta = \frac{7}{24}$ (2θ in Quadrant I), $\cos 2\theta = \frac{7}{25}$.

$$\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \sqrt{\frac{1-\frac{7}{25}}{2}} = \sqrt{\frac{18}{50}} = \frac{3}{5}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \sqrt{\frac{1+\frac{7}{25}}{2}} = \sqrt{\frac{32}{50}} = \frac{4}{5}$$

The rotation formulas are:

$$x = x'\cos\theta - y'\sin\theta = \frac{4}{5}x' - \frac{3}{5}y'$$

$$y = x'\sin\theta + y'\cos\theta = \frac{3}{5}x' + \frac{4}{5}y'$$

Since $9x^2-24xy+16y^2=(3x-4y)^2$ and $-80x-60y=-20(4x+3y)$, the equation becomes $(3x-4y)^2-20(4x+3y)+100=0$.

Substituting the rotation formulas:

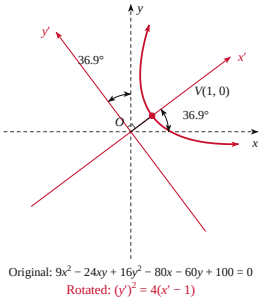
$$3x-4y = 3\left(\frac{4}{5}x' - \frac{3}{5}y'\right) - 4\left(\frac{3}{5}x' + \frac{4}{5}y'\right)$$
$$= \frac{12}{5}x' - \frac{9}{5}y' - \frac{12}{5}x' - \frac{16}{5}y'$$
$$= -5y'$$

$$4x+3y = 4\left(\frac{4}{5}x' - \frac{3}{5}y'\right) + 3\left(\frac{3}{5}x' + \frac{4}{5}y'\right)$$
$$= \frac{16}{5}x' - \frac{12}{5}y' + \frac{9}{5}x' + \frac{12}{5}y'$$
$$= 5x'$$

So the equation becomes:

$$(-5y')^2 - 20(5x') + 100 = 0$$
$$25(y')^2 - 100x' + 100 = 0$$
$$(y')^2 - 4x' + 4 = 0$$
$$(y')^2 = 4x' - 4$$
$$\mathbf{(y')^2 = 4(x'-1)}$$

The new conic is a parabola that opens right with vertex (1,0) in the x′y′-system. Since $\sin\theta = \frac{3}{5}$, $\theta \approx 36.9°$.



3. Rotate the coordinate axes to remove the xy-term. Then sketch the graph. $x^2-2xy+y^2-20x-20y+100=0$

SOLUTION

$A=1,\ B=-2,\ C=1$

$$\cot 2\theta = \frac{A-C}{B} = \frac{1-1}{-2} = 0 \rightarrow 2\theta = 90° \rightarrow \theta = 45°$$

$$\therefore \sin\theta = \cos\theta = \frac{\sqrt{2}}{2}$$

The rotation formulas are:

$$x = x'\cos\theta - y'\sin\theta = \frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y'$$

$$y = x'\sin\theta + y'\cos\theta = \frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y'$$

Since $x^2-2xy+y^2=(x-y)^2$ and $-20x-20y=-20(x+y)$, the equation becomes $(x-y)^2-20(x+y)+100=0$.

Substituting the rotation formulas:

$$x-y = \left(\frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y'\right) - \left(\frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y'\right)$$
$$= -\sqrt{2}y'$$

$$x+y = \left(\frac{\sqrt{2}}{2}x' - \frac{\sqrt{2}}{2}y'\right) + \left(\frac{\sqrt{2}}{2}x' + \frac{\sqrt{2}}{2}y'\right)$$
$$= \sqrt{2}x'$$

So the equation becomes:

$(-\sqrt{2}y')^2-20(\sqrt{2}x')+100 = 0$

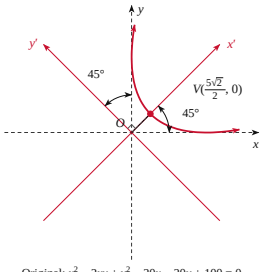
$2(y')^2-20\sqrt{2}x'+100 = 0$

$(y')^2-10\sqrt{2}x'+50 = 0$

$(y')^2 = 10\sqrt{2}x'-50$

$(y')^2 = 10\sqrt{2}(x'-\frac{5\sqrt{2}}{2})$

The new conic is a parabola that opens right with vertex $(\frac{5\sqrt{2}}{2},0)$ in the x′y′-system, with θ=45°.



Original: $x^2 - 2xy + y^2 - 20x - 20y + 100 = 0$
Rotated: $(y')^2 = 10\sqrt{2}(x' - \frac{5\sqrt{2}}{2})$

4. Rotate the coordinate axes to remove the xy-term. Then sketch the graph.$16x^2+24xy+9y^2-130x+90y=0$

SOLUTION

A=16, B=24, C=9

$\cot 2\theta = \frac{A-C}{B}$

$= \frac{16-9}{24}$

$= \frac{7}{24}$

Since $\cot 2\theta = \frac{7}{24}$ (2θ in Quadrant I), $\cos 2\theta = \frac{7}{25}$.

$\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}}$

$= \sqrt{\frac{1-\frac{7}{25}}{2}}$

$= \sqrt{\frac{18}{50}}$

$= \frac{3}{5}$

$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}}$

$= \sqrt{\frac{1+\frac{7}{25}}{2}}$

$= \sqrt{\frac{32}{50}}$

$= \frac{4}{5}$

The rotation formulas are:

$x = x'\cos\theta - y'\sin\theta$

$= \frac{4}{5}x' - \frac{3}{5}y'$

$y = x'\sin\theta + y'\cos\theta$

$= \frac{3}{5}x' + \frac{4}{5}y'$

Since $16x^2+24xy+9y^2=(4x+3y)^2$, the equation becomes $(4x+3y)^2-130x+90y=0$.

Substituting the rotation formulas:

$4x+3y = 4(\frac{4}{5}x' - \frac{3}{5}y') + 3(\frac{3}{5}x' + \frac{4}{5}y')$

$= \frac{16}{5}x' - \frac{12}{5}y' + \frac{9}{5}x' + \frac{12}{5}y'$

$= 5x'$

$-130x = -130(\frac{4}{5}x' - \frac{3}{5}y')$

$= -104x' + 78y'$

$90y = 90(\frac{3}{5}x' + \frac{4}{5}y')$

$= 54x' + 72y'$

$-130x+90y = (-104x'+78y') + (54x'+72y')$

$= -50x' + 150y'$

So the equation becomes:

$(5x')^2-50x'+150y' = 0$

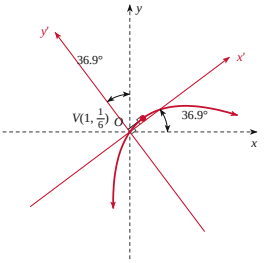
$25(x')^2-50x'+150y' = 0$

$(x')^2-2x'+6y' = 0$

$(x')^2-2x'+1 = -6y'+1$

$(x'-1)^2 = -6(y'-\frac{1}{6})$

The new conic is a parabola that opens down with vertex $(1,\frac{1}{6})$ in the x′y′-system. Since $\sin\theta = \frac{3}{5}$, θ≈36.9°.



Original: $16x^2 + 24xy + 9y^2 - 130x + 90y = 0$
Rotated: $(x' - 1)^2 = -6(y' - \frac{1}{6})$

5. Rotate the coordinate axes to remove the xy-term. Then sketch the graph.$3x^2+2\sqrt{3}xy+y^2-8x+8\sqrt{3}y=0$

SOLUTION

A=3, B=$2\sqrt{3}$, C=1

$\cot 2\theta = \frac{A-C}{B}$

$= \frac{3-1}{2\sqrt{3}}$

$= \frac{1}{\sqrt{3}}$

$\therefore 2\theta=60° \rightarrow \theta=30°$

$\therefore \sin\theta=\frac{1}{2}, \cos\theta=\frac{\sqrt{3}}{2}$

The rotation formulas are:

$x = x'\cos\theta - y'\sin\theta$

$= \frac{\sqrt{3}}{2}x' - \frac{1}{2}y'$

$y = x'\sin\theta + y'\cos\theta$

$= \frac{1}{2}x' + \frac{\sqrt{3}}{2}y'$

Since $3x^2+2\sqrt{3}xy+y^2=(\sqrt{3}x+y)^2$ and $-8x+8\sqrt{3}y=-8(x-\sqrt{3}y)$, the equation becomes $(\sqrt{3}x+y)^2-8(x-\sqrt{3}y)=0$.

Substituting the rotation formulas:

$$\sqrt{3}x+y = \sqrt{3}\left(\frac{\sqrt{3}}{2}x' - \frac{1}{2}y'\right) + \left(\frac{1}{2}x' + \frac{\sqrt{3}}{2}y'\right)$$

$$= \frac{3}{2}x' - \frac{\sqrt{3}}{2}y' + \frac{1}{2}x' + \frac{\sqrt{3}}{2}y'$$

$$= 2x'$$

$$x-\sqrt{3}y = \left(\frac{\sqrt{3}}{2}x' - \frac{1}{2}y'\right) - \sqrt{3}\left(\frac{1}{2}x' + \frac{\sqrt{3}}{2}y'\right)$$

$$= \frac{\sqrt{3}}{2}x' - \frac{1}{2}y' - \frac{\sqrt{3}}{2}x' - \frac{3}{2}y'$$

$$= -2y'$$

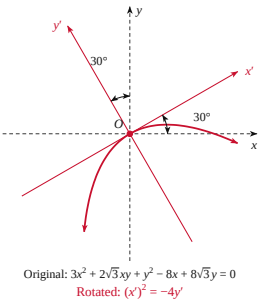
So the equation becomes:

$$(2x')^2 - 8(-2y') = 0$$

$$4(x')^2 + 16y' = 0$$

$$\mathbf{(x')^2 = -4y'}$$

The new conic is a parabola that opens down with vertex (0,0) in the x′y′-system. Since $\theta=30°$.



Original: $3x^2 + 2\sqrt{3}xy + y^2 - 8x + 8\sqrt{3}y = 0$
Rotated: $(x')^2 = -4y'$

CH.07 Trigonometric Functions

Notes · Examples · Exercises

NOTES

Periodic Function and Period

If f(x) = f(x + p), where p is a positive real number, then f is called a **periodic function**. If p is the smallest such number, then p is the **period** of the function f.

The sine function y = sin x is periodic with period 2π, since 2π is the smallest positive real number such that sin x = sin(x + 2π).

Amplitude

The **amplitude** of a periodic function is half the difference between the maximum and minimum values.

So the amplitude of y = sin x is $\frac{1-(-1)}{2} = 1$.

Inverse Trigonometric Functions

Trigonometric functions are not one-to-one, so the domain of each is restricted to make it one-to-one. In these restricted domains the inverse functions are defined:

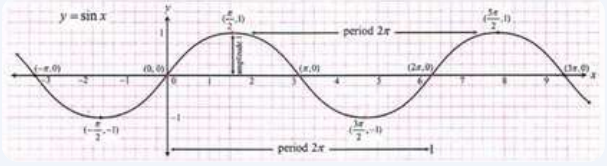
Function	Definition	Conditions
$y = \sin^{-1} x$	sin y = x	$-1 \le x \le 1, \frac{-\pi}{2} \le y \le \frac{\pi}{2}$
$y = \cos^{-1} x$	cos y = x	$-1 \le x \le 1, 0 \le y \le \pi$
$y = \tan^{-1} x$	tan y = x	$x \in R, \frac{-\pi}{2} < y < \frac{\pi}{2}$
$y = \cot^{-1} x$	cot y = x	$x \in R, 0 < y < \pi$
$y = \sec^{-1} x$	sec y = x	$\lvert x\rvert \ge 1, 0 \le y \le \pi, y \ne \frac{\pi}{2}$
$y = \csc^{-1} x$	csc y = x	$\lvert x\rvert \ge 1, \frac{-\pi}{2} \le y \le \frac{\pi}{2}, y \ne 0$

Graph of y = sin x

Domain: R Range: {y | −1 ≤ y ≤ 1} Period: 2π

Amplitude: 1

Five key points: x-intercepts (0, 0), (π, 0), (2π, 0); maximum point $(\frac{\pi}{2}, 1)$; minimum point $(\frac{3\pi}{2}, -1)$.



Graph of y = a sin bx (a > 0, b > 0)

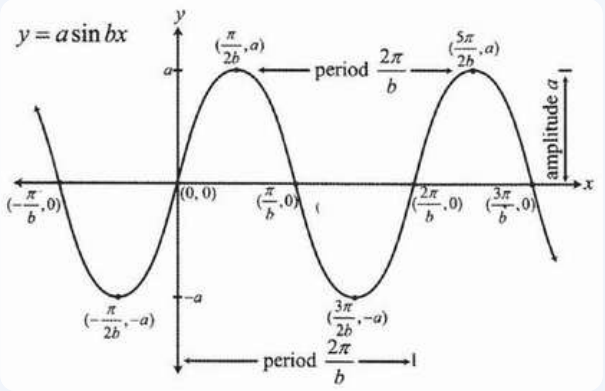
Obtained from y = sin x by vertical scaling (scale factor a) and horizontal scaling (scale factor $\frac{1}{b}$), in either order:

$(x, y) \to (\frac{x}{b}, ay)$

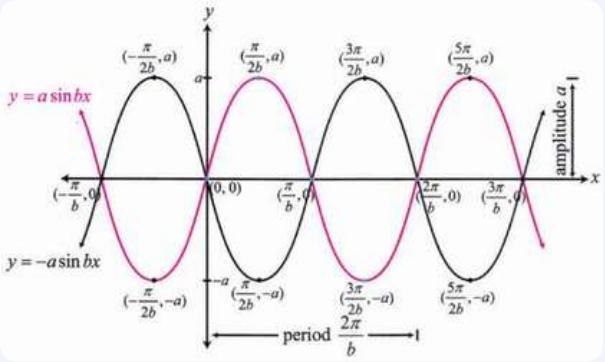
Domain: R Range: {y | −a ≤ y ≤ a} Period: $\frac{2\pi}{b}$

Amplitude: a

Five key points: x-intercepts (0, 0), $(\frac{\pi}{b}, 0)$, $(\frac{2\pi}{b}, 0)$; maximum point $(\frac{\pi}{2b}, a)$; minimum point $(\frac{3\pi}{2b}, -a)$.



y = −a sin bx: reflect y = a sin bx in the x-axis.



Note: a sin(−b)x = −a sin bx, so there is no need to consider b < 0.

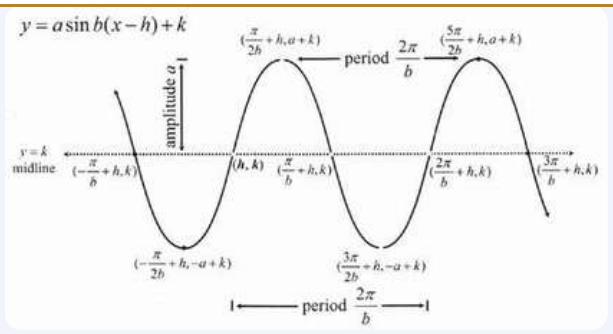
Graph of y = a sin b(x − h) + k

y = sin x → y = a sin bx → y = a sin b(x − h) + k

$(x, y) \to (\frac{x}{b}, ay) \to (\frac{x}{b} + h, ay + k)$

y = sin x	y = a sin bx	y = a sin b(x − h) + k
(0, 0)	(0, 0)	(h, k)
$(\frac{\pi}{2}, 1)$	$(\frac{\pi}{2b}, a)$	$(\frac{\pi}{2b} + h, a + k)$
(π, 0)	$(\frac{\pi}{b}, 0)$	$(\frac{\pi}{b} + h, k)$
$(\frac{3\pi}{2}, -1)$	$(\frac{3\pi}{2b}, -a)$	$(\frac{3\pi}{2b} + h, -a + k)$
(2π, 0)	$(\frac{2\pi}{b}, 0)$	$(\frac{2\pi}{b} + h, k)$

The midline is y = k, amplitude a, period $\frac{2\pi}{b}$.

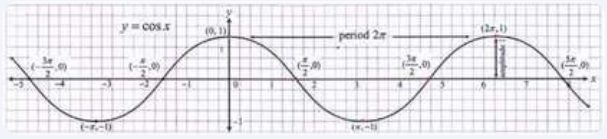


Graph of y = cos x

Domain: R Range: {y | −1 ≤ y ≤ 1} Period: 2π
Amplitude: 1

Five key points: x-intercepts $(\frac{\pi}{2}, 0)$, $(\frac{3\pi}{2}, 0)$; maximum points (0, 1), (2π, 1); minimum point (π, −1).

Note that $\sin(x + \frac{\pi}{2}) = \cos x$: translate y = sin x horizontally by $\frac{-\pi}{2}$ units to get y = cos x.

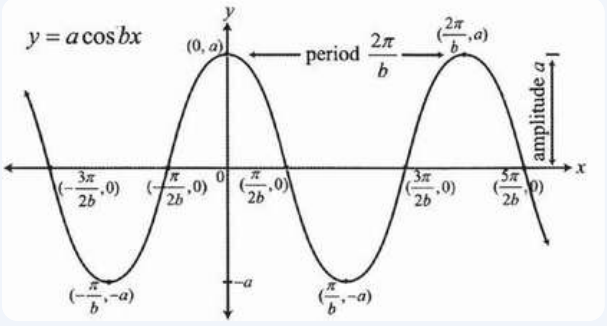


Graph of y = a cos bx and y = a cos b(x − h) + k

From y = cos x: (x, y) → $(\frac{x}{b}, ay)$ gives y = a cos bx (a > 0, b > 0).

Domain: R Range: {y | −a ≤ y ≤ a} Period: $\frac{2\pi}{b}$
Amplitude: a

Five key points: x-intercepts $(\frac{\pi}{2b}, 0)$, $(\frac{3\pi}{2b}, 0)$; maximum points (0, a), $(\frac{2\pi}{b}, a)$; minimum point $(\frac{\pi}{b}, -a)$.

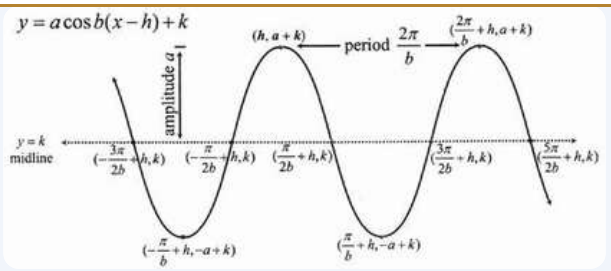


y = −a cos bx: reflect y = a cos bx in the x-axis. Note: a cos(−b)x = a cos bx.

y = a cos b(x − h) + k:

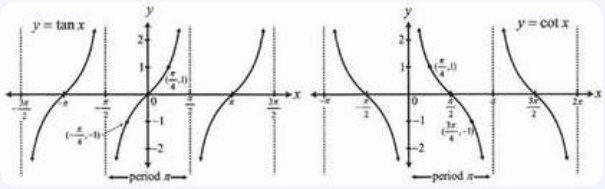
(x, y) → $(\frac{x}{b}, ay)$ → $(\frac{x}{b} + h, ay + k)$

Five key points: points on the midline $(\frac{\pi}{2b} + h, k)$, $(\frac{3\pi}{2b} + h, k)$; maximum points $(h, a + k)$, $(\frac{2\pi}{b} + h, a + k)$; minimum point $(\frac{\pi}{b} + h, -a + k)$.

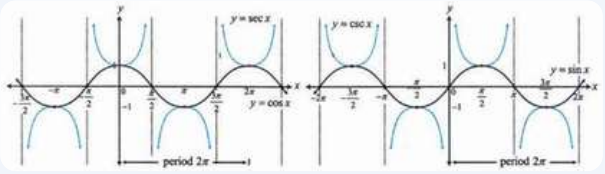


Graphs of tan x, cot x, sec x, csc x

	y = tan x	y = cot x
Domain	$x \neq \frac{\pm\pi}{2}, \frac{\pm3\pi}{2}, \ldots$	x ≠ 0, ±π, ±2π, ...
Range	R	R
Period	π	π
x-intercepts	x = 0, ±π, ±2π, ...	$x = \frac{\pm\pi}{2}, \frac{\pm3\pi}{2}, \ldots$
Asymptotes	$x = \frac{\pm\pi}{2}, \frac{\pm3\pi}{2}, \ldots$	x = 0, ±π, ±2π, ...

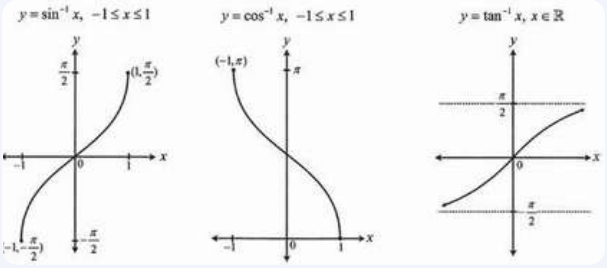


	y = sec x	y = csc x
Domain	$x \neq \frac{\pm\pi}{2}, \frac{\pm3\pi}{2}, \ldots$	x ≠ 0, ±π, ±2π, ...
Range	y ≤ −1 or y ≥ 1	y ≤ −1 or y ≥ 1
Period	2π	2π
Asymptotes	$x = \frac{\pm\pi}{2}, \frac{\pm3\pi}{2}, \ldots$	x = 0, ±π, ±2π, ...



Transforming with (x, y) → $(\frac{x}{b}, ay)$ gives y = a tan bx, y = a cot bx, y = a sec bx and y = a csc bx.

Properties of Inverse Trigonometric Functions



1. $\sin(\sin^{-1} x) = x$ for $-1 \le x \le 1$ (but $\sin^{-1}(\sin x) = x$ is not true in general)
2. $\cos(\cos^{-1} x) = x$ for $-1 \le x \le 1$
3. $\sin^{-1}(-x) = -\sin^{-1} x$ ($y = \sin^{-1} x$ is an odd function)
4. $\cos^{-1}(-x) = \pi - \cos^{-1} x$, i.e. $\cos^{-1} x + \cos^{-1}(-x) = \pi$

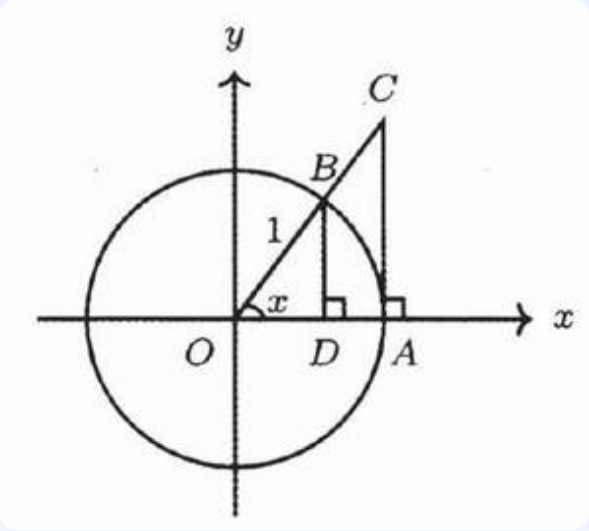
Important Limit

$\lim_{x \to 0} \frac{\sin x}{x} = 1$

Proof: In the unit circle, OA = OB = 1, ∠AOB = x radians, and BD < arc AB < AC. In right △OBD, BD = sin x; in right △OAC, AC = tan x; arc AB = x. So $\sin x < x < \tan x$

$1 < \frac{x}{\sin x} < \frac{1}{\cos x} \Rightarrow 1 > \frac{\sin x}{x} > \cos x$

When $x \to 0$, $\cos x \to 1$. By the squeeze theorem, $\lim_{x\to 0} \frac{\sin x}{x} = 1$.



Derivative of sin x, cos x, tan x (derivations)

Derivative of sin x. Let y = sin x. Then $\delta y = \sin(x + \delta x) - \sin x = 2\cos(x + \frac{\delta x}{2}) \cdot \sin(\frac{\delta x}{2})$, using $\sin\alpha - \sin\beta = 2\cos(\frac{\alpha+\beta}{2})\sin(\frac{\alpha-\beta}{2})$.

$$\frac{dy}{dx} = \lim_{\delta x\to 0} \cos(x + \frac{\delta x}{2}) \cdot \lim_{\delta x\to 0} \frac{\sin(\frac{\delta x}{2})}{\frac{\delta x}{2}} = \cos x \times 1$$

$$\frac{d}{dx}\sin x = \cos x$$

Derivative of cos x. Since $\cos x = \sin(\frac{\pi}{2} + x)$,

$$\frac{d}{dx}\cos x = \cos(\frac{\pi}{2} + x) \cdot 1 = -\sin x$$

Derivative of tan x. Since $\tan x = \frac{\sin x}{\cos x}$, by the quotient rule

$$\frac{d}{dx}\tan x = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

Derivatives of Trigonometric Functions

	Basic	Chain rule form (u = u(x))
1	$\frac{d}{dx}\sin x = \cos x$	$\frac{d}{dx}\sin u = \cos u \cdot \frac{du}{dx}$
2	$\frac{d}{dx}\cos x = -\sin x$	$\frac{d}{dx}\cos u = -\sin u \cdot \frac{du}{dx}$
3	$\frac{d}{dx}\tan x = \sec^2 x$	$\frac{d}{dx}\tan u = \sec^2 u \cdot \frac{du}{dx}$
4	$\frac{d}{dx}\cot x = -\csc^2 x$	$\frac{d}{dx}\cot u = -\csc^2 u \cdot \frac{du}{dx}$
5	$\frac{d}{dx}\sec x = \sec x \cdot \tan x$	$\frac{d}{dx}\sec u = \sec u \cdot \tan u \cdot \frac{du}{dx}$
6	$\frac{d}{dx}\csc x = -\csc x \cdot \cot x$	$\frac{d}{dx}\csc u = -\csc u \cdot \cot u \cdot \frac{du}{dx}$

Section 7.1

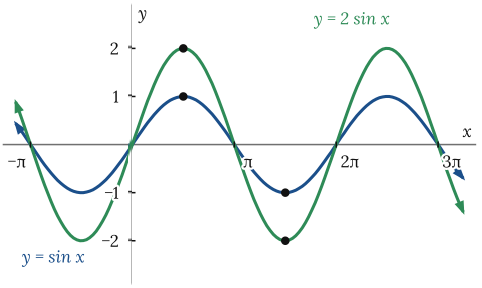
EXAMPLE 1 From the graph of y = sin x, draw step-by-step transformation graphs to get the graph of $y = 2\sin(\frac{\pi}{2}x)$.

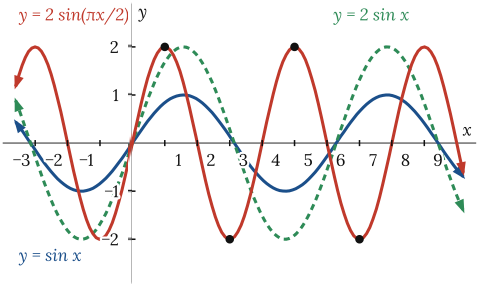
SOLUTION

Method 1

$y = \sin x \longrightarrow y = 2\sin x \longrightarrow y = 2\sin(\frac{\pi}{2}x)$

(vertical scaling, scale factor 2; then horizontal scaling, scale factor $\frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$)

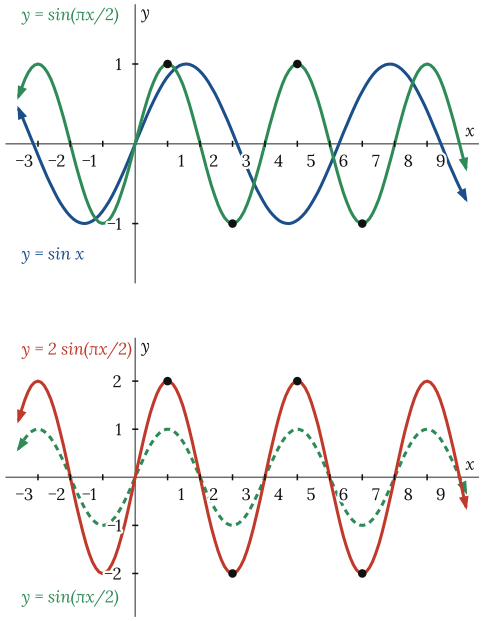




Method 2

$y = \sin x \longrightarrow y = \sin(\frac{\pi}{2}x) \longrightarrow y = 2\sin(\frac{\pi}{2}x)$

(horizontal scaling, scale factor $\frac{2}{\pi}$; then vertical scaling, scale factor 2)



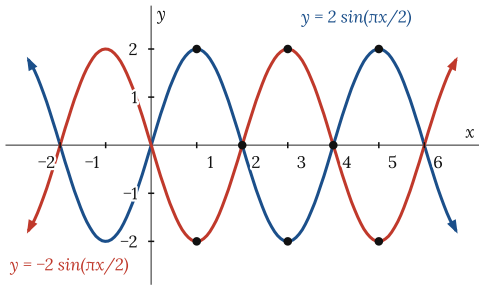
EXAMPLE 2 Draw the graphs of $y = 2\sin(\frac{\pi}{2}x)$ and $y = -2\sin(\frac{\pi}{2}x)$.

SOLUTION

For $y = 2\sin(\frac{\pi}{2}x)$: amplitude 2, period $2\pi \div (\frac{\pi}{2}) = 4$.

Five key points: (0, 0), (1, 2), (2, 0), (3, −2), (4, 0).

$y = -2\sin(\frac{\pi}{2}x)$ is its reflection in the x-axis: (0, 0), (1, −2), (2, 0), (3, 2), (4, 0).

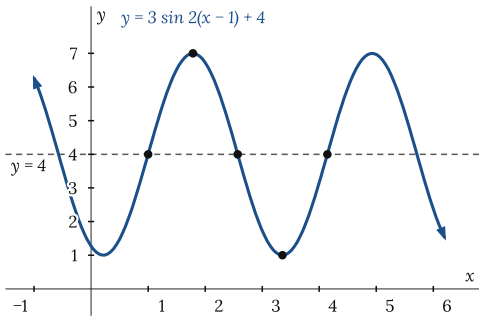


EXAMPLE 3 Draw the graph of $y = 3\sin 2(x - 1) + 4$.

SOLUTION

Midline: **$y = 4$, Amplitude: 3, Period: π**

Five key points: $(1, 4)$, $(\frac{\pi}{4} + 1, 7)$, $(\frac{\pi}{2} + 1, 4)$, $(\frac{3\pi}{4} + 1, 1)$, $(\pi + 1, 4)$

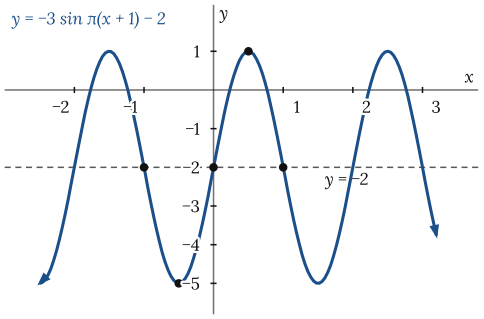


EXAMPLE 4 Draw the graph of $y = -3\sin \pi(x + 1) - 2$.

SOLUTION

Midline: **$y = -2$, Amplitude: 3, Period: 2**

Five key points: $(-1, -2)$, $(\frac{-1}{2}, -5)$, $(0, -2)$, $(\frac{1}{2}, 1)$, $(1, -2)$



EXERCISE 7.1

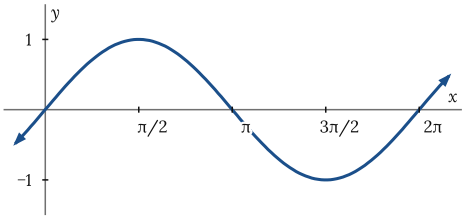
1. From the graph of $y = \sin x$, draw step-by-step transformation graphs to get the graph of $y = -3\sin(\frac{\pi}{3}x)$.

SOLUTION

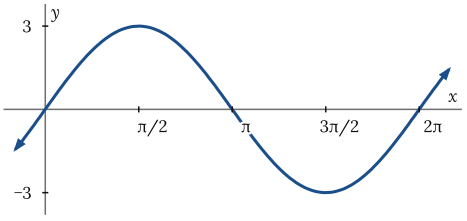
Answer:

Steps: vertical scaling (factor 3), horizontal scaling (factor $\frac{3}{\pi}$), reflection in the x-axis. Amplitude 3, period 6. Key points: $(0, 0)$, $(\frac{3}{2}, -3)$, $(3, 0)$, $(\frac{9}{2}, 3)$, $(6, 0)$

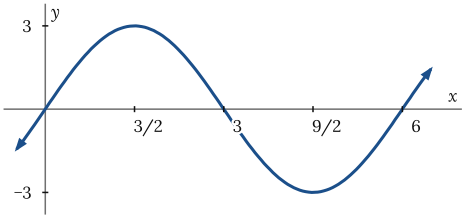
1) $y = \sin x$



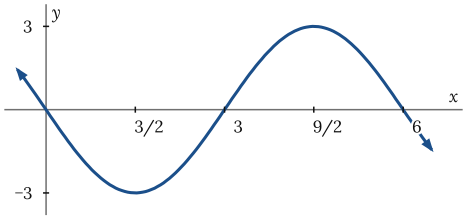
2) $y = 3\sin x$ (vertical scaling, factor 3)



3) $y = 3\sin(\pi x/3)$ (horizontal scaling)



4) y = −3 sin(πx/3) (reflection in the x-axis)



2. Draw the graph of

a. $y = \frac{1}{2}\sin x$

b. $y = \sin 4x$

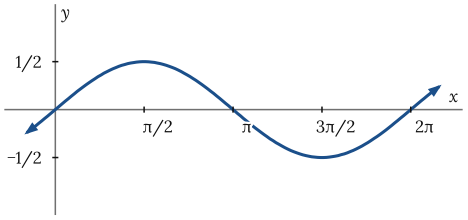
SOLUTION

Answer:

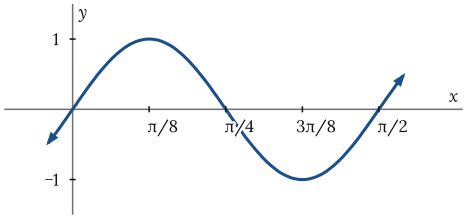
(a) Amplitude $\frac{1}{2}$, period 2π. Key points: (0, 0), $(\frac{\pi}{2}, \frac{1}{2})$, (π, 0), $(\frac{3\pi}{2}, \frac{-1}{2})$, (2π, 0)

(b) Amplitude 1, period $\frac{\pi}{2}$. Key points: (0, 0), $(\frac{\pi}{8}, 1)$, $(\frac{\pi}{4}, 0)$, $(\frac{3\pi}{8}, -1)$, $(\frac{\pi}{2}, 0)$

(a) y = ½ sin x



(b) y = sin 4x



3. Draw the graph of

a. $y = 2\sin(\frac{\pi}{4}x)$

b. $y = -\sin \pi x$.

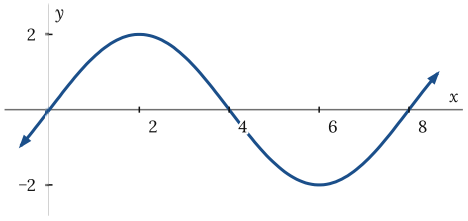
SOLUTION

Answer:

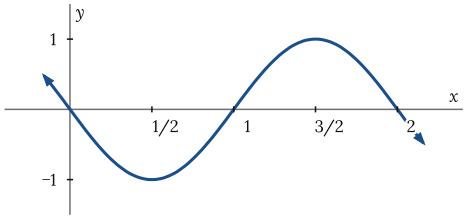
(a) Amplitude 2, period 8. Key points: (0, 0), (2, 2), (4, 0), (6, −2), (8, 0)

(b) Amplitude 1, period 2. Key points: (0, 0), $(\frac{1}{2}, -1)$, (1, 0), $(\frac{3}{2}, 1)$, (2, 0)

(a) y = 2 sin(πx/4)



(b) y = −sin πx



4. Draw the graph of

a. $y = 2\sin\frac{\pi}{3}(x - 2) + 1$

b. $y = -2\sin\frac{1}{2}(x + 1) + 2$.

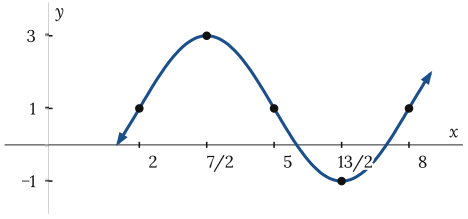
SOLUTION

Answer:

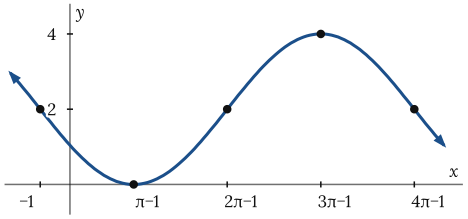
(a) Midline y = 1, amplitude 2, period 6. Key points: (2, 1), $(\frac{7}{2}, 3)$, (5, 1), $(\frac{13}{2}, -1)$, (8, 1)

(b) Midline y = 2, amplitude 2, period 4π. Key points: (−1, 2), (π − 1, 0), (2π − 1, 2), (3π − 1, 4), (4π − 1, 2)

(a) y = 2 sin(π/3 (x − 2)) + 1



(b) y = −2 sin(½(x + 1)) + 2



5. Show that y = a sin bx is an odd function.

SOLUTION

Answer:

Let f(x) = a sin bx. Then f(−x) = a sin(−bx) = −a sin bx = −f(x). Hence f is an odd function.

Section 7.2

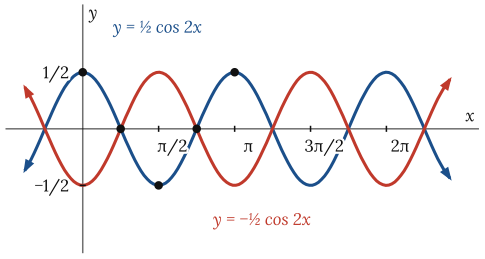
EXAMPLE 5 Draw the graphs of $y = \frac{1}{2}\cos 2x$ and $y = -\frac{1}{2}\cos 2x$.

SOLUTION

For $y = \frac{1}{2}\cos 2x$: amplitude $\frac{1}{2}$, period $\frac{2\pi}{2} = \pi$.

Five key points: $(0, \frac{1}{2}), (\frac{\pi}{4}, 0), (\frac{\pi}{2}, \frac{-1}{2}), (\frac{3\pi}{4}, 0), (\pi, \frac{1}{2})$.

$y = -\frac{1}{2}\cos 2x$ is its reflection in the x-axis: $(0, \frac{-1}{2}), (\frac{\pi}{4}, 0), (\frac{\pi}{2}, \frac{1}{2}), (\frac{3\pi}{4}, 0), (\pi, \frac{-1}{2})$.



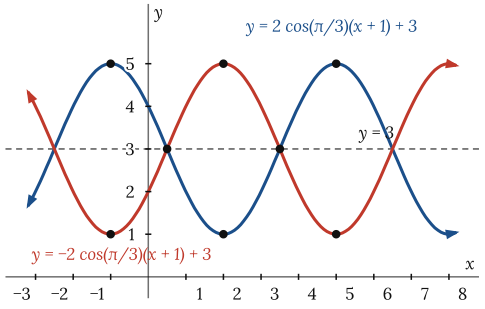
EXAMPLE 6 Draw the graphs of $y = 2\cos(\frac{\pi}{3})(x + 1) + 3$ and $y = -2\cos(\frac{\pi}{3})(x + 1) + 3$.

SOLUTION

Midline: $y = 3$, Amplitude: 2, Period: 6

Five key points for $y = 2\cos(\frac{\pi}{3})(x + 1) + 3$: (−1, 5), (0.5, 3), (2, 1), (3.5, 3), (5, 5)

Five key points for $y = -2\cos(\frac{\pi}{3})(x + 1) + 3$: (−1, 1), (0.5, 3), (2, 5), (3.5, 3), (5, 1)



EXERCISE 7.2

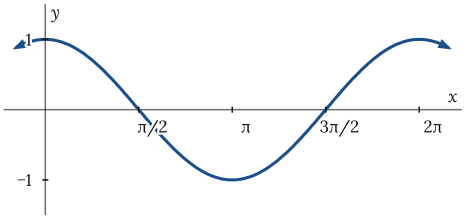
1. From the graph of $y = \cos x$, draw step-by-step transformation graphs to get the graph of $y = -3\cos(\frac{\pi}{2}x)$.

SOLUTION

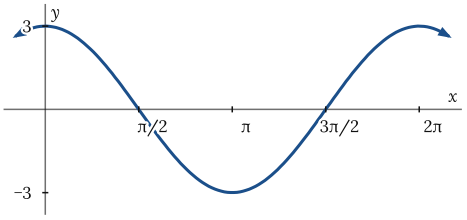
Answer:

Amplitude 3, period 4. Key points: (0, −3), (1, 0), (2, 3), (3, 0), (4, −3)

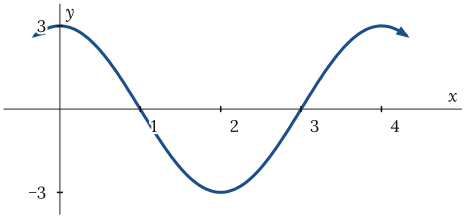
1) y = cos x



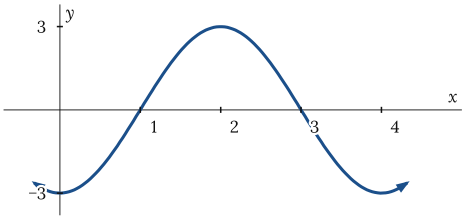
2) y = 3 cos x (vertical scaling, factor 3)



3) y = 3 cos(πx/2) (horizontal scaling)



4) y = −3 cos(πx/2) (reflection in the x-axis)



2. Draw the graph of

a. $y = \frac{1}{2}\cos x$

b. $y = \cos 4x$.

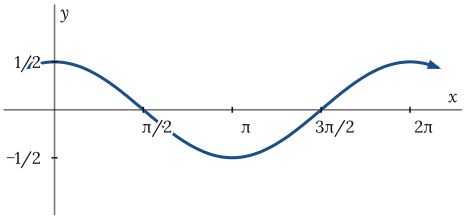
SOLUTION

Answer:

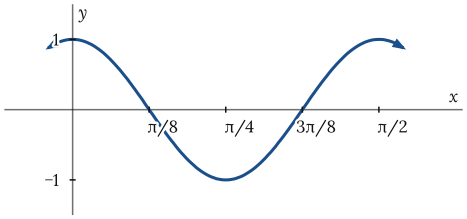
(a) Amplitude $\frac{1}{2}$, period 2π. Key points: $(0, \frac{1}{2}), (\frac{\pi}{2}, 0), (\pi, \frac{-1}{2}), (\frac{3\pi}{2}, 0), (2\pi, \frac{1}{2})$

(b) Amplitude 1, period $\frac{\pi}{2}$. Key points: $(0, 1), (\frac{\pi}{8}, 0), (\frac{\pi}{4}, -1), (\frac{3\pi}{8}, 0), (\frac{\pi}{2}, 1)$

(a) y = ½ cos x



(b) y = cos 4x



3. Draw the graph of

a. $y = 2\cos(\frac{\pi}{4}x)$

b. $y = -\cos \pi x$.

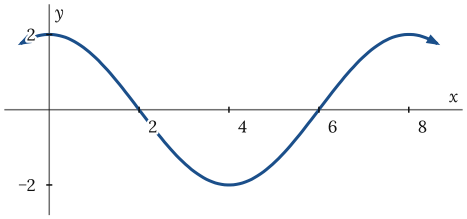
SOLUTION

Answer:

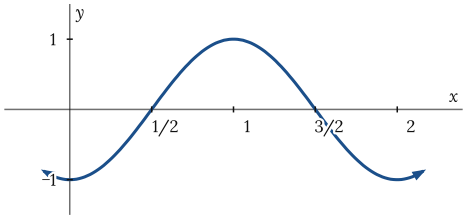
(a) Amplitude 2, period 8. Key points: (0, 2), (2, 0), (4, −2), (6, 0), (8, 2)

(b) Amplitude 1, period 2. Key points: (0, −1), $(\frac{1}{2}, 0)$, (1, 1), $(\frac{3}{2}, 0)$, (2, −1)

(a) y = 2 cos(πx/4)



(b) y = −cos πx



4. Draw the graph of

a. $y = 2\cos \pi(x - 2) + 1$

b. $y = -2\cos \frac{1}{2}(x + 1) + 2$.

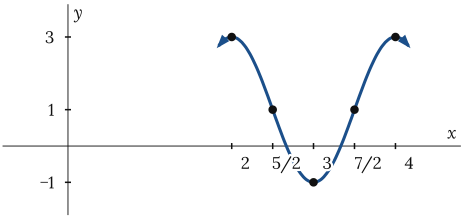
SOLUTION

Answer:

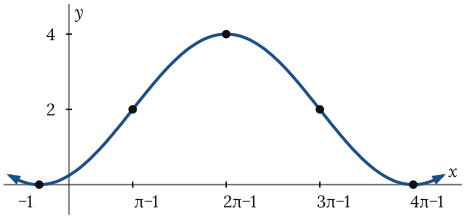
(a) Midline y = 1, amplitude 2, period 2. Key points: (2, 3), $(\frac{5}{2}, 1)$, (3, −1), $(\frac{7}{2}, 1)$, (4, 3)

(b) Midline y = 2, amplitude 2, period 4π. Key points: (−1, 0), (π − 1, 2), (2π − 1, 4), (3π − 1, 2), (4π − 1, 0)

(a) y = 2 cos π(x − 2) + 1



(b) y = −2 cos(½(x + 1)) + 2



5. Show that y = a cos bx is an even function.

SOLUTION

Answer:

Let f(x) = a cos bx. Then f(−x) = a cos(−bx) = a cos bx = f(x). Hence f is an even function.

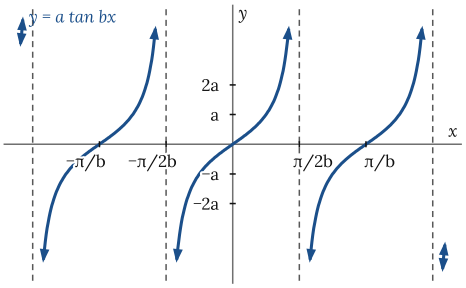
Section 7.3

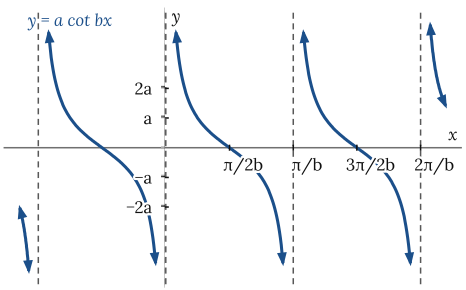
EXAMPLE 7 Draw the graphs of y = a tan bx and y = a cot bx for a, b > 0.

SOLUTION

From y = tan x and y = cot x, apply (x, y) → $(\frac{x}{b}, ay)$:

	y = a tan bx	y = a cot bx
Period	$\frac{\pi}{b}$	$\frac{\pi}{b}$
x-intercepts	$x = 0, \pm\frac{\pi}{b}, \pm\frac{2\pi}{b}, \ldots$	$x = \pm\frac{\pi}{2b}, \pm\frac{3\pi}{2b}, \ldots$
Asymptotes	$x = \pm\frac{\pi}{2b}, \pm\frac{3\pi}{2b}, \ldots$	$x = 0, \pm\frac{\pi}{b}, \pm\frac{2\pi}{b}, \ldots$





EXERCISE 7.3

1. Draw the graph of

a. $y = \frac{1}{2}\tan \pi x$

b. $y = 2\tan \frac{1}{2}x$

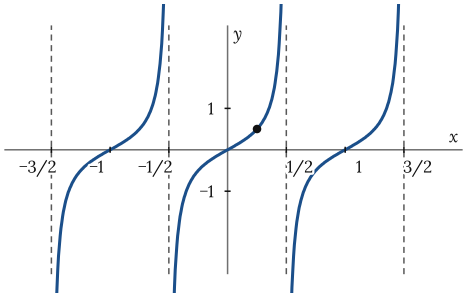
SOLUTION

Answer:

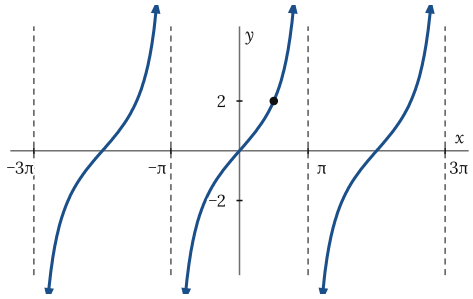
(a) Period 1; asymptotes $x = \frac{\pm 1}{2}, \frac{\pm 3}{2}, \ldots$; x-intercepts $x = 0, \pm 1, \ldots$; passes through $(\frac{1}{4}, \frac{1}{2})$.

(b) Period 2π; asymptotes $x = \pm\pi, \pm 3\pi, \ldots$; x-intercepts $x = 0, \pm 2\pi, \ldots$; passes through $(\frac{\pi}{2}, 2)$.

(a) y = ½ tan πx



(b) y = 2 tan(x/2)



2. Draw the graph of

a. $y = 2\cot \frac{\pi}{3}x$

b. $y = 3\cot 2x$.

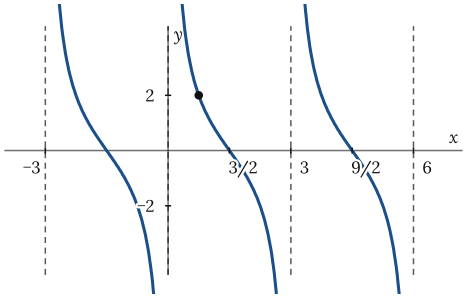
SOLUTION

Answer:

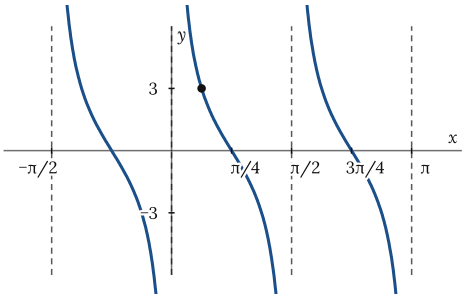
(a) Period 3; asymptotes $x = 0, \pm 3, \pm 6, \ldots$; x-intercepts $x = \frac{\pm 3}{2}, \frac{\pm 9}{2}, \ldots$; passes through $(\frac{3}{4}, 2)$.

(b) Period $\frac{\pi}{2}$; asymptotes $x = 0, \frac{\pm\pi}{2}, \ldots$; x-intercepts $x = \frac{\pm\pi}{4}, \frac{\pm 3\pi}{4}, \ldots$; passes through $(\frac{\pi}{8}, 3)$.

(a) y = 2 cot(πx/3)



(b) y = 3 cot 2x



3. Draw the graph of

a. $y = 2\sec \pi x$

b. $y = 2\csc \frac{1}{2}x$

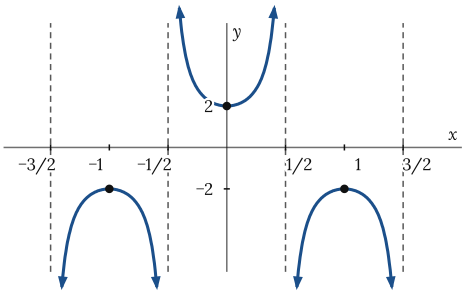
SOLUTION

Answer:

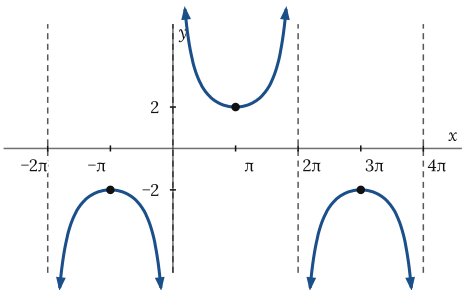
(a) Period 2; asymptotes $x = \frac{\pm 1}{2}, \frac{\pm 3}{2}, \ldots$; range $y \le -2$ or $y \ge 2$; passes through (0, 2) and (1, −2).

(b) Period 4π; asymptotes $x = 0, \pm 2\pi, \ldots$; range $y \le -2$ or $y \ge 2$; passes through $(\pi, 2)$ and $(3\pi, -2)$.

(a) y = 2 sec πx



(b) y = 2 csc(x/2)



4. Show that $y = a\tan bx$, $y = a\cot bx$ and $y = a\csc bx$ are odd functions.

SOLUTION

Answer:

$\tan(-bx) = -\tan bx$, $\cot(-bx) = -\cot bx$ and $\csc(-bx) = \frac{1}{\sin(-bx)} = -\csc bx$. So $f(-x) = -f(x)$ for each function, i.e. they are odd functions.

5. Show that $y = a\sec bx$ is an even function.

SOLUTION

Answer:

$\sec(-bx) = \frac{1}{\cos(-bx)} = \frac{1}{\cos bx} = \sec bx$. So $f(-x) = f(x)$, i.e. $y = a\sec bx$ is an even function.

Section 7.4

EXAMPLE 8 Evaluate

a. $\sin^{-1}(\frac{1}{2})$

b. $\cos^{-1}(-\frac{1}{2})$

c. $\tan^{-1}(-\sqrt{3})$

d. $\sin^{-1}(\sin\frac{5\pi}{6})$

SOLUTION

(a) Since $\sin(\frac{\pi}{6}) = \frac{1}{2}$ and $\frac{-\pi}{2} \le \frac{\pi}{6} \le \frac{\pi}{2}$, we have $\sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$.

(b) Since $\cos(\frac{2\pi}{3}) = \frac{-1}{2}$ and $0 \le \frac{2\pi}{3} \le \pi$, we have $\cos^{-1}(\frac{-1}{2}) = \frac{2\pi}{3}$.

(c) Since $\tan(\frac{-\pi}{3}) = -\sqrt{3}$ and $\frac{-\pi}{2} \le \frac{-\pi}{3} \le \frac{\pi}{2}$, we have $\tan^{-1}(-\sqrt{3}) = \frac{-\pi}{3}$.

(d) $\sin^{-1}(\sin\frac{5\pi}{6}) = \sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$.

EXERCISE 7.4

1. Evaluate

a. $\csc(\sin^{-1} 0.3)$

b. $\cot(\tan^{-1} 5)$

c. $\sec(\cos^{-1}(-0.75))$.

SOLUTION

Answer:

(a) $\frac{10}{3}$; (b) $\frac{1}{5}$; (c) $\frac{-4}{3}$

2. Evaluate

a. $\sin^{-1}\frac{\sqrt{3}}{2}$

b. $\cos^{-1}\frac{\sqrt{2}}{2}$

c. $\tan^{-1}\frac{\sqrt{3}}{3}$.

SOLUTION

Answer:

(a) $\frac{\pi}{3}$; (b) $\frac{\pi}{4}$; (c) $\frac{\pi}{6}$

3. Evaluate

a. $\sin(\cos^{-1}\frac{\sqrt{3}}{2})$

b. $\tan(\sin^{-1}\frac{\sqrt{3}}{2})$

c. $\csc(\tan^{-1}(-\frac{\sqrt{3}}{3}))$.

SOLUTION

Answer:

(a) $\frac{1}{2}$; (b) $\sqrt{3}$; (c) -2

4. Evaluate

a. $\sin^{-1}(\sin\frac{2\pi}{3})$

b. $\cos^{-1}(\cos(-\frac{1}{2}))$

c. $\tan^{-1}(\tan(-\sqrt{3}))$.

SOLUTION

Answer:

(a) $\frac{\pi}{3}$; (b) $\frac{1}{2}$; (c) $\pi - \sqrt{3}$ (since $-\sqrt{3} \approx -1.73$ is outside $(\frac{-\pi}{2}, \frac{\pi}{2})$)

5. Prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

SOLUTION

Answer:

Let $\theta = \sin^{-1} x$, so $\sin\theta = x$ and $\frac{-\pi}{2} \le \theta \le \frac{\pi}{2}$. Then $\cos(\frac{\pi}{2} - \theta) = \sin\theta = x$, and $0 \le \frac{\pi}{2} - \theta \le \pi$. Hence $\cos^{-1} x = \frac{\pi}{2} - \theta$, i.e. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

Section 7.5

EXAMPLE 9 Differentiate the following functions with respect to x.

a. $\sin 5x$

b. $\cos(7x^2 - 2)$

c. $\tan(6x + 7)$

d. $5\sec(3x + 1)$

e. $\frac{\cot(1 - 2x)}{3}$

f. $-2\csc 3x$

SOLUTION

(a) $\frac{d}{dx}\sin 5x = \cos 5x \cdot \frac{d}{dx}(5x) = 5\cos 5x$

(b) $\frac{d}{dx}\cos(7x^2 - 2) = -\sin(7x^2 - 2) \cdot 14x$

(c) $\frac{d}{dx}\tan(6x + 7) = \sec^2(6x + 7) \cdot 6$

(d) $\frac{d}{dx}5\sec(3x + 1) = 5\sec(3x + 1)\tan(3x + 1) \cdot 3 = 15\sec(3x + 1)\tan(3x + 1)$

(e) $\frac{d}{dx}[\frac{\cot(1 - 2x)}{3}] = -\frac{1}{3}\csc^2(1 - 2x) \cdot (-2) = \frac{2}{3}\csc^2(1 - 2x)$

(f) $\frac{d}{dx}(-2\csc 3x) = (-2)(-\csc 3x \cot 3x) \cdot 3 = 6\csc 3x \cot 3x$

EXAMPLE 10 Find $\frac{dy}{dx}$.

a. $y = \sin^2 x$

b. $y = \cos\sqrt{x}$

c. $y = \tan^2(x^2)$

d. $y = \sin 2x - x\cos x$

e. $y = \sin x \cdot \cos^2 x$

f. $y = \dfrac{x}{\tan x}$

g. $y = \sqrt{x + \sin x}$

SOLUTION

(a) $y = \sin^2 x \rightarrow \dfrac{dy}{dx} = 2\sin x \cos x$

(b) $y = \cos x^{1/2} \rightarrow \dfrac{dy}{dx} = -\sin\sqrt{x} \cdot \dfrac{1}{2}x^{-1/2}$

$= -\dfrac{1}{2\sqrt{x}} \sin\sqrt{x}$

(c) $y = \tan^2(x^2) \rightarrow \dfrac{dy}{dx} = 2\tan(x^2) \cdot \sec^2(x^2) \cdot 2x$

(d) $y = \sin 2x - x\cos x$

$\dfrac{dy}{dx} = (\cos 2x)(2) - (x \cdot (-\sin x) + \cos x)$

$= 2\cos 2x + x\sin x - \cos x$

(e) $y = \sin x \cdot \cos^2 x$

$\dfrac{dy}{dx} = \sin x \cdot 2\cos x \cdot (-\sin x) + \cos^2 x \cdot \cos x$

$= -2\sin^2 x \cos x + \cos^3 x$

(f) $y = \dfrac{x}{\tan x} \rightarrow \dfrac{dy}{} = \dfrac{\tan x - x \cdot \sec^2 x}{\tan^2 x}$

(g) $y = (x + \sin x)^{1/2} \rightarrow \dfrac{dy}{dx} = \dfrac{1}{2}(x + \sin x)^{-1/2}(1 + \cos x)$

EXAMPLE 11 Given that $x + \sin y = \cos(xy)$, find $\dfrac{dy}{dx}$.

SOLUTION

Differentiate with respect to x:

$1 + \cos y \cdot \dfrac{dy}{dx} = -\sin(xy) \cdot (x \cdot \dfrac{dy}{dx} + y)$

$(\cos y + x\sin(xy)) \dfrac{dy}{dx} = -(1 + y\sin(xy))$

$\dfrac{dy}{dx} = -\dfrac{1 + y\sin(xy)}{\cos y + x\sin(xy)}$

EXAMPLE 12 Given that $y = x\sin x$, find $\dfrac{d^2y}{dx^2}$.

SOLUTION

$y = x\sin x$

$\dfrac{dy}{dx} = x\cos x + \sin x$

$d^2\dfrac{y}{dx}^2 = x(-\sin x) + \cos x + \cos x$

$= 2\cos x - x\sin x$

EXERCISE 7.5

1. Differentiate the following functions with respect to x.

a. $\sin(2x + 3)$

b. $\cos\dfrac{3}{x}$

c. $x^3\cos 2x$

d. $\cos 7x + \sin 3x$

e. $\sin x \cdot \cos 2x$

f. $\cos^2(5x)$

g. $\tan^3\sqrt{x}$

h. $\sin(\cos x)$

i. $\dfrac{\sin x}{1 + \tan x}$

j. $\sqrt{\sin x + \cos x}$

SOLUTION

(a) $\dfrac{d}{dx}\sin(2x + 3) = \cos(2x + 3) \cdot \dfrac{d}{dx}(2x + 3)$

$= \mathbf{2\cos(2x + 3)}$

(b) Write $\dfrac{3}{x} = 3x^{-1}$, so $\dfrac{d}{dx}(\dfrac{3}{x}) = -3x^{-2} = -\dfrac{3}{x^2}$

$\dfrac{d}{dx}\cos\dfrac{3}{x} = -\sin\dfrac{3}{x} \cdot (-\dfrac{3}{x^2})$

$= \mathbf{\dfrac{3}{x^2}\sin\dfrac{3}{x}}$

(c) Product rule with $u = x^3$, $v = \cos 2x$: $u' = 3x^2$, $v' = -2\sin 2x$

$\dfrac{d}{dx}(x^3\cos 2x) = x^3(-2\sin 2x) + \cos 2x \cdot 3x^2$

$= \mathbf{-2x^3\sin 2x + 3x^2\cos 2x}$

(d) $\dfrac{d}{dx}\cos 7x = -\sin 7x \cdot 7 = -7\sin 7x$

$\dfrac{d}{dx}\sin 3x = \cos 3x \cdot 3 = 3\cos 3x$

Therefore $\dfrac{d}{dx}(\cos 7x + \sin 3x) = \mathbf{-7\sin 7x + 3\cos 3x}$

(e) Product rule with $u = \sin x$, $v = \cos 2x$: $u' = \cos x$, $v' = -2\sin 2x$

$\dfrac{d}{dx}(\sin x \cdot \cos 2x) = \sin x(-2\sin 2x) + \cos 2x \cdot \cos x$

$= \mathbf{-2\sin x\sin 2x + \cos x\cos 2x}$

(f) $\cos^2(5x) = (\cos 5x)^2$. By the chain rule:

$\dfrac{d}{dx}(\cos 5x)^2 = 2\cos 5x \cdot (-\sin 5x) \cdot 5$

$= \mathbf{-10\sin 5x\cos 5x}$

(g) $\tan^3\sqrt{x} = (\tan\sqrt{x})^3$, and $\dfrac{d}{dx}\sqrt{x} = \dfrac{1}{2\sqrt{x}}$

$\dfrac{d}{dx}(\tan\sqrt{x})^3 = 3\tan^2\sqrt{x} \cdot \sec^2\sqrt{x} \cdot \dfrac{1}{2\sqrt{x}}$

$= \mathbf{\dfrac{3}{2\sqrt{x}}\tan^2\sqrt{x}\sec^2\sqrt{x}}$

(h) $\dfrac{d}{dx}\sin(\cos x) = \cos(\cos x) \cdot \dfrac{d}{dx}(\cos x) = \cos(\cos x) \cdot (-\sin x)$

$= \mathbf{-\sin x\cos(\cos x)}$

(i) Quotient rule with $u = \sin x$, $v = 1 + \tan x$: $u' = \cos x$, $v' = \sec^2 x$

$$\frac{d}{dx}\frac{\sin x}{1+\tan x} = \frac{\cos x(1+\tan x) - \sin x \cdot \sec^2 x}{(1+\tan x)^2}$$

$$= \frac{\cos x + \cos x \tan x - \sin x \sec^2 x}{(1+\tan x)^2}$$

(j) $\sqrt{\sin x + \cos x} = (\sin x + \cos x)^{1/2}$

$$\frac{d}{dx}(\sin x + \cos x)^{1/2} = \frac{1}{2}(\sin x + \cos x)^{-1/2}(\cos x - \sin x)$$

$$= \frac{\cos x - \sin x}{2\sqrt{\sin x + \cos x}}$$

---

2. Find $\frac{dy}{dx}$.

a. $y = \sin(1 - x^2)$

b. $y = 2\pi x + 2\cos \pi x$

c. $y = \sin^2 x \cdot \cos 3x$

d. $y = x^2 \sin(\frac{1}{x})$

e. $3x^2 + 2\sin y = y^2$

f. $\sin x \cdot \cos y = 2y$

SOLUTION

(a) $\frac{dy}{dx} = \cos(1 - x^2) \cdot (-2x)$

$= -2x\cos(1 - x^2)$

(b) $\frac{dy}{dx} = 2\pi + 2 \cdot (-\sin \pi x) \cdot \pi$

$= 2\pi - 2\pi \sin \pi x$

(c) Product rule with $u = \sin^2 x$, $v = \cos 3x$: $u' = 2\sin x \cos x$, $v' = -3\sin 3x$

$$\frac{dy}{dx} = \sin^2 x(-3\sin 3x) + \cos 3x \cdot 2\sin x \cos x$$

$= -3\sin^2 x \sin 3x + 2\sin x \cos x \cos 3x$

(d) Product rule with $u = x^2$, $v = \sin(\frac{1}{x})$: $u' = 2x$, $v' = \cos(\frac{1}{x}) \cdot (-\frac{1}{x^2})$

$$\frac{dy}{dx} = x^2 \cdot \cos(\frac{1}{x}) \cdot (-\frac{1}{x^2}) + 2x\sin(\frac{1}{x})$$

$= -\cos(\frac{1}{x}) + 2x\sin(\frac{1}{x})$

(e) $3x^2 + 2\sin y = y^2$. Differentiate both sides with respect to x:

$$6x + 2\cos y \cdot \frac{dy}{dx} = 2y \cdot \frac{dy}{dx}$$

$$6x = (2y - 2\cos y)\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{3x}{y - \cos y}$$

(f) $\sin x \cdot \cos y = 2y$. Differentiate both sides with respect to x (product rule on the left):

$$\cos x \cos y + \sin x(-\sin y)\frac{dy}{dx} = 2\frac{dy}{dx}$$

$$\cos x \cos y = (2 + \sin x \sin y)\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{\cos x \cos y}{2 + \sin x \sin y}$$

---

3. Given that $y = \cos^2 x$, prove that $\frac{d^2y}{dx^2} + 4y = 2$.

SOLUTION

$y = \cos^2 x$

$\frac{dy}{dx} = 2\cos x \cdot (-\sin x) = -2\sin x \cos x = -\sin 2x$

$\frac{d^2y}{dx^2} = -\cos 2x \cdot 2 = -2\cos 2x$

Then $\frac{d^2y}{dx^2} + 4y = -2\cos 2x + 4\cos^2 x$

Using $\cos 2x = 2\cos^2 x - 1$:

$= -2(2\cos^2 x - 1) + 4\cos^2 x = -4\cos^2 x + 2 + 4\cos^2 x = 2$. ✓

---

4. Given that $y = \frac{1}{3}\cos^3 x - \cos x$, prove that $\frac{dy}{dx} = \sin^3 x$.

SOLUTION

$y = \frac{1}{3}\cos^3 x - \cos x$

$\frac{dy}{dx} = \frac{1}{3} \cdot 3\cos^2 x \cdot (-\sin x) + \sin x$

$= -\sin x \cos^2 x + \sin x = \sin x(1 - \cos^2 x)$

$= \sin x \cdot \sin^2 x = \sin^3 x$. ✓

CH.08 Logarithmic and Exponential Functions

Notes · Examples · Exercises

NOTES

Logarithmic Function

For a number $b > 0$, $b \neq 1$, the **logarithmic function** $y = \log_b x$ is defined as: y is the number such that $x = b^y$, for $x > 0$.

Exponential Function

A function of the form $y = b^x$, where $b > 0$, $b \neq 1$, is called an **exponential function** of x. $y = \log_b x$ and $y = b^x$ are inverse of each other.

Exponential Growth and Decay

For $b > 1$, $y = b^x$ is an **exponential growth** function, such as $y = 2^x$ and $y = 3^x$.

For $0 < b < 1$, $y = b^x$ is an **exponential decay** function, such as $y = 2^{-x} = (\frac{1}{2})^x$ and $y = 3^{-x} = (\frac{1}{3})^x$.

Exponential Number e

The limit $\lim_{t \to 0} (1 + t)^{1/t} = 2.7183\ldots$ is denoted by **e**, called the exponential number. e is an irrational number.

Natural (Napierian) Logarithm

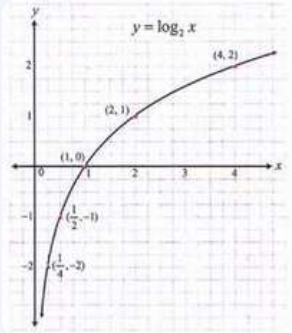
The logarithm of x to the base e is called the **natural logarithm** (Napierian logarithm), denoted $\log_e x = \ln x$.

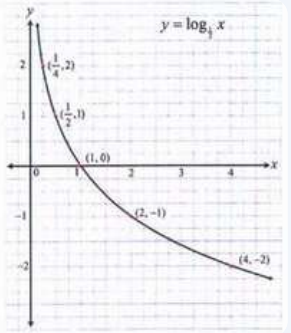
Logarithmic Function

Domain of $y = \log_b x$: $x > 0$ Range: R Asymptote: y-axis ($x = 0$)

For $b > 1$, $y = \log_b x$ is an **increasing** function (if $x_1 > x_2$ then $\log_b x_1 > \log_b x_2$), e.g. $y = \log_2 x$.

For $0 < b < 1$, $y = \log_b x$ is a **decreasing** function (if $x_1 > x_2$ then $\log_b x_1 < \log_b x_2$), e.g. $y = \log_{1/2} x$.





b > 1	0 < b < 1
$\log_b x < 0$ if $0 < x < 1$ $\log_b x = 0$ if $x = 1$ $\log_b x > 0$ if $x > 1$	$\log_b x > 0$ if $0 < x < 1$ $\log_b x = 0$ if $x = 1$ $\log_b x < 0$ if $x > 1$

Transformations of Logarithmic Graphs

Since $\log_{1/b} x = -\log_b x$:

$y = \log_b x \rightarrow$ (reflection on x-axis) $\rightarrow y = \log_{1/b} x$

For $b > 1$: $\log_b x = \dfrac{\log_2 x}{\log_2 b}$, so the graph of $y = \log_b x$ is a vertical scaling of $y = \log_2 x$ with scale factor $\dfrac{1}{\log_2 b}$.

For $0 < b < 1$: $\log_b x = \dfrac{\log_{1/2} x}{\log_{1/2} b}$, a vertical scaling of $y = \log_{1/2} x$ with scale factor $\dfrac{1}{\log_{1/2} b}$.

So every graph of $y = \log_b x$ is a vertical scaling of the graph of $y = \log_2 x$ or $y = \log_{1/2} x$.

Derivative of Logarithmic Functions

$$\frac{d}{dx} \log_b x = \frac{1}{x} \log_b e$$

$$\frac{d}{dx} \log_b u(x) = \frac{1}{u(x)} \cdot \log_b e \cdot \frac{du}{dx}$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \ln u(x) = \frac{1}{u(x)} \cdot \frac{du}{dx}$$

Derivation. Let $y = \log_b x$ and let δx be a small increment. Then $\delta y = \log_b(x + \delta x) - \log_b x = \log_b \frac{x + \delta x}{x}$.

$$\frac{\delta y}{\delta x} = \frac{1}{\delta x} \log_b\left(1 + \frac{\delta x}{x}\right) = \frac{1}{x} \log_b (1 + t)^{1/t}, \quad t = \frac{\delta x}{x}$$

When $\delta x \to 0$, $t \to 0$, so $\dfrac{dy}{dx} = \dfrac{1}{x} \log_b \lim_{t \to 0} (1 + t)^{1/t} = \dfrac{1}{x} \log_b e$.

t	1	0.5	0.25	0.10	0.01	0.001	0.0001	0.00001	0.000001
$\frac{1}{t}$	1	2	4	10	100	1000	10000	100000	1000000
$(1 + t)^{1/t}$	2	2.25	2.4414	2.5937	2.7048	2.7169	2.7181	2.7183	2.7183

The table shows $(1 + t)^{1/t} \to 2.7183$ as $t \to 0$; this limit is denoted by e. Since $\log_e e = 1$, $\dfrac{d}{dx} \ln x = \dfrac{1}{x}$.

Logarithmic Differentiation

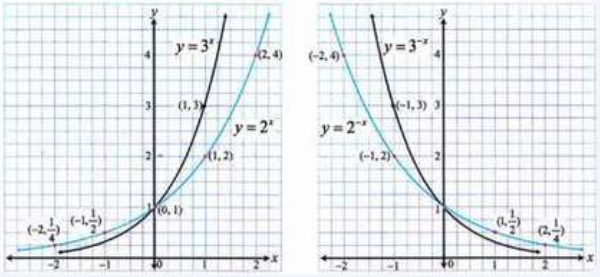
The derivatives of positive functions can be found by taking the natural logarithm of both sides before differentiating:

1. Take ln of both sides: $\ln y = \ln f(x)$, and simplify using the laws of logarithms.

2. Differentiate both sides with respect to x: $\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}[\ln f(x)]$.
3. Multiply by y: $\frac{dy}{dx} = y \cdot \frac{d}{dx}[\ln f(x)]$, and replace y by f(x).

Exponential Function

Domain of $y = b^x$: R Range: $y > 0$ Asymptote: x-axis ($y = 0$)



$y = b^x$ → (reflection on y-axis) → $y = b^{-x}$

From $y = b^x$: $y = ab^x$ (vertical scaling, factor $a > 0$); $y = ab^{x-h} + k$ (horizontal translation h units, vertical translation k units).

Key Points of y = ab^(x − h) + k

Points (0, 1) and (1, b) on $y = b^x$ correspond to
(0, 1) → (0, a) → (h, a + k)
(1, b) → (1, ab) → (1 + h, ab + k)

Asymptote of $y = ab^{x-h} + k$: $y = k$.
Range: if $a > 0$, $\{y : y > k\}$; if $a < 0$, $\{y : y < k\}$.

Derivative of Exponential Functions

$$\frac{d}{dx} b^x = b^x \ln b$$

$$\frac{d}{dx} b^{u(x)} = b^{u(x)} \cdot \ln b \cdot \frac{du}{dx}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} e^{u(x)} = e^{u(x)} \cdot \frac{du}{dx}$$

Derivation. Let $y = b^x$, so $x = \log_b y$. Differentiate both sides with respect to x: $1 = \frac{1}{y} \log_b e \cdot \frac{dy}{dx}$, so

$$\frac{dy}{dx} = y \cdot \frac{1}{\log_b e} = \frac{b^x}{\log_e b} = b^x \ln b$$

Section 8.1

EXAMPLE 1 From the graph of $y = \log_2 x$, draw step-by-step transformation graphs to get the graph of

a. $y = \log_2(x - 1) + 2$

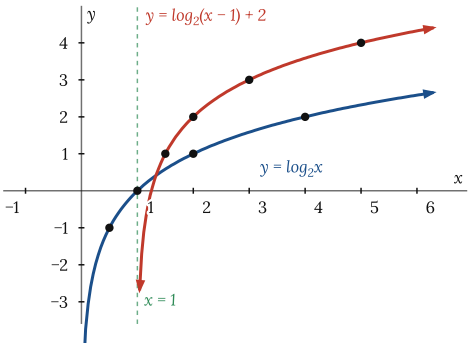
b. $y = \log_{1/2}(x + 2) - 1$

SOLUTION

(a)

$y = \log_2 x$ → (vertical translation 2 units) → (horizontal translation 1 unit) → $y = \log_2(x - 1) + 2$

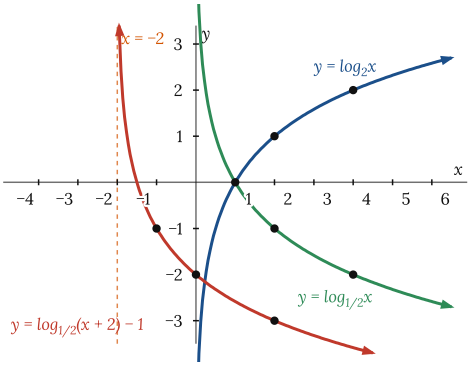
Note: asymptote x = 1.



(b)

$y = \log_2 x$ → (reflection on x-axis) → $y = \log_{1/2} x$ → (horizontal translation −2 units) → (vertical translation −1 unit) → $y = \log_{1/2}(x + 2) - 1$

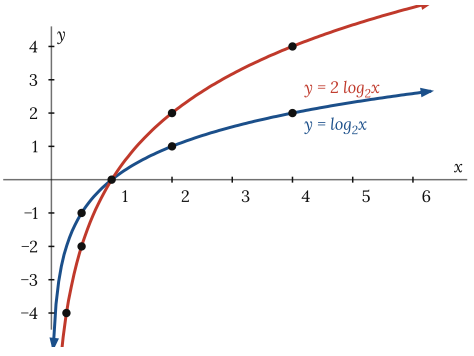
Note: asymptote x = −2.



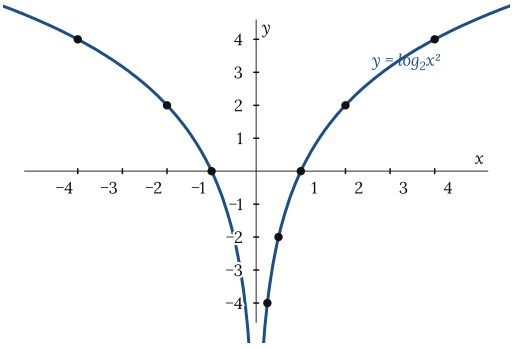
EXAMPLE 2 Draw the graphs $y = 2\log_2 x$ and $y = \log_2 x^2$.

SOLUTION

The graph of $y = 2\log_2 x$ can be drawn directly, or as the vertical scaling with scale factor 2 of the graph of $y = \log_2 x$. Note: asymptote y-axis (x = 0).



The domain of $y = \log_2 x^2$ is $R \setminus \{0\}$ and $y = \log_2 x^2$ is an even function. Note: asymptote y-axis (x = 0).



Note that $2\log_2 x = \log_2 x^2$ for $x > 0$.

EXERCISE 8.1

1. Draw the graph of

a. $y = \log_2(x - 2) + 1$

b. $y = \log_2(x + 1) - 2$

c. $y = \log_{1/2}(x + 1) + 2$

d. $y = \log_{1/2}(x - 2) - 1$.

SOLUTION

Answer:

a. Asymptote $x = 2$; points $(\frac{5}{2}, 0)$, $(3, 1)$, $(4, 2)$, $(6, 3)$

b. Asymptote $x = -1$; points $(0, -2)$, $(1, -1)$, $(3, 0)$, $(7, 1)$

c. Asymptote $x = -1$; points $(\frac{-1}{2}, 3)$, $(0, 2)$, $(1, 1)$, $(3, 0)$

d. Asymptote $x = 2$; points $(\frac{9}{4}, 1)$, $(\frac{5}{2}, 0)$, $(3, -1)$, $(4, -2)$

2. Draw the graph of

a. $y = \log_{1/2} x^2$

b. $y = 2\log_{1/2} x$

SOLUTION

Answer:

a. Even function ($y = -2\log_2 |x|$); asymptote $x = 0$; points $(\frac{\pm 1}{2}, 2)$, $(\pm 1, 0)$, $(\pm 2, -2)$, $(\pm 4, -4)$

b. Equals $-2\log_2 x$ for $x > 0$; asymptote $x = 0$; points $(\frac{1}{2}, 2)$, $(1, 0)$, $(2, -2)$, $(4, -4)$

3. Draw the graph of

a. $y = \log_2(-x)$

b. $y = \log_{1/2}(-x)$.

SOLUTION

Answer:

a. Reflection of $y = \log_2 x$ in the y-axis (domain $x < 0$); asymptote $x = 0$; points $(\frac{-1}{2}, -1)$, $(-1, 0)$, $(-2, 1)$, $(-4, 2)$

b. Reflection of $y = \log_{1/2} x$ in the y-axis (domain $x < 0$); asymptote $x = 0$; points $(\frac{-1}{2}, 1)$, $(-1, 0)$, $(-2, -1)$, $(-4, -2)$

4. Draw the graph of

a. $y = \log_2 |x|$

b. $y = \log_{1/2} |x|$.

SOLUTION

Answer:

a. Even function; asymptote $x = 0$; points $(\frac{\pm 1}{2}, -1)$, $(\pm 1, 0)$, $(\pm 2, 1)$, $(\pm 4, 2)$

b. Even function; asymptote $x = 0$; points $(\frac{\pm 1}{2}, 1)$, $(\pm 1, 0)$, $(\pm 2, -1)$, $(\pm 4, -2)$

Section 8.2

EXAMPLE 3 Differentiate the following functions with respect to x.

a. $\log_{10} x^3$

b. $\log_2 x^3$

c. $\ln x^3$

d. $\ln \sqrt{x^2 + 5}$

e. $\ln \sin 2x$

f. $\ln x \cdot \log_{10} x$

g. $\ln \frac{x}{\sqrt{x^2 + 2}}$

h. $\frac{x^2}{\log_{10} x}$

SOLUTION

(a) $\frac{d}{dx} \log_{10} x^3 = \frac{1}{x^3} \cdot \log_{10} e \cdot \frac{d}{dx} x^3$

$= \frac{1}{x^3} \cdot \log_{10} e \cdot 3x^2$

$= \frac{3}{x} \log_{10} e$

(b) $\frac{d}{dx} \log_2 x^3 = \frac{1}{x^3} \cdot \log_2 e \cdot 3x^2$

$= \frac{3}{x} \log_2 e$

(c) $\frac{d}{dx} \ln x^3 = \frac{1}{x^3} \cdot 3x^2$

$= \frac{3}{x}$

(d) $\frac{d}{dx} \ln \sqrt{x^2 + 5} = \frac{1}{\sqrt{x^2 + 5}} \cdot \frac{1}{2\sqrt{x^2 + 5}} \cdot 2x$

$= \frac{x}{x^2 + 5}$

(e) $\frac{d}{dx} \ln \sin 2x = \frac{1}{\sin 2x} \cdot \frac{d}{dx} \sin 2x$

$= \frac{1}{\sin 2x} \cdot 2\cos 2x$

$= 2 \cot 2x$

(f) $\frac{d}{dx} (\ln x \cdot \log_{10} x)$

$= \ln x \cdot \frac{d}{dx} \log_{10} x + \log_{10} x \cdot \frac{d}{dx} \ln x$

$= \ln x \cdot \frac{1}{x} \log_{10} e + \log_{10} x \cdot \frac{1}{x}$

(g) $\frac{d}{dx} \ln \frac{x}{\sqrt{x^2 + 2}}$

$= \frac{\sqrt{x^2 + 2}}{x} \cdot \frac{\sqrt{x^2 + 2} - x \cdot \frac{2x}{2\sqrt{x^2 + 2}}}{x^2 + 2}$

$= \frac{\sqrt{x^2+2}}{x} \cdot \frac{(x^2+2)-x^2}{\sqrt{x^2+2}(x^2+2)}$

$= \frac{2}{x(x^2+2)}$

(h) $\frac{d}{dx}\left(\frac{x^2}{\log_{10} x}\right) = \frac{\log_{10} x \cdot \frac{d}{dx} x^2 - x^2 \cdot \frac{d}{dx} \log_{10} x}{(\log_{10} x)^2}$

$= \frac{2x \log_{10} x - x \log_{10} e}{(\log_{10} x)^2}$

EXAMPLE 4 Find $\frac{dy}{dx}$ if $y = \sqrt{(x^2+1)(x-1)^2}$.

SOLUTION

$y = \sqrt{(x^2+1)(x-1)^2}$

$\ln y = \ln \sqrt{(x^2+1)(x-1)^2}$

$= \frac{1}{2}[\ln(x^2+1) + \ln(x-1)^2]$

Differentiate both sides with respect to x:

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{2}\left[\frac{1}{x^2+1} \cdot 2x + \frac{1}{(x-1)^2} \cdot 2(x-1)\right]$

$= \frac{x}{x^2+1} + \frac{1}{x-1}$

$\frac{dy}{dx} = y\left(\frac{x}{x^2+1} + \frac{1}{x-1}\right) = \sqrt{(x^2+1)(x-1)^2}\left(\frac{x}{x^2+1} + \frac{1}{x-1}\right)$

EXAMPLE 5 Find $\frac{dy}{dx}$ if $y = \frac{x \sin x}{\sqrt{\sec x}}$, $0 < x < \frac{\pi}{2}$.

SOLUTION

$y = \frac{x \sin x}{\sqrt{\sec x}}$

$\ln y = \ln x + \ln \sin x - \frac{1}{2} \ln \sec x$

Differentiate both sides with respect to x:

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x} + \frac{1}{\sin x} \cdot \cos x - \frac{1}{2} \cdot \frac{1}{\sec x} \cdot \sec x \tan x$

$= \frac{1}{x} + \cot x - \frac{1}{2} \tan x$

$\frac{dy}{dx} = y\left(\frac{1}{x} + \cot x - \frac{1}{2}\tan x\right) = \frac{x \sin x}{\sqrt{\sec x}}\left(\frac{1}{x} + \cot x - \frac{1}{2}\tan x\right)$

EXERCISE 8.2

1. Differentiate the following functions with respect to x.

a. $\ln(2x^2+3)$

b. $\ln|x|$

c. $x^2 \log_2 x$

d. $\sin 3x \cdot \log_{10}(x+1)$

e. $\ln\sqrt{5x-4}$

f. $\ln(\ln x)$

SOLUTION

Answer:

a. $\frac{4x}{2x^2+3}$

b. $\frac{1}{x}$

c. $2x \log_2 x + x \log_2 e$

d. $3\cos 3x \cdot \log_{10}(x+1) + \frac{\sin 3x}{x+1} \log_{10} e$

e. $\frac{5}{2(5x-4)} = \frac{5}{10x-8}$

f. $\frac{1}{x \ln x}$

2. Use logarithmic differentiation to find the derivative of y with respect to x.

a. $y = \sqrt{x(x+1)}$

b. $y = x(x+1)(x+2)$

c. $y = \frac{x\sqrt{x^2+1}}{(x+1)^{2/3}}$

SOLUTION

Answer:

a. $\frac{dy}{dx} = \frac{2x+1}{2\sqrt{x(x+1)}}$

b. $\frac{dy}{dx} = 3x^2 + 6x + 2$

c. $\frac{dy}{dx} = \frac{x\sqrt{x^2+1}}{(x+1)^{2/3}}\left[\frac{1}{x} + \frac{x}{x^2+1} - \frac{2}{3(x+1)}\right]$

Section 8.3

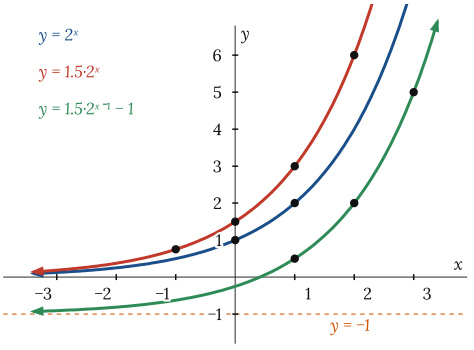
EXAMPLE 6 Draw the graphs of $y = 1.5 \cdot 2^{x-1} - 1$ from $y = 2^x$, and $y = -2^{-(x+1)} + 2$ from $y = -2^{-x}$.

SOLUTION

(a)

$y = 2^x \rightarrow$ (vertical scaling, scale factor 1.5) $y = 1.5 \cdot 2^x \rightarrow$ (horizontal translation 1 unit, vertical translation −1 unit) $y = 1.5 \cdot 2^{x-1} - 1$

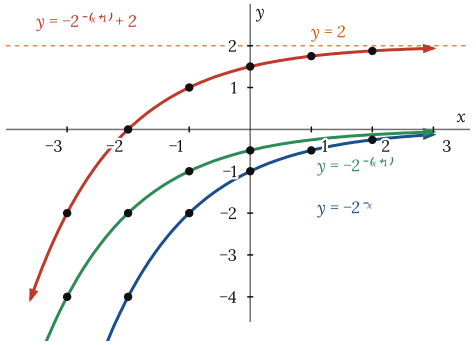
Note: asymptote $y = -1$.



(b)

$y = 2^{-x} \rightarrow$ (reflection on x-axis) $y = -2^{-x} \rightarrow$ (horizontal translation −1 unit) $\rightarrow$ (vertical translation 2 units) $y = -2^{-(x+1)} + 2$

Note: asymptote $y = 2$.



EXAMPLE 7 Points (0, 1) and (1, b) are on the graph of $y = b^x$. Find the corresponding points on the graphs of $y = ab^x$ and $y = ab^{x-h} + k$. What is the asymptote of $y = ab^{x-h} + k$? Find the range of $y = ab^{x-h} + k$ if $a > 0$ and if $a < 0$.

SOLUTION

$y = b^x \rightarrow$ (vertical scaling, factor a) $y = ab^x \rightarrow$ (vertical translation k units, horizontal translation h units) $y = ab^{x-h} + k$

$(0, 1) \rightarrow (0, a) \rightarrow (h, a + k)$

$(1, b) \rightarrow (1, ab) \rightarrow (1 + h, ab + k)$

The asymptote of $y = b^x$ is $y = 0$ (x-axis), and the asymptote of $y = ab^x$ is also $y = 0$. After vertical translation of k units, the asymptote of $y = ab^{x-h} + k$ is **$y = k$**.

If $a > 0$, the range of $y = ab^{x-h} + k$ is $\{y : y > k\}$, i.e. all y above the asymptote $y = k$.

If $a < 0$, the range is $\{y : y < k\}$, i.e. all y below the asymptote $y = k$.

EXERCISE 8.3

1. Draw the graph of

a. $y = 2 \cdot 3^{x+1} - 2$

b. $y = -2^{x-1} + 3$.

SOLUTION

Answer:

a. Asymptote $y = -2$; range $y > -2$; y-intercept 4; points $(-1, 0)$, $(0, 4)$, $(1, 16)$

b. Asymptote $y = 3$; range $y < 3$; y-intercept $\frac{5}{2}$; points $(1, 2)$, $(2, 1)$, $(3, -1)$

2. Draw the graph of

a. $y = 2^{|x|}$

b. $y = 2^{-|x|}$

SOLUTION

Answer:

a. Even function; y-intercept 1 (lowest point); range $y \geq 1$; points $(0, 1)$, $(\pm 1, 2)$, $(\pm 2, 4)$

b. Even function; y-intercept 1 (highest point); asymptote $y = 0$; range $0 < y \leq 1$; points $(0, 1)$, $(\pm 1, \frac{1}{2})$, $(\pm 2, \frac{1}{4})$

3. Find the y-intercept, asymptote and the range of

a. $y = 3e^{x-1} + 2$

b. $y = -2e^{-x+1} + 3$.

SOLUTION

Answer:

a. y-intercept $3e^{-1} + 2 = \frac{3}{e} + 2$; asymptote $y = 2$; range $y > 2$

b. y-intercept $3 - 2e$; asymptote $y = 3$; range $y < 3$

Section 8.4

EXAMPLE 8 Differentiate the following functions with respect to x.

a. e^{3x}

b. e^{1-x^2}

c. $e^{\sin x}$

d. x^2e^{3x}

e. $e^{2x} \sin 3x$

f. $(e^x + e^{-x})^2$

g. $\frac{3e^{2x}}{1 - 2x}$

SOLUTION

(a) $\frac{d}{dx} e^{3x} = e^{3x} \cdot \frac{d}{dx} 3x = 3e^{3x}$

(b) $\frac{d}{dx} e^{1-x^2} = e^{1-x^2} \cdot \frac{d}{dx} (1 - x^2) = e^{1-x^2}(-2x)$

(c) $\frac{d}{dx} e^{\sin x} = e^{\sin x} \cdot \frac{d}{dx} \sin x = e^{\sin x} \cos x$

(d) $\frac{d}{dx} (x^2e^{3x}) = x^2 \cdot \frac{d}{dx} e^{3x} + e^{3x} \cdot \frac{d}{dx} x^2 = x^2 \cdot e^{3x} \cdot 3 + e^{3x} \cdot 2x$

(e) $\frac{d}{dx} (e^{2x} \sin 3x) = e^{2x} \cdot \frac{d}{dx} \sin 3x + \sin 3x \cdot \frac{d}{dx} e^{2x}$
$= e^{2x} \cdot \cos 3x \cdot 3 + \sin 3x \cdot e^{2x} \cdot 2$

(f) $\frac{d}{dx} (e^x + e^{-x})^2 = 2(e^x + e^{-x}) \cdot \frac{d}{dx} (e^x + e^{-x}) = 2(e^x + e^{-x})(e^x - e^{-x})$

(g) $\frac{d}{dx} \frac{3e^{2x}}{1 - 2x} = \frac{3(1 - 2x) \cdot \frac{d}{dx} e^{2x} - e^{2x} \cdot \frac{d}{dx} (1 - 2x)}{(1 - 2x)^2}$
$= \frac{3(1 - 2x) \cdot 2e^{2x} - e^{2x}(-2)}{(1 - 2x)^2}$
$= \frac{3(4 - 4x)e^{2x}}{(1 - 2x)^2}$

EXAMPLE 9 Find $\frac{dy}{dx}$.

a. $y = e^x \ln x$

b. $y = \log_{10} e^{x^2}$

c. $y = \log_3(\sin x + e^x)$

d. $xe^y + \ln(xy) = \sin x$

SOLUTION

(a) $y = e^x \ln x$

$$\frac{dy}{dx} = e^x \cdot \frac{d}{dx} \ln x + \ln x \cdot \frac{d}{dx} e^x$$

$$= e^x \cdot \frac{1}{x} + \ln x \cdot e^x$$

(b) $y = \log_{10} e^{x^2}$

$$= x^2 \log_{10} e$$

$$\frac{dy}{dx} = \log_{10} e \cdot \frac{d}{dx} x^2$$

$$= 2x \log_{10} e$$

(c) $y = \log_3(\sin x + e^x)$

$$\frac{dy}{dx} = \frac{1}{\sin x + e^x} \cdot \log_3 e \cdot \frac{d}{dx} (\sin x + e^x)$$

$$= \frac{\cos x + e^x}{\sin x + e^x} \log_3 e$$

(d) $xe^y + \ln(xy) = \sin x$. Differentiate both sides with respect to x:

$$x \cdot e^y \cdot \frac{dy}{dx} + e^y + \frac{1}{xy}\left(x \frac{dy}{dx} + y\right) = \cos x$$

$$x e^y \frac{dy}{dx} + \frac{1}{y} \frac{dy}{dx} = \cos x - e^y - \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{\cos x - e^y - \frac{1}{x}}{x e^y + \frac{1}{y}}$$

EXAMPLE 10 Differentiate $y = x^x$, $x > 0$.

SOLUTION

$$y = x^x$$

$$= e^{\ln x^x}$$

$$= e^{x \ln x}$$

$$\frac{dy}{dx} = \frac{d}{dx} e^{x \ln x}$$

$$= e^{x \ln x} \cdot \frac{d}{dx} (x \ln x)$$

$$= e^{x \ln x}\left(x \cdot \frac{1}{x} + \ln x \cdot 1\right)$$

$$= x^x(1 + \ln x)$$

(Alternatively, by logarithmic differentiation: $\ln y = x \ln x$, so $\frac{1}{y}\frac{dy}{dx} = 1 + \ln x$.)

EXERCISE 8.4

1. Differentiate the following functions with respect to x.

a. $(5 + 3x)e^{-2x}$

b. $3^x x^3$

c. $2^x \log_2 x$

d. $10^x \log_{10}(x + 1)$

e. $\frac{x^2 + \tan 3x}{e^x}$

f. $x \ln y + e^{xy} = 2$

SOLUTION

Answer:

a. $(-7 - 6x)e^{-2x}$

b. $3^x \cdot 3x^2 + x^3 \cdot 3^x \ln 3$

c. $2^x \frac{1}{x} \log_2 e + \log_2 x \cdot 2^x \ln 2$

d. $10^x \frac{1}{x+1} \log_{10} e + \log_{10}(x + 1) \cdot 10^x \ln 10$

e. $\frac{2x + 3\sec^2 3x - x^2 - \tan 3x}{e^x}$

f. $\frac{dy}{dx} = -\frac{\ln y + y e^{xy}}{\frac{x}{y} + x e^{xy}}$

2. Given that $y = e^{3x} \sin 2x$, prove that $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$.

SOLUTION

Answer:

$$\frac{dy}{dx} = e^{3x}(2 \cos 2x + 3 \sin 2x)$$

$$\frac{d^2}{dx}\left(\frac{y}{dx}\right)^2 = e^{3x}(5 \sin 2x + 12 \cos 2x)$$

Hence $\frac{d^2}{dx}\left(\frac{y}{dx}\right)^2 - 6\frac{dy}{dx} + 13y = e^{3x}[(5 \sin 2x + 12 \cos 2x) - 6(2 \cos 2x + 3 \sin 2x) + 13 \sin 2x] = e^{3x}(0) = 0.$

3. Use logarithmic differentiation to find the derivative of y with respect to x.

a. $y = (\sqrt{x})^x$

b. $y = x^{\cos x}$

c. $y = (\ln x)^{\ln x}$

SOLUTION

Answer:

a. $\frac{dy}{dx} = (\sqrt{x})^x \cdot \frac{1 + \ln x}{2}$

b. $\frac{dy}{dx} = x^{\cos x}\left(\frac{\cos x}{x} - \sin x \ln x\right)$

c. $\frac{dy}{dx} = (\ln x)^{\ln x} \cdot \frac{\ln(\ln x) + 1}{x}$

CH.09 Application of Differentiation

Notes · Examples · Exercises

NOTES

Inclination of a line

The **inclination** of a line that intersects the x-axis is the measure of the smallest nonnegative angle θ which the line makes with the positive end of the x-axis. The slope of the line is the tangent of the inclination:

$m = \tan\theta$

A vertical line has inclination 90° but no slope; a horizontal line has inclination 0° and slope 0.

Normal line

The line which is perpendicular to the tangent at (x_1, y_1) is called the **normal** to the curve at (x_1, y_1). Its gradient is $\frac{-1}{y'}(x_1)$, where $y'(x_1) \neq 0$.

Increasing function

A function f is **increasing** on an interval (a, b) if for any two numbers x_1 and x_2 in (a, b),

$x_2 > x_1$ implies $f(x_2) > f(x_1)$

namely, an increase in x produces an increase in y.

Decreasing function

A function f is **decreasing** on an interval (a, b) if for any two numbers x_1 and x_2 in (a, b),

$x_2 > x_1$ implies $f(x_2) < f(x_1)$

namely, an increase in x produces a decrease in y.

Critical point

A **critical point** of a function f(x) is a point in the domain of the function such that either $f'(x) = 0$ or $f'(x)$ is undefined.

Global (absolute) maximum and minimum

A point where f(x) has the minimum value on the entire domain is a **global minimum**; a point where f(x) has the maximum value on the entire domain is a **global maximum**.

Local maximum and minimum

A turning point on the graph where f(x) has a maximum value compared to the neighboring points is a **local maximum**; a turning point where f(x) has a minimum value compared to the neighboring points is a **local minimum**.

Concave up / Concave down

Let f be differentiable on an open interval I. The graph of f is

1. **concave up** on I if f' is increasing on the interval.
2. **concave down** on I if f' is decreasing on the interval.

Point of inflection

A point where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**. At a point of inflection (c, f(c)), either $f''(c) = 0$ or $f''(c)$ is undefined.

Remark. We have an exceptional case that a point c is not an inflection point even if $f''(c) = 0$. For example, $f(x) = (x - c)^{2k}$ where the integer $k \geq 2$.

Linearization

If f is differentiable at x = a, then the approximation function

$L(x) = f(a) + f'(a)(x - a)$

is the **linearization** of f at a. The approximation $f(x) \approx L(x)$ is the **standard linear approximation** of f at a. The point x = a is the **center** of the approximation. The difference of f(x) and L(x) is the **error** of the linearization.

Slope and inclination

$m = \tan\theta$

For two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ on a line:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan\theta$$

Equation of tangent at (x1, y1)

For the curve y = f(x), the gradient of the tangent at (x_1, y_1) is $y'(x_1)$, so

$y - y_1 = y'(x_1)(x - x_1)$

Equation of normal at (x1, y1)

The gradient of the normal is $\frac{-1}{y'}(x_1)$ where $y'(x_1) \neq 0$, so

$$y - y_1 = -\frac{1}{y'(x_1)}(x - x_1)$$

Test for increasing and decreasing functions

Let y = f(x) be differentiable on the open interval (a, b).

1. If $f'(x) > 0$ for each $x \in (a, b)$, then f is increasing on (a, b).
2. If $f'(x) < 0$ for each $x \in (a, b)$, then f is decreasing on (a, b).

Steps: increasing / decreasing on an interval

1. Find $f'(x)$.

2. Find the critical points of f.
3. Find the open intervals determined by the critical points.
4. Test the signs of $f'(x)$ for each open interval.
5. Decide whether f is increasing or decreasing on each interval.

Critical point

$f'(x) = 0$ or $f'(x)$ is undefined (at a point in the domain of f).

Test for global maximum and minimum on a closed interval

Let f(x) be a continuous function on a closed interval.

1. Find the critical points of f.
2. Find f(x) at all critical points and endpoints of the interval.
3. Take the smallest value as the global minimum and the largest value as the global maximum.

First derivative test (local maximum / minimum)

Let c be a critical point of a continuous function f.

1. If $f'(x)$ changes from positive to negative at c, then f has a local maximum at $x = c$.
2. If $f'(x)$ changes from negative to positive at c, then f has a local minimum at $x = c$.
3. If $f'(x)$ does not change sign at c, then f has neither a local maximum nor a local minimum at $x = c$.

Test for concavity

Let f be twice differentiable on an open interval I.

1. If $f''(x) > 0$ for all x in I, then f is concave up on I.
2. If $f''(x) < 0$ for all x in I, then f is concave down on I.

Point of inflection

A point (c, f(c)) where the concavity changes: either $f''(c) = 0$ or $f''(c)$ is undefined, and f'' changes sign at c.

Second derivative test

Suppose f'' is continuous on an open interval that contains $x = c$.

1. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

Linearization (standard linear approximation)

$$y = f(a) + f'(a)(x - a)$$

$$L(x) = f(a) + f'(a)(x - a)$$

$f(x) \approx L(x)$

Error = true value − approximate value.

Section 9.1

EXAMPLE 1 Find the equations of tangent and normal line to the curve $y = \ln x$ at the point $y = -1$.

SOLUTION

$y = \ln x$

$y' = \frac{1}{x}$

When $y = -1$, $\ln x = -1$, so

$x = e^{-1}$

$= \frac{1}{e}$

At the point $(\frac{1}{e}, -1)$, the gradient of tangent is $y'(\frac{1}{e}) = e$, and the gradient of normal is $-\frac{1}{e}$.

The equation of tangent at $(\frac{1}{e}, -1)$ is

$y - (-1) = e(x - \frac{1}{e})$

$y = ex - 2$

The equation of normal at $(\frac{1}{e}, -1)$ is

$y - (-1) = -\frac{1}{e}(x - \frac{1}{e})$

$y = -\frac{1}{e}x + \frac{1}{e^2} - 1$

EXAMPLE 2 Find the equation of tangent to $y = \sin x$ at the origin.

SOLUTION

$y = \sin x$

$y' = \cos x$

At (0, 0), the gradient of tangent is $y'(0) = \cos 0 = 1$.

The equation of tangent is

$y - 0 = 1(x - 0)$

$y = x$

EXAMPLE 3 Consider the curve $f(x) = x^2 + ax + b$ where a and b are constants. The tangent to this curve at the point $x = 2$ is $y = 2x + 1$. Find the values of a and b.

SOLUTION

$f(x) = x^2 + ax + b$

$f'(x) = 2x + a$

At $x = 2$, $f(2) = 2a + b + 4$ and $f'(2) = a + 4$.

By problem, the gradient of tangent line is 2, so $a + 4 = 2$ which gives **$a = -2$**.

The tangent line equation at $(2, 2a + b + 4)$ is

$y - (2a + b + 4) = (a + 4)(x - 2)$

Substitute the value a = −2 in the above equation, then

$y - b = 2(x - 2)$

$y = 2x + b - 4$

Since the tangent line equation at x = 2 is y = 2x + 1, it follows that b − 4 = 1, so b = 5.

Therefore a = −2 and b = 5.

EXERCISE 9.1

1. Find the equations of tangent and normal line to the curve y = 2 ln x at (1, 0).

SOLUTION

$y = 2 \ln x$

$y = 2 \ln x$

$y' = 2 \cdot \frac{1}{x}$

$= \frac{2}{x}$

At the point (1, 0):

The gradient of tangent is $y'(1) = 2$.

The gradient of normal is $-\frac{1}{y'(1)} = -\frac{1}{2}$.

The equation of tangent at (1, 0) is

$y - y_1 = y'(x_1)(x - x_1)$

$y - 0 = 2(x - 1)$

$y = 2x - 2$

The equation of normal at (1, 0) is

$y - y_1 = -\frac{1}{y'(x_1)}(x - x_1)$

$y - 0 = -\frac{1}{2}(x - 1)$

$2y = -x + 1$

Tangent: y = 2x − 2; Normal: 2y = −x + 1.

2. Find the equation of a line which passes through the origin and tangent to $y = e^x$.

SOLUTION

$y = e^x$

$y' = e^x$

Let (a, e^a) be any point of the tangent line to the curve.

$y'(a) = e^a$

The equation of tangent at (a, e^a) is

$y - y_1 = y'(x_1)(x - x_1)$

$y - e^a = e^a(x - a)$

Since the tangent line passes through the origin (0, 0),

$0 - e^a = e^a(0 - a)$

$-e^a = -ae^a$

$a = 1$

At the point (1, e):

$y - e = e(x - 1)$

$y - e = ex - e$

$y = ex$

The required line is y = ex.

3. Find the points of contact where horizontal tangent meet the curve $y = x^4 - 2x^2 + 2$.

SOLUTION

$y = x^4 - 2x^2 + 2$

$y' = 4x^3 - 4x$

Since the tangent is a horizontal line, $y' = 0$:

$4x^3 - 4x = 0$

$x^3 - x = 0$

$x(x^2 - 1) = 0$

$x = 0$ or $x^2 - 1 = 0$, that is $x = 0$ or $x = \pm 1$.

If x = 0, y = 2.

If x = 1, y = 1 − 2 + 2 = 1.

If x = −1, y = 1 − 2 + 2 = 1.

The points of contact are (0, 2), (1, 1) and (−1, 1).

4. Consider the curve $y = a\sqrt{x} + \frac{b}{\sqrt{x}}$ where a and b are constants. The normal to this curve at the point at x = 4 is 4x + y = 22. Find the values of a and b.

SOLUTION

$y = f(x) = a\sqrt{x} + \frac{b}{\sqrt{x}}$

At x = 4, $f(4) = a\sqrt{4} + \frac{b}{\sqrt{4}} = 2a + \frac{b}{2}$.

The point is $(4, 2a + \frac{b}{2})$.

$f'(x) = \frac{1}{2}ax^{-1/2} + (-\frac{1}{2})bx^{-3/2}$

$= \frac{a}{2\sqrt{x}} - \frac{b}{2x\sqrt{x}}$

$f'(4) = \frac{a}{2\sqrt{4}} - \frac{b}{2(4)\sqrt{4}}$

$= \frac{a}{4} - \frac{b}{16}$

$= \frac{4a - b}{16}$

The gradient of normal at x = 4 is

$-\frac{1}{f'(4)} = -\frac{1}{\frac{4a - b}{16}}$

$= \frac{16}{b - 4a}$

The normal 4x + y = 22 is y = −4x + 22, so its gradient is −4. Hence

$\frac{16}{b - 4a} = -4$

$4a - b = 4$ (1)

The point $(4, 2a + \frac{b}{2})$ lies on the normal line $4x + y = 22$:

$$4(4) + 2a + \frac{b}{2} = 22$$

$$2a + \frac{b}{2} = 6$$

$$4a + b = 12 \quad (2)$$

From (1) and (2): $8a = 16$, so $a = 2$.
Substituting $a = 2$ in (2): $4(2) + b = 12$, so $b = 4$.
$a = 2, b = 4$

Section 9.2

EXAMPLE 4 Show that the function $f(x) = x^2 + 1$ is decreasing on the interval $x < 0$ and increasing on the interval $x > 0$.

SOLUTION

$f(x) = x^2 + 1$

$f'(x) = 2x$

On $x < 0$, $f'(x) = 2x < 0$. Thus f is decreasing on $x < 0$.
On $x > 0$, $f'(x) = 2x > 0$. Thus f is increasing on $x > 0$.

EXAMPLE 5 Consider the function $f(x) = x^3 - 3x + 1$.

a. Find the open intervals on which f is increasing and those on which f is decreasing.
b. Find the points at which the graph has a horizontal tangent.

SOLUTION

(a) $f(x) = x^3 - 3x + 1$

$f'(x) = 3x^2 - 3$

$= 3(x - 1)(x + 1)$

So $f'(x) > 0$ when $x - 1$ and $x + 1$ have the same sign. Similarly, $f'(x) < 0$ when $x - 1$ and $x + 1$ have the opposite signs.

Interval	$x < -1$	$-1 < x < 1$	$x > 1$
$x + 1$	−	+	+
$x - 1$	−	−	+
sign of $f'(x)$	+	−	+
Behavior of f	increasing	decreasing	increasing

Thus f(x) is increasing on $x < -1$ and $x > 1$, and decreasing on $-1 < x < 1$.

(b) The graph has a horizontal tangent when $f'(x) = 0$.

$3(x - 1)(x + 1) = 0$

$x = 1$ or $x = -1$

So $f(-1) = 3$, $f(1) = -1$. The horizontal tangents are at the points $(-1, 3)$ and $(1, -1)$.

EXAMPLE 6 Find the critical points of:

a. $f(x) = x^3 - 3x + 2$
b. $f(x) = (x - 1)^{2/3}$
c. $f(x) = \frac{1}{x}$

SOLUTION

(a) $f(x) = x^3 - 3x + 2$

$f'(x) = 3x^2 - 3$

For critical points, $f'(x) = 0$ so $3x^2 - 3 = 0$. Therefore $x = \pm 1$.

When $x = 1$, $f(1) = 1^3 - 3(1) + 2 = 0$.

When $x = -1$, $f(-1) = (-1)^3 - 3(-1) + 2 = 4$.

Therefore the critical points are (1, 0) and (−1, 4).

(b) $f(x) = (x - 1)^{2/3}$

$f'(x) = \frac{2}{3}(x - 1)^{-1/3}$

$= \frac{2}{3(x - 1)^{1/3}}$

It is observed that $f'(x)$ is undefined when $x = 1$. Therefore the critical point is (1, 0).

(c) $f(x) = \frac{1}{x}$

$f'(x) = -\frac{1}{x^2}$

It is observed that $f'(x)$ is undefined when $x = 0$. However, $x = 0$ is not in the domain of the function. Therefore, there are no critical points.

EXAMPLE 7 Find the open intervals on which the following functions are increasing or decreasing:

a. $f(x) = 2x^3 - 6x + 5$
b. $f(x) = (x^2 - 9)^{4/5}$
c. $f(x) = \frac{2x - 3}{x - 1}$

SOLUTION

(a) $f(x) = 2x^3 - 6x + 5$

$f'(x) = 6(x^2 - 1)$

$= 6(x + 1)(x - 1)$

For critical points, $f'(x) = 0$, namely $x = \pm 1$.

Interval	$x < -1$	$-1 < x < 1$	$x > 1$
$x + 1$	−	+	+
$x - 1$	−	−	+
sign of $f'(x)$	+	−	+
Behavior of f	increasing	decreasing	increasing

Therefore, on $x < -1$ and $x > 1$, the given function is increasing, and on $-1 < x < 1$ the given function is decreasing.

(b) $f(x) = (x^2 - 9)^{4/5}$

$f'(x) = \frac{8}{5} \cdot \frac{x}{(x^2 - 9)^{1/5}}$

For critical points, $f'(x) = 0$, namely $x = 0$. $f'(x)$ is undefined, namely $x = -3$ and $x = 3$.

Interval	x < −3	−3 < x < 0	0 < x < 3	x > 3
x + 3	−	+	+	+
x	−	−	+	+
x − 3	−	−	−	+
sign of f′(x)	−	+	−	+
Behavior of f	decreasing	increasing	decreasing	increasing

Therefore, on x < −3 and 0 < x < 3, the given function is decreasing, and on −3 < x < 0 and x > 3 the given function is increasing.

(c) $f(x) = \dfrac{2x-3}{x-1}$

$f'(x) = \dfrac{1}{(x-1)^2}$

It is observed that x = 1 does not belong to the domain of function.

Interval	x < 1	x > 1
$(x-1)^2$	+	+
sign of f′(x)	+	+
Behavior of f	increasing	increasing

Therefore, on both x < 1 and x > 1, the function is increasing.

EXAMPLE 8 Find the global minimum and global maximum values of each of the following functions:

a. $f(x) = x^2 + 5x - 3$ on $[-4, 5]$

b. $f(x) = \dfrac{x}{x^2+1}$ on $[-3, 3]$

SOLUTION

(a) $f(x) = x^2 + 5x - 3$ on $[-4, 5]$

$f'(x) = 2x + 5$

For critical points, f′(x) = 0, namely $x = -\dfrac{5}{2}$.

Therefore, at the critical point $f(-\dfrac{5}{2}) = -9\dfrac{1}{4}$.

At the endpoints f(−4) = −7 and f(5) = 47.

Hence, the global minimum value is $-9\dfrac{1}{4}$ and the global maximum value is 47.

(b) $f(x) = \dfrac{x}{x^2+1}$ on $[-3, 3]$

For critical point, f′(x) = 0,

$\dfrac{1-x^2}{(x^2+1)^2} = 0$

$1 - x^2 = 0$

Thus x = ±1.

At the critical points, $f(1) = \dfrac{1}{2}$ and $f(-1) = -\dfrac{1}{2}$.

At the endpoints, $f(-3) = -\dfrac{3}{10}$ and $f(3) = \dfrac{3}{10}$.

Hence, the global minimum value is $-\dfrac{1}{2}$ and global maximum value is $\dfrac{1}{2}$.

EXAMPLE 9 Find the critical points of $f(x) = x^3 - 3x + 1$. Find the open intervals on which the function is increasing or decreasing. Identify the function's local maximum and minimum.

SOLUTION

$f(x) = x^3 - 3x + 1$

$f'(x) = 3x^2 - 3$

$= 3(x + 1)(x - 1)$

For critical points, f′(x) = 0, namely x = ±1. So f(−1) = 3 and f(1) = −1; the critical points are (−1, 3) and (1, −1).

Interval	x < −1	−1 < x < 1	x > 1
sign of f′(x)	+	−	+
Behavior of f	increasing	decreasing	increasing

f′ changes from positive to negative at x = −1, so f has a local maximum at x = −1, with f(−1) = 3.

f′ changes from negative to positive at x = 1, so f has a local minimum at x = 1, with f(1) = −1.

Therefore (−1, 3) is a local maximum and (1, −1) is a local minimum.

EXAMPLE 10 Show that $f(x) = x^3$ is increasing on both sides of critical point.

SOLUTION

$f(x) = x^3$

$f'(x) = 3x^2$

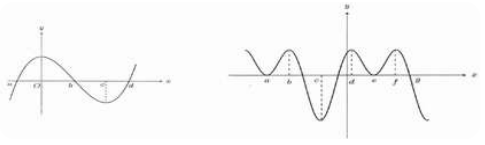
For critical point, f′(x) = 0, namely x = 0.

Interval	x < 0	x > 0
x^2	+	+
sign of f′(x)	+	+
Behavior of f	increasing	increasing

Thus the function is increasing on both sides of the critical point, so (0, 0) is neither a local maximum nor a local minimum.

EXERCISE 9.2

1. Consider the following graphs.



a. State all open intervals where the function is increasing.

b. State all open intervals where the function is decreasing.

SOLUTION

Answer:

For left figure: (a, 0), (c, d) → increasing; (0, c) → decreasing.

For right figure: (a, b), (c, d), (e, f) → increasing; (b, c), (d, e), (f, g) → decreasing.

2.(a) Find the critical points:$f(x) = x^2 - 2x + 5$

SOLUTION

$f(x) = x^2 - 2x + 5$

$f'(x) = 2x - 2$

For critical points, $f'(x) = 0$:

$2x - 2 = 0$

$x = 1$

At $x = 1$, $f(1) = (1)^2 - 2(1) + 5 = 4$.

∴ The critical point is (1, 4).

2.(b) Find the critical points:$f(x) = x^3 + 6x^2 + 3x + 10$

SOLUTION

$f(x) = x^3 + 6x^2 + 3x + 10$

$f'(x) = 3x^2 + 12x + 3$

For critical points, $f'(x) = 0$:

$3x^2 + 12x + 3 = 0$

$x^2 + 4x + 1 = 0$

$x^2 + 4x = -1$

$x^2 + 2x(2) + 2^2 = -1 + 2^2$

$(x + 2)^2 = 3$

$x + 2 = \pm\sqrt{3}$

$x = -2 \pm \sqrt{3}$

$x = -0.27$ or $x = -3.73$

At $x = -0.27$, $f(-0.27) = (-0.27)^3 + 6(-0.27)^2 + 3(-0.27) + 10 = 9.6$.

At $x = -3.73$, $f(-3.73) = (-3.73)^3 + 6(-3.73)^2 + 3(-3.73) + 10 = 30.4$.

∴ The critical points are (−0.27, 9.6) and (−3.73, 30.4).

2.(c) Find the critical points:$f(x) = \dfrac{1}{x - 3}$

SOLUTION

$f(x) = \dfrac{1}{x - 3}$, $x \neq 3$

$y = (x - 3)^{-1}$

$y' = -(x - 3)^{-2}$

$= -\dfrac{1}{(x - 3)^2}$

$f'(x) = -\dfrac{1}{(x - 3)^2}$

$f'(x)$ is undefined when $x = 3$. Since $x = 3$ is not in the domain of the function, **there are no critical points.**

2.(d) Find the critical points:$f(x) = (x^2 - 2x)^{2/3}$

SOLUTION

$f(x) = (x^2 - 2x)^{2/3}$

$f'(x) = \dfrac{2}{3}(x^2 - 2x)^{-1/3}(2x - 2)$

$= \dfrac{2(2x - 2)}{3(x^2 - 2x)^{1/3}}$

For critical point, $f'(x) = 0$:

$\dfrac{2(2x - 2)}{3(x^2 - 2x)^{1/3}} = 0$

$2x - 2 = 0$

$x = 1$

$f(1) = (1^2 - 2)^{2/3} = (-1)^{2/3} = 1$.

$f'(x)$ is undefined when $x^2 - 2x = 0$, that is $x = 0$ and $x = 2$.

At $x = 0$, $f(0) = ((0)^2 - 2(0))^{2/3} = 0$. At $x = 2$, $f(2) = (2^2 - 2(2))^{2/3} = 0$.

∴ The critical points are (0, 0), (2, 0) and (1, 1).

2.(e) Find the critical points:$f(x) = \dfrac{1}{x^2 - 2x + 1}$

SOLUTION

$f(x) = \dfrac{1}{x^2 - 2x + 1} = \dfrac{1}{(x - 1)^2}$, $x \neq 1$

$f'(x) = -2(x - 1)^{-3}$

$= -\dfrac{2}{(x - 1)^3}$

$f'(x)$ is undefined when $x = 1$. Since $x = 1$ is not in the domain of the function, **there are no critical points.**

2.(f) Find the critical points:$f(x) = \dfrac{x^2}{x^2 + 2}$

SOLUTION

$f(x) = \dfrac{x^2}{x^2 + 2}$

$f'(x) = \dfrac{(x^2 + 2)\dfrac{d}{dx}x^2 - x^2\dfrac{d}{dx}(x^2 + 2)}{(x^2 + 2)^2}$

$= \dfrac{(x^2 + 2)2x - x^2(2x)}{(x^2 + 2)^2}$

$= \dfrac{2x^3 + 4x - 2x^3}{(x^2 + 2)^2}$

$= \dfrac{4x}{(x^2 + 2)^2}$

For critical points, $f'(x) = 0$:

$\dfrac{4x}{(x^2 + 2)^2} = 0$

$4x = 0$, $x = 0$

At $x = 0$, $f(0) = \dfrac{0^2}{0^2 + 2} = 0$.

∴ The critical point is (0, 0).

3.(a) Find the global minimum and global maximum values of the function:$f(x) = 3x - x^3 + 2$ on $[-5, 5]$

SOLUTION

$f(x) = 3x - x^3 + 2$ on $[-5, 5]$

$f'(x) = 3 - 3x^2$

For critical points, $f'(x) = 0$:

$$3 - 3x^2 = 0$$
$$3x^2 = 3$$
$$x^2 = 1$$
$$x = \pm 1$$

At the critical points, $f(1) = 3(1) - 1^3 + 2 = 4$ and $f(-1) = 3(-1) - (-1)^3 + 2 = -3 + 1 + 2 = 0$.

At the endpoints, $f(-5) = 3(-5) - (-5)^3 + 2 = -15 + 125 + 2 = 112$ and $f(5) = 3(5) - 5^3 + 2 = 15 - 125 + 2 = -108$.

The global minimum value is −108 and the global maximum value is 112.

3.(b) Find the global minimum and global maximum values of the function: $f(x) = \sin 2x$ on $[-\frac{\pi}{4}, \frac{\pi}{4}]$

SOLUTION

$f(x) = \sin 2x$ on $[-\frac{\pi}{4}, \frac{\pi}{4}]$

$f'(x) = 2\cos 2x$

For critical points, $f'(x) = 0$:

$$2\cos 2x = 0$$
$$\cos 2x = 0$$
$$2x = \frac{\pi}{2} \text{ or } -\frac{\pi}{2}$$
$$x = \frac{\pi}{4} \text{ or } -\frac{\pi}{4}$$

(since $\cos(\pm\frac{\pi}{2}) = 0$)

At the critical points, $f(\frac{\pi}{4}) = \sin 2(\frac{\pi}{4}) = \sin\frac{\pi}{2} = 1$ and $f(-\frac{\pi}{4}) = \sin 2(-\frac{\pi}{4}) = \sin(-\frac{\pi}{2}) = -1$.

At the end points, $f(-\frac{\pi}{4}) = -1$ and $f(\frac{\pi}{4}) = 1$.

The global minimum value is −1 and the global maximum value is 1.

4.(a) Find the open intervals on which the following function is increasing or decreasing. Identify the function's local maximum and minimum. $f(x) = x^3 - 3x^2 + 5$

SOLUTION

$f(x) = x^3 - 3x^2 + 5$

$f'(x) = 3x^2 - 6x$

For critical points, $f'(x) = 0$:

$$3x^2 - 6x = 0$$
$$3x(x - 2) = 0$$
$$x = 0 \text{ and } x = 2$$

Interval	$x < 0$	$0 < x < 2$	$x > 2$
x	−	+	+
x − 2	−	−	+
sign of f′(x)	+	−	+
Behavior of f	increasing	decreasing	increasing

∴ On $x < 0$ and $x > 2$, the given function is increasing and on $0 < x < 2$, the given function is decreasing.

The local maximum at x = 0 and the local minimum at x = 2.

4.(b) Find the open intervals on which the following function is increasing or decreasing. Identify the function's local maximum and minimum. $f(x) = (x^2 - 2x)^{2/5}$

SOLUTION

$f(x) = (x^2 - 2x)^{2/5}$

$$f'(x) = \frac{2}{5}(x^2 - 2x)^{-3/5}(2x - 2)$$
$$= \frac{4(x - 1)}{5(x^2 - 2x)^{3/5}}$$

For critical points, $f'(x) = 0$:

$$\frac{4(x - 1)}{5(x^2 - 2x)^{3/5}} = 0$$
$$4(x - 1) = 0$$
$$x = 1$$

$f'(x)$ is undefined when $x = 0$ and $x = 2$.

Interval	$x < 0$	$0 < x < 1$	$1 < x < 2$	$x > 2$
x	−	+	+	+
x − 1	−	−	+	+
x − 2	−	−	−	+
sign of f′(x)	−	+	−	+
Behavior of f	decreasing	increasing	decreasing	increasing

∴ On $0 < x < 1$ and $x > 2$, the given function is increasing and on $x < 0$ and $1 < x < 2$, the given function is decreasing.

The local maximum at x = 1 and the local minimum at x = 0 and x = 2.

4.(c) Find the open intervals on which the following function is increasing or decreasing. Identify the function's local maximum and minimum. $f(x) = \frac{x^2 + 5}{x}$

SOLUTION

$$f(x) = \frac{x^2 + 5}{x} = x + 5x^{-1}, \quad x \neq 0$$

$$f'(x) = 1 - \frac{5}{x^2}$$
$$= \frac{x^2 - 5}{x^2}$$

For critical points, $f'(x) = 0$:

$$\frac{x^2 - 5}{x^2} = 0$$
$$x^2 - 5 = 0$$

$x = \pm\sqrt{5}$

$f'(x)$ is undefined when $x = 0$ (not in the domain).

Interval	$x < -\sqrt{5}$	$-\sqrt{5} < x < 0$	$0 < x < \sqrt{5}$	$x > \sqrt{5}$
$x - \sqrt{5}$	−	−	−	+
x^2	+	+	+	+
$x + \sqrt{5}$	−	+	+	+
sign of $f'(x)$	+	−	−	+
Behavior of f	increasing	decreasing	decreasing	increasing

∴ On $x < -\sqrt{5}$ and $x > \sqrt{5}$, the given function is increasing and on $-\sqrt{5} < x < 0$ and $0 < x < \sqrt{5}$, the given function is decreasing.

The local maximum at $x = -\sqrt{5}$ and the local minimum at $x = \sqrt{5}$.

4.(d) Find the open intervals on which the following function is increasing or decreasing. Identify the function's local maximum and minimum.$f(x) = -x^3 - 6x^2 + 12$

SOLUTION

$$f(x) = -x^3 - 6x^2 + 12$$
$$f'(x) = -3x^2 - 12x$$
$$= -3x(x + 4)$$

For critical points, $f'(x) = 0$:

$$-3x^2 - 12x = 0$$
$$-3x(x + 4) = 0$$
$$x = 0 \text{ or } x = -4$$

Interval	$x < -4$	$-4 < x < 0$	$x > 0$
$-x$	+	+	−
$x + 4$	−	+	+
sign of $f'(x)$	−	+	−
Behavior of f	decreasing	increasing	decreasing

∴ On $x < -4$ and $x > 0$, the given function is decreasing and on $-4 < x < 0$, the given function is increasing.

The local maximum at $x = 0$ and the local minimum at $x = -4$.

4.(e) Find the open intervals on which the following function is increasing or decreasing. Identify the function's local maximum and minimum.$f(x) = \dfrac{x - 3}{x - 1}$

SOLUTION

$$f(x) = \frac{x - 3}{x - 1}, \quad x \neq 1$$
$$f'(x) = \frac{(x - 1)\cdot 1 - (x - 3)}{(x - 1)^2}$$
$$= \frac{x - 1 - x + 3}{(x - 1)^2}$$
$$= \frac{2}{(x - 1)^2}$$

$x = 1$ does not belong to the domain of the function.

Interval	$x < 1$	$x > 1$
$(x - 1)^2$	+	+
sign of $f'(x)$	+	+
Behavior of f	increasing	increasing

∴ On both $x < 1$ and $x > 1$, the function is increasing, then we cannot decide the local maximum and local minimum.

4.(f) Find the open intervals on which the following function is increasing or decreasing. Identify the function's local maximum and minimum.$f(x) = x^{2/3}(x^2 - 16)$

SOLUTION

$$f(x) = x^{2/3}(x^2 - 16)$$
$$= x^{8/3} - 16x^{2/3}$$
$$f'(x) = \frac{8}{3}x^{5/3} - 16\left(\frac{2}{3}\right)x^{-1/3}$$
$$= \frac{8}{3}x^{5/3} - \frac{32}{3x^{1/3}}$$
$$= \frac{8x^2 - 32}{3x^{1/3}}$$
$$= \frac{8(x^2 - 4)}{3x^{1/3}}$$

For critical points, $f'(x) = 0$:

$$\frac{8(x^2 - 4)}{3x^{1/3}} = 0$$
$$x^2 - 4 = 0$$
$$(x + 2)(x - 2) = 0$$
$$x = -2 \text{ or } x = 2$$

$f'(x)$ is undefined when $x = 0$.

Interval	$x < -2$	$-2 < x < 0$	$0 < x < 2$	$x > 2$
$x + 2$	−	+	+	+
x	−	−	+	+
$x - 2$	−	−	−	+
sign of $f'(x)$	−	+	−	+
Behavior of f	decreasing	increasing	decreasing	increasing

∴ On $x < -2$ and $0 < x < 2$, the given function is decreasing and on $-2 < x < 0$ and $x > 2$, the given function is increasing.

The local maximum at $x = 0$ and the local minimum at $x = -2$ and $x = 2$.

Section 9.3

EXAMPLE 11 Determine the open intervals on which the graph of the function $f(x) = \dfrac{1}{x^2 + 1}$ is concave up or concave down.

SOLUTION

$$f(x) = \frac{1}{x^2 + 1}$$
$$f'(x) = -\frac{2x}{(x^2 + 1)^2}$$

We calculate that

$$f''(x) = \frac{-2(x^2+1)^2 + 2x(2)(x^2+1)(2x)}{(x^2+1)^4}$$
$$= \frac{-2(x^2+1)^2 + 8x^2(x^2+1)}{(x^2+1)^4}$$
$$= \frac{2(3x^2-1)}{(x^2+1)^3}$$

$f''(x)$ is defined for all real numbers and $f''(x) = 0$.

Hence

$$\frac{2(3x^2-1)}{(x^2+1)^3} = 0$$
$$2(3x^2-1) = 0$$
$$(x + \frac{1}{\sqrt{3}})(x - \frac{1}{\sqrt{3}}) = 0$$

Therefore $x = \pm\frac{1}{\sqrt{3}}$.

Interval	$x < -\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$	$x > \frac{1}{\sqrt{3}}$
$x + \frac{1}{\sqrt{3}}$	−	+	+
$x - \frac{1}{\sqrt{3}}$	−	−	+
sign of $f''(x)$	+	−	+
Behavior of f	concave up	concave down	concave up

Therefore the graph is concave up on $x < -\frac{1}{\sqrt{3}}$ and $x > \frac{1}{\sqrt{3}}$, and concave down on $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$.

EXAMPLE 12 Determine the open intervals on which the graph of following functions

a. $f(x) = x^{1/3}$

b. $f(x) = \frac{9}{88}x^{11/3}$

is concave up or concave down. State their inflection points.

SOLUTION

(a) $f(x) = x^{1/3}$

$$f'(x) = \frac{1}{3}x^{-2/3}$$
$$f''(x) = -\frac{2}{9}x^{-5/3}$$
$$= -\frac{2}{9x^{5/3}}$$

Interval	x < 0	x = 0	x > 0
sign of $f''(x)$	+	undefined	−
Behavior of f	concave up	inflection point	concave down

Therefore on x < 0 the graph of function is concave up and on x > 0 the graph of function is concave down. The point (0, 0) is an inflection point.

(b) $f(x) = \frac{9}{88}x^{11/3}$

$$f'(x) = \frac{3}{8}x^{8/3}$$
$$f''(x) = x^{5/3}$$

Interval	x < 0	x = 0	x > 0
sign of $f''(x)$	−	$f''(0) = 0$	+
Behavior of f	concave down	inflection point	concave up

Therefore on x < 0 the graph of function is concave down and on x > 0 the graph of function is concave up. The point (0, 0) is an inflection point.

EXAMPLE 13 Show that x = 0 is not an inflection point of $f(x) = x^6$.

SOLUTION

$$f(x) = x^6$$
$$f'(x) = 6x^5$$

For critical points, $f'(x) = 0$, namely $x = 0$.

Next, we consider $f''(x) = 30x^4$. At $x = 0$, $f''(0) = 0$.

We observe that $f''(x) > 0$ for $x > 0$, and $f''(x) > 0$ for $x < 0$, so the second derivative does not change sign at $x = 0$.

Therefore, x = 0 is not an inflection point of $f(x) = x^6$.

EXAMPLE 14 Find and classify the critical points of $f(x) = x^4 - 4x^3 + 5$.

SOLUTION

$$f(x) = x^4 - 4x^3 + 5$$
$$f'(x) = 4x^3 - 12x^2$$
$$= 4x^2(x - 3)$$

For critical points, $f'(x) = 0$, namely $x = 0$ and $x = 3$.

When $x = 0$, $f(0) = 5$, and when $x = 3$, $f(3) = -22$.

Therefore, the critical points are $(0, 5)$ and $(3, -22)$.

Next,

$$f''(x) = 12x^2 - 24x$$
$$= 12x(x - 2)$$

At $x = 0$, $f''(0) = 0$. The second derivative test fails.

It is observed that $f'(x) < 0$ when $x < 0$ and $f'(x) < 0$ when $0 < x < 3$.

Therefore (0, 5) is a point of inflection.

At $x = 3$, $f''(3) = 36$. Therefore $(3, -22)$ is a local minimum.

EXAMPLE 15 Find the range of $f(x) = \ln x - x$.

SOLUTION

$$f(x) = \ln x - x$$
$$f'(x) = \frac{1}{x} - 1$$
$$f''(x) = -\frac{1}{x^2}$$

For critical point, $f'(x) = 0$,

$$\frac{1}{x} - 1 = 0$$

Therefore x = 1, and f(1) = ln 1 − 1 = −1.
At x = 1, $f''(1) = -1 < 0$. Thus (1, −1) is a maximum point.
The range of f(x) is {y : y ≤ −1}.

EXAMPLE 16 Find the least amount of material needed to build an open cylindrical vessel with a capacity of $400\pi\ cm^3$.

SOLUTION

Let r = radius and h = height of cylindrical vessel.
Then volume of vessel is

$$\pi r^2 h = 400\pi$$
$$h = \frac{400}{r^2}$$

Since vessel is open top, area of material for vessel is

$$A = \pi r^2 + 2\pi rh$$
$$= \pi r^2 + 2\pi r \cdot \frac{400}{r^2}$$
$$= \pi r^2 + \frac{800\pi}{r}, \ (r > 0)$$

Differentiating both sides with respect to r, we get

$$\frac{dA}{dr} = 2\pi r - \frac{800\pi}{r^2}$$

For critical point, $\frac{dA}{dr} = 0$

$$2\pi r - \frac{800\pi}{r^2} = 0$$
$$r^3 = 400$$
$$r = \sqrt[3]{400}$$

Next,

$$\frac{d^2A}{dr^2} = 2\pi + \frac{1600\pi}{r^3}$$

When $r = \sqrt[3]{400}$, $\frac{d^2A}{dr^2} = 2\pi + 4\pi > 0$.

Therefore the area of material is the least when $r = \sqrt[3]{400}$.

Therefore the least amount of material = $\pi r^2 + \frac{800\pi}{r} = \frac{1200\pi}{\sqrt[3]{400}}\ cm^2$.

EXERCISE 9.3

1.(a) Determine the open intervals on which the graph of the following functions are concave up or concave down. State their points of inflection.$f(x) = (x-1)^5$

SOLUTION

$$f(x) = (x-1)^5$$
$$f'(x) = 5(x-1)^4$$
$$f''(x) = 20(x-1)^3$$

f″(x) is defined for all real numbers and f″(x) = 0:

$$20(x-1)^3 = 0$$
$$x = 1$$

Interval	x < 1	x = 1	x > 1
$(x-1)^3$	−	0	+
sign of f″(x)	−	f″(1) = 0	+
Behavior of f	concave down	inflection point	concave up

∴ On x < 1 the graph of the function is concave down and on x > 1 the graph of the function is concave up.
The point (1, 0) is an inflection point.

1.(b) Determine the open intervals on which the graph of the following functions are concave up or concave down. State their points of inflection.$g(x) = x^4 - 2x^3$

SOLUTION

$$g(x) = x^4 - 2x^3$$
$$g'(x) = 4x^3 - 6x^2$$
$$g''(x) = 12x^2 - 12x$$
$$= 12x(x-1)$$

g″(x) is defined for all real numbers and g″(x) = 0:

$$12x(x-1) = 0$$
$$x = 0 \text{ and } x = 1$$

Interval	x < 0	x = 0	0 < x < 1	x = 1	x > 1
x	−	0	+	+	+
x − 1	−	−	−	0	+
sign of g″(x)	+	g″(0) = 0	−	g″(1) = 0	+
Behavior of g	concave up	inflection point	concave down	inflection point	concave up

∴ On x < 0 and x > 1 the graph of the function is concave up and on 0 < x < 1 the graph of the function is concave down.
The points (0, 0) and (1, −1) are inflection points.

1.(c) Determine the open intervals on which the graph of the following functions are concave up or concave down. State their points of inflection.$h(x) = \sin x$ on $[-\pi, \pi]$

SOLUTION

$$h(x) = \sin x \text{ on } [-\pi, \pi]$$
$$h'(x) = \cos x$$
$$h''(x) = -\sin x$$

h″(x) is defined for all real numbers and h″(x) = 0:

$$-\sin x = 0$$
$$x = -\pi,\ x = 0 \text{ and } x = \pi$$

Interval	$-\pi < x < 0$	x = 0	$0 < x < \pi$
−sin x	+	0	−
sign of h″(x)	+	h″(0) = 0	−
Behavior of h	concave up	inflection point	concave down

∴ On $-\pi < x < 0$ the graph of the function is concave up and on $0 < x < \pi$ the graph of the function is concave down.
The point (0, 0) is an inflection point.

2.(a) Find and classify the critical points of the following functions.$f(x) = x^4 - 4x^3 + 5$

SOLUTION

curve: $f(x) = x^4 - 4x^3 + 5$

$f'(x) = 4x^3 - 12x^2$

$= 4x^2(x - 3)$

For critical points, $f'(x) = 0$:

$4x^2(x - 3) = 0$

$x = 0$ and $x = 3$

At $x = 0$, $f(0) = 0^4 - 4(0)^3 + 5 = 5$. At $x = 3$, $f(3) = 3^4 - 4(3)^3 + 5 = -22$.

$\therefore$ The critical points are $(0, 5)$ and $(3, -22)$.

$f''(x) = 12x^2 - 24x$

$= 12x(x - 2)$

At $x = 0$, $f''(0) = 12(0)^2 - 24(0) = 0$. The second derivative test fails.

$f'(x) < 0$ when $x < 0$ and $f'(x) < 0$ when $0 < x < 3$, so f' does not change sign at $x = 0$.

$\therefore (0, 5)$ is a point of inflection.

At $x = 3$, $f''(3) = 12(3)(3 - 2) = 36 > 0$.

$\therefore$ (3, −22) is a local minimum.

2.(b) Find and classify the critical points of the following functions.$g(x) = x^5 - 5x + 5$

SOLUTION

curve: $g(x) = x^5 - 5x + 5$

$g'(x) = 5x^4 - 5$

$= 5(x^4 - 1)$

For critical points, $g'(x) = 0$:

$5(x^4 - 1) = 0$

$x = -1$ and $x = 1$

At $x = -1$, $g(-1) = (-1)^5 - 5(-1) + 5 = 9$. At $x = 1$, $g(1) = (1)^5 - 5(1) + 5 = 1$.

$\therefore$ The critical points are $(-1, 9)$ and $(1, 1)$.

$g''(x) = 20x^3$

At $x = -1$, $g''(-1) = -20 < 0$. At $x = 1$, $g''(1) = 20 > 0$.

$\therefore$ (−1, 9) is a local maximum and (1, 1) is a local minimum.

2.(c) Find and classify the critical points of the following functions.$h(x) = xe^{-x}$

SOLUTION

curve: $h(x) = xe^{-x}$

$h'(x) = -xe^{-x} + e^{-x}$

$= (-x + 1)e^{-x}$

For critical points, $h'(x) = 0$:

$(-x + 1)e^{-x} = 0$

$-x + 1 = 0$

$x = 1$

At $x = 1$, $h(1) = 1 \cdot e^{-1} = \frac{1}{e}$.

The critical point is $(1, \frac{1}{e})$.

$h''(x) = (-x + 1)e^{-x}(-1) + e^{-x}(-1)$

$= xe^{-x} - e^{-x} - e^{-x}$

$= (x - 2)e^{-x}$

At $x = 1$, $h''(1) = (1 - 2)e^{-1} = -e^{-1} = -\frac{1}{e} < 0$.

$\therefore (1, \frac{1}{e})$ is a local maximum.

---

3. Find the range of$f(x) = x - e^x$

SOLUTION

$f(x) = x - e^x$

$f'(x) = 1 - e^x$

$f''(x) = -e^x$

For critical point, $f'(x) = 0$:

$1 - e^x = 0$

$e^x = 1$

$= e^0$

$x = 0$

At $x = 0$, $f(0) = 0 - e^0 = -1$. The critical point is $(0, -1)$.

At $x = 0$, $f''(0) = -e^0 < 0$.

$\therefore (0, -1)$ is a maximum point.

The range of f(x) is { y : y ≤ −1 }.

---

4. If a piece of string of fixed length is made to enclose a rectangle, show that the enclosed area is the greatest when the rectangle is a square.

SOLUTION

Let the length of string = L, the length and the breadth of the rectangle be x and y.

$2x + 2y = L$

$y = \frac{L}{2} - x$

The area,

$A = xy$

$= x(\frac{L}{2} - x)$

$= \frac{L}{2}x - x^2$

$\frac{dA}{dx} = \frac{L}{2} - 2x$

For critical point, $\frac{dA}{dx} = 0$:

$\frac{L}{2} - 2x = 0$

$2x = \frac{L}{2}$

$x = \frac{L}{4}$

When $x = \frac{L}{4}$, $y = \frac{L}{2} - \frac{L}{4} = \frac{L}{4}$.

Since the length and breadth are the same, the rectangle is a square.

$\frac{dA}{dx} = \frac{L}{2} - 2x$

$\frac{d^2A}{dx^2} = -2$

When $x = \frac{L}{4}$, $\frac{d^2A}{dx^2} = -2 < 0$.

∴ The enclosed area is the greatest when the rectangle is a square.

5. Find the minimum value of the sum of a positive number and its reciprocal.

SOLUTION

Let the positive number and its reciprocal be x and $\frac{1}{x}$.

the sum $S = x + \frac{1}{x}$

$\frac{dS}{dx} = 1 - \frac{1}{x^2}$

For critical point, $\frac{dS}{dx} = 0$:

$1 - \frac{1}{x^2} = 0$

$\frac{1}{x^2} = 1$

$x = \pm 1$

∴ Since the number is positive, x = 1.

$\frac{d^2S}{dx^2} = \frac{2}{x^3}$

At $x = 1$, $\frac{d^2S}{dx^2} = \frac{2}{1} = 2 > 0$.

∴ The sum is minimum when x = 1.

The minimum value of the sum = $x + \frac{1}{x} = 1 + 1 = 2$.

6. A rectangular field is surrounded by a fence on three of its sides and a straight hedge on the fourth side. If the length of the fence is 320 meters, find the maximum area of the field enclosed.

SOLUTION

Let the length and the breadth of the rectangular field be x m and y m.

The length of the fence = 320 m

$x + 2y = 320$

$y = (160 - \frac{x}{2})$ m

The area,

$A = xy$

$= x(160 - \frac{x}{2})$

$= 160x - \frac{x^2}{2}$

$\frac{dA}{dx} = 160 - x$

For critical point, $\frac{dA}{dx} = 0$:

$160 - x = 0$

$x = 160$

$\frac{d^2A}{dx^2} = -1$

When $x = 160$, $\frac{d^2A}{dx^2} = -1 < 0$.

∴ The area is the maximum when x = 160.

The maximum area of the field enclosed is $A = 160x - \frac{x^2}{2} = 160(160) - \frac{(160)^2}{2} = 12800\ m^2$.

7. A rectangular box has a square base of side x cm. If the sum of one side of the square and height is 15 cm, express the volume of the box in terms of x. Use this expression to determine the maximum volume of the box.

SOLUTION

Let the height be y cm.

$x + y = 15$

$y = (15 - x)$ cm

The volume,

$V = x \cdot x \cdot y$

$= x^2(15 - x)$

$= (15x^2 - x^3)\ cm^3$

$\frac{dV}{dx} = 30x - 3x^2$

For the critical point, $\frac{dV}{dx} = 0$:

$30x - 3x^2 = 0$

$3x(10 - x) = 0$

$x = 0$ and $x = 10$

x = 0 is omitted because x is considered as the side of a rectangle, so x = 10 [x > 0].

$\frac{d^2V}{dx^2} = 30 - 6x$

When $x = 10$, $\frac{d^2V}{dx^2} = 30 - 6(10) = -30 < 0$.

∴ The volume is the maximum when x = 10.

The maximum volume of the box $V = x^2(15 - x) = (10)^2(15 - 10) = 500\ cm^3$.

8. Two sides of triangle have lengths x and y, and the angle between them is θ. What value of θ will maximize the triangle's area?

SOLUTION

Area of the triangle is

$$A = \frac{1}{2}xy\sin\theta$$

$$\frac{dA}{d\theta} = \frac{1}{2}xy\cos\theta$$

For the critical point, $\frac{dA}{d\theta} = 0$:

$$\frac{1}{2}xy\cos\theta = 0$$

$$\cos\theta = 0$$

$$\theta = \frac{\pi}{2} \quad [0 < \theta < \pi]$$

$$\frac{d^2A}{d\theta^2} = -\frac{1}{2}xy\sin\theta$$

At $\theta = \frac{\pi}{2}$, $\frac{d^2A}{d\theta^2} = -\frac{1}{2}xy\sin\frac{\pi}{2} = -\frac{1}{2}xy(1) < 0$ [x > 0, y > 0].

Thus $\theta = \frac{\pi}{2}$ will give the maximum area $A = \frac{1}{2}xy$.

Section 9.4

EXAMPLE 17 Find the linearization of $f(x) = x^2 + 2x$ at x = 2. Compare the approximate value and true value at x = 2.5, x = 2.05 and x = 2.005.

SOLUTION

$f(x) = x^2 + 2x$

$f'(x) = 2x + 2$

Since f(2) = 8 and f′(2) = 6, the linearization is

$L(x) = f(2) + f'(2)(x - 2)$

$= 8 + 6(x - 2)$

$= 6x - 4$

That is, $x^2 + 2x \approx 6x - 4$.

At x = 2.5, the true value gives $x^2 + 2x = (2.5)^2 + 2(2.5) = 11.25$ and the linearization gives $6x - 4 = 6(2.5) - 4 = 11$. Therefore, the approximate value differs from the true value by less than 0.25.

At x = 2.05, the true value gives $(2.05)^2 + 2(2.05) = 8.3025$ and the linearization gives $6(2.05) - 4 = 8.3$. Therefore, the approximate value differs from the true value by less than 0.25×10^{-2}.

At x = 2.005, the true value gives $(2.005)^2 + 2(2.005) = 8.030025$ and the linearization gives $6(2.005) - 4 = 8.03$. Therefore, the approximate value differs from the true value by less than 0.25×10^{-4}.

Note: From the above example, we see that how accurate the approximation is for some values of x near 2. It turns out that, if the variable x moves a small increment then the true value and approximate value become close.

EXAMPLE 18 If the radius of a circle increases from r = 5 cm to 5.01 cm, find the approximate increase in the area. Estimate the area of the enlarged circle and find the error.

SOLUTION

Let A be the area of the circle of radius r.

$A(r) = \pi r^2$

$A'(r) = 2\pi r$

Since $A(5) = 25\pi$ and $A'(5) = 10\pi$, the linearization is

$L(r) = A(5) + A'(5)(r - 5)$

$= 10\pi r - 25\pi$

That is, $\pi r^2 \approx 10\pi r - 25\pi$.

At r = 5.01, $L(r) = L(5.01) = 10\pi(5.01) - 25\pi = 25.1\pi$.

The area of circle of radius 5.01 cm is approximately 25.1π cm^2. Hence the approximate increase is 0.1π cm^2.

The true area is

$A(5.01) = \pi(5.01)^2$

$= 25.1001\pi$ cm^2

Therefore

Error = True area − Approximate area

$= 0.0001\pi$ cm^2

EXAMPLE 19 Given that $f(x) = x^{1/2}$, determine the approximate value for $\sqrt{101}$ by using approximation.

SOLUTION

$f(x) = x^{1/2}$

When x = 100, $f(100) = 100^{1/2} = 10$, we have to approximate $\sqrt{101}$.

$f'(x) = \frac{1}{2\sqrt{x}}$

$f'(100) = \frac{1}{20}$

The linearization is

$L(x) = f(100) + f'(100)(x - 100)$

$= \frac{x}{20} + 5$

That is, $\sqrt{x} \approx \frac{x}{20} + 5$.

At x = 101, L(x) = L(101) = 10.05.

Hence, $\sqrt{101} \approx 10.05$.

EXERCISE 9.4

1.(a) Find the linearization of the function. Compare the approximate value and true value at x = 3.2, x = 3.02 and x = 3.002.f(x) = $\sqrt{1+x}$ at x = 3

SOLUTION

$f(x) = \sqrt[2]{1+x} = (1+x)^{1/2}$ at $x = 3$

$f'(x) = \frac{1}{2\sqrt{1+x}}$

$f(3) = \sqrt{1+3}$

$= 2$

$f'(3) = \frac{1}{2\sqrt{1+3}}$

$= \frac{1}{4}$

The linearization is

$L(x) = f(3) + f'(3)(x-3)$

$= 2 + \frac{1}{4}(x-3)$

$\mathbf{L(x) = \frac{1}{4}(x+5)}$

$\sqrt{1+x} \approx \frac{1}{4}(x+5)$

x	True value f(x)	Linearization L(x)
3.2	2.04939	2.05
3.02	2.00499	2.005
3.002	2.0005	2.0005

At $x = 3.2$, the approximate value differs from the true value by less than 0.001 [10^{-3}].

At $x = 3.02$, it differs by less than 0.0001 [10^{-4}].

At $x = 3.002$, it differs by less than 0.00001 [10^{-5}].

1.(b) Find the linearization of the function. Compare the approximate value and true value at $x = 1$, $x = 1.01$ and $x = 1.001$.$f(x) = \frac{x}{x+1}$ at $x = 1$

SOLUTION

$f(x) = \frac{x}{x+1}$ at $x = 1$

$f'(x) = \frac{(x+1)\cdot 1 - x(1)}{(x+1)^2}$

$= \frac{1}{(x+1)^2}$

$f(1) = \frac{1}{1+1}$

$= \frac{1}{2}$

$f'(1) = \frac{1}{(1+1)^2}$

$= \frac{1}{4}$

The linearization is

$L(x) = f(a) + f'(a)(x-a)$

$= f(1) + f'(1)(x-1)$

$= \frac{1}{2} + \frac{1}{4}(x-1)$

$\mathbf{L(x) = \frac{1}{4}(x+1)}$

$\frac{x}{x+1} \approx \frac{1}{4}(x+1)$

x	True value f(x)	Linearization L(x)
1	0.5	0.5
1.01	0.50249	0.5025
1.001	0.50025	0.50025

At $x = 1$, the approximate value is the same as the true value (0.5).

At $x = 1.01$, the approximate value differs from the true value by less than 0.0001 [10^{-4}].

At $x = 1.001$, the approximate value differs from the true value by less than 0.00001 [10^{-5}].

2. Show that the linearization of $f(x) = (1+x)^k$ at $x = 0$ is $L(x) = 1 + kx$.

SOLUTION

$f(x) = (1+x)^k$ at $x = 0$

$= 0$

$f'(x) = k(1+x)^{k-1}$

$f(0) = (1+0)^k$

$= 1^k$

$= 1$

$f'(0) = k(1+0)^{k-1}$

$= k(1)^{k-1}$

$= k$

The linearization is

$L(x) = f(0) + f'(0)(x-0)$

$= 1 + kx$

$\therefore \mathbf{L(x) = 1 + kx}$

3. Find the approximate change in the volume of a sphere when its radius decreases from 5 cm to 4.97 cm. Estimate the volume of the contracted sphere and find the error.

SOLUTION

Let V be the volume of a sphere of radius r.

$V(r) = \frac{4}{3}\pi r^3$

$V'(r) = \frac{4}{3}\cdot 3\pi r^2$

$= 4\pi r^2$

At $r = 5$,

$V(5) = \frac{4}{3}\pi(5)^3$

$= \frac{500}{3}\pi$

$= 166.667\pi$

$V'(5) = 4\pi(5)^2$

$= 100\pi$

The linearization is $L(r) = V(5) + V'(5)(r-5) = 100\pi(r - \frac{10}{3})$.

At r = 4.97, $L(4.97) = 100\pi(4.97 - \frac{10}{3}) = 163.667\pi$ cm^3.

The volume of sphere of radius 4.97 cm is approximately 163.667π cm^3.

The approximate change $= L(4.97) - V(5)$
$= 163.667\pi - 166.667\pi$
$= -3\pi$

True volume, $V(4.97) = \frac{4}{3}\pi(4.97)^3 = 163.6846\pi$ cm^3.

Error $= V(4.97) - L(4.97)$
$= 163.6846\pi - 163.6667\pi$
$= 0.0179\pi$ cm^3

Approximate change = -3π cm^3; volume ≈ 163.667π cm^3; error = 0.0179π cm^3.

---

4. If $y = 4\sqrt{x} + 3x^2$, find the approximate change in y when x changes from 9 to 8.98.

SOLUTION

$y(x) = 4\sqrt{x} + 3x^2$

$y'(x) = \frac{2}{\sqrt{x}} + 6x$

$y(9) = 4\sqrt{9} + 3(9)^2$
$= 4 \times 3 + 81 \times 3$
$= 255$

$y'(9) = \frac{2}{\sqrt{9}} + 6(9)$
$= \frac{2}{3} + 54$
$= \frac{164}{3}$

The linearization is

$L(x) = y(9) + y'(9)(x - 9)$
$= 255 + \frac{164}{3}(x - 9)$
$= \frac{164}{3}x - 237$

At x = 8.98, $L(8.98) = \frac{164}{3}(8.98) - 237 = 253.91$.

∴ The approximate change = L(8.98) − y(9) = 253.91 − 255 = −1.09.

5.(i) If $y = 3\sqrt{72 + x^2}$, find the approximate change in y when x increases from 3 to 3.01. Find the error.

SOLUTION

Let $f(x) = y = 3\sqrt{72 + x^2}$

$f'(x) = 3 \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{72 + x^2}} \cdot 2x$
$= \frac{3x}{\sqrt{72 + x^2}}$

$f(3) = 3\sqrt{72 + 3^2}$
$= 27$

$f'(3) = \frac{3(3)}{\sqrt{72 + 3^2}}$
$= \frac{9}{9}$
$= 1$

The linearization,

$L(x) = f(3) + f'(3)(x - 3)$
$= 27 + 1(x - 3)$
$= x + 24$

$3\sqrt{72 + x^2} \approx x + 24$

At x = 3.01, L(3.01) = 3.01 + 24 = 27.01.

∴ The approximate change = L(3.01) − f(3) = 27.01 − 27 = 0.01.

The true value is $f(3.01) = 3\sqrt{72 + (3.01)^2} = 27.01001$.

Error = True value − Approximate value
$= 27.01001 - 27.01$
$= 0.00001$

Approximate change = 0.01; error = 0.00001.

5.(ii) If $y = 3\sqrt{72 + x^2}$, find the approximate change in y when x decreases from 3 to 2.98. Find the error.

SOLUTION

Using the linearization L(x) = x + 24 (from part (i)):

At x = 2.98, L(2.98) = 2.98 + 24 = 26.98.

∴ The approximate change = L(2.98) − f(3) = 26.98 − 27 = −0.02.

The true value is $f(2.98) = 3\sqrt{72 + (2.98)^2} = 26.98006$.

Error = True value − Approximate value
$= 26.98006 - 26.98$
$= 0.00006$

Approximate change = −0.02; error = 0.00006.

6.(i) Use linearization to approximate the value of $\sqrt{80}$.

SOLUTION

Let $f(x) = x^{1/2}$. Then $f'(x) = \frac{1}{2\sqrt{x}}$.

When x = 81, $f(81) = (81)^{1/2} = 9$ and $f'(81) = \frac{1}{2\sqrt{81}} = \frac{1}{18}$.

The linearization is

$L(x) = f(81) + f'(81)(x - 81)$
$= 9 + \frac{1}{18}(x - 81)$
$= \frac{x}{18} + \frac{9}{2}$

$\sqrt{x} \approx \frac{x}{18} + \frac{9}{2}$

At x = 80, $L(80) = \frac{80}{18} + \frac{9}{2} = 8.944$.

$\therefore \sqrt{80} \approx 8.944$

6.(ii) Use linearization to approximate the value of $\sqrt[3]{65}$.

SOLUTION

Let $f(x) = x^{1/3}$. Then $f'(x) = \frac{1}{3x^{2/3}}$.

When $x = 64$, $f(64) = 64^{1/3} = 4$ and $f'(64) = \frac{1}{3(64)^{2/3}} = \frac{1}{48}$.

The linearization is

$$L(x) = f(64) + f'(64)(x - 64)$$
$$= 4 + \frac{1}{48}(x - 64)$$
$$= \frac{x}{48} + \frac{8}{3}$$

$$\sqrt[3]{x} \approx \frac{x}{48} + \frac{8}{3}$$

At $x = 65$, $L(65) = \frac{65}{48} + \frac{8}{3} = 4.0208$.

$\therefore \sqrt[3]{65} \approx 4.0208$

6.(iii) Use linearization to approximate the value of $\frac{1}{\sqrt{1.22}}$.

SOLUTION

Let $f(x) = x^{-1/2} = \frac{1}{\sqrt{x}}$. Then $f'(x) = -\frac{1}{2}x^{-3/2}$.

Take $a = 1.21$ (so that $\sqrt{a} = 1.1$):

$$f(1.21) = \frac{1}{\sqrt{1.21}}$$
$$= \frac{1}{1.1}$$
$$= \frac{10}{11}$$

$$f'(1.21) = -\frac{1}{2(1.21)^{3/2}}$$
$$= -\frac{1}{2(1.331)}$$
$$= -\frac{500}{1331}$$

The linearization is

$$L(x) = f(1.21) + f'(1.21)(x - 1.21)$$
$$= \frac{10}{11} - \frac{500}{1331}(x - 1.21)$$

At $x = 1.22$, $L(1.22) = \frac{10}{11} - \frac{500}{1331}(0.01) = \frac{1205}{1331} = 0.9053$.

$\therefore \frac{1}{\sqrt{1.22}} \approx 0.9053$

CH.10 Integration

Notes · Examples · Exercises

NOTES

Antiderivative

If F(x) is a function such that $F'(x) = f(x)$ for all x in some interval, then F(x) is called an **antiderivative** of f(x). Antiderivatives are not unique: if F(x) is an antiderivative of f(x), so is F(x) + C for any constant C.

Indefinite Integral

The set of all antiderivatives of f(x) is called the **indefinite integral** of f(x) with respect to x, written $\int f(x)\,dx = F(x) + C$, where $F'(x)=f(x)$.

Constant of Integration

The arbitrary constant C that appears in every indefinite integral is called the **constant of integration**. It arises because the derivative of any constant is zero, so infinitely many functions share the same derivative.

Integrand

In $\int f(x)\,dx$, the function f(x) is called the **integrand**, and dx indicates that the integration is taking place with respect to the variable x.

Rules of Integration

Suppose f(x) and g(x) are continuous functions and $k \in \mathbb{R}$.

1. $\int k\,dx = kx + C$
2. $\int k\,f(x)\,dx = k\int f(x)\,dx$
3. $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

Power Rule

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1)$$

Linear Substitution Rule

If F(x) is an antiderivative of f(x) (so $F'(x)=f(x)$), and a, b are constants with $a \neq 0$, then:

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$$

Special case — for a derivative $f'(ax+b)$ already in this form:

$$\int f'(ax+b)\,dx = \frac{1}{a}f(ax+b) + C$$

This shortcut avoids a full substitution when the integrand is a simple function of a linear expression ax+b. Common cases:

Integral	Result
$\int (ax+b)^n\,dx\ (n\neq -1)$	$\frac{(ax+b)^{n+1}}{a(n+1)} + C$
$\int \frac{1}{ax+b}\,dx$	$\frac{1}{a}\ln\lvert ax+b\rvert + C$
$\int e^{ax+b}\,dx$	$\frac{1}{a}e^{ax+b} + C$
$\int \sin(ax+b)\,dx$	$-\frac{1}{a}\cos(ax+b) + C$
$\int \cos(ax+b)\,dx$	$\frac{1}{a}\sin(ax+b) + C$
$\int \sec^2(ax+b)\,dx$	$\frac{1}{a}\tan(ax+b) + C$

Standard Integral Forms

Integral	Result
$\int \frac{1}{x}\,dx$	$\ln\lvert x\rvert + C$
$\int e^x\,dx$	$e^x + C$
$\int a^x\,dx\ (a>0, a\neq 1)$	$\frac{a^x}{\ln a} + C$
$\int \sin x\,dx$	$-\cos x + C$
$\int \cos x\,dx$	$\sin x + C$
$\int \sec^2 x\,dx$	$\tan x + C$
$\int \csc^2 x\,dx$	$-\cot x + C$
$\int \sec x \tan x\,dx$	$\sec x + C$
$\int \csc x \cot x\,dx$	$-\csc x + C$
$\int (1+\tan^2 x)\,dx = \int \sec^2 x\,dx$	$\tan x + C$

Additional Results

$\int \sec x\,dx = \ln\lvert\sec x + \tan x\rvert + C$
$\int \csc x\,dx = -\ln\lvert\csc x + \cot x\rvert + C$
$\int \tan x\,dx = \ln\lvert\sec x\rvert + C$
$\int \cot x\,dx = \ln\lvert\sin x\rvert + C$

Trigonometric Formulas (useful for Integration)

Pythagorean Identities

$\sin^2 x + \cos^2 x = 1$
$1 + \tan^2 x = \sec^2 x$
$1 + \cot^2 x = \csc^2 x$

Double-Angle Formulas

$\sin 2x = 2\sin x \cos x$
$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$
$\tan 2x = \frac{2\tan x}{1-\tan^2 x}$

Power-Reducing (Half-Angle) Formulas — used to integrate $\sin^2 x$ and $\cos^2 x$:

$\sin^2 x = \frac{1-\cos 2x}{2}$
$\cos^2 x = \frac{1+\cos 2x}{2}$

Product-to-Sum Formulas — used to integrate products such as $\sin 3x \cos 2x$:

$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

$\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$

$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$

$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

Substitution Method — Steps

We use the substitution method when the given integral can be transformed to a simpler integral by a change of variable.

1. Select a substitution $u = g(x)$.
2. Differentiate the substitution and arrange to write dx in terms of du.
3. Substitute and transform the entire integral from x-variable to u-variable form.
4. Integrate with respect to u.
5. Rewrite the answer in terms of x.

Integration by Parts Formula

$\int u\,dv = uv - \int v\,du$

ILATE Rule (Choosing u)

When the integrand is a product of two different types of functions, ILATE gives the priority order for choosing u (the earlier a function's type appears in ILATE, the more likely it should be chosen as u):

Letter	Function type	Example
I	Inverse trigonometric	$\sin^{-1} x$, $\tan^{-1} x$
L	Logarithmic	$\ln x$
A	Algebraic	x, x^2, $\sqrt{x}$
T	Trigonometric	$\sin x$, $\cos x$
E	Exponential	e^x, a^x

The remaining factor (along with dx) is taken as dv. For example, in $\int x \ln x\,dx$, L comes before A, so $u = \ln x$ and $dv = x\,dx$.

Tabular Method (Repeated Integration by Parts)

For a product where one factor (usually a polynomial) eventually differentiates to 0 — e.g. $\int x^2 e^x\,dx$ — repeated integration by parts can be organized in a table instead of applying the formula over and over:

1. Make two columns: D (successive derivatives of the polynomial factor, down to 0) and I (successive integrals of the other factor).
2. Pair each D-row with the I-row directly below it, and attach alternating signs +, −, +, −, ... starting with +.
3. The answer is the sum of each signed D×I product (stop once the D column reaches 0), plus C.

D (differentiate)	I (integrate)
+ x^2	e^x
− $2x$	e^x
+ 2	e^x
0	e^x

$\int x^2e^x\,dx = x^2e^x - 2xe^x + 2e^x + C$ — read off by multiplying diagonally (each D entry by the I entry one row below it) and adding with the shown signs.

Partial Fraction Method — Approach

We use the partial fraction method to integrate rational functions by decomposing them into simpler fractional components.

1. Factor the denominator completely.
2. Write the rational function as a sum of simpler fractions with unknown constants (A, B, ...) over each factor — a repeated linear factor $(x-a)^2$ gets two terms, $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$.
3. Clear denominators and solve for the unknown constants, either by equating coefficients or by substituting convenient values of x.
4. Integrate each simpler fraction separately using standard forms such as $\int \frac{1}{x-a}\,dx = \ln|x-a| + C$.

Cover-Up Method (Partial Fractions)

A quick shortcut for finding the constants when the denominator factors into distinct linear factors, e.g. $\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$:

1. To find A: cover up the factor $(x-a)$ in the original fraction, and substitute $x=a$ into what remains.
2. To find B: cover up the factor $(x-b)$, and substitute $x=b$ into what remains.

Example: $\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$:

$A = \frac{5(1)-1}{1+2} = \frac{4}{3}$ (cover $x-1$, put $x=1$)

$B = \frac{5(-2)-1}{(-2)-1} = \frac{-11}{-3} = \frac{11}{3}$ (cover $x+2$, put $x=-2$)

This only works directly for non-repeated linear factors; repeated or irreducible quadratic factors still need the equate-coefficients approach above.

Section 10.1

EXAMPLE 1 Find the antiderivative of

a. $\sqrt{x}$

b. $3x^5$

c. e^{3x}

d. $\frac{1}{x^3}$

SOLUTION

(a) $\frac{d}{dx}x^{3/2} = \frac{3}{2}x^{1/2}$

$\frac{d}{dx}\left(\frac{2}{3}x^{3/2}\right) = x^{1/2}$

$= \sqrt{x}$

$\therefore$ the antiderivative of $\sqrt{x}$ is $\frac{2}{3}x^{3/2}$.

(b) $\frac{d}{dx}x^6 = 6x^5$

$\frac{d}{dx}(\frac{1}{2}x^6) = 3x^5$

∴ the antiderivative of $3x^5$ is $\frac{1}{2}x^6$.

(c) $\frac{d}{dx}e^{3x} = 3e^{3x}$

$\frac{d}{dx}(\frac{1}{3}e^{3x}) = e^{3x}$

∴ the antiderivative of e^{3x} is $\frac{1}{3}e^{3x}$.

(d) $\frac{d}{dx}x^{-2} = -2x^{-3}$

$\frac{d}{dx}(-\frac{1}{2x^2}) = \frac{1}{x^3}$

∴ the antiderivative of $\frac{1}{x^3}$ is $-\frac{1}{2x^2}$.

EXAMPLE 2 Evaluate each of the following integrals.

a. $\int (4x^5+1)\,dx$

b. $\int (2x^6-\frac{1}{3}x^3+\frac{3}{x})\,dx$

c. $\int 5x\sqrt{x}\,dx$

d. $\int \frac{(x-1)^2}{\sqrt{x}}\,dx$

SOLUTION

(a) $\int (4x^5+1)\,dx = 4\int x^5\,dx+\int 1\,dx$

$=(\frac{4}{6}x^6+C_1)+(x+C_2)$

$=\frac{2}{3}x^6+x+C$

(b) $\int (2x^6-\frac{1}{3}x^3+\frac{3}{x})\,dx$

$= 2\int x^6\,dx-\frac{1}{3}\int x^3\,dx+3\int \frac{1}{x}\,dx$

$=\frac{2}{7}x^7-\frac{1}{12}x^4+3\ln|x|+C$

(c) $\int 5x\sqrt[4]{x}\,dx = 5\int x\cdot x^{1/2}\,dx$

$= 5\int x^{3/2}\,dx$

$=5(\frac{2}{5}x^{5/2})+C$

$=2x^{5/2}+C$

(d) $\int \frac{(x-1)^2}{\sqrt{x}}\,dx = \int \frac{x^2-2x+1}{x^{1/2}}\,dx$

$=\int (x^{3/2}-2x^{1/2}+x^{-1/2})\,dx$

$=\frac{2}{5}x^{5/2}-\frac{4}{3}x^{3/2}+2x^{1/2}+C$

EXAMPLE 3 Find the following integrals.

a. $\int \frac{e^{-x}+1}{e^{-x}}\,dx$

b. $\int (e^x+2x)\,dx$

c. $\int (e^x\log 5+\frac{1}{x^2})\,dx$

d. $\int 3^x\ln 3\,dx$

SOLUTION

(a) $\int \frac{e^{-x}+1}{e^{-x}}\,dx = \int (1+e^x)\,dx$

$=\int 1\,dx+\int e^x\,dx$

$= x+e^x+C$

(b) $\int (e^x+2x)\,dx = \int e^x\,dx+\int 2x\,dx$

$=e^x+x^2+C$

(c) $\int (e^x\log 5+\frac{1}{x^2})\,dx = \int e^x\log 5\,dx+\int x^{-2}\,dx$

$=e^x\log 5-\frac{1}{x}+C$

(d) $\int 3^x\ln 3\,dx = \frac{1}{\ln 3}\cdot 3^x\ln 3+C$

$=3^x+C$

EXAMPLE 4 Evaluate.

a. $\int (3\cos x-5\sin x)\,dx$

b. $\int (e^x+2\sin x)\,dx$

c. $\int \frac{e^x-\sqrt{x}}{2}\,dx$

d. $\int (1+\tan^2 x)\,dx$

SOLUTION

(a) $\int (3\cos x-5\sin x)\,dx = 3\int \cos x\,dx-5\int \sin x\,dx$

$=3\sin x+5\cos x+C$

(b) $\int (e^x+2\sin x)\,dx = \int e^x\,dx+2\int \sin x\,dx$

$=e^x-2\cos x+C$

(c) $\int \frac{e^x-\sqrt{x}}{2}\,dx = \frac{1}{2}\int e^x\,dx-\frac{1}{2}\int x^{1/2}\,dx$

$=\frac{1}{2}e^x-\frac{1}{3}x^{3/2}+C$

(d) $\int (1+\tan^2 x)\,dx = \int \sec^2 x\,dx$

$=\tan x+C$

EXAMPLE 5 Evaluate the following integrals.

a. $\int \frac{1}{3-4x}\,dx$

b. $\int \sqrt{8x-7}\,dx$

c. $\int e^{2x+1}\,dx$

d. $\int 2^{3x+1}\,dx$

e. $\int \cos^2 x\,dx$

f. $\int \sin 4x\cos 3x\,dx$

SOLUTION

(a)

$\int \frac{1}{3-4x}\,dx = -\frac{1}{4}\ln|3-4x|+C$

(b) $\int \sqrt[3]{8x-7}\,dx = \frac{1}{8}\cdot\frac{2}{3}(8x-7)^{3/2}+C$

$=\frac{1}{12}(8x-7)^{3/2}+C$

(c)

$\int e^{2x+1}\,dx = \frac{1}{2}e^{2x+1}+C$

(d)

$\int 2^{3x+1}\,dx = \frac{1}{3}\cdot\frac{1}{\ln 2}\cdot 2^{3x+1}+C$

(e) Since $\cos^2 x=\frac{1}{2}(1+\cos 2x)$,

$\int \cos^2 x\,dx = \frac{1}{2}\int(1+\cos 2x)\,dx$

$=\frac{1}{2}(x+\frac{1}{2}\sin 2x)+C$

$=\frac{1}{2}x+\frac{1}{4}\sin 2x+C$

(f) Since $\sin 4x\cos 3x=\frac{1}{2}[\sin(4x+3x)+\sin(4x-3x)]=\frac{1}{2}(\sin 7x+\sin x)$,

$\int \sin 4x\cos 3x\,dx = \frac{1}{2}\int(\sin 7x+\sin x)\,dx$

$=\frac{1}{2}(-\frac{1}{7}\cos 7x-\cos x)+C$

$=-\frac{1}{14}\cos 7x-\frac{1}{2}\cos x+C$

EXERCISE 10.1

1.(a) Evaluate the following integrals: $\int 4x^8\,dx$

SOLUTION

$\int 4x^8\,dx = 4\cdot\frac{x^9}{9}$

$=\frac{4}{9}x^9+C$

1.(b) Evaluate the following integrals: $\int \frac{3}{2}x^2x^{1/3}\,dx$

SOLUTION

$\int \frac{3}{2}x^2\sqrt[3]{x}\,dx = \int \frac{3}{2}x^{7/3}\,dx$

$=\frac{3}{2}\cdot\frac{x^{10/3}}{\frac{10}{3}}$

$=\frac{9}{20}x^{10/3}+C$

1.(c) Evaluate the following integrals: $\int (5x+2)\,dx$

SOLUTION

$\int (5x+2)\,dx = \frac{5}{2}x^2+2x+C$

1.(d) Evaluate the following integrals: $\int \sin^2 x\,dx$

SOLUTION

$\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx$

$=\frac{1}{2}x-\frac{1}{4}\sin 2x+C$

1.(e) Evaluate the following integrals: $\int \frac{x+3}{\sqrt{x}}\,dx$

SOLUTION

$\int \frac{x+3}{\sqrt{x}}\,dx = \int (x^{1/2}+3x^{-1/2})\,dx$

$=\frac{2}{3}x^{3/2}+6x^{1/2}+C$

1.(f) Evaluate the following integrals: $\int (\frac{1}{2x}+5)\,dx$

SOLUTION

$\int (\frac{1}{2x}+5)\,dx = \frac{1}{2}\ln|x|+5x+C$

1.(g) Evaluate the following integrals: $\int (e^x+\frac{2}{x})\,dx$

SOLUTION

$\int (e^x+\frac{2}{x})\,dx = e^x+2\ln|x|+C$

1.(h) Evaluate the following integrals: $\int (\frac{1}{x^5}+4e^x)\,dx$

SOLUTION

$\int (\frac{1}{x^5}+4e^x)\,dx = \int (x^{-5}+4e^x)\,dx$

$=\frac{x^{-4}}{-4}+4e^x$

$=-\frac{1}{4x^4}+4e^x+C$

1.(i) Evaluate the following integrals: $\int (\frac{3}{x}+e^x+10)\,dx$

SOLUTION

$\int (\frac{3}{x}+e^x+10)\,dx = 3\ln|x|+e^x+10x+C$

1.(j) Evaluate the following integrals: $\int \cos^2 3x\,dx$

SOLUTION

$\int \cos^2 3x\,dx = \int \frac{1+\cos 6x}{2}\,dx$

$=\frac{1}{2}x+\frac{1}{12}\sin 6x+C$

1.(k) Evaluate the following integrals: $\int \sin 2x\sin 5x\,dx$

SOLUTION

$\int \sin 2x\sin 5x\,dx = \int \frac{1}{2}[\cos(2x-5x)-\cos(2x+5x)]\,dx$

$=\frac{1}{2}\int[\cos(-3x)-\cos(7x)]\,dx$

$=\frac{1}{2}[\frac{\sin 3x}{3}-\frac{\sin 7x}{7}]+C$

$=\frac{1}{6}\sin 3x-\frac{1}{14}\sin 7x+C$

1.(l) Evaluate the following integrals: $\int \cos 4x\cos 7x\,dx$

SOLUTION

$\int \cos 4x\cos 7x\,dx = \int \frac{1}{2}[\cos(4x-7x)+\cos(4x+7x)]\,dx$

$=\frac{1}{2}\int[\cos(-3x)+\cos(11x)]\,dx$

$= \frac{1}{2}\left[\frac{\sin 3x}{3} + \frac{\sin 11x}{11}\right] + C$

$= \frac{1}{6}\sin 3x + \frac{1}{22}\sin 11x + C$

2.(a) Evaluate the following integrals: $\int (1-2x)^3\,dx$

SOLUTION

Let $u = 1-2x$, then $du = -2\,dx$.

$$\int (1-2x)^3\,dx = -\frac{1}{2}\int u^3\,dx$$

$$= -\frac{1}{2}\cdot\frac{u^4}{4}$$

$$= -\frac{1}{8}(1-2x)^4 + C$$

2.(b) Evaluate the following integrals: $\int \sin(2\pi x+7)\,dx$

SOLUTION

$$\int \sin(2\pi x+7)\,dx = -\frac{1}{2\pi}\cos(2\pi x+7) + C$$

2.(c) Evaluate the following integrals: $\int \cos(3x-7)\,dx$

SOLUTION

$$\int \cos(3x-7)\,dx = \frac{1}{3}\sin(3x-7) + C$$

2.(d) Evaluate the following integrals: $\int 3^{5x-2}\,dx$

SOLUTION

$$\int 3^{5x-2}\,dx = \frac{3^{5x-2}}{5\ln 3} + C$$

2.(e) Evaluate the following integrals: $\int \frac{1}{7x-6}\,dx$

SOLUTION

$$\int \frac{1}{7x-6}\,dx = \frac{1}{7}\ln|7x-6| + C$$

2.(f) Evaluate the following integrals: $\int \frac{\sin 2x}{\sin x}\,dx$

SOLUTION

$$\int \frac{\sin 2x}{\sin x}\,dx = \int \frac{2\sin x\cos x}{\sin x}\,dx$$

$$= \int 2\cos x\,dx$$

$$= 2\sin x + C$$

2.(g) Evaluate the following integrals: $\int \sec^2(2x+3)\,dx$

SOLUTION

$$\int \sec^2(2x+3)\,dx = \frac{1}{2}\tan(2x+3) + C$$

2.(h) Evaluate the following integrals: $\int e^{7x-3}\,dx$

SOLUTION

$$\int e^{7x-3}\,dx = \frac{1}{7}e^{7x-3} + C$$

2.(i) Evaluate the following integrals: $\int (1+\tan^2 2\pi x)\,dx$

SOLUTION

$$\int (1+\tan^2 2\pi x)\,dx = \int \sec^2(2\pi x)\,dx$$

$$= \frac{1}{2\pi}\tan(2\pi x) + C$$

Section 10.2

EXAMPLE 6 Evaluate each of the following integrals.

a. $\int xe^{x^2}\,dx$

b. $\int \sin^4 x\cos x\,dx$

c. $\int x^2\left(\frac{x^3}{3}+1\right)^3\,dx$

d. $\int \sqrt{1-5x}\,dx$

e. $\int 3x\sqrt{x^2+5}\,dx$

f. $\int \cos^3 x\,dx$

SOLUTION

(a) Let $u = x^2$. Then $du = 2x\,dx$, so $\frac{1}{2}du = x\,dx$.

$$\int xe^{x^2}\,dx = \int e^{x^2}\cdot x\,dx$$

$$= \int e^u\cdot\frac{1}{2}\,dx$$

$$= \frac{1}{2}e^u + C$$

$$= \frac{1}{2}e^{x^2} + C$$

(b) Let $u = \sin x$. Then $du = \cos x\,dx$.

$$\int \sin^4 x\cos x\,dx = \int u^4\,dx$$

$$= \frac{u^5}{5} + C$$

$$= \frac{\sin^5 x}{5} + C$$

(c) Let $u = \frac{x^3}{3}+1$. Then $du = x^2dx$.

$$\int x^2\left(\frac{x^3}{3}+1\right)^3\,dx = \int u^3\,dx$$

$$= \frac{u^4}{4} + C$$

$$= \frac{1}{4}\left(\frac{x^3}{3}+1\right)^4 + C$$

(d) Let $u = 1-5x$. Then $du = -5\,dx$, so $-\frac{1}{5}du = dx$.

$$\int \sqrt[4]{1-5x}\,dx = \int (1-5x)^{1/2}\,dx$$

$$= \int u^{1/2}\left(-\frac{1}{5}\right)dx$$

$$= -\frac{1}{5}\cdot\frac{2}{3}u^{3/2} + C$$

$$= -\frac{2}{15}(1-5x)^{3/2} + C$$

(e) Let $u = x^2+5$. Then $du = 2x\,dx$, so $\frac{1}{2}du = x\,dx$.

$$\int 3x\sqrt{x^2+5}\,dx = 3\int (x^2+5)^{1/2}\cdot x\,dx$$

$$= 3\int u^{1/2}\cdot\frac{1}{2}\,dx$$

$$= \frac{3}{2}\cdot\frac{2}{3}u^{3/2} + C$$

$$= u^{3/2} + C$$

$$= (x^2+5)^{3/2} + C$$

(f) Use $\cos^2 x = 1-\sin^2 x$, then

$$\int \cos^3 x\,dx = \int (1-\sin^2 x)\cos x\,dx$$

Let $u=\sin x$, then $du=\cos x\,dx$.

$$\int \cos^3 x\,dx = \int (1-u^2)\,dx$$
$$= u-\frac{u^3}{3}+C$$
$$= \sin x-\frac{1}{3}\sin^3 x+C$$

EXAMPLE 7 Evaluate the following integrals.

a. $\int \frac{2x-1}{x^2-x-6}\,dx$

b. $\int \frac{x^2+2x-1}{x^2-1}\,dx$

c. $\int \frac{1}{1+e^x}\,dx$

SOLUTION

(a) Let $u=x^2-x-6$. Then $du=(2x-1)dx$.

$$\int \frac{2x-1}{x^2-x-6}\,dx = \int \frac{1}{u}\,du$$
$$= \ln|u|+C$$
$$= \ln|x^2-x-6|+C$$

(b) Rewrite as

$$\int \frac{x^2+2x-1}{x^2-1}\,dx = \int \left(1+\frac{2x}{x^2-1}\right)dx$$

Let $u=x^2-1$. Then $du=2x\,dx$.

$$\int \frac{x^2+2x-1}{x^2-1}\,dx = \int 1\,dx+\int \frac{2x}{x^2-1}\,dx$$
$$= x+\int \frac{1}{u}\,du$$
$$= x+\ln|u|+C$$
$$= x+\ln|x^2-1|+C$$

(c) Multiply and divide by e^{-x}:

$$\int \frac{1}{1+e^x}\,dx = \int \frac{1}{1+e^x}\cdot\frac{e^{-x}}{e^{-x}}\,dx = \int \frac{e^{-x}}{e^{-x}+1}\,dx$$

Let $u=e^{-x}+1$. Then $du=-e^{-x}dx$.

$$\int \frac{1}{1+e^x}\,dx = -\int \frac{1}{u}\,du$$
$$= -\ln|u|+C$$
$$= -\ln(e^{-x}+1)+C$$

Additional Results:

$\int \sec x\,dx = \ln|\sec x + \tan x| + C$

$\int \csc x\,dx = -\ln|\csc x + \cot x| + C$

EXERCISE 10.2

1.(a) Integrate the following functions using the given substitutions: $\int 4x^3\sqrt{x^4-1}\,dx$

SOLUTION

Let $u=x^4-1$. Then $du=4x^3\,dx$.

Substitute:

$$\int 4x^3\sqrt{x^4-1}\,dx = \int \sqrt{u}\,dx$$
$$= \frac{2}{3}u^{3/2}+C$$
$$= \frac{2}{3}(x^4-1)^{3/2}+C$$

1.(b) Integrate the following functions using the given substitutions: $\int \cos^3 x \sin x\,dx$

SOLUTION

Let $u=\cos x$. Then $du=-\sin x\,dx$.

Substitute:

$$\int \cos^3 x \sin x\,dx = -\int u^3\,dx$$
$$= -\frac{1}{4}u^4+C$$
$$= -\frac{1}{4}\cos^4 x+C$$

1.(c) Integrate the following functions using the given substitutions: $\int \frac{1}{x\ln|x|}\,dx$

SOLUTION

Let $u=\ln|x|$. Then $du=\frac{1}{x}\,dx$.

Substitute:

$$\int \frac{1}{x\ln|x|}\,dx = \int \frac{1}{u}\,du$$
$$= \ln|u|+C$$
$$= \ln|\ln|x||+C$$

1.(d) Integrate the following functions using the given substitutions: $\int \sin^5 x \cos x\,dx$

SOLUTION

Let $u=\sin x$. Then $du=\cos x\,dx$.

Substitute:

$$\int \sin^5 x \cos x\,dx = \int u^5\,dx$$
$$= \frac{1}{6}u^6+C$$
$$= \frac{1}{6}\sin^6 x+C$$

1.(e) Integrate the following functions using the given substitutions: $\int \frac{\ln x}{x}\,dx$ $(x>0)$

SOLUTION

Let $u=\ln x$. Then $du=\frac{1}{x}\,dx$.

Substitute:

$$\int \frac{\ln x}{x}\,dx = \int u\,dx$$
$$= \frac{1}{2}u^2+C$$
$$= \frac{1}{2}(\ln x)^2+C$$

1.(f) Integrate the following functions using the given substitutions: $\int x^3 e^{x^4}\,dx$

SOLUTION

Let $u=x^4$. Then $du=4x^3\,dx$.
Substitute:

$$\int x^3e^{x^4}\,dx = \frac{1}{4}\int e^u\,dx$$
$$=\frac{1}{4}e^u+C$$
$$=\frac{1}{4}e^{x^4}+C$$

2.(a) Use the substitution method to evaluate the following integrals: $\int x\sqrt{1-x}\,dx$

SOLUTION

Let $u=1-x$, so $x=1-u$. Then $dx=-du$.
Substitute:

$$\int x\sqrt{1-x}\,dx = \int (1-u)\sqrt{u}(-du)$$
$$=\int (u^{1/2}-u^{3/2})\,dx$$
$$=\frac{2}{3}u^{3/2}-\frac{2}{5}u^{5/2}+C$$
$$=\frac{2}{3}(1-x)^{3/2}-\frac{2}{5}(1-x)^{5/2}+C$$

2.(b) Use the substitution method to evaluate the following integrals: $\int (2x+1)(x^2+x)^7\,dx$

SOLUTION

Let $u=x^2+x$. Then $du=(2x+1)\,dx$.
Substitute:

$$\int (2x+1)(x^2+x)^7\,dx = \int u^7\,dx$$
$$=\frac{1}{8}u^8+C$$
$$=\frac{1}{8}(x^2+x)^8+C$$

2.(c) Use the substitution method to evaluate the following integrals: $\int \sin^3 x\,dx$

SOLUTION

Let $u=\cos x$ (using $\sin^3 x=\sin x(1-\cos^2 x)$). Then $du=-\sin x\,dx$.
Substitute:

$$\int \sin^3 x\,dx = \int (1-\cos^2 x)\sin x\,dx$$
$$=\int (1-u^2)(-du)\,dx$$
$$=\int (u^2-1)\,dx$$
$$=\frac{1}{3}u^3-u+C$$
$$=\frac{1}{3}\cos^3 x-\cos x+C$$

2.(d) Use the substitution method to evaluate the following integrals: $\int x^2\sqrt{x^3-2}\,dx$

SOLUTION

Let $u=x^3-2$. Then $du=3x^2\,dx$.
Substitute:

$$\int x^2\sqrt{x^3-2}\,dx = \frac{1}{3}\int u^{1/2}\,dx$$
$$=\frac{1}{3}\cdot\frac{2}{3}u^{3/2}+C$$
$$=\frac{2}{9}(x^3-2)^{3/2}+C$$

2.(e) Use the substitution method to evaluate the following integrals: $\int \frac{\sec^2 x}{\tan x}\,dx$

SOLUTION

Let $u=\tan x$. Then $du=\sec^2 x\,dx$.
Substitute:

$$\int \frac{\sec^2 x}{\tan x}\,dx = \int \frac{1}{u}\,du$$
$$=\ln|u|+C$$
$$=\ln|\tan x|+C$$

2.(f) Use the substitution method to evaluate the following integrals: $\int \frac{x}{\sqrt{x+1}}\,dx$

SOLUTION

Let $u=x+1$, so $x=u-1$. Then $dx=du$.
Substitute:

$$\int \frac{x}{\sqrt{x+1}}\,dx = \int \frac{u-1}{\sqrt{u}}\,du$$
$$=\int (u^{1/2}-u^{-1/2})\,dx$$
$$=\frac{2}{3}u^{3/2}-2u^{1/2}+C$$
$$=\frac{2}{3}(x+1)^{3/2}-2\sqrt{x+1}+C$$

3 Evaluate the integral $\int \frac{x}{(x^2+1)\ln(x^2+1)}\,dx$

SOLUTION

Let $u=\ln(x^2+1)$. Then $du=\frac{2x}{x^2+1}\,dx$.
Substitute:

$$\int \frac{x}{(x^2+1)\ln(x^2+1)}\,dx = \frac{1}{2}\int \frac{1}{u}\,du$$
$$=\frac{1}{2}\ln|u|+C$$
$$=\frac{1}{2}\ln|\ln(x^2+1)|+C$$

Section 10.3

EXAMPLE 8 Evaluate the following integrals.

a. $\int xe^x\,dx$
b. $\int \ln x\,dx$
c. $\int x\cos x\,dx$
d. $\int x\sin x\,dx$
e. $\int e^x\sin x\,dx$
f. $\int x^2\ln x\,dx$

SOLUTION

(a) Let $u=x$, $dv=e^x dx$.
Then $du=dx$, $v=e^x$.

$\int xe^x\,dx = xe^x - \int e^x\,dx$

$= xe^x - e^x + C$

(b) Let u=ln x, dv=dx.

Then $du = \frac{1}{x}\,dx$, v=x.

$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx$

$= x\ln x - x + C$

(c) Let u=x, dv=cos x dx.

Then du=dx, v=sin x.

$\int x\cos x\,dx = x\sin x - \int \sin x\,dx$

$= x\sin x + \cos x + C$

(d) Let u=x, dv=sin x dx.

Then du=dx, v=−cos x.

$\int x\sin x\,dx = -x\cos x - \int(-\cos x)\,dx$

$= -x\cos x + \sin x + C$

(e) Let $u=e^x$, dv=sin x dx.

Then $du=e^x dx$, v=−cos x.

$\int e^x\sin x\,dx = -e^x\cos x - \int(-\cos x)e^x\,dx$

$= -e^x\cos x + \int e^x\cos x\,dx$

Let $I = \int e^x\sin x\,dx$. Then $I = -e^x\cos x + \int e^x\cos x\,dx$.

Now for $\int e^x\cos x\,dx$, let $u=e^x$, dv=cos x dx.

Then $du=e^x dx$, v=sin x.

$\int e^x\cos x\,dx = e^x\sin x - \int e^x\sin x\,dx$

Substituting back:

$I = -e^x\cos x + e^x\sin x - I$

$2I = e^x(\sin x - \cos x)$

$I = \frac{1}{2}e^x(\sin x - \cos x) + C$

(f) Let u=ln x, $dv=x^2dx$.

Then $du = \frac{1}{x}\,dx$, $v = \frac{x^3}{3}$.

$\int x^2\ln x\,dx = \frac{x^3}{3}\ln x - \int \frac{x^3}{3}\cdot\frac{1}{x}\,dx$

$= \frac{x^3}{3}\ln x - \frac{1}{3}\int x^2\,dx$

$= \frac{x^3\ln x}{3} - \frac{x^3}{9} + C$

EXERCISE 10.3

1.(a) Use integration by parts to evaluate the following integrals: $\int se^{-2s}\,ds$

SOLUTION

Let u=s, $dv=e^{-2s}ds$.

Then du=ds, $v=\int e^{-2s}ds = -\frac{1}{2}e^{-2s}$.

$\int se^{-2s}ds = -\frac{1}{2}se^{-2s} - \int(-\frac{1}{2}e^{-2s})ds$

$= -\frac{1}{2}se^{-2s} + \frac{1}{2}\int e^{-2s}ds$

$= -\frac{1}{2}se^{-2s} - \frac{1}{4}e^{-2s} + C$

1.(b) Use integration by parts to evaluate the following integrals: $\int \ln(x+1)\,dx$

SOLUTION

Let u=ln(x+1), dv=dx.

Then $du = \frac{1}{x+1}\,dx$, $v=\int dx\,dx = x$.

$\int \ln(x+1)\,dx = x\ln(x+1) - \int x\cdot\frac{1}{x+1}\,dx$

$= x\ln(x+1) - \int \frac{x}{x+1}\,dx$

$= x\ln(x+1) - \int(1 - \frac{1}{x+1})\,dx$

$= x\ln(x+1) - x + \ln|x+1| + C$

1.(c) Use integration by parts to evaluate the following integrals: $\int t\sin 2t\,dx$

SOLUTION

Let u=t, dv=sin 2t dt.

Then du=dt, $v=\int \sin 2t\,dt = -\frac{1}{2}\cos 2t$.

$\int t\sin 2t\,dt = -\frac{1}{2}t\cos 2t - \int(-\frac{1}{2}\cos 2t)\,dt$

$= -\frac{1}{2}t\cos 2t + \frac{1}{2}\int \cos 2t\,dt$

$= -\frac{1}{2}t\cos 2t + \frac{1}{4}\sin 2t + C$

1.(d) Use integration by parts to evaluate the following integrals: $\int x2^x\,dx$

SOLUTION

Let u=x, $dv=2^x dx$.

Then du=dx, $v=\int 2^x dx = \frac{2^x}{\ln 2}$.

$\int x2^x\,dx = x\cdot\frac{2^x}{\ln 2} - \int \frac{2^x}{\ln 2}\,dx$

$= \frac{x2^x}{\ln 2} - \frac{1}{\ln 2}\int 2^x\,dx$

$= \frac{x2^x}{\ln 2} - \frac{1}{\ln 2}\cdot\frac{2^x}{\ln 2} + C$

$= \frac{x2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2} + C$

1.(e) Use integration by parts to evaluate the following integrals: $\int x\cos 5x\,dx$

SOLUTION

Let u=x, dv=cos 5x dx.

Then du=dx, $v=\int \cos 5x\,dx = \frac{1}{5}\sin 5x$.

$\int x\cos 5x\,dx = \frac{1}{5}x\sin 5x - \int \frac{1}{5}\sin 5x\,dx$

$= \frac{1}{5}x\sin 5x - \frac{1}{5}\int \sin 5x\,dx$

$= \frac{1}{5}x\sin 5x + \frac{1}{25}\cos 5x + C$

1.(f) Use integration by parts to evaluate the following integrals: $\int e^x \cos x\,dx$

SOLUTION

Let $u=\cos x$, $dv=e^x dx$.

Then $du=-\sin x\,dx$, $v=\int e^x dx=e^x$.

$$\int e^x\cos x\,dx = e^x\cos x-\int e^x(-\sin x)dx$$
$$= e^x\cos x+\int e^x\sin x\,dx$$

For $\int e^x\sin x\,dx$: let $u=\sin x$, $dv=e^x dx$. Then $du=\cos x\,dx$, $v=e^x$.

$$\int e^x\sin x\,dx = e^x\sin x-\int e^x\cos x\,dx$$

So,

$$\int e^x\cos x\,dx = e^x\cos x+e^x\sin x-\int e^x\cos x\,dx$$
$$2\int e^x\cos x\,dx = e^x(\sin x+\cos x)$$
$$\int e^x\cos x\,dx = \frac{1}{2}e^x(\sin x+\cos x)+C$$

Section 10.4

EXAMPLE 9 Evaluate the integral $\int \frac{1}{x^2+2x-3}\,dx$.

SOLUTION

$$\frac{1}{x^2+2x-3} = \frac{1}{(x+3)(x-1)}$$
$$= \frac{A}{x+3}+\frac{B}{x-1}$$
$$1 = A(x-1)+B(x+3)$$

Equating coefficients:

$$1 = (A+B)x+(-A+3B)$$
$$A+B=0,\quad -A+3B=1$$

Solving: $A=-\frac{1}{4}$, $B=\frac{1}{4}$.

$$\frac{1}{x^2+2x-3} = \frac{-\frac{1}{4}}{x+3}+\frac{\frac{1}{4}}{x-1}$$

$$\int \frac{1}{x^2+2x-3}\,dx = -\frac{1}{4}\int\frac{1}{x+3}\,dx+\frac{1}{4}\int\frac{1}{x-1}\,dx$$
$$= -\frac{1}{4}\ln|x+3|+\frac{1}{4}\ln|x-1|+C$$
$$= \frac{1}{4}\ln\left|\frac{x-1}{x+3}\right|+C$$

EXAMPLE 10 Evaluate the integral $\int \frac{1}{2x^2+3x+1}\,dx$.

SOLUTION

$$\frac{1}{2x^2+3x+1} = \frac{1}{(2x+1)(x+1)}$$
$$= \frac{A}{2x+1}+\frac{B}{x+1}$$
$$1 = A(x+1)+B(2x+1)$$

When $x=-1$: $1=B(-1) \to B=-1$.

When $x=-\frac{1}{2}$: $1=A(\frac{1}{2}) \to A=2$.

$$\int \frac{1}{2x^2+3x+1}\,dx = \int\frac{2}{2x+1}\,dx+\int\frac{-1}{x+1}\,dx$$
$$=\ln|2x+1|-\ln|x+1|+C$$
$$=\ln\left|\frac{2x+1}{x+1}\right|+C$$

EXAMPLE 11 Evaluate the integral $\int \frac{x^3+x^2-4x}{x^2-4}\,dx$.

SOLUTION

$$\int \frac{x^3+x^2-4x}{x^2-4}\,dx = \int\left(x+1+\frac{4}{x^2-4}\right)dx$$

$$\frac{4}{x^2-4} = \frac{4}{(x+2)(x-2)}$$
$$= \frac{A}{x+2}+\frac{B}{x-2}$$
$$4 = A(x-2)+B(x+2)$$

When $x=2$: $4=B(4) \to B=1$.

When $x=-2$: $4=A(-4) \to A=-1$.

$$\int \frac{x^3+x^2-4x}{x^2-4}\,dx = \int(x+1)\,dx+\int\frac{-1}{x+2}\,dx+\int\frac{1}{x-2}\,dx$$
$$= \frac{x^2}{2}+x-\ln|x+2|+\ln|x-2|+C$$
$$= \frac{x^2}{2}+x+\ln\left|\frac{x-2}{x+2}\right|+C$$

EXAMPLE 12 Evaluate the integral $\int \frac{2x+1}{(x-1)^2}\,dx$.

SOLUTION

$$\frac{2x+1}{(x-1)^2} = \frac{A}{x-1}+\frac{B}{(x-1)^2}$$
$$2x+1 = A(x-1)+B$$
$$= Ax+(-A+B)$$

Equating coefficients:

$$A=2,\quad -A+B=1 \to B=3$$

$$\frac{2x+1}{(x-1)^2} = \frac{2}{x-1}+\frac{3}{(x-1)^2}$$

$$\int \frac{2x+1}{(x-1)^2}\,dx = \int\frac{2}{x-1}\,dx+\int\frac{3}{(x-1)^2}\,dx$$
$$=2\ln|x-1|-\frac{3}{x-1}+C$$

EXAMPLE 13 Find the function $f(x)$ satisfying the equation $f'(x)=\frac{x-1}{\sqrt{x}}$ with $f(1)=0$.

SOLUTION

$$f(x) = \int\frac{x-1}{\sqrt{x}}\,dx$$
$$= \int(x^{1/2}-x^{-1/2})\,dx$$
$$= \frac{2}{3}x^{3/2}-2x^{1/2}+C$$

Since $f(1)=0$:

$$f(1) = \frac{2}{3}(1)^{3/2}-2(1)^{1/2}+C$$
$$=0$$
$$\frac{2}{3}-2+C=0 \rightarrow C$$
$$=\frac{4}{3}$$

Therefore,

$$f(x)=\frac{2}{3}x^{3/2}-2x^{1/2}+\frac{4}{3}$$

EXAMPLE 14 Find the function f(x) satisfying the equation $f''(x)=\frac{e^x-e^{-x}}{2}$ with $f'(0)=1$ and $f(0)=1$.

SOLUTION

$$f''(x)=\frac{e^x-e^{-x}}{2}$$

First integration:

$$f'(x) = \int \frac{e^x-e^{-x}}{2}\,dx$$
$$=\frac{1}{2}(e^x+e^{-x})+C_1$$

Apply $f'(0)=1$:

$$f'(0) = \frac{1}{2}(e^0+e^0)+C_1$$
$$=1$$
$$1+C_1=1 \rightarrow C_1$$
$$=0$$

So:

$$f'(x)=\frac{1}{2}(e^x+e^{-x})$$

Second integration:

$$f(x) = \int \frac{e^x+e^{-x}}{2}\,dx$$
$$=\frac{1}{2}(e^x-e^{-x})+C_2$$

Apply $f(0)=1$:

$$f(0) = \frac{1}{2}(e^0-e^0)+C_2$$
$$=1$$
$$0+C_2=1 \rightarrow C_2$$
$$=1$$

Therefore,

$$f(x)=\frac{1}{2}(e^x-e^{-x})+1$$

EXERCISE 10.4

1.(a) Use the partial fractions method to evaluate the following integrals: $\int \frac{1}{2x^2+5x+3}\,dx$

SOLUTION

Factor: $2x^2+5x+3=(2x+3)(x+1)$

Partial fractions:

$$\frac{1}{(2x+3)(x+1)} = \frac{A}{2x+3}+\frac{B}{x+1}$$

Multiply both sides by $(2x+3)(x+1)$:

$1 = A(x+1)+B(2x+3)$

Set $x=-1$:

$$1 = A(0)+B(2(-1)+3)$$
$$1 = B(1)$$
$$B = 1$$

Set $x=-\frac{3}{2}$:

$$1 = A(-\frac{3}{2}+1)+B(0)$$
$$1 = A(-\frac{1}{2})$$
$$A = -2$$

Therefore:

$$\int \frac{1}{2x^2+5x+3}\,dx = \int (\frac{-2}{2x+3}+\frac{1}{x+1})\,dx$$
$$=-\ln|2x+3|+\ln|x+1|+C$$
$$\int \frac{1}{2x^2+5x+3}\,dx = \ln|\frac{x+1}{2x+3}|+C$$

1.(b) Use the partial fractions method to evaluate the following integrals: $\int \frac{2x-1}{(x-3)^2}\,dx$

SOLUTION

Partial fractions:

$$\frac{2x-1}{(x-3)^2} = \frac{A}{x-3}+\frac{B}{(x-3)^2}$$

Multiply both sides by $(x-3)^2$:

$2x-1 = A(x-3)+B$

Set $x=3$:

$$2(3)-1 = A(0)+B$$
$$5 = B$$

Set $x=0$:

$$2(0)-1 = A(0-3)+5$$
$$-1 = -3A+5$$
$$-6 = -3A$$
$$A = 2$$

Therefore:

$$\int \frac{2x-1}{(x-3)^2}\,dx = \int (\frac{2}{x-3}+\frac{5}{(x-3)^2})\,dx$$
$$\int \frac{2x-1}{(x-3)^2}\,dx = 2\ln|x-3|-\frac{5}{x-3}+C$$

1.(c) Use the partial fractions method to evaluate the following integrals: $\int \frac{x+1}{(2x+5)(x+4)}\,dx$

SOLUTION

Partial fractions:

$$\frac{x+1}{(2x+5)(x+4)} = \frac{A}{2x+5}+\frac{B}{x+4}$$

Multiply both sides by $(2x+5)(x+4)$:

$x+1 = A(x+4)+B(2x+5)$

Set $x=-4$:

$$-4+1 = A(0)+B(2(-4)+5)$$

$-3 = B(-3)$

$B = 1$

Set $x=-\frac{5}{2}$:

$-\frac{5}{2}+1 = A(-\frac{5}{2}+4)+B(0)$

$-\frac{3}{2} = A(\frac{3}{2})$

$A = -1$

Therefore:

$$\int \frac{x+1}{(2x+5)(x+4)}\,dx = \int \left(\frac{-1}{2x+5}+\frac{1}{x+4}\right)dx$$

$$=-\frac{1}{2}\ln|2x+5|+\ln|x+4|+C$$

$$\int \frac{x+1}{(2x+5)(x+4)}\,dx = \ln\left|\frac{x+4}{\sqrt{2x+5}}\right|+C$$

1.(d) Use the partial fractions method to evaluate the following integrals: $\int \frac{2x^2-1}{x^2-1}\,dx$

SOLUTION

Divide first:

$$\frac{2x^2-1}{x^2-1} = 2+\frac{1}{x^2-1}$$

Partial fractions:

$$\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{A}{x-1}+\frac{B}{x+1}$$

Multiply both sides by $(x-1)(x+1)$:

$1 = A(x+1)+B(x-1)$

Set $x=1$:

$1 = A(2)+B(0)$

$A = \frac{1}{2}$

Set $x=-1$:

$1 = A(0)+B(-2)$

$1 = -2B$

$B = -\frac{1}{2}$

Therefore:

$$\int \frac{2x^2-1}{x^2-1}\,dx = \int \left(2+\frac{\frac{1}{2}}{x-1}-\frac{\frac{1}{2}}{x+1}\right)dx$$

$$=2x+\frac{1}{2}\ln|x-1|-\frac{1}{2}\ln|x+1|+C$$

$$\int \frac{2x^2-1}{x^2-1}\,dx = 2x+\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right|+C$$

2 Find the function $f(x)$ satisfying $f'(x)=\sin 4x\cos 2x$ with $f(\frac{\pi}{2})=0$

SOLUTION

Use product-to-sum:

$$f'(x) = \sin 4x\cos 2x = \frac{1}{2}[\sin(4x+2x)+\sin(4x-2x)]$$

$$= \frac{1}{2}[\sin 6x+\sin 2x]$$

Integrate:

$$f(x) = \int \frac{1}{2}(\sin 6x+\sin 2x)\,dx$$

$$= \frac{1}{2}\left(-\frac{1}{6}\cos 6x-\frac{1}{2}\cos 2x\right)+C$$

$$=-\frac{1}{12}\cos 6x-\frac{1}{4}\cos 2x+C$$

Apply $f(\frac{\pi}{2})=0$:

$$0 = -\frac{1}{12}\cos 3\pi-\frac{1}{4}\cos\pi+C$$

$$0 = -\frac{1}{12}(-1)-\frac{1}{4}(-1)+C$$

$$0 = \frac{1}{12}+\frac{1}{4}+C$$

$$0 = \frac{1}{3}+C$$

$$C = -\frac{1}{3}$$

Therefore:

$$f(x) = -\frac{1}{12}\cos 6x-\frac{1}{4}\cos 2x-\frac{1}{3}$$

3 Find the function $g(x)$ satisfying $g'(x)=x^2e^{x^3}$ with $g(0)=-\frac{2}{3}$

SOLUTION

Let $u=x^3$. Then $du=3x^2dx$.

$$g(x) = \int x^2e^{x^3}\,dx = \frac{1}{3}\int e^u\,du = \frac{1}{3}e^u+C = \frac{1}{3}e^{x^3}+C$$

Apply $g(0)=-\frac{2}{3}$:

$$-\frac{2}{3} = \frac{1}{3}e^0+C$$

$$-\frac{2}{3} = \frac{1}{3}+C$$

$$C = -1$$

Therefore:

$$g(x) = \frac{1}{3}e^{x^3}-1$$

4 Find the function $h(x)$ satisfying $h'(x)=\frac{x}{x^2-1}$ with $h(2)=\frac{1}{2}$

SOLUTION

Let $u=x^2-1$. Then $du=2x\,dx$.

$$h(x) = \int \frac{x}{x^2-1}\,dx$$
$$= \frac{1}{2}\int \frac{1}{u}\,du$$
$$= \frac{1}{2}\ln|x^2-1|+C$$

Apply $h(2)=\frac{1}{2}$:

$$\frac{1}{2} = \frac{1}{2}\ln|4-1|+C$$
$$\frac{1}{2} = \frac{1}{2}\ln 3+C$$
$$C = \frac{1}{2}-\frac{1}{2}\ln 3$$

Therefore:

$$\mathbf{h(x) = \frac{1}{2}\ln|x^2-1|+\frac{1}{2}-\frac{1}{2}\ln 3}$$

5 Find the function f(x) satisfying$f''(x)=2x-1$ with $f(0)=-1$ and $f'(1)=2$

SOLUTION

First integration:

$$f'(x) = \int (2x-1)\,dx$$
$$= x^2-x+C_1$$

Apply $f'(1)=2$:

$$2 = 1^2-1+C_1$$
$$2 = C_1$$
$$C_1 = 2$$

So:

$f'(x)=x^2-x+2$

Second integration:

$$f(x) = \int (x^2-x+2)\,dx$$
$$= \frac{x^3}{3}-\frac{x^2}{2}+2x+C_2$$

Apply $f(0)=-1$:

$$-1 = 0-0+0+C_2$$
$$C_2 = -1$$

Therefore:

$$\mathbf{f(x) = \frac{x^3}{3}-\frac{x^2}{2}+2x-1}$$

6 Find the function g(x) satisfying$g''(x)=x\sin x$ with $g(\frac{\pi}{2})=0$ and $g'(0)=0$

SOLUTION

First integration (by parts):

Let $u=x$, $dv=\sin x\,dx$. Then $du=dx$, $v=-\cos x$.

$$g'(x) = \int x\sin x\,dx$$
$$= -x\cos x-\int(-\cos x)\,dx$$
$$= -x\cos x+\sin x+C_1$$

Apply $g'(0)=0$:

$$0 = -0\cdot\cos 0+\sin 0+C_1$$
$$0 = C_1$$
$$C_1 = 0$$

So:

$g'(x) = -x\cos x+\sin x$

Second integration:

$$g(x) = \int(-x\cos x+\sin x)\,dx$$
$$= -\int x\cos x\,dx+\int \sin x\,dx$$

For $\int x\cos x\,dx$: let $u=x$, $dv=\cos x\,dx$. Then $du=dx$, $v=\sin x$.

$$\int x\cos x\,dx = x\sin x-\int \sin x\,dx$$
$$= x\sin x+\cos x$$

So:

$$g(x) = -(x\sin x+\cos x)-\cos x+C_2$$
$$= -x\sin x-2\cos x+C_2$$

Apply $g(\frac{\pi}{2})=0$:

$$0 = -\frac{\pi}{2}\sin\frac{\pi}{2}-2\cos\frac{\pi}{2}+C_2$$
$$0 = -\frac{\pi}{2}(1)-0+C_2$$
$$C_2 = \frac{\pi}{2}$$

Therefore:

$$\mathbf{g(x) = -x\sin x-2\cos x+\frac{\pi}{2}}$$

CH.11 Application of Integration

Notes · Examples · Exercises

NOTES

Definite Integral

If $F'(x) = f(x)$, where $f(x)$ is continuous on the interval $a \le x \le b$, then the **definite integral** of $f(x)$ on $[a, b]$ is defined by

$\int_a^b f(x)\,dx = F(b) - F(a)$

We write $F(b) - F(a) = [F(x)]_a^b$, or $F(x)\Big|_a^b$. The constant of integration C cancels out in the subtraction, so it is omitted.

Limits of Integration

In $\int_a^b f(x)\,dx$, the numbers a and b are called the **limits of integration**. Here a is the **lower limit** and b is the **upper limit**.

The symbol $\int_a^b f(x)\,dx$ is read "the integral from x = a to x = b of f(x) with respect to x".

Cross-section of a Solid

A **cross-section** of a solid is the plane region generated by intersecting the solid with a plane. If A(x) is the (integrable) cross-section area at x, for x from a to b, the volume of the solid is found by integrating A(x).

Solid of Revolution

If the region under the curve $y = f(x)$ for $a \le x \le b$ is rotated through 360° about the x-axis, a three-dimensional solid is formed. This solid is called a **solid of revolution**.

Definite Integral

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

where $F'(x) = f(x)$.

Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$. Then for each $x \in [a, b]$,

$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$

Let F' be continuous on $[a, b]$. Then for each $x \in [a, b]$,

$\int_a^x F'(t)\,dt = F(x) - F(a)$

Rules for Definite Integration

Suppose f and g are continuous on $[a, b]$ and $k \in \mathbb{R}$.

1. $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$
2. $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
3. $\int_a^a f(x)\,dx = 0$
4. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
5. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where $c \in (a, b)$

Area between the Curve and x-axis

If $f(x)$ is positive and continuous on $[a, b]$, the area bounded by $y = f(x)$, the x-axis and the vertical lines $x = a$ and $x = b$ is

Area $= \int_a^b f(x)\,dx$

To find the area between the graph of y = f(x) and the x-axis over [a, b]:

1. Find the intersection point(s) of the graph $y = f(x)$ and the x-axis.
2. Divide the interval $[a, b]$ at the zeros of f.
3. Integrate f over each subinterval.
4. Add the absolute values of the integrals.

Area between Two Curves

If $f(x) \ge g(x)$ on $[a, b]$, the area between $y = f(x)$ and $y = g(x)$ is

Area $= \int_a^b [f(x) - g(x)]\,dx$

To find the area between y = f(x) and y = g(x) over [a, b]:

1. Find the intersection points of the graphs $y = f(x)$ and $y = g(x)$.
2. Divide the interval $[a, b]$ at the intersection points.
3. Investigate which function is on the top in each subinterval.
4. Integrate $(f - g)$ if f is on the top, or $(g - f)$ if g is on the top, over each subinterval.

Volume by Method of Slicing

If A(x) is the integrable cross-section area from $x = a$ to $x = b$, then

Volume $= \int_a^b A(x)\,dx$

Volume of Revolution (Disc Method)

For the curve $y = f(x)$, $a \le x \le b$, rotated through 360° about the x-axis:

Volume $= \pi\int_a^b [f(x)]^2\,dx$

Volume of Revolution (Washer Method)

If the region bounded by $y = f(x)$ (on top), $y = g(x)$ and the lines $x = a$, $x = b$ is revolved about the x-axis, then

Volume $= \pi\int_a^b ([f(x)]^2 - [g(x)]^2)\,dx$

Steps:

1. Find the intersection points of the graphs $y = f(x)$ and $y = g(x)$.
2. Investigate which function is on the top.

3. Integrate $f^2 - g^2$ if f is on the top, or $g^2 - f^2$ if g is on the top, on the interval.

Section 11.2

EXAMPLE 1 Evaluate the following definite integrals.

a. $\int_{-1}^{2} x^3\,dx$

b. $\int_{0}^{\frac{\pi}{2}} \sin x\,dx$

c. $\int_{-1}^{1} \sqrt{x+2}\,dx$

d. $\int_{\frac{\pi}{4}}^{\pi} |\cos x|\,dx$

SOLUTION

(a) $\int_{-1}^{2} x^3\,dx = \left[\frac{x^4}{4}\right]_{-1}^{2}$

$= \frac{2^4}{4} - \frac{(-1)^4}{4}$

$= 4 - \frac{1}{4}$

$= \frac{15}{4}$

$= 3\frac{3}{4}$

(b) $\int_{0}^{\frac{\pi}{2}} \sin x\,dx = [-\cos x]_{0}^{\frac{\pi}{2}}$

$= -\cos\frac{\pi}{2} + \cos 0$

$= -0 + 1$

$= 1$

(c) $\int_{-1}^{1} \sqrt{x+2}\,dx = \int_{-1}^{1} (x+2)^{1/2}\,dx$

$= \left[\frac{2}{3}(x+2)^{3/2}\right]_{-1}^{1}$

$= \frac{2}{3}[(3)^{3/2} - (1)^{3/2}]$

$= \frac{2}{3}(3\sqrt{3} - 1)$

(d) When $\frac{\pi}{4} \le x \le \frac{\pi}{2}$, $|\cos x| = \cos x$. When $\frac{\pi}{2} \le x \le \pi$, $|\cos x| = -\cos x$.

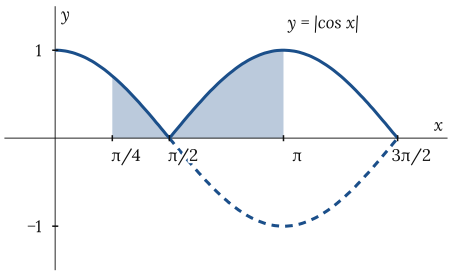
$\int_{\frac{\pi}{4}}^{\pi} |\cos x|\,dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x\,dx + \int_{\frac{\pi}{2}}^{\pi} (-\cos x)\,dx$

$= [\sin x]_{\frac{\pi}{4}}^{\frac{\pi}{2}} - [\sin x]_{\frac{\pi}{2}}^{\pi}$

$= (\sin\frac{\pi}{2} - \sin\frac{\pi}{4}) - (\sin\pi - \sin\frac{\pi}{2})$

$= (1 - \frac{\sqrt{2}}{2}) - (0 - 1)$

$= 2 - \frac{\sqrt{2}}{2}$



EXAMPLE 2 Evaluate the integral $\int_{1}^{2} x\sqrt{2-x}\,dx$.

SOLUTION

Method 1

Let $u = 2 - x$. Then $du = -dx$ and $x = 2 - u$.

$\int x\sqrt{2-x}\,dx = -\int (2-u)u^{1/2}\,du$

$= -\int (2u^{1/2} - u^{3/2})\,du$

$= -(\frac{4}{3}u^{3/2} - \frac{2}{5}u^{5/2})$

$= -\frac{4}{3}(2-x)^{3/2} + \frac{2}{5}(2-x)^{5/2}$

Next,

$\int_{1}^{2} x\sqrt{2-x}\,dx = [-\frac{4}{3}(2-x)^{3/2} + \frac{2}{5}(2-x)^{5/2}]_{1}^{2}$

$= 0 - (-\frac{4}{3} + \frac{2}{5})$

$= \frac{14}{15}$

Method 2

Let $u = 2 - x$. Then $du = -dx$. Changing the variable from x to u (and the limits as well):

x	1	2
u	1	0

$\int_{1}^{2} x\sqrt{2-x}\,dx = -\int_{1}^{0} (2-u)u^{1/2}\,du$

$= \int_{0}^{1} (2u^{1/2} - u^{3/2})\,du$

$= [\frac{4}{3}u^{3/2} - \frac{2}{5}u^{5/2}]_{0}^{1}$

$= \frac{4}{3} - \frac{2}{5}$

$= \frac{14}{15}$

EXERCISE 11.1

1. Evaluate the following definite integrals.

a. $\int_{0}^{1} \frac{e^x}{e^x+1}\,dx$

b. $\int_{-1}^{1} x\sqrt{x+2}\,dx$

c. $\int_{0}^{6} (\frac{1}{3}x - 1)^3\,dx$

d. $\int_{2}^{4} \frac{x+5}{x^2-6x+5}\,dx$

e. $\int_{0}^{2} e^{-x}\cos 2x\,dx$

f. $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \tan 2x\,dx$

SOLUTION

(a)

Let $u = e^x + 1$, so $du = e^x\,dx$.

When $x = 0$, $u = e^0 + 1 = 2$; when $x = 1$, $u = e + 1$.

x	0 → 1
u	2 → e + 1

$$\int_0^1 \frac{e^x}{e^x+1}\,dx = \int_2^{e+1} \frac{1}{u}\,du$$
$$= [\ln|u|]_2^{e+1}$$
$$= \ln|e+1| - \ln|2|$$
$$= \ln\frac{e+1}{2}$$

(b)

Let $u = x + 2$, so $x = u - 2$ and $du = dx$.

When $x = -1$, $u = 1$; when $x = 1$, $u = 3$.

x	−1 → 1
u	1 → 3

$$\int_{-1}^1 x\sqrt[3]{x+2}\,dx = \int_1^3 (u-2)\sqrt{u}\,du$$
$$= \int_1^3 (u^{3/2} - 2u^{1/2})\,du$$
$$= [\frac{2}{5}u^{5/2} - 2\cdot\frac{2}{3}u^{3/2}]_1^3$$
$$= [\frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2}]_1^3$$
$$= \frac{2}{5}(3)^{5/2} - \frac{4}{3}(3)^{3/2} - \frac{2}{5}(1)^{5/2} + \frac{4}{3}($$
$$= \frac{18\sqrt{3}}{5} - 4\sqrt{3} + \frac{14}{15}$$
$$= -\frac{2\sqrt{3}}{5} + \frac{14}{15}$$

Use $\int (ax+b)^n\,dx = \frac{1}{a}\frac{(ax+b)^{n+1}}{n+1}$.

(c) $\int_0^6 (\frac{1}{3}x - 1)^3\,dx = \frac{1}{\frac{1}{3}}\cdot\frac{1}{4}[(\frac{1}{3}x-1)^4]_0^6$

$$= \frac{3}{4}[(\frac{1}{3}x-1)^4]_0^6$$
$$= \frac{3}{4} - \frac{3}{4}$$
$$= 0$$

Partial fractions:

(d) $$\frac{x+5}{x^2-6x+5} = \frac{x+5}{(x-5)(x-1)}$$
$$= \frac{A}{x-5} + \frac{B}{x-1}$$
$$x + 5 = A(x-1) + B(x-5)$$

When $x = 1$: $6 = B(-4)$, so $B = -\frac{3}{2}$. When $x = 5$: $10 = A(4)$, so $A = \frac{5}{2}$.

$$\int_2^4 \frac{x+5}{x^2-6x+5}\,dx = \int_2^4 (\frac{\frac{5}{2}}{x-5} + \frac{\frac{-3}{2}}{x-1})\,dx$$
$$= [\frac{5}{2}\ln|x-5| - \frac{3}{2}\ln|x-1|]_2^4$$
$$= \frac{5}{2}\ln 1 - \frac{3}{2}\ln 3 - \frac{5}{2}\ln 3 + \frac{3}{2}\ln 1$$
$$= -\frac{3}{2}\ln 3 - \frac{5}{2}\ln 3$$
$$= -4\ln 3$$

Integrate by parts twice. Let $u = e^{-x}$, $dv = \cos 2x\,dx$; then $du = -e^{-x}\,dx$, $v = \frac{1}{2}\sin 2x$.

(e) $I = \int e^{-x}\cos 2x\,dx$

$$= \frac{1}{2}e^{-x}\sin 2x + \frac{1}{2}\int e^{-x}\sin 2x\,dx \quad (1)$$

For $\int e^{-x}\sin 2x\,dx$, let $u = e^{-x}$, $dv = \sin 2x\,dx$; then $du = -e^{-x}\,dx$, $v = -\frac{1}{2}\cos 2x$.

$$\int e^{-x}\sin 2x\,dx = -\frac{1}{2}e^{-x}\cos 2x - \frac{1}{2}\int e^{-x}\cos 2x\,dx \quad (2)$$

Substituting (2) into (1):

$$I = \frac{1}{2}e^{-x}\sin 2x - \frac{1}{4}e^{-x}\cos 2x - \frac{1}{4}I$$
$$\frac{5I}{4} = \frac{1}{2}e^{-x}\sin 2x - \frac{1}{4}e^{-x}\cos 2x$$
$$I = \frac{2}{5}e^{-x}\sin 2x - \frac{1}{5}e^{-x}\cos 2x + C$$

$$\int_0^2 e^{-x}\cos 2x\,dx = [\frac{2}{5}e^{-x}\sin 2x - \frac{1}{5}e^{-x}\cos 2x]_0^2$$
$$= \frac{2}{5}e^{-2}\sin 4 - \frac{1}{5}e^{-2}\cos 4 - \frac{2}{5}e^0\sin 0 + \frac{1}{5}e^0\cos 0$$
$$= \frac{2}{5}e^{-2}\sin 4 - \frac{1}{5}e^{-2}\cos 4 + \frac{1}{5} \approx 0.1767$$

(f) $\int_{\pi/8}^{\pi/6} \tan 2x\,dx = [-\frac{1}{2}\ln|\cos 2x|]_{\pi/8}^{\pi/6}$

$$= -\frac{1}{2}[\ln|\cos\frac{\pi}{3}| - \ln|\cos\frac{\pi}{4}|]$$
$$= -\frac{1}{2}[\ln\frac{1}{2} - \ln\frac{\sqrt{2}}{2}]$$
$$= -\frac{1}{2}(\ln\frac{1}{2} - \ln\frac{\sqrt{2}}{2})$$
$$= \ln\sqrt[4]{2} \approx 0.1733$$

2. Find the integral of $\int_0^{2\pi} \cos mx \sin nx\,dx$ for (a) $m = n$ and (b) $m \neq n$ (m, n integers).

SOLUTION

(a) For m = n:

$$\int_0^{2\pi} \cos mx \sin nx\,dx = \int_0^{2\pi} \cos nx \sin nx\,dx$$
$$= \int_0^{2\pi} \frac{1}{2}\sin 2nx\,dx$$
$$= \frac{1}{2}[-\frac{1}{2n}\cos 2nx]_0^{2\pi}$$
$$= -\frac{1}{4n}[\cos(4n\pi) - \cos 0]$$
$$= -\frac{1}{4n}(1-1)$$

$= 0$

(using $\sin 2x = 2 \sin x \cos x$)

(b) For $m \neq n$: use $\sin\alpha \cos\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$ with $\alpha = nx$, $\beta = mx$.

$\cos mx \sin nx = \frac{1}{2}[\sin(n+m)x + \sin(n-m)x]$. Then

$$\int_0^{2\pi} \cos mx \sin nx \, dx$$

$$= \frac{1}{2}\int_0^{2\pi} [\sin(m+n)x - \sin(m-n)x] \, dx$$

$$= \frac{1}{2}\left[-\frac{1}{m+n}\cos(m+n)x + \frac{1}{m-n}\cos(m-n)x\right]_0^{2\pi}$$

$$= \frac{1}{2}\left[-\frac{1}{m+n}\cos((m+n)2\pi) + \frac{1}{m-n}\cos((m-n)2\pi) - \left(-\frac{1}{m+n}\cos 0 + \frac{1}{m-n}\cos 0\right)\right]$$

$$= \frac{1}{2}\left[-\frac{1}{m+n} + \frac{1}{m-n} + \frac{1}{m+n} - \frac{1}{m-n}\right]$$

$$= \frac{1}{2}(0)$$

$$= 0$$

(since cos of any integer multiple of 2π is 1)

Answer: (a) 0; (b) 0

3. Evaluate the following integrals.

a. $\int_0^4 |\sqrt{2x} - 2| \, dx$

b. $\int_{-1}^1 |e^x - 1| \, dx$

c. $\int_0^{2\pi} |\sin x| \, dx$

SOLUTION

(a) $|\sqrt{2x} - 2| = -(\sqrt{2x} - 2)$ when $0 \le x \le 2$, and $= \sqrt{2x} - 2$ when $2 \le x \le 4$.

$$\int_0^4 |\sqrt{2x} - 2| \, dx = \int_0^2 [-(\sqrt{2x} - 2)] \, dx + \int_2^4 (\sqrt{2x} - 2) \, dx$$

$$= -\left[\frac{2\sqrt{2}}{3}x^{3/2} - 2x\right]_0^2 + \left[\frac{2\sqrt{2}}{3}x^{3/2} - 2\right]$$

$$= -\left[\frac{2\sqrt{2}}{3}(2)^{3/2} - 4 - 0\right] + \left[\frac{2\sqrt{2}}{3}(4)^{3/2} - 8 - \frac{2\sqrt{2}}{3}(2)^{3/2} + 4\right]$$

$$= -\frac{8}{3} + 4 + \frac{16\sqrt{2}}{3} - 8 - \frac{8}{3} + 4$$

$$= \frac{16\sqrt{2} - 16}{3}$$

(b) $|e^x - 1| = -(e^x - 1)$ when $-1 \le x \le 0$, and $= e^x - 1$ when $0 \le x \le 1$.

$$\int_{-1}^1 |e^x - 1| \, dx = \int_{-1}^0 [-(e^x - 1)] \, dx + \int_0^1 (e^x - 1) \, dx$$

$$= -[e^x - x]_{-1}^0 + [e^x - x]_0^1$$

$$= -[e^0 - 0 - e^{-1} - 1] + [e - 1 - e^0 + 0]$$

$$= -1 + e^{-1} + 1 + e - 1 - 1$$

$$= \frac{1}{e} + e - 2$$

(c) $|\sin x| = \sin x$ when $0 \le x \le \pi$, and $= -\sin x$ when $\pi \le x \le 2\pi$.

$$\int_0^{2\pi} |\sin x| \, dx = \int_0^{\pi} \sin x \, dx + \int_{\pi}^{2\pi} (-\sin x) \, dx$$

$$= [-\cos x]_0^{\pi} + [\cos x]_{\pi}^{2\pi}$$

$$= -\cos\pi + \cos 0 + \cos 2\pi - \cos\pi$$

$$= 1 + 1 + 1 + 1$$

$$= 4$$

Answer: (a) $\frac{16\sqrt{2} - 16}{3}$; (b) $\frac{1}{e} + e - 2$; (c) 4

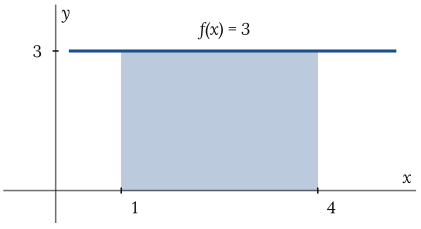
Section 11.3 & 11.4

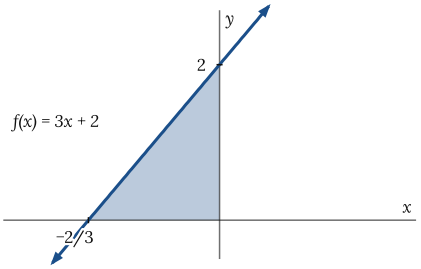
EXAMPLE 3 Find the area enclosed between the graph of

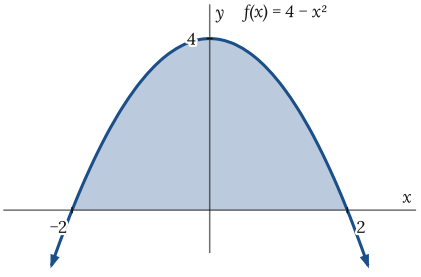
a. $f(x) = 3$ and the x-axis over $[1, 4]$,

b. $f(x) = 3x + 2$ and the x-axis over $[-\frac{2}{3}, 0]$,

c. $f(x) = -x^2 + 4$ and the x-axis over $[-2, 2]$.







SOLUTION

(a) $\int_1^4 3 \, dx = [3x]_1^4$

$= 12 - 3$

$= 9$

Area $= 9$ unit^2

(b) $\int_{-\frac{2}{3}}^0 (3x + 2) \, dx = \left[\frac{3x^2}{2} + 2x\right]_{-\frac{2}{3}}^0$

$= 0 - \left[\frac{3}{2} \cdot \frac{4}{9} - \frac{4}{3}\right]$

$= \frac{2}{3}$

Area $= \frac{2}{3}$ unit^2

(c) $\int_{-2}^{2} (4 - x^2)\,dx = [4x - \frac{x^3}{3}]_{-2}^{2}$

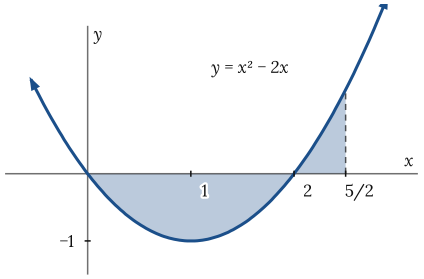
$= (8 - \frac{8}{3}) - (-8 + \frac{8}{3})$

$= \frac{32}{3}$

$= 10\frac{2}{3}$

Area $= 10\frac{2}{3}$ unit^2

EXAMPLE 4 (a) Compute the definite integral $\int_{0}^{\frac{5}{2}} (x^2 - 2x)\,dx$.(b) Find the total area enclosed between the graph of $f(x) = x^2 - 2x$ and the x-axis over $[0, \frac{5}{2}]$.



SOLUTION

(a) $\int_{0}^{\frac{5}{2}} (x^2 - 2x)\,dx = [\frac{x^3}{3} - x^2]_{0}^{\frac{5}{2}}$

$= \frac{1}{3}(\frac{5}{2})^3 - (\frac{5}{2})^2$

$= \frac{125}{24} - \frac{25}{4}$

$= -\frac{25}{24}$

$= -1\frac{1}{24}$

(b) To find the intersection points of the x-axis and the curve, $f(x) = 0$:

$x^2 - 2x = 0$

$x(x - 2) = 0$

$x = 0, 2$

The graph of f cuts the x-axis at 0 and 2, so we divide the domain into two intervals: $[0, 2]$ and $[2, \frac{5}{2}]$.

$\int_{0}^{2} (x^2 - 2x)\,dx = [\frac{x^3}{3} - x^2]_{0}^{2}$

$= \frac{8}{3} - 4 - 0$

$= -\frac{4}{3}$

On $[0, 2]$, $f \le 0$, so the value of the integral is negative.

$\int_{2}^{\frac{5}{2}} (x^2 - 2x)\,dx = [\frac{x^3}{3} - x^2]_{2}^{\frac{5}{2}}$

$= (\frac{125}{24} - \frac{25}{4}) - (\frac{8}{3} - 4)$

$= \frac{7}{24}$

On $[2, \frac{5}{2}]$, $f \ge 0$, so the value of the integral is positive. Therefore,

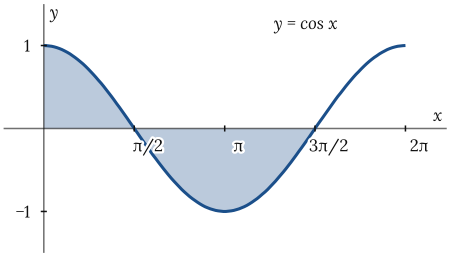
Area $= |-\frac{4}{3}| + \frac{7}{24}$

$= \frac{32}{24} + \frac{7}{24}$

$= \frac{13}{8}$

$= 1\frac{5}{8}$ unit^2

EXAMPLE 5 The figure shows the graph of the function $f(x) = \cos x$.



Find the area enclosed between the graph of $f(x) = \cos x$ and the lines

a. $x = 0$ and $x = \frac{\pi}{2}$

b. $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$

c. $x = 0$ and $x = \frac{3\pi}{2}$.

d. Find the definite integral $\int_{0}^{\frac{3\pi}{2}} \cos x\,dx$.

SOLUTION

(a) $\int_{0}^{\frac{\pi}{2}} \cos x\,dx = [\sin x]_{0}^{\frac{\pi}{2}}$

$= \sin\frac{\pi}{2} - \sin 0$

$= 1$

Area $= 1$ unit^2

(b) $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos x\,dx = [\sin x]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$

$= \sin\frac{3\pi}{2} - \sin\frac{\pi}{2}$

$= -2$

Area $= |-2|$

$= 2$ unit^2

(c) The area between the graph f(x) = cos x and the lines x = 0 and x = $\frac{3\pi}{2}$ is 1 + 2 = 3 unit^2.

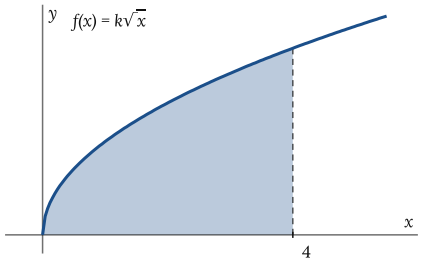
(d) $\int_0^{\frac{3\pi}{2}} \cos x\,dx = [\sin x]_0^{\frac{3\pi}{2}}$

$= \sin\frac{3\pi}{2} - \sin 0$

$= -1$

(Note that the definite integral (−1) and the area (3) are different when the graph goes below the x-axis.)

EXAMPLE 6 The shaded area is 16 unit^2. Find the value of k.



(The curve is f(x) = k$\sqrt{x}$, shaded from x = 0 to x = 4.)

SOLUTION

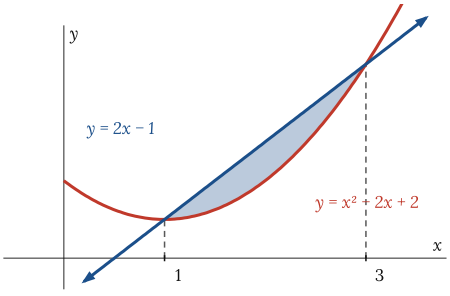
$\int_0^4 k\sqrt{x}\,dx = [\frac{2k}{3}x^{3/2}]_0^4$

$= \frac{2k}{3}\cdot 4^{3/2}$

$= \frac{16k}{3}$

By the problem, $\frac{16k}{3} = 16$, so **k = 3**.

EXAMPLE 7 Find the area between y = x^2 − 2x + 2 and y = 2x − 1.



SOLUTION

To find the x-coordinates of the intersection points,

$x^2 - 2x + 2 = 2x - 1$

$x^2 - 4x + 3 = 0$

$(x - 1)(x - 3) = 0$

$x = 1, 3$

Area $= \int_1^3 [(2x - 1) - (x^2 - 2x + 2)]\,dx$

$= \int_1^3 (-x^2 + 4x - 3)\,dx$

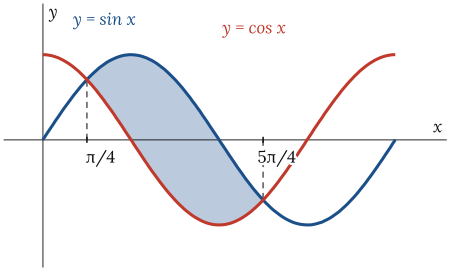
$= [-\frac{x^3}{3} + \frac{4x^2}{2} - 3x]_1^3$

$= 0 - (-\frac{4}{3})$

$= \frac{4}{3}$

$= 1\frac{1}{3}$ unit^2

EXAMPLE 8 Find the area between y = sin x and y = cos x for $\frac{\pi}{4} \le x \le \frac{5\pi}{4}$.



SOLUTION

From the figure, we see that sin x ≥ cos x when $\frac{\pi}{4} \le x \le \frac{5\pi}{4}$.

Area $= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x)\,dx$

$= [-\cos x - \sin x]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$

$= (-\cos\frac{5\pi}{4} - \sin\frac{5\pi}{4}) - (-\cos\frac{\pi}{4} - \sin\frac{\pi}{4})$

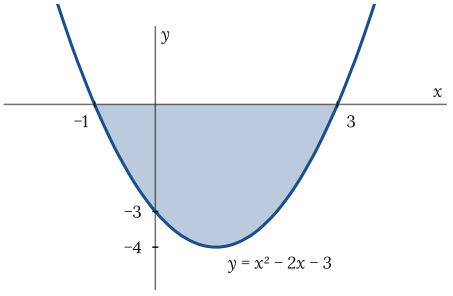
$= (\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}) - (-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2})$

$= \sqrt{2} + \sqrt{2}$

$= 2\sqrt{2}$ unit^2

EXERCISE 11.2

1(a). Find the shaded areas.**(a)**



SOLUTION

(a) y = x^2 − 2x − 3 = (x − 3)(x + 1) = 0, so x = −1, 3. The region lies below the x-axis, so the integral is negative.

$\int_{-1}^3 (x^2 - 2x - 3)\,dx = [\frac{x^3}{3} - x^2 - 3x]_{-1}^3$

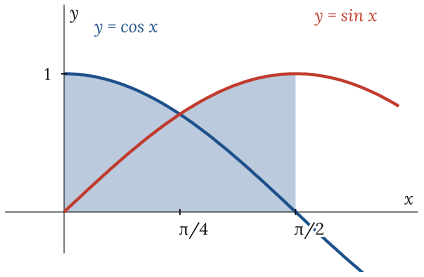
$= (9 - 9 - 9) - (-\frac{1}{3} - 1 + 3)$

$= -9 - \frac{5}{3}$

$= -10\frac{2}{3}$

The shaded area $= |-10\frac{2}{3}| = \mathbf{10\frac{2}{3}\ unit^2}$.

1(b). Find the shaded areas.**(b)**



SOLUTION

(b) $\sin x = \cos x \rightarrow \tan x = 1 \rightarrow x = \frac{\pi}{4}$. cos x is on top on $[0, \frac{\pi}{4}]$, sin x is on top on $[\frac{\pi}{4}, \frac{\pi}{2}]$.

$$\text{Area} = \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx$$

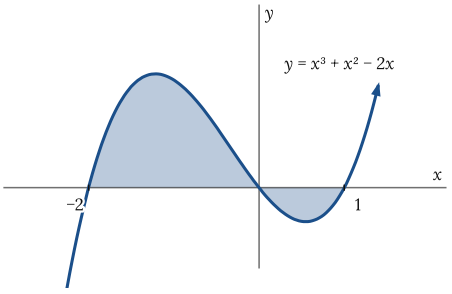
$$= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2}$$

$$= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - 0 - 1 - 0 - 1 + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}$$

$$= 2(\sqrt{2} - 1)$$

The shaded area $= \mathbf{2(\sqrt{2} - 1)\ unit^2}$.

1(c). Find the shaded areas.**(c)**



SOLUTION

(c) $y = x^3 + x^2 - 2x = x(x + 2)(x - 1)$; the curve cuts the x-axis at −2, 0, 1.

$$\int_{-2}^{0} (x^3 + x^2 - 2x)\,dx = [\frac{x^4}{4} + \frac{x^3}{3} - x^2]_{-2}^{0}$$

$$= 0 - (4 - \frac{8}{3} - 4)$$

$$= \frac{8}{3}$$

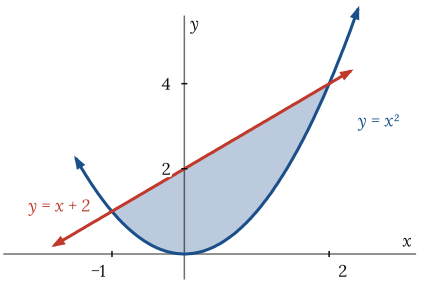
$$\int_0^1 (x^3 + x^2 - 2x)\,dx = [\frac{x^4}{4} + \frac{x^3}{3} - x^2]_0^1$$

$$= \frac{1}{4} + \frac{1}{3} - 1$$

$$= -\frac{5}{12}$$

The shaded area $= |\frac{8}{3}| + |-\frac{5}{12}| = \frac{37}{12} = \mathbf{3\frac{1}{12}\ unit^2}$.

1(d). Find the shaded areas.**(d)**



SOLUTION

(d) $x^2 = x + 2 \rightarrow (x - 2)(x + 1) = 0 \rightarrow x = -1, 2$. The line $y = x + 2$ is above the parabola.

$$\text{Area} = \int_{-1}^{2} (x + 2 - x^2)\,dx$$

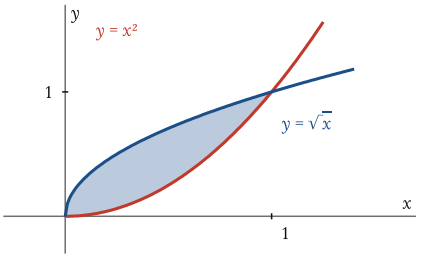
$$= [\frac{x^2}{2} + 2x - \frac{x^3}{3}]_{-1}^{2}$$

$$= (2 + 4 - \frac{8}{3}) - (\frac{1}{2} - 2 + \frac{1}{3})$$

$$= \frac{9}{2}$$

The shaded area $= \mathbf{4\frac{1}{2}\ unit^2}$.

1(e). Find the shaded areas.**(e)**



SOLUTION

(e) $x^2 = \sqrt{x} \rightarrow x^4 = x \rightarrow x(x^3 - 1) = 0 \rightarrow x = 0, 1$. $\sqrt{x}$ is above x^2 on $[0, 1]$.

$$\text{Area} = \int_0^1 (\sqrt{x} - x^2)\,dx$$

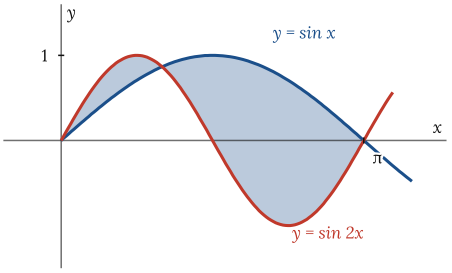
$$= [\frac{2}{3}x^{3/2} - \frac{x^3}{3}]_0^1$$

$$= (\frac{2}{3} - \frac{1}{3}) - 0$$

$$= \frac{1}{3}$$

The shaded area $= \mathbf{\frac{1}{3}\ unit^2}$.

1(f). Find the shaded areas.**(f)**



SOLUTION

(f) $\sin x = \sin 2x = 2 \sin x \cos x \rightarrow \sin x (2 \cos x - 1) = 0 \rightarrow \sin x = 0$ or $\cos x = \frac{1}{2}$, so $x = 0, \frac{\pi}{3}$ (and π) on $[0, \pi]$. $\sin 2x$ is on top on $[0, \frac{\pi}{3}]$; $\sin x$ is on top on $[\frac{\pi}{3}, \pi]$.

$$\int_0^{\pi/3} (\sin 2x - \sin x)\, dx = \left[-\frac{1}{2}\cos 2x + \cos x\right]_0^{\pi/3}$$
$$= -\frac{1}{2}\cos\frac{2\pi}{3} + \cos\frac{\pi}{3} + \frac{1}{2}\cos 0 - \cos 0$$
$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{2} - 1$$
$$= \frac{1}{4}$$

$$\int_{\pi/3}^{\pi} (\sin x - \sin 2x)\, dx = \left[-\cos x + \frac{1}{2}\cos 2x\right]_{\pi/3}^{\pi}$$
$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4}$$
$$= \frac{9}{4}$$

The shaded area $= \frac{1}{4} + \frac{9}{4} = \frac{5}{2}$ **unit²**.

2(a). Find the area under each curve between the given x-values:**(a)** $f(x) = \ln x$ from $x = 1$ to $x = e$.

SOLUTION

(a) First $\int \ln x\, dx$ by parts: $u = \ln x$, $dv = dx$, $du = \frac{1}{x} dx$, $v = x$.

$$\int \ln x\, dx = x \ln x - \int x \cdot \frac{1}{x}\, dx$$
$$= x \ln x - x + C$$
$$\int_1^e \ln x\, dx = [x \ln x - x]_1^e$$
$$= e \ln e - e - \ln 1 + 1$$
$$= 1$$

The area under the curve = 1 unit².

2(b). Find the area under each curve between the given x-values:**(b)** $f(x) = 12 - 3x^2$ from $x = -1$ to $x = 2$.

SOLUTION

(b) $\int_{-1}^{2} (12 - 3x^2)\, dx = [12x - x^3]_{-1}^{2}$
$= 24 - 8 + 12 - 1$
$= 27$

The area under the curve = 27 unit².

2(c). Find the area under each curve between the given x-values:**(c)** $f(x) = xe^{x^2}$ from $x = 0$ to $x = 1$.

SOLUTION

(c) Let $u = x^2$, $du = 2x\, dx$, so $x\, dx = \frac{1}{2} du$. When $x = 0$, $u = 0$; when $x = 1$, $u = 1$.

$$\int_0^1 xe^{x^2}\, dx = \frac{1}{2}\int_0^1 e^u\, du$$
$$= \frac{1}{2}[e^u]_0^1$$
$$= \frac{1}{2}(e - 1)$$

The area under the curve = $\frac{1}{2}(e - 1)$ unit².

2(d). Find the area under each curve between the given x-values:**(d)** $f(x) = \sin x$ from $x = 0$ to $x = \frac{3\pi}{2}$.

SOLUTION

(d) $\sin x = 0 \rightarrow x = \pi$ between 0 and $\frac{3\pi}{2}$.

$$\int_0^{\pi} \sin x\, dx = [-\cos x]_0^{\pi}$$
$$= -\cos\pi + \cos 0$$
$$= 2$$
$$\int_{\pi}^{3\pi/2} \sin x\, dx = [-\cos x]_{\pi}^{3\pi/2}$$
$$= -(\cos\frac{3\pi}{2} - \cos\pi)$$
$$= -(0 - (-1))$$
$$= -1$$

The area under the curve = |2| + |−1| = 3 unit².

2(e). Find the area under each curve between the given x-values:**(e)** $f(x) = (1 + \cos x)\sin x$ from $x = 0$ to $x = \pi$.

SOLUTION

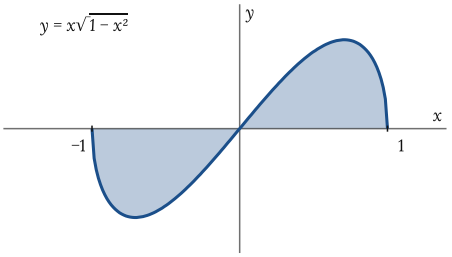
(e) Let $u = 1 + \cos x$, $du = -\sin x\, dx$. When $x = 0$, $u = 2$; when $x = \pi$, $u = 0$.

$$\int_0^{\pi} (1 + \cos x)\sin x\, dx = -\int_2^0 u\, du$$
$$= -[\frac{u^2}{2}]_2^0$$
$$= 0 + \frac{4}{2}$$
$$= 2$$

The area under the curve = 2 unit².

---

3. (a) Compute $\int_{-1}^{1} x\sqrt{1 - x^2}\, dx$.(b) Find the area enclosed between the graph of $f(x) = x\sqrt{1 - x^2}$ and the x-axis over $[-1, 1]$.



SOLUTION

(a) Let $u = 1 - x^2$, $du = -2x\,dx$, so $x\,dx = -\frac{1}{2}du$.

$$\int x\sqrt{1-x^2}\,dx = -\frac{1}{2}\int \sqrt{u}\,du$$
$$= -\frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C$$
$$= -\frac{1}{3}(1-x^2)^{3/2} + C$$

$$\int_{-1}^{1} x\sqrt{1-x^2}\,dx = \left[-\frac{1}{3}(1-x^2)^{3/2}\right]_{-1}^{1}$$
$$= -\frac{1}{3}(0-0)$$
$$= 0$$

(b) The graph is below the x-axis on $[-1, 0]$ and above it on $[0, 1]$.

$$\int_{-1}^{0} x\sqrt{1-x^2}\,dx = \left[-\frac{1}{3}(1-x^2)^{3/2}\right]_{-1}^{0}$$
$$= -\frac{1}{3}(1-0)$$
$$= -\frac{1}{3}$$

$$\int_{0}^{1} x\sqrt{1-x^2}\,dx = \left[-\frac{1}{3}(1-x^2)^{3/2}\right]_{0}^{1}$$
$$= -\frac{1}{3}(0-1)$$
$$= \frac{1}{3}$$

The enclosed area $= \left|-\frac{1}{3}\right| + \left|\frac{1}{3}\right| = \mathbf{\frac{2}{3}\ unit^2}$.

4. Find the area enclosed between the graphs of $y = x^3 - 3x$ and $y = x$ for $0 \le x \le 2$.

SOLUTION

Let $f(x) = x^3 - 3x$, $g(x) = x$. On $0 \le x \le 2$ the graph of g is above f ($g - f = 4x - x^3 = x(4 - x^2) \ge 0$).

$$\int_0^2 (g(x) - f(x))\,dx = \int_0^2 (x - x^3 + 3x)\,dx$$
$$= \int_0^2 (-x^3 + 4x)\,dx$$
$$= \left[-\frac{x^4}{4} + 2x^2\right]_0^2$$
$$= -\frac{16}{4} + 8 - 0$$
$$= 4$$

The enclosed area = **4 unit²**.

5. Find the area enclosed by the graphs of $y = \ln(3x - 2)$ and $y = x^2 - 2x + 1$ for $1 \le x \le 2$.

SOLUTION

First $\int \ln(3x-2)\,dx$ by parts: $u = \ln(3x-2)$, $dv = dx$;

$du = \frac{3}{3x-2}dx$, $v = x$.

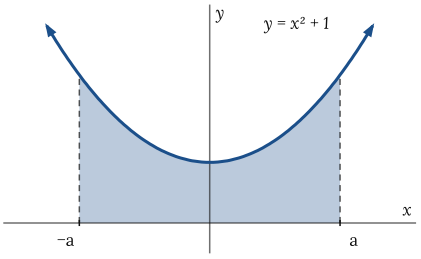
$$\int \ln(3x-2)\,dx = x\ln(3x-2) - \int \frac{3x}{3x-2}\,dx$$
$$= x\ln(3x-2) - \int \left(1 + \frac{2}{3x-2}\right)dx$$
$$= x\ln(3x-2) - x - \frac{2}{3}\ln|3x-2| + C$$

The enclosed area:

$$\int_1^2 [\ln(3x-2) - (x^2 - 2x + 1)]\,dx$$
$$= \left[x\ln(3x-2) - x - \frac{2}{3}\ln|3x-2| - \frac{x^3}{3} + x^2 - x\right]_1^2$$
$$= \left[x\ln(3x-2) - \frac{2}{3}\ln|3x-2| - \frac{x^3}{3} + x^2 - 2x\right]_1^2$$
$$= \left(2\ln 4 - \frac{2}{3}\ln 4 - \frac{8}{3} + 4 - 4\right) - \left(\ln 1 - \frac{2}{3}\ln 1 - \frac{1}{3} + 1 - 2\right)$$
$$= \frac{4}{3}\ln 4 - \frac{8}{3} + \frac{4}{3}$$
$$= \frac{4}{3}(\ln 4 - 1)$$

The enclosed area = $\mathbf{\frac{4}{3}(\ln 4 - 1)\ unit^2} \approx 0.515\ \text{unit}^2$.

6. The shaded area is $8a$ unit². Find the value of a.



SOLUTION

The shaded region lies under $y = x^2 + 1$ from $x = -a$ to $x = a$.

$$\int_{-a}^{a} (x^2 + 1)\,dx = \left[\frac{x^3}{3} + x\right]_{-a}^{a}$$
$$= \frac{a^3}{3} + a + \frac{a^3}{3} + a$$
$$= \frac{2a^3}{3} + 2a$$

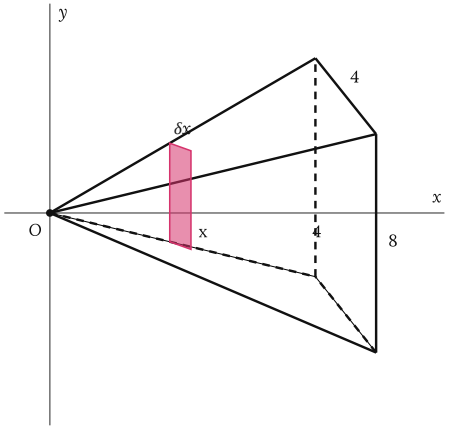
By given, area = $8a$:

$$\frac{2a^3}{3} + 2a = 8a$$
$$2a^3 + 6a = 24a$$
$$2a^3 - 18a = 0$$
$$a^3 - 9a = 0$$
$$a(a^2 - 9) = 0$$

a = 0 or a = ±3. Since a > 0 for the region, **a = 3**.

Section 11.5

EXAMPLE 9 A pyramid with the height of 4 m has a rectangular base as shown in the figure. Find the volume of the pyramid.



SOLUTION

The cross-section area at x is $A(x) = 2x^2$.

$V = \int_0^4 A(x)\,dx$

$= \int_0^4 2x^2\,dx$

$= [\frac{2x^3}{3}]_0^4$

$= \frac{128}{3}$ m^3

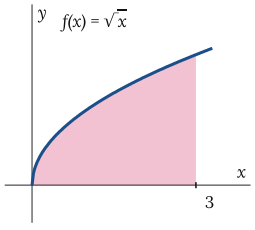
EXAMPLE 10 Find the volume of revolution formed when the curve

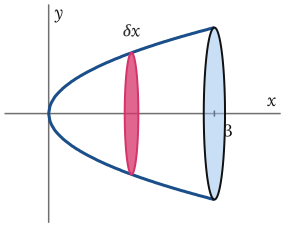
a. $y = \sqrt{x}$ for $0 < x \le 3$

b. $y = x^2$ for $0 < x < 1$

are rotated through 360° about the x-axis.

SOLUTION



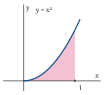


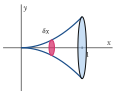
(a) Volume of revolution $= \pi \int_0^3 (\sqrt{x})^2\,dx$

$= \pi \int_0^3 x\,dx$

$= \pi[\frac{x^2}{2}]_0^3$

$= \frac{9\pi}{2}$ unit^3





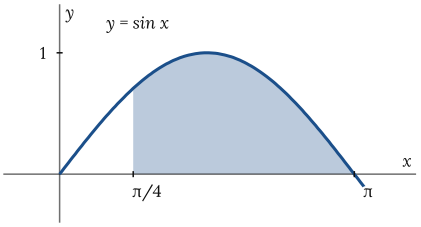
(b) Volume of revolution $= \pi \int_0^1 (x^2)^2\,dx$

$= \pi \int_0^1 x^4\,dx$

$= \pi[\frac{x^5}{5}]_0^1$

$= \frac{\pi}{5}$ unit^3

EXAMPLE 11 Find the volume of revolution formed when the curve $y = \sin x$ for $\frac{\pi}{4} \le x \le \pi$ is rotated through 360° about the x-axis.



SOLUTION

Volume of revolution $= \pi \int_{\frac{\pi}{4}}^{\pi} \sin^2 x\,dx$

$= \pi \int_{\frac{\pi}{4}}^{\pi} \frac{1 - \cos 2x}{2}\,dx$

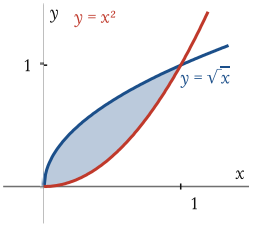
$= \pi[\frac{x}{2} - \frac{\sin 2x}{4}]_{\frac{\pi}{4}}^{\pi}$

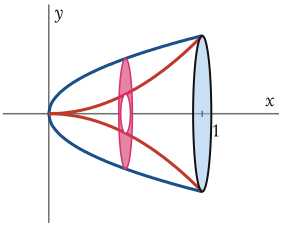
$= \pi(\frac{\pi}{2} - \frac{\sin 2\pi}{4}) - \pi(\frac{\pi}{8} - \frac{\sin(\frac{\pi}{2})}{4})$

$= \pi(\frac{\pi}{2} - 0) - \pi(\frac{\pi}{8} - \frac{1}{4})$

$= \frac{3\pi^2}{8} + \frac{\pi}{4}$ unit^3

EXAMPLE 12 Find the volume of revolution generated by revolving the region between $y = x^2$ and $y = \sqrt{x}$ about the x-axis.





SOLUTION

The two graphs cut each other at x = 0 and x = 1.

The curve $y = \sqrt{x}$ is bounded above.

$V = \pi \int_0^1 [(\sqrt{x})^2 - (x^2)^2]\,dx$

$= \pi \int_0^1 (x - x^4)\,dx$

$= \pi[\frac{x^2}{2} - \frac{x^5}{5}]_0^1$

$= \pi(\frac{1}{2} - \frac{1}{5})$

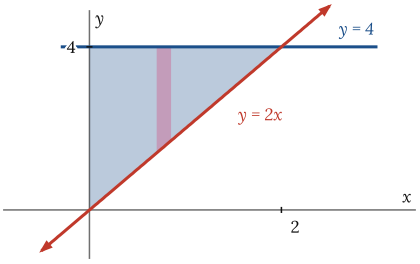
$= \frac{3\pi}{10}$ **unit^3**

EXERCISE 11.3

1. Find the volume of solid by using the slicing method in Exercise 1–3.The base of the solid is the region bounded by the graphs of y = 2x, y = 4 and x = 0. The cross-sections perpendicular to the x-axis are

a. rectangles of height 8.

b. rectangles of perimeter 10.



SOLUTION

The base: y = 2x meets y = 4 when 2x = 4, so x = 2. The region is bounded by x = 0 to x = 2, with vertical strips of length 4 − 2x.

(a) Rectangles of height 8: cross-section area at x is

$A(x) = 8(4 - 2x)$

$= 32 - 16x$

$V = \int_0^2 (32 - 16x)\,dx$

$= [32x - 8x^2]_0^2$

$= 32(2) - 8(2)^2 - 0$

$= 64 - 32$

$= 32$

Volume = 32 unit^3.

(b) Rectangles of perimeter 10: one side is 4 − 2x, so the other side b satisfies (4 − 2x) + b = 5 (half the perimeter), giving b = 1 + 2x.

$A(x) = (1 + 2x)(4 - 2x)$

$= 4 - 2x + 8x - 4x^2$

$= 4 + 6x - 4x^2$

$V = \int_0^2 (4 + 6x - 4x^2)\,dx$

$= [4x + 3x^2 - \frac{4}{3}x^3]_0^2$

$= 8 + 12 - \frac{32}{3}$

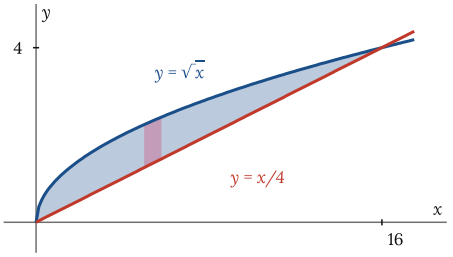
$= \frac{28}{3}$

Volume = $\frac{28}{3}$ unit^3.

2. The base of the solid is the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{4}$. The cross-sections perpendicular to the x-axis are

a. isosceles triangles of height 4.

b. semicircles with diameters running across the base of the solid.



SOLUTION

Intersection: $\sqrt{x} = \frac{x}{4} \rightarrow 4\sqrt{x} = x \rightarrow 16x = x^2 \rightarrow x(x - 16) = 0$, so x = 0, 16. $\sqrt{x}$ is above $\frac{x}{4}$ on [0, 16].

(a) Isosceles triangles of height 4: base = $\sqrt{x} - \frac{x}{4}$.

$A(x) = \frac{1}{2}(\sqrt{x} - \frac{x}{4})(4)$

$= 2(\sqrt{x} - \frac{x}{4})$

$V = \int_0^{16} 2(\sqrt{x} - \frac{x}{4})\,dx$

$= 2[\frac{2}{3}x^{3/2} - \frac{x^2}{8}]_0^{16}$

$= 2[\frac{2}{3}(16)^{3/2} - \frac{(16)^2}{8}]$

$= 2[\frac{128}{3} - 32]$

$= 2 \times \frac{32}{3}$

$= \frac{64}{3}$

Volume = $\frac{64}{3}$ unit^3.

(b) Semicircles with diameter running across the base: the diameter is $\sqrt{x} - \frac{x}{4}$, so the radius is r = $\frac{1}{2}(\sqrt{x} - \frac{x}{4})$.

$A(x) = \frac{1}{2}\pi r^2$

$= \frac{\pi}{2}[\frac{1}{2}(\sqrt{x} - \frac{x}{4})]^2$

$= \frac{\pi}{8}(x - \frac{1}{2}x^{3/2} + \frac{1}{16}x^2)$

$V = \frac{\pi}{8}\int_0^{16} (x - \frac{1}{2}x^{3/2} + \frac{1}{16}x^2)\,dx$

$= \frac{\pi}{8}[\frac{x^2}{2} - \frac{1}{5}x^{5/2} + \frac{1}{48}x^3]_0^{16}$

$= \frac{\pi}{8}(128 - \frac{1024}{5} + \frac{256}{3})$

$= \frac{\pi}{8} \times \frac{128}{15}$

$= \frac{16\pi}{15}$

Volume = $\frac{16\pi}{15}$ unit^3.

3. The solid lies between the planes perpendicular to the x-axis at x = −1 and x = 1. The cross-sections perpendicular to the x-axis are circular disks whose diameters run from $y = -x^2 + 4$ to $y = x^2 + 2$.

SOLUTION

Let $f(x) = -x^2 + 4$, $g(x) = x^2 + 2$. They meet when $-x^2 + 4 = x^2 + 2$, so x = ±1. At x = 0: f(0) = 4 > g(0) = 2, so f is on top.

diameter $d = f(x) - g(x)$

$= (-x^2 + 4) - (x^2 + 2)$

$= -2x^2 + 2$

radius $= \frac{-2x^2 + 2}{2}$

$= 1 - x^2$

$A(x) = \pi r^2$

$= \pi(1 - x^2)^2$

$V = \int_{-1}^{1} A(x)\,dx$

$= \pi\int_{-1}^{1} (1 - 2x^2 + x^4)\,dx$

$= \pi[x - \frac{2x^3}{3} + \frac{x^5}{5}]_{-1}^{1}$

$= \pi[(1 - \frac{2}{3} + \frac{1}{5}) - (-1 + \frac{2}{3} - \frac{1}{5})]$

$= \pi \cdot \frac{16}{15}$

Volume = $\frac{16\pi}{15}$ unit^3.

4. By using the disc method, find the volume of revolution formed when the curve y = f(x) for a ≤ x ≤ b is rotated through 360° about the x-axis.

a. $f(x) = 3x + 2$; $a = \frac{1}{2}$, b = 4.

b. $f(x) = \sqrt{\cos x}$; a = 0, $b = \frac{\pi}{2}$.

c. $f(x) = \frac{1}{x - 1}$; a = 3, b = 5.

d. $f(x) = \sin x \cos x$; a = 0, $b = \frac{\pi}{2}$.

e. $f(x) = \sqrt{x}^4\sqrt{1 + x^2}$; a = 2, b = 3.

SOLUTION

Disc method: $V = \pi\int_a^b [f(x)]^2\,dx$.

(a) $f(x) = 3x + 2$, $a = \frac{1}{2}$, b = 4

$V = \pi\int_{1/2}^{4} (3x + 2)^2\,dx$

$= \pi\int_{1/2}^{4} (9x^2 + 12x + 4)\,dx$

$= \pi[3x^3 + 6x^2 + 4x]_{1/2}^{4}$

$= \pi[3(4)^3 + 6(4)^2 + 4(4) - 3(\frac{1}{2})^3 - 6(\frac{1}{2})^2 - 4(\frac{1}{2})]$

$= \pi[304 - \frac{31}{8}]$

$= \frac{2401\pi}{8}$

V = $\frac{2401\pi}{8}$ unit^3

(b) $f(x) = \sqrt{\cos x}$, a = 0, $b = \frac{\pi}{2}$

$V = \pi\int_0^{\pi/2} (\sqrt{\cos x})^2\,dx$

$= \pi\int_0^{\pi/2} \cos x\,dx$

$= \pi[\sin x]_0^{\pi/2}$

$= \pi(\sin \frac{\pi}{2} - \sin 0)$

$= \pi(1 - 0)$

$= \pi$

V = π unit^3

(c) $f(x) = \frac{1}{x - 1}$, a = 3, b = 5

$V = \pi\int_3^5 (\frac{1}{x - 1})^2\,dx$

$= \pi\int_3^5 \frac{1}{(x - 1)^2}\,dx$

$= \pi[-\frac{1}{x - 1}]_3^5$

$= \pi(-\frac{1}{4} + \frac{1}{2})$

$= \frac{\pi}{4}$

V = $\frac{\pi}{4}$ unit^3

(d) $f(x) = \sin x \cos x$, a = 0, $b = \frac{\pi}{2}$. Use $\sin^2 x \cos^2 x = \frac{1 - \cos 4x}{8}$.

$V = \pi\int_0^{\pi/2} (\sin x \cos x)^2\,dx$

$= \pi\int_0^{\pi/2} \sin^2 x \cos^2 x\,dx$

$= \frac{\pi}{8}\int_0^{\pi/2} (1 - \cos 4x)\,dx$

$= \frac{\pi}{8}[x - \frac{1}{4}\sin 4x]_0^{\pi/2}$

$= \frac{\pi}{8}[(\frac{\pi}{2} - \frac{1}{4}\sin 2\pi) - (0 - \frac{1}{4}\sin 0)]$

$= \frac{\pi}{8} \cdot \frac{\pi}{2}$

$= \frac{\pi^2}{16}$

V = $\frac{\pi^2}{16}$ unit^3

(e) $f(x) = \sqrt{x}\sqrt[4]{1+x^2}$, $a = 2$, $b = 3$

$V = \pi\int_2^3 (\sqrt{x}\sqrt[4]{1+x^2})^2\,dx$

$= \pi\int_2^3 x\sqrt{1+x^2}\,dx$

Let $u = 1 + x^2$, $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$.

$\int x\sqrt{1+x^2}\,dx = \frac{1}{2}\int \sqrt{u}\,du$

$= \frac{1}{2}\cdot\frac{2}{3}u^{3/2}$

$= \frac{1}{3}(1+x^2)^{3/2}$

$V = \pi[\frac{1}{3}(1+x^2)^{3/2}]_2^3$

$= \frac{\pi}{3}[(1+3^2)^{3/2} - (1+2^2)^{3/2}]$

$= \frac{\pi}{3}[(10)^{3/2} - (5)^{3/2}]$

$= \frac{\pi}{3}(10\sqrt{10} - 5\sqrt{5}) \approx \frac{20.44\pi}{3}$

$V = \frac{\pi}{3}(10\sqrt{10} - 5\sqrt{5})$ unit^3 (≈ 21.41 unit^3)

5. By using the washer method, find the volume of revolution generated by revolving the regions between $y = f(x)$ and $y = g(x)$ for $a \le x \le b$ about the x-axis.

a. $f(x) = x^2$, $g(x) = 2x$; $a = 0$, $b = 2$

b. $f(x) = \tan x$, $g(x) = \frac{x}{2}$; $a = 0$, $b = \frac{\pi}{4}$

c. $f(x) = 2\sqrt{x}$, $g(x) = 2$; $a = 0$, $b = 1$.

SOLUTION

Washer method: $V = \pi\int_a^b ([\text{top}]^2 - [\text{bottom}]^2)\,dx$.

(a) $f(x) = x^2$, $g(x) = 2x$, $a = 0$, $b = 2$. At $x = 1$: $f(1) = 1$, $g(1) = 2$, so g is on top.

$V = \pi\int_0^2 [(2x)^2 - (x^2)^2]\,dx$

$= \pi\int_0^2 (4x^2 - x^4)\,dx$

$= \pi[\frac{4}{3}x^3 - \frac{1}{5}x^5]_0^2$

$= \pi[\frac{32}{3} - \frac{32}{5}]$

$= \frac{64\pi}{15}$

$V = \frac{64\pi}{15}$ unit^3

(b) $f(x) = \tan x$, $g(x) = \frac{x}{2}$, $a = 0$, $b = \frac{\pi}{4}$. ($\tan x \ge \frac{x}{2}$ on this interval.)

$V = \pi\int_0^{\pi/4} (\tan^2 x - (\frac{x}{2})^2)\,dx$

$= \pi\int_0^{\pi/4} (\sec^2 x - 1 - \frac{x^2}{4})\,dx$

$= \pi[\tan x - x - \frac{x^3}{12}]_0^{\pi/4}$

$= \pi[\tan\frac{\pi}{4} - \frac{\pi}{4} - \frac{1}{12}\cdot\frac{\pi^3}{64}]$

$= \pi(1 - \frac{\pi}{4} - \frac{\pi^3}{768})$

$V = \pi(1 - \frac{\pi}{4} - \frac{\pi^3}{768}) = (\pi - \frac{\pi^2}{4} - \frac{\pi^4}{768})$ unit^3

(c) $f(x) = 2\sqrt{x}$, $g(x) = 2$, $a = 0$, $b = 1$. On $[0, 1]$, $g(x) = 2$ is above $f(x) = 2\sqrt{x}$.

$V = \pi\int_0^1 [2^2 - (2\sqrt{x})^2]\,dx$

$= \pi\int_0^1 (4 - 4x)\,dx$

$= \pi[4x - 2x^2]_0^1$

$= \pi(4 - 2)$

$= 2\pi$

$V = 2\pi$ unit^3